



RA8876

Character/Graphic TFT LCD Controller

Datasheet

June 5, 2025

RAiO Technology Inc.

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Revise History		
Version	Date	Description
1.0	October 07, 2015	Preliminary Version
1.1	October 21, 2015	Modify Section 2-5
	November 04, 2015	Remove POR feature
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1. Introduction

This is the Hardware Functional Specification for the RA8876 TFT LCD Controller. RA8876 supports CMOS type interface (MIPI DPI-2). Including in this document are system block diagrams, Pin information, AC/DC characteristics, each block's function description, detail register descriptions, and power mode control.

1.1 Overview Description

The RA8876 is a low power color LCD Controller with support for up to 512M-bits external SDRAM memory. The RA8876 supports an 8/16-bit asynchronous parallel host bus while providing high performance bandwidth into the external display memory allowing for fast screen updates. The RA8876 also provides support for multiple display buffers, Picture-in-Picture, Opacity control, and display rotation/mirror ... etc.

1.2 System Diagram & Chip Diagram

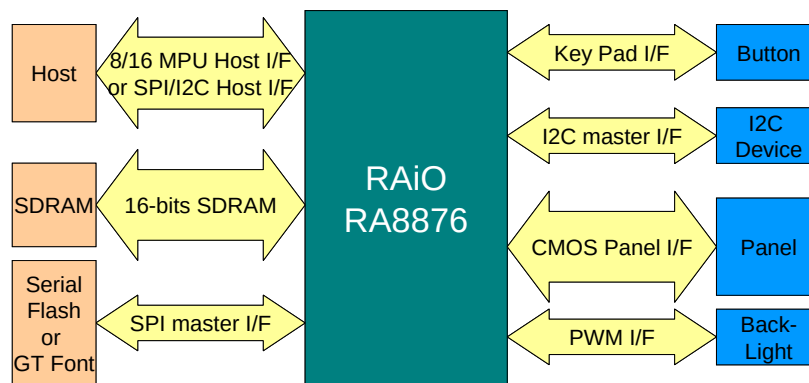


Figure 1-1 : System Diagram

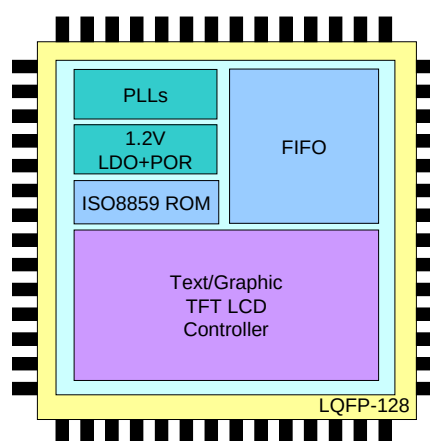


Figure 1-2 : Chip Diagram

2. Features

2.1 Frame Buffer

- Supported SDRAM density: 16Mb, 32Mb, 64Mb, 128Mb, 256Mb or 512Mb
- Supported SDRAM configuration: x16
- 16-bit SDRAM bus, maximum frame buffer: 256MB/512MB

2.2 Host Interface

- Support 8080/6800 8/16-bit asynchronous parallel bus interface (MIPI DBI Type A)
 - Provide xwait event to extend MPU cycle
- Support serial host Interface. Ex. IIC, 3/4-wire SPI
- Mirror and rotation functions are available for image data writes.

2.3 Display Input Data Formats

- 1bpp: monochrome data (1-bit/pixel)
- 8bpp: RGB 3:3:2 (1-byte/pixel)
- 16bpp: RGB 5:6:5 (2-byte/pixel)
- 24bpp: RGB 8:8:8 (3-byte/pixel or 4-byte/pixel)
 - Index 2:6 (64 index colors/pixel with opacity attribute)
 - αRGB 4:4:4:4 (4096 colors/pixel with opacity attribute)

2.4 Display Mode

- Configurable digital TFT output: 24-bits TFT output / 18-bits TFT output / 16-bits TFT output

2.5 Support Various Panel Resolution

- Support 16/18/24-bit CMOS interface type panel or MIPI DPI-2
- Support panel's resolution up-to 2048 dots by 2048 dots. (*Note :The real panel resolution is based on the limitations of pixel clock and color depth.)
 - QVGA: 320 x 240 x 16/18/24-bit LCD panel
 - WQVGA: 480 x 272 x 16/18/24-bit LCD panel
 - VGA: 640 x 480 x 16/18/24-bit LCD panel
 - WVGA: 800 x 480 x 16/18/24-bit LCD panel
 - SVGA: 800 x 600 x 16/18/24-bit LCD panel
 - QHD: 960 x 540 x 16/18/24-bit LCD panel
 - WSVGA: 1024 x 600 x 16/18/24-bit LCD panel
 - XGA: 1024 x 768 x 16/18/24-bit LCD panel
 - WXGA: 1280 x 768 x 16/18/24-bit LCD panel
 - WXGA: 1280 x 800 x 16/18/24-bit LCD panel
 - WXGA: 1366 x 768 x 16/18/24-bit LCD panel

2.6 Display Features

- Provide 4 User-defined 32x32 pixels Graphic Cursor
- Display Window

The display window is defined by the size of the LCD display. Complete or partial updates to the display window are done through canvas image's setting. The active window size and start position are specified in 8 pixel resolution (horizontal) and 1 line resolution (vertical). Window coordinates are referenced to top left corner of the display window (even when flip is enabled or rotate text, no host side translation is required).

- Virtual display
Virtual display is available to show an image which is larger than LCD panel size. The image may scroll easily in any direction.
- Picture-in-Picture (PIP) display
Two PIP windows are supported. Enabled PIP windows are always displayed on top of Main window. The PIP windows sizes and start positions are specified in 4 pixel resolution (horizontal) and 1 line resolution (vertical). Image scrolling can be performed by changing the start address of a PIP window. The PIP1 window is always on top of PIP2 window.
- Multi Buffer
Multi buffering allows the main display window to be switched among buffers. The number of buffers depends on the external SDRAM size and the desired size of the write buffers. Multi buffering allows a simple animation display to be performed by switching the buffers.
- Wake-up display
Wake-up display is available to show the display data quickly which data is stored in SDRAM. This feature is used when returning from the Standby mode or Suspend mode.
- Vertical Flip display
Vertical Flip display functions are available for image data reads. PIP window will be disabled if flip display function enable.
- Color Bar Display
It could display color bar on panel and need not SDRAM. Default resolution is 640 dots by 480 dots.

2.7 Initial Display

- Embed a tiny processor and use to show display data which stored in the serial flash and need not external MPU participate. It will auto execute after power-on, until program execute complete then handover control rights to external MPU. It supports 12 instructions. They are:
 - EXIT: Exit instruction (00h/FFh) -- one byte instruction
 - NOP: NOP instruction (AAh) -- one byte instruction
 - EN4B: Enter 4-Byte mode instruction (B7h) -- one byte instruction
 - EX4B: Exit 4-Byte mode instruction (E9h) -- one byte instruction
 - STSR: Status read instruction (10h) -- two bytes instruction
 - CMDW: Command write instruction (11h) -- two bytes instruction
 - DATR: Data read instruction (12h) -- two bytes instruction
 - DATW: Data write instruction (13h) -- two bytes instruction
 - REPT: Load repeat counter instruction (20h) -- two bytes instruction
 - ATTR: Fetch Attribute instruction (30h) -- two bytes instruction
 - JUMP: Jump instruction (80h) -- five bytes instruction
 - DJNZ: Decrement & Jump instruction (81h) -- five bytes instruction

2.8 Block Transfer Engine (BTE)

- 2D BitBLT Engine
- Copy with ROP & color expansion
- Solid fill & Pattern fill
 - Provide User-defined Patterns with 8x8 pixels or 16x16 pixels
- Opacity (Alpha-Blend) control
It allows two images to be blended to create a new image which can **then** be displayed using a PIP window. The processing speed of Alpha-blend function varies depending on the image size. Optionally, a single input image can be processed.
 - Chroma-keying function: Mixes images with applying the specified RGB color according to transparency rate.
 - Window Alpha-blending function: Mixes two images according to transparency rate in the specified region (fade-in and fade-out functions are available).
 - Dot Alpha-blending function: Mixes images according to transparency rate when the target is a graphics image in the RGB format.

2.9 Geometric Drawing Engine

- Draw dot, Line, Curve, Circle, Ellipse, Triangle, Square & Circular Square

2.10 SPI Master Interface

2.10.1 Text Features

- Embedded 8x16,12x24,16x32 Character Sets of ISO/IEC 8859-1/2/4/5.
- Supporting Genitop Inc. UNICODE/BIG5/GB etc. Serial Character ROM with 16x16/24x24/32x32 dots Font Size. The supporting product numbers are GT21L16T1W, GT30L16U2W, GT30L24T3Y, GT30L24M1Z, and GT30L32S4W, GT20L24F6Y, GT21L24S1W.
- User-defined Characters support half size (8x16/12x24/16x32) & full size
- Programmable Text Cursor for Writing with Character
- Character Enlargement Function X1, X2, X3, X4 for Horizontal/Vertical Direction
- Support Character 90 degree Rotation

2.10.2 DMA function

- Support direct data transfer from external serial flash to frame buffer

2.10.3 General SPI master

- Compatible with Motorola's SPI specifications
- 16 bytes entries deep read FIFO
- 16 bytes entries deep write FIFO
- Interrupt generation after Tx FIFO empty and SPI Tx/Rx engine idle

2.11 IIC Interface

- IIC master interface
 - For the expand I/O device, external touch screen controller for panel control
 - Support Standard mode (100kbps) and Fast mode (400kbps)

2.12 PWM Timer

- Two 16-bit timers
- One 8-bit pre-scalars & One 4-bit divider
- Programmable duty control of output waveform (PWM)
- Auto reload mode or one-shot pulse mode
- Dead-zone generator

2.13 Key-scan Interface

- Support up-to 5x5 key matrix (share with the GPIO pin)
- Programmable scan period
- Support long Key & repeat key
- Support up to 2 keys are pressed simultaneously
- **Note:** Restricted support 3-keys are pressed simultaneously (3-keys cannot form 90°)
- Support Key-Scan Wakeup function

2.14 Power Saving

- Support 3 kind of power saving mode
 - Standby mode, Suspend mode & Sleep mode
- It may wakeup by host, key & external event

2.15 Clock Source

- Embedded programmable PLL for system core clock, LCD panel scan clock and the SDRAM clock
- Single crystal clock input: (XI/XO: 10-15MHz)
- Internal system clock (Maximum 120MHz)
- SDRAM clock (Maximum 166MHz)
- LCD panel scan clock (Maximum 100MHz)

2.16 Reset

- Accept external hardware reset to synchronize with system
- Software command reset

2.17 Power Supply

- I/O voltage: 3.3V +/- 0.3V
- Embedded 1.2V LDO for core power

2.18 Package

- LQFP-128
- Operation temperature: -40°C ~ 85°C

3. Symbol and Package

3.1 RA8876 Symbol & Pin Assignment

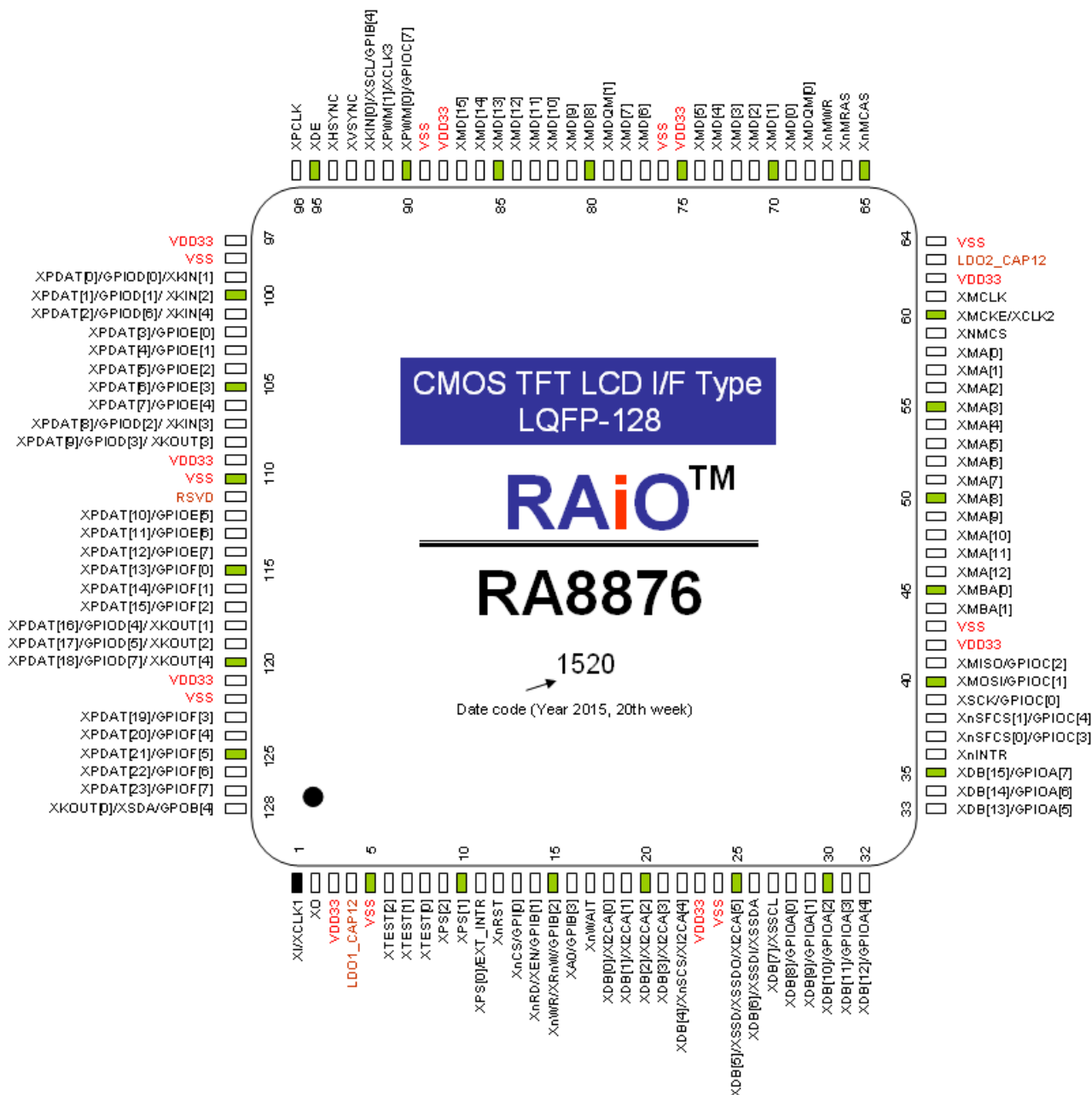


Figure 3-1

4. Signal Description

4.1 Parallel Host Interface (25 signals)

Pin Name	Dir/Drv.	Pin Description
XDB[15:0]	IO (8mA)	Data Bus These are data bus for data transfer between parallel host and RA8876. XDB[15:8] will become GPIO (GPIO-A[7:0]) if parallel host 8080/6800 16-bits data bus mode doesn't set. XDB[7:0] are multiplex with serial host signals if serial host mode set. Please refer to serial host interface section.
XA0	I	Command / Data Select Input The pin is used to select command/data cycle. XA0 = 0, status read / command write cycle is selected. XA0 = 1, data read / Write cycle is selected.
XnCS	I	Chip Select Input Low active chip select pin. If host I/F set as serial host mode then this pin can be read from GPI-B0. With internal pull-high with resistor.
XnRD (XEN)	I	Enable/Read Enable When MPU interface (I/F) is 8080 series, this pin is used as XnRD signal (Data Read) , active low. When MPU I/F is 6800 series, this pin is used as XEN signal (Enable), active high. If host I/F set as serial host mode then this pin can be read from GPI-B1. With internal pull-high with resistor.
XnWR (XRnW)	I	Write/Read-Write When MPU I/F is 8080 series, this pin is used as XnWR signal (data write) , active low. When MPU I/F is 6800 series, this pin is used as XRnW signal (data read/write control). Active high for read and active low for write. If host I/F set as serial host mode then this pin can be read from GPI-B2. With internal pull-high with resistor.
XnINTR	O (8mA)	Interrupt Signal Output The interrupt output for host to indicate the status.
XnWAIT	O (8mA)	Wait Signal Output When high, it indicates that the RA8876 is ready to transfer data. When low, then microprocessor is in wait state.
XPS[2:0]	I	Parallel /Serial Host I/F Select 00X: (parallel host) 8080 interface with 8/16-bits data bus 01X: (parallel host) 6800 interface with 8/16-bits data bus 100: (serial host) 3-Wire SPI 101: (serial host) 4-Wire SPI 11x: (serial host) IIC Note: If host I/F set as parallel host mode, then XPS[0] pin is external interrupt pin.

4.2 Serial Host Interface (Multiplex with Parallel Host Interface)

Pin Name	Dir/Drv.	Pin Description
XSSCL (XDB[7])	I	SPI or IIC Clock XSSCL, 3-wire, 4-wire Serial or IIC I/F clock.
XSSDI XSSDA (XDB[6])	I	IIC data /4-wire SPI Data Input 3-wire SPI I/F: NC, please connect it to GND. 4-wire SPI I/F: XSSDI, Data input for serial I/F. IIC I/F: XSSDA, Bi-direction data for serial I/F
XSSD XSSDO (XDB[5])	IO	3-wire SPI Data /4-wire SPI Data Output/IIC Slave Address Select 3-wire SPI I/F: XSSD, Bi-direction data for serial I/F 4-wire SPI I/F: XSSDO, Data output for serial I/F. IIC I/F: XIICA[5], IIC device address bit [5]
XnSCS (XDB[4])	I	SPI Chip Select/IIC Slave Address Select XnSCS, Chip select pin for 3-wire or 4-wire serial I/F. IIC I/F : XIICA[4], IIC device address bit [4].
XIICA[3:0] (XDB[3:0])	I	IIC I/F: IIC Slave Address Select. XIICA[3:0], 3 4-wire SPI I/F: NC, please connect it to GND. IIC I/F : IIC device address bit [3:0]

4.3 SDR SDRAM Interface (39 signals)

Pin Name	Dir/Drv.	Pin Description
XMCKE (XCLK2)	IO (8mA)	Clock enable / Clock 2 input (memory clock) When XTEST[0] set low, this pin is SDR memory clock enable When XTEST[0] set high, this pin is external clock 2 input for SDR access.
XMCLK	IO (8mA)	SDR memory Clock out It derives from MPLL or XCLK2
XnMCS	O (8mA)	Chip select
XnMRAS	O (8mA)	Command outputs: XnMRAS, XnMCAS and XnMWR (along with XnMCS) define the command being entered
XnMCAS	O (8mA)	Command outputs
XnMWR	O (8mA)	Command outputs
XMBA[1:0]	O (8mA)	Bank address
XMA[12:0]	O (8mA)	Address
XMD[15:0]	I/O (8mA)	Data bus.
XMDQM[1:0]	O (8mA)	Input/Output mask

4.4 Serial Flash or SPI master Interface (5 signals)

Pin Name	Dir/Drv.	Pin Description
XnSFCS0	IO (8mA)	Chip Select 0 for External Serial Flash/ROM or SPI device SPI Chip select pin #0 for serial Flash/ROM or SPI device. * If SPI master I/F is disabled then it can be programmed as GPIO (GPIO-C3); default is GPIO-C3 input function.
XnSFCS1	IO (8mA)	Chip Select 1 for External Serial Flash/ROM or SPI device SPI Chip select pin #1 for serial Flash/ROM or SPI device. * If SPI master I/F is disabled then it can be programmed as GPIO (GPIO-C4); default is GPIO-C4 input function. *auto pull-high in reset period if xtest[2:1] is not equal to 01b.
XSCK	IO (8mA)	SPI Serial Clock Serial clock output for serial Flash/ROM or SPI device. * If SPI master I/F is disabled then it can be programmed as GPIO (GPIO-C0); default is GPIO-C0 input function.
XMOSI (XSIO0)	IO (8mA)	Master Output Slave Input Single mode: Data input of serial Flash/ROM or SPI device. For RA8876, it is output. Dual mode: The signal is used as bi-direction data #0(SIO0). Only valid in serial flash DMA mode. * If SPI master I/F is disabled then it can be programmed as GPIO (GPIO-C1); default is GPIO-C1 input function.
XMISO (XSIO1)	IO (8mA)	Master Input Slave Output Single mode: Data output of serial Flash/ROM or SPI device. For RA8876, it is input. Dual mode: The signal is used as bi-direction data #1(SIO1). Only valid in serial flash DMA mode. * If SPI master I/F is disabled then it can be programmed as GPIO (GPIO-C2); default is GPIO-C2 input function.

4.5 PWM Interface (2 signals)

Pin Name	Dir/Drv.	Pin Description
XPWM0	IO (8mA)	PWM signal output 1 / Initial Display Enable Pull-high this pin will enable Initial Display function. This pin has internal pull-down in reset period to disable Initial Display function by default. i.e. after reset complete, internal pull-down resistor will be disabled. XPWM 0 output mode is decided by configuration register. If PWM function disabled then it can be programmed as GPIO (GPIO-C7), default is GPIO-C7 input function, or output core clock.
XPWM1 (XCLK3)	IO (8mA)	PWM signal output 2 / Clock 3 input (panel scan clock) When XTEST[0] set low: XPWM1 set as output mode & output function is decided by configuration register. It may normal XPWM1 function, oscillator clock output or error flag for Scan bandwidth insufficient or Memory access out of range. (or Iso clock output) When XTEST[0] set high: XPWM1 pin is external panel scan clock input

4.6 KEYSKAN Interface (10 signals)

Pin Name	Dir/Drv.	Pin Description
XKIN[4:0]	I	Keypad Data Line or GPIs (General Purpose Input) Keypad data inputs (Default), with internal pull-up resistor. XKIN[0] also has IIC master's XSCL function. In RA8876, XKIN [4:1] are share with XPDAT & GPIO-D.
XKOUT[4:0]	O (2mA)	Keypad Strobe Line or GPOs (General Purpose Output) Keypad matrix strobe lines outputs with open-drain. (Default). XKOUT[0] also has IIC master's XSDA function. In RA8876, XKOUT [4:1] are share with XPDAT & GPIO-D.

4.7 LCD Panel Digital Interface (28 signals)

Pin Name	Dir/Drv.	Pin Description																																																																																																																																	
XPCLK	O (8mA)	Panel scan Clock Generic TFT interface signal for panel scan clock. It derives from SPLL.																																																																																																																																	
XVSYNC	O (4mA)	VSYNC Pulse Generic TFT interface signal for vertical synchronous pulse.																																																																																																																																	
XHSYNC	O (4mA)	HSYNC Pulse Generic TFT interface signal for horizontal synchronous pulse.																																																																																																																																	
XDE	O (4mA)	Data Enable Generic TFT interface signal for data valid or data enable.																																																																																																																																	
XPDAT [23:0]	IO (4mA)	LCD Panel Data Bus TFT LCD data bus output for source driver. RA8876 supports 65K/262K/16.7M color depth by register setting; user can connect corresponding RGB bus for different setting.																																																																																																																																	
		Pin Name	Digital TFT Interface				TFT output Setting	11b (GPIO)	10b (16-bits)	01b (18-bits)	00b (24-bits)	XPDAT[0]	GPIO-D0/ XKIN[1]			B0	XPDAT[1]	GPIO-D1/ XKIN[2]			B1	XPDAT[2]	GPIO-D6/ XKIN[4]		B0	B2	XPDAT[3]	GPIO-E0	B0	B1	B3	XPDAT[4]	GPIO-E1	B1	B2	B4	XPDAT[5]	GPIO-E2	B2	B3	B5	XPDAT[6]	GPIO-E3	B3	B4	B6	XPDAT[7]	GPIO-E4	B4	B5	B7	XPDAT[8]	GPIO-D2/ XKIN[3]			G0	XPDAT[9]	GPIO-D3/ XKOUT[3]			G1	XPDAT[10]	GPIO-E5	G0	G0	G2	XPDAT[11]	GPIO-E6	G1	G1	G3	XPDAT[12]	GPIO-E7	G2	G2	G4	XPDAT[13]	GPIO-F0	G3	G3	G5	XPDAT[14]	GPIO-F1	G4	G4	G6	XPDAT[15]	GPIO-F2	G5	G5	G7	XPDAT[16]	GPIO-D4/ XKOUT[1]			R0	XPDAT[17]	GPIO-D5/ XKOUT[2]			R1	XPDAT[18]	GPIO-D7/ XKOUT[4]		R0	R2	XPDAT[19]	GPIO-F3	R0	R1	R3	XPDAT[20]	GPIO-F4	R1	R2	R4	XPDAT[21]	GPIO-F5	R2	R3	R5	XPDAT[22]	GPIO-F6	R3	R4	R6	XPDAT[23]	GPIO-F7	R4	R5	R7
		Pin Name	Digital TFT Interface																																																																																																																																
		TFT output Setting	11b (GPIO)	10b (16-bits)	01b (18-bits)	00b (24-bits)																																																																																																																													
		XPDAT[0]	GPIO-D0/ XKIN[1]			B0																																																																																																																													
		XPDAT[1]	GPIO-D1/ XKIN[2]			B1																																																																																																																													
		XPDAT[2]	GPIO-D6/ XKIN[4]		B0	B2																																																																																																																													
		XPDAT[3]	GPIO-E0	B0	B1	B3																																																																																																																													
		XPDAT[4]	GPIO-E1	B1	B2	B4																																																																																																																													
		XPDAT[5]	GPIO-E2	B2	B3	B5																																																																																																																													
		XPDAT[6]	GPIO-E3	B3	B4	B6																																																																																																																													
		XPDAT[7]	GPIO-E4	B4	B5	B7																																																																																																																													
		XPDAT[8]	GPIO-D2/ XKIN[3]			G0																																																																																																																													
		XPDAT[9]	GPIO-D3/ XKOUT[3]			G1																																																																																																																													
		XPDAT[10]	GPIO-E5	G0	G0	G2																																																																																																																													
		XPDAT[11]	GPIO-E6	G1	G1	G3																																																																																																																													
		XPDAT[12]	GPIO-E7	G2	G2	G4																																																																																																																													
		XPDAT[13]	GPIO-F0	G3	G3	G5																																																																																																																													
		XPDAT[14]	GPIO-F1	G4	G4	G6																																																																																																																													
		XPDAT[15]	GPIO-F2	G5	G5	G7																																																																																																																													
		XPDAT[16]	GPIO-D4/ XKOUT[1]			R0																																																																																																																													
		XPDAT[17]	GPIO-D5/ XKOUT[2]			R1																																																																																																																													
		XPDAT[18]	GPIO-D7/ XKOUT[4]		R0	R2																																																																																																																													
		XPDAT[19]	GPIO-F3	R0	R1	R3																																																																																																																													
		XPDAT[20]	GPIO-F4	R1	R2	R4																																																																																																																													
		XPDAT[21]	GPIO-F5	R2	R3	R5																																																																																																																													
		XPDAT[22]	GPIO-F6	R3	R4	R6																																																																																																																													
		XPDAT[23]	GPIO-F7	R4	R5	R7																																																																																																																													
				*unused pins can be programmed as GPIO-D/E/F(default) or XKIN/XOUT. Default is 18bpp function mode, so XPDAT[17:16/8:9/1:0] are default at GPI mode.																																																																																																																															

4.8 Clock, Reset & Test Mode (6 signals)

Pin Name	Dir/Drv.	Pin Description
XI (XCLK1)	I	Crystal input/Clock 1 input The recommended frequency range of the external crystal must be 10MHz. When the XTEST[0] pin is set to low level, the XI (XCLK1) pin is provided to the internal PLL circuit for use. Therefore, in such application conditions, the XI (XCLK1) pin must be connected to an external crystal to generate the relevant clock signals required by RA8876. On the contrary, when the XTEST[0] pin is set to high level, the XI (XCLK1) pin will be used as the input pin of the external clock.
XO	O	Crystal Output The XO pin is the output pin of the internal PLL circuit. The XO pin should be connected to an external crystal.
XnRST	I/OC	Reset Signal input To avoid noise interfere XnRST signal and cause fake reset behavior, external XnRST level will be admitted only if it keep its signal level at least 256 OSC clocks.
XTEST[0]	I	Clock Test Mode Internal pull down. For chip test function, should be connected to GND for normal operation. 0: Normal mode, Use internal PLL clock. 1: bypass internal PLL clock and instead them with CLK1I, CLK2I & CLK3I.
XTEST[2:1]	I	Chip Test Mode 00: normal mode 01: Force SPI master I/F pin floating (for in-system-programming) 1X: RESERVED

4.9 Power and Ground

Pin Name	Dir/Drv.	Pin Description
LDO1_CAP12 LDO2_CAP12	P	Loading Capacitor for each LDO Connect a 1uF capacitor to ground.
VDD33	P	IO VDD 3.3V IO power input.
VSS	P	GND IO Cell/Core ground signal
RSVD	P	Reserved Suggest to connect a 1uF capacitor to ground.

5. AC/DC Characteristics

5.1 Maximum Absolute Limit

Table 5-1 : Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Supply Voltage Range	V _{DD33}	-0.3 ~ 4.0	V
Input Voltage Range	V _{IN}	-0.3 ~ V _{DD33} +0.3	V
Operation Temperature Range	T _{OPR}	-40 ~ 85	°C
Power Dissipation	P _D	≤300	mW
Storage Temperature	T _{Storage}	-45 ~ 125	°C
Soldering Temperature	T _{Solder}	260	°C

Note :

1. The humidity resistance of the flat package may be reduced if the package is immersed in solder. Use a soldering technique that does not heat stress the package.
2. If the power supply has a high impedance, a large voltage differential can occur between the input and supply voltages. Take appropriate care with the power supply and the layout of the supply lines.

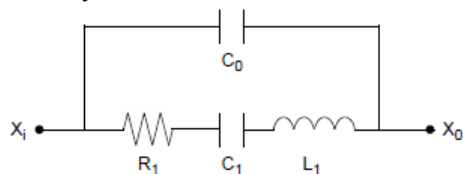
5.2 DC Characteristic

Table 5-2 : DC Characteristic Table

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
System Voltage	V _{DD33}	3.0	3.3	3.6	V	
Loading capacitor	Cap _{Vdd}	1	-	10	uF	
Operation Current	I _{OPR}		60		mA	Note 3
Standby mode	I _{Stby}		30		mA	Note 3
Suspend mode	I _{Susp}		10		mA	Note 3
Sleep Mode	I _{SLP}		7		mA	Note 3
Power ramp up time	T _{ramp}	3.5		35	ms	V _{DD33} Ramp up to 3.3 V
OSC/PLL						
Oscillator Clock	F _{OSC}	10		15	MHz	V _{DD33} = 3.3 V, Note 1
Clock Period Jitter	T _{jitter}	-2.5		2.5	%	
Lock Time	T _{Lock}		1024		OSC	Note 2
MPLL Output Clock (MCLK)	Freq _{mclk}			166	MHz	V _{DD33} = 3.3 V
CPLL Output Clock (CCLK)	Freq _{cclk}			133	MHz	V _{DD33} = 3.3 V Without enable internal ISO-8859 font feature
CPLL Output Clock (CCLK)	Freq _{cclk}			120	MHz	V _{DD33} = 3.3 V When enable internal ISO-8859 font feature
SPLL Output Clock (PCLK)	Freq _{pclk}			100	MHz	V _{DD33} = 3.3 V
Serial MPU I/F						
SPI Input clock	Freq _{xssck}			50	MHz	
Input/Output (CMOS 3-state Output pad with Schmitt Trigger Input, Pull-Up/Down)						
Input High Voltage	V _{IH}	2		3.6	V	
Input Low Voltage	V _{IL}	-0.3		0.8	V	
Output High Voltage	V _{OH}	2.4			V	
Output Low Voltage	V _{OL}			0.4	V	
Pull up resistance	R _{PU}	20		80	Kohm	

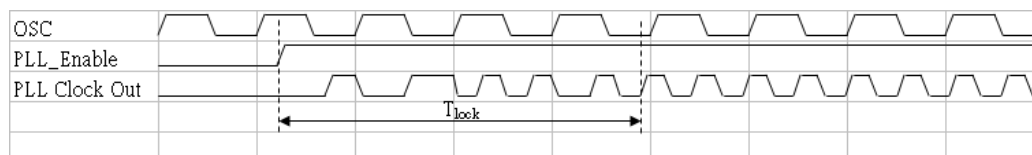
Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Pull down resistance	R_{PD}	20		80	Kohm	
Schmitt trigger low to high threshold	V_{T+}	1.5		2.1	V	
Schmitt trigger high to low threshold	V_{T-}	0.8		1.3	V	
Hysterisis	V_{hys}	200			mV	
Input Leakage Current	I_{leak}	-10		+10	μA	
Rise/fall slew rate	Slew		1.5		V/ns	

Note 1: Parasitics Used in the Crystal Oscillator



Typical: $R_1 = 50\Omega$ (25-100 Ω), $L_1 = 3.4mH$, $C_1 = 13fF$, $C_0 = 2.8pF$

Note 2:



Note 3: Measured on tester with 8 bit MPU interface and without extra load.

6. Clock & Reset

6.1 Clock

This chip embeds 3 PLL macros to handle different functional blocks. For example, CPLL provide clock (CCLK) for MPU I/F, BTE engine, Geometric engine & Font_DMA engine; MPLL provide clock (MCLK) for DRAM operation; SPLL provide clock (SCLK) for LCD panel scan operation.

6.1.1 Clock Scheme

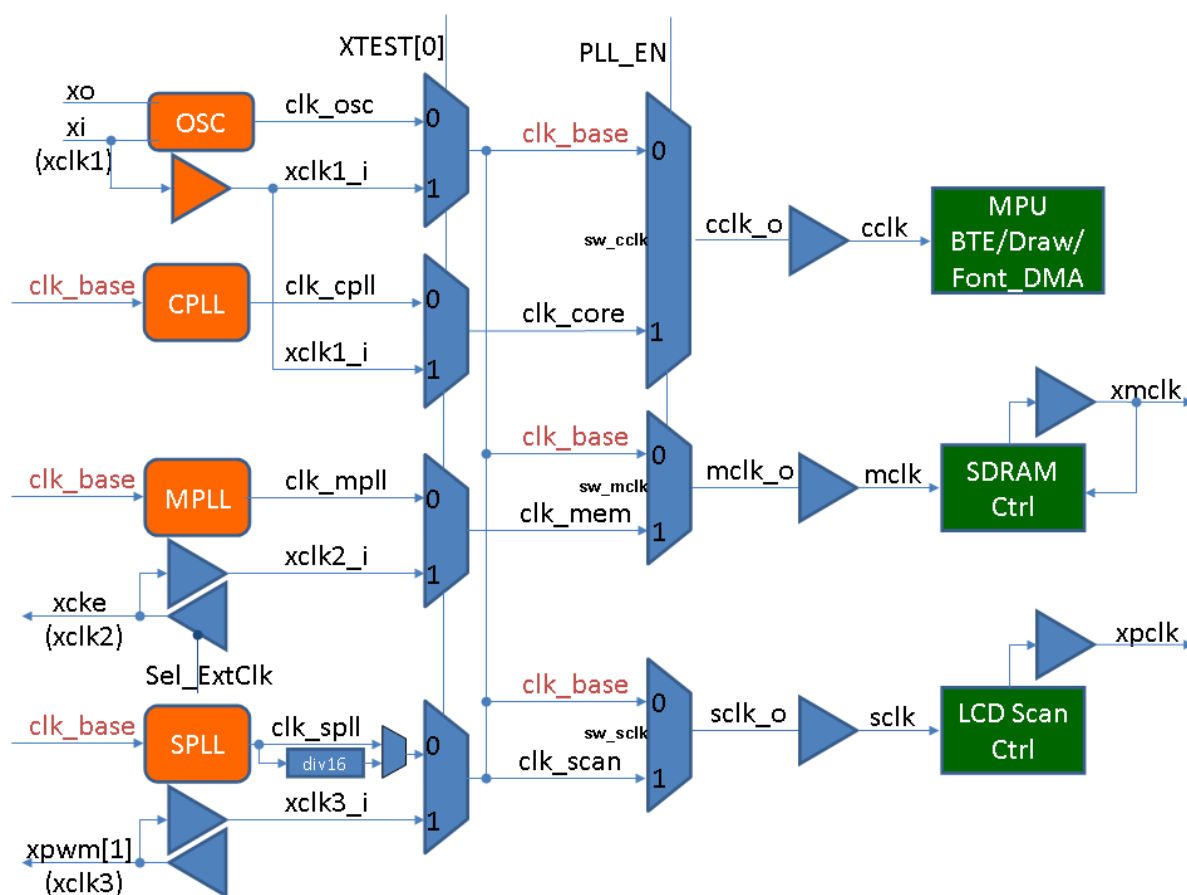


Figure 6-1

XTEST[0] to control major clock sources come from internal PLL or external pins. Set XTEST[0] low will select sources of core clock, memory clock and panel scan clock comes from CPLL, MPLL and SPLL. Set XTEST[0] high will configure sources of core clock, memory clock and panel scan clock comes from primary IO pins, XI(XCLK1), XCKE(XCLK2) and XPWM[1](XCLK3).

Clock frequency must fulfill following criteria:

- Color depth = 16bpp:
 $\text{Freq}_{\text{MCLK}} \geq \text{Freq}_{\text{CCLK}} \geq \text{Freq}_{\text{SCLK}} * 1.5$
- Color depth = 24bpp:
 $\text{Freq}_{\text{MCLK}} \geq \text{Freq}_{\text{CCLK}} \geq \text{Freq}_{\text{SCLK}} * 2$

6.1.2 PLL Setting

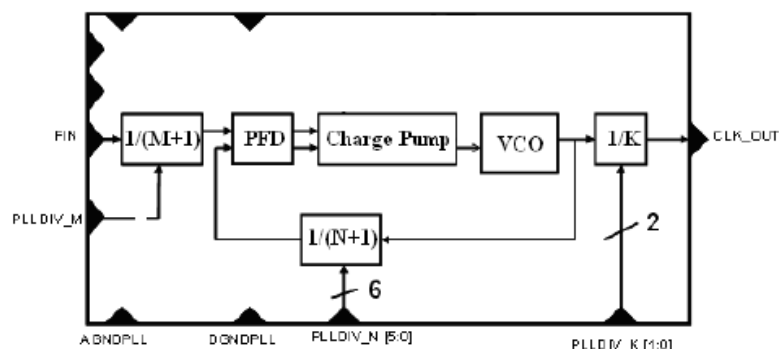


Figure 6-2

To a phase-locked loop frequency synthesizer, the output frequency can be programmed by PLLDIVM, PLLDIVN and PLLDIVK. The formula of output frequency is

$$xCLK = \frac{\left(\frac{Fin}{2^{xPLLDIVM}} \right) \times (xPLLDIVN + 1)}{2^{xPLLDIVK}}$$

Divider range:

- i. PLLDIVM : 0 or 1
- ii. PLLDIVN[5:0] : 1~63 (minimum ≥ 1)
- iii. PLLDIVK[1:0] : 0~3

Note :

1. PLL's parameters can be changed only when PLL disabled.
2. After REG[05h] ~ REG[0Ah] is programmed, a lock time (< 30us) must be kept to guarantee the stability of the PLL output.
3. The input OSC frequency (F_{IN}) must greater & PLLDIVM has following restriction:

$$10MHz \leq Fin \leq 15MHz$$

&

$$10MHz \leq \frac{Fin}{2^{PLLDIVM}} \leq 40MHz$$

4. The internal multiplied clock frequency $F_{VCO} = \frac{Fin}{2^{PLLDIVM}} \times (PLLDIVN + 1)$ must be equal to or greater than 250 MHz and small than 500MHz. i.e,

$$250MHz \leq F_{VCO} \leq 500MHz$$

6.2 Reset

6.2.1 External Reset

This chip has capability to receive external reset event (low active) to synchronize with external system. Removal external reset event will be admitted when it stable at least 256 OSC clocks.

Before start access this chip, MPU should check status register bit [1], operation mode status bit, and make sure it is in "Normal operation state".

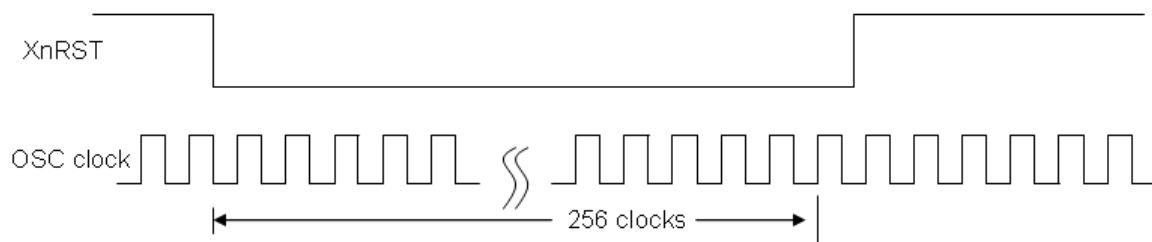


Figure 6-3 : External Reset must be greater than 256 OSC clock

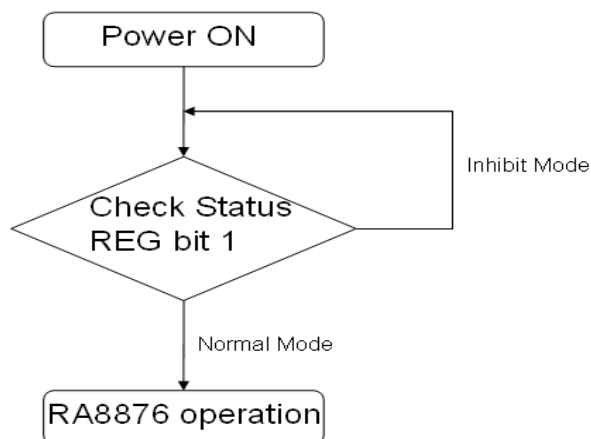


Figure 6-4 : After Power On , User must be check Status REG bit 1

7. Host Interface

7.1 Indirect Interface

The RA8876 supports parallel host interface (ex. 8080/6800 series MPU interface) and serial host interface (ex. IIC, 3-Wire/4-Wire SPI). The type of MPU interface is decided by hardware pin XPS[2:0]. Accessing the RA8876 through the asynchronous host interface is a multiple step process. All Registers and Memory are accessed through the register space. Basically, MIPI DBI-2 type A is similar to parallel host 6800 series MPU interface.

Note --

All Register accesses are 8-bit only, except for the Memory Data Port. Even if the Host interface is 16-bits wide, the LSB (XDB[7:0]) are used for all registers except the Memory Data Port (REG[04h]). For the Memory Data Port (REG[04h]), full 16-bits is used when the host interface type bit (REG[01h], bit[0]) set as 16-bits wide and only low 8-bits is used when it set as 8-bits wide.

Table 7-1 : Parallel /Serial Host I/F Select Pin

XPS[2:0]	Host Mode
00X	(parallel host) 8080 interface with 8/16-bits data bus
01X	(parallel host) 6800 interface with 8/16-bits data bus
100	(serial host) 3-Wire SPI
101	(serial host) 4-Wire SPI
11X	(serial host) IIC

To access a configuration register. First, perform a single “Address Write” to setup the register address. Next, perform a “Data Read/Write” to specify the data to be stored or read from the registers or memory specified in the “Address Write” cycle. Subsequent data read/writes without an Address Write to change the register address, **will not** automatically “auto” increment the register address, but the internal memory address **will** automatically “auto” increment if accessing the Memory Data Port (REG[04h]).

To write display data to a Window Aperture, specify the Window coordinates followed by burst data writes to the Memory Data Port to fill the window. In this sequence, the internal memory addressing is automatic.

7.1.1 Register Write Procedure

1. Perform an address write to setup register address bits 7-0.
2. Perform a data write to update the register's contain.

7.1.2 Register Read Procedure

1. Perform an address write to setup register address bits 7-0.
2. Perform a data read to get the register's value.

7.1.3 Memory Write Procedure

The RA8876 has a special procedure to minimize setup accesses when access image data.

1. Set the active window registers before writing any image data.
2. Perform an register write to Graphic R/W position Register 0, REG[5Fh]).
3. Repeat step 2 until setup all the active window & Graphic R/W position coordinates.
4. Perform an address write to point to Memory Data Port Register (REG[04h])
5. Perform data writes to fill the window. Each write to the Memory Data Port will auto-increment the internal memory address.

7.2 Parallel Host

7.2.1 Parallel Host Interface

Connect 8080 and 6800 series MPU interface, Please refer to the Figure 7-1 and Figure 7-2.

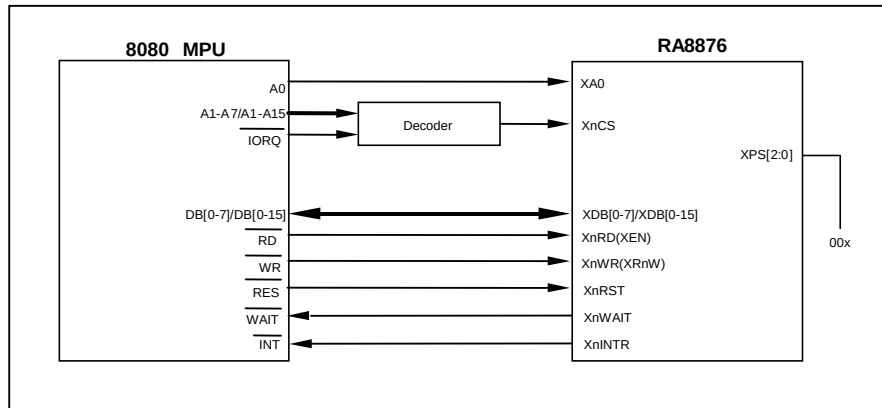


Figure 7-1 : 8080 MPU Interface

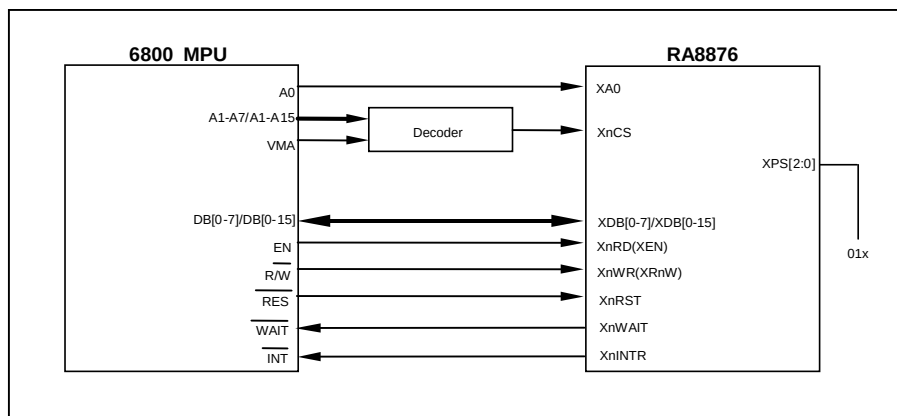


Figure 7-2 : 6800 MPU Interface

7.2.2 Parallel host I/F Protocol

The following timing charts are used to describe the timing specification of the standard 8080 and 6800 interfaces.

8080 – 8/16-bit Interface

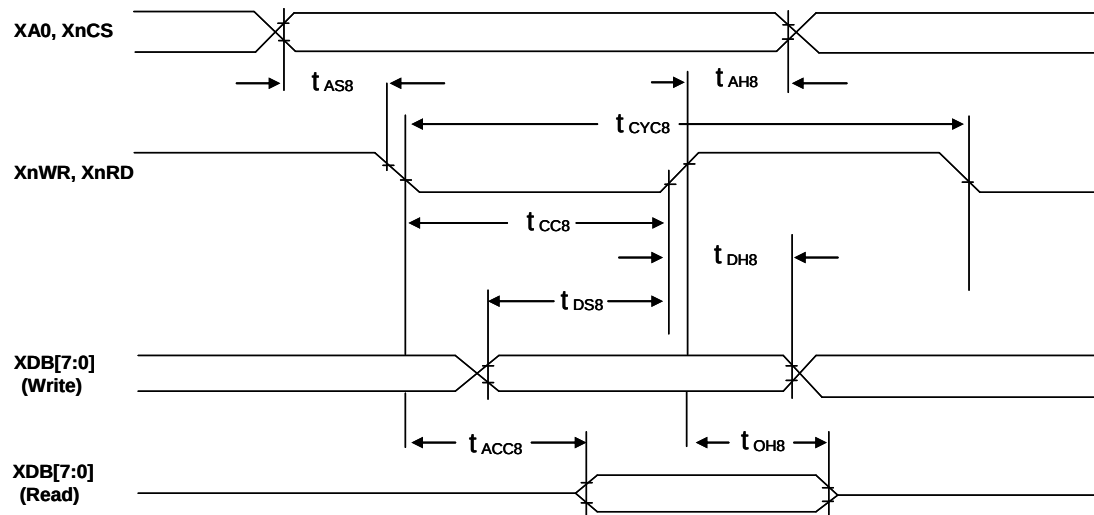
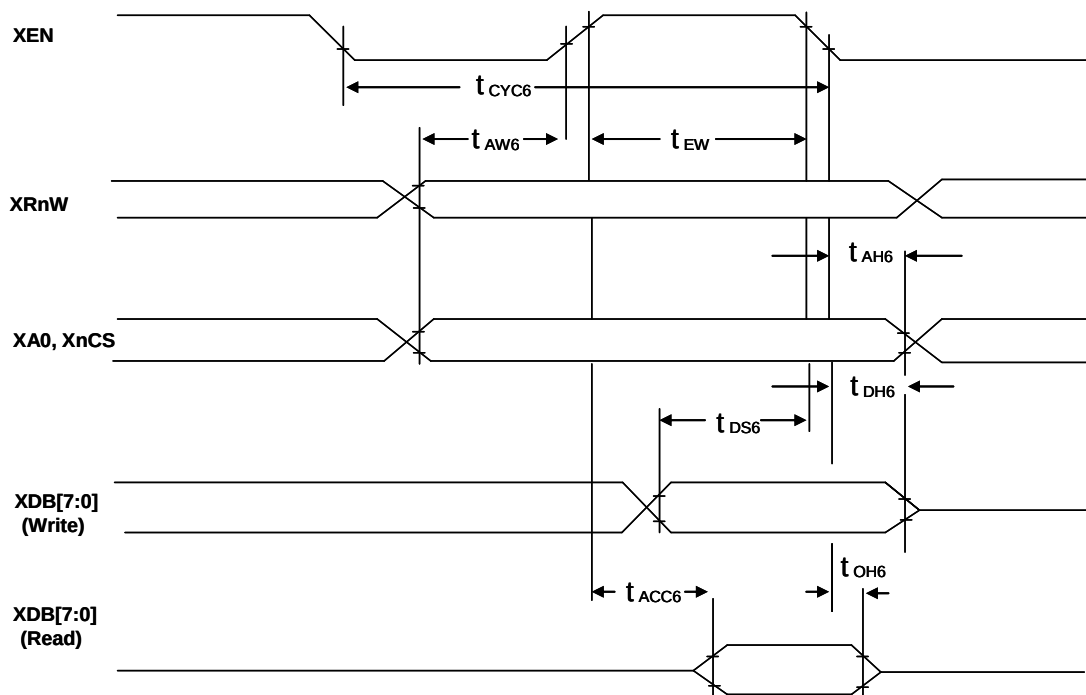


Figure 7-3 : 8080 Waveform

Table 7-2 : 8080 MPU I/F Timing

Symbol	Parameter	Rating		Unit	Symbol
		Min.	Max.		
t_{CYC8}	Cycle time	50	--	ns	tc is one system clock period: tc = 1/SYS_CLK
t_{CC8}	Strobe Pulse width	20	--	ns	
t_{AS8}	Address setup time	0	--	ns	
t_{AH8}	Address hold time	10	--	ns	
t_{DS8}	Data setup time	20	--	ns	
t_{DH8}	Data hold time	10	--	ns	
t_{ACC8}	Data output access time	0	20	ns	
t_{OH8}	Data output hold time	0	20	ns	

6800 – 8/16-bit Interface

Figure 7-4 : 6800 MPU Waveform
Table 7-3 : 6800 MPU I/F Timing

Symbol	Parameter	Rating		Unit	Symbol
		Min.	Max.		
t_{CYC6}	Cycle time	50	--	ns	t_c is one system clock period: $t_c = 1/SYS_CLK$
t_{EW}	Strobe Pulse width	20	--	ns	
t_{AW6}	Address setup time	0	--	ns	
t_{AH6}	Address hold time	10	--	ns	
t_{DS6}	Data setup time	20	--	ns	
t_{DH6}	Data hold time	10	--	ns	
t_{ACC6}	Data output access time	0	20	ns	
t_{OH6}	Data output hold time	0	20	ns	

The continuous data write speed determines the display update speed. The cycle-to-cycle interval must be larger than 5 of system clock period if user without adopt XnWait to insert wait state. Over the specification may cause the data lose or function fail if xnwait mechanism does not use. Please refer to Figure 7-5 and Figure 7-6 for waveform detail.

In order to reduce the transmission interference between MPU interface and RA8876, It is suggested that a small capacitor to the GND should be added at the signal of XnCS, XnRD_EN, XnWR_RnW. If using cable to connect MPU and RA8876, please keep the cable length less than 20cm. Otherwise it's suggested to add 1~10Kohm pull-up resistors on pins XnCS, XnRD_EN, XnWR_RnW and XA0.

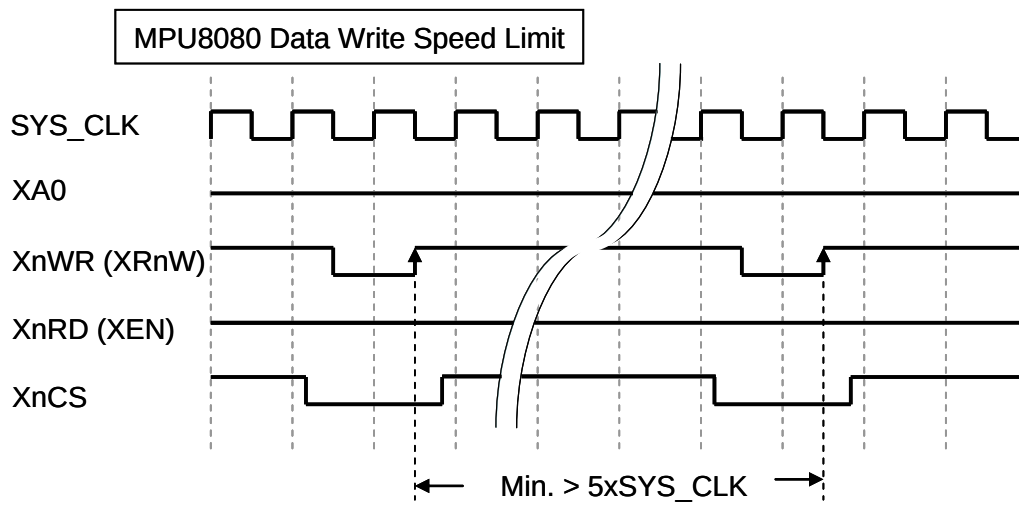


Figure 7-5 : 8080 I/F Continuous Data Write Cycle Waveform

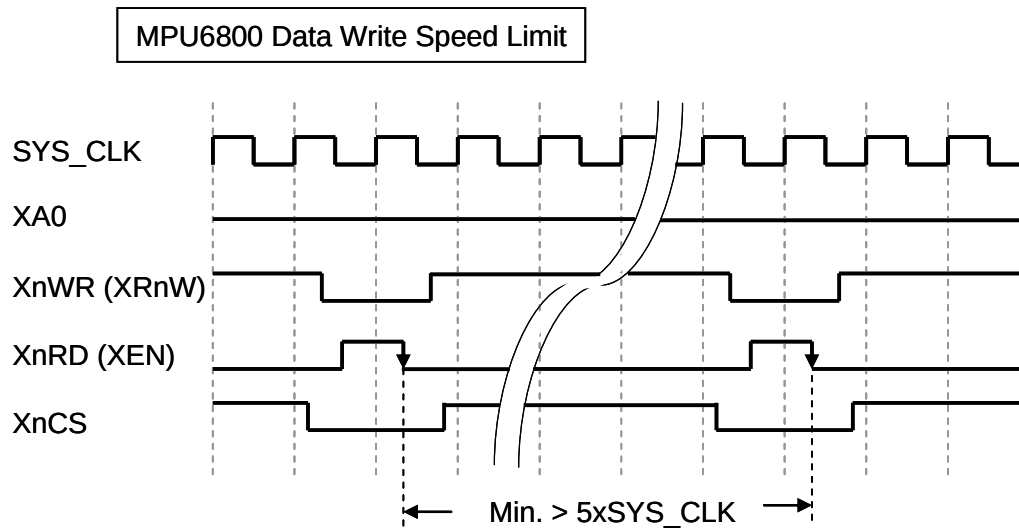


Figure 7-6 : 6800 I/F Continuous Data Write Cycle Waveform

7.3 Serial Host

7.3.1 3-Wire SPI Interface

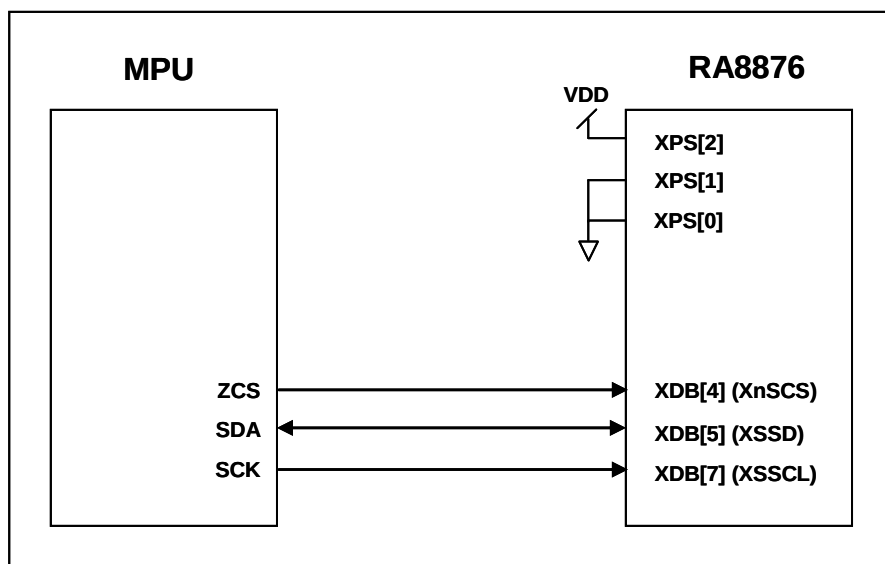


Figure 7-7 : The MPU Interface Diagram of 3-Wire SPI

RA8876 provides a SPI slave controller (Mode 3). The SPI I/F are available through the chip select line (XnSCS), serial transfer clock line (XSSCL) and serial input/output line (XSSD). XSSCL is driven by the master controller, which is used to latch the XSSD signal when XnSCS is active. The SPI can be configured in command/data write mode or status/data read mode by setting MSB two bits of first byte of protocol. Before a data transmission begins, low active XnSCS must be set to low, and keep low until the transmission is finished. When the SPI module is in command/data write mode (Figure 7-9, Figure 7-11), the 2nd byte of the protocol is write data asserted by the master controller via XSSD pin. When the SPI module is in status/data read mode (Figure 7-8, Figure 7-10), the 2nd byte is the read data or status byte which is sent from RA8876 to the controller via XSSD according to the activation of XSSCL from the master controller. Maximum XSSCL frequency is 50MHz.

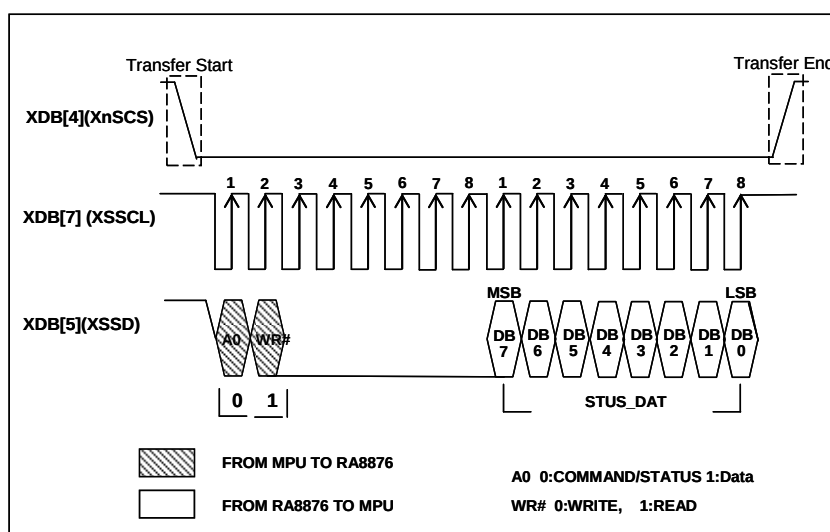
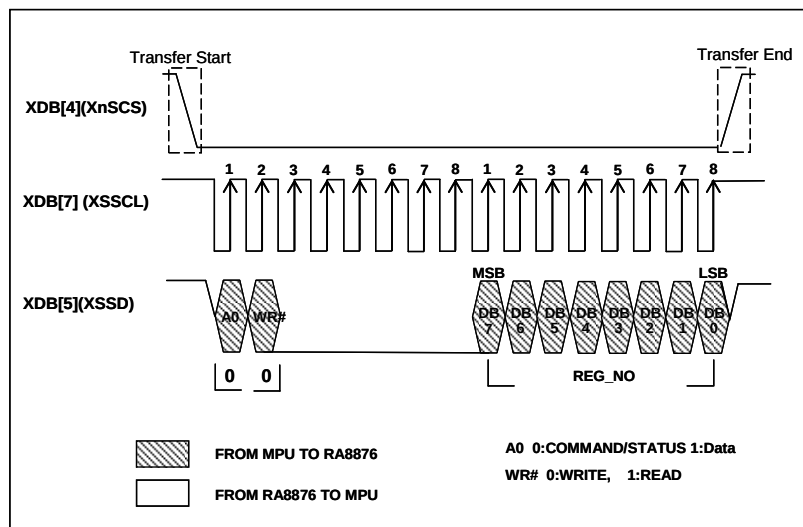
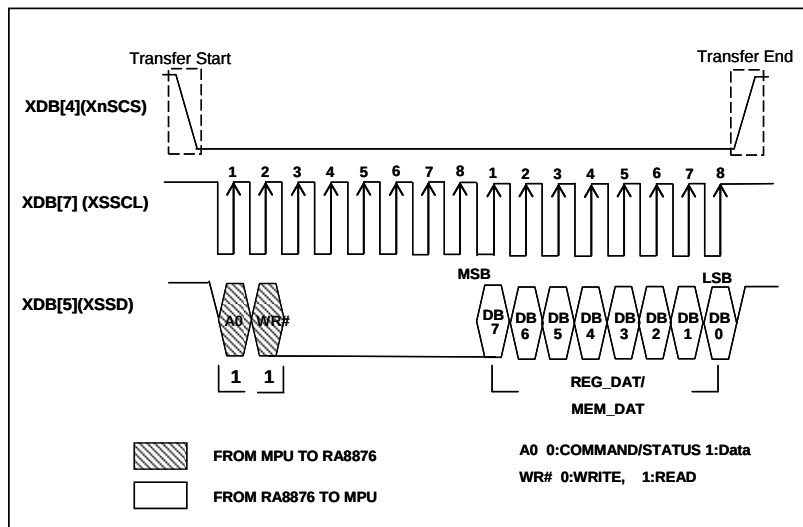
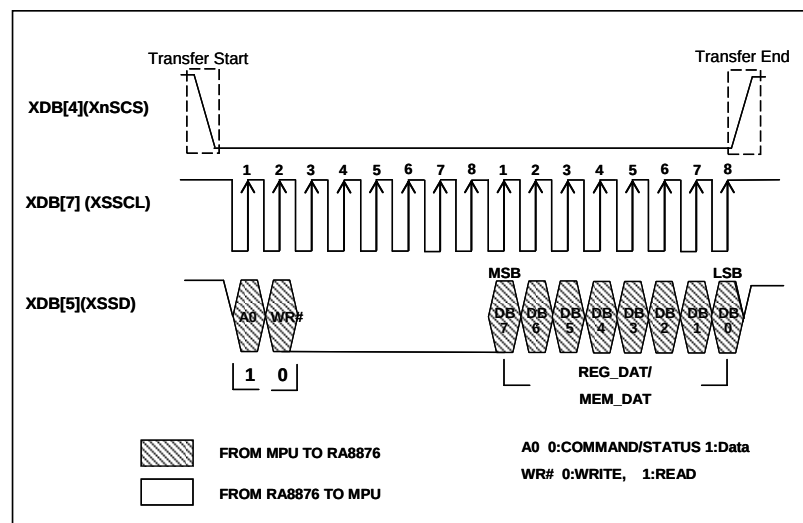


Figure 7-8 : Status Read on 3-Wire SPI-Bus


Figure 7-9 : CMD Write on 3-Wire SPI-Bus

Figure 7-10 : Data Read on 3-Wire SPI-Bus

Figure 7-11 : Date Write on 3-Wire SPI-Bus

The following timing charts are used to describe the timing specification of the 3-Wire SPI Interface.

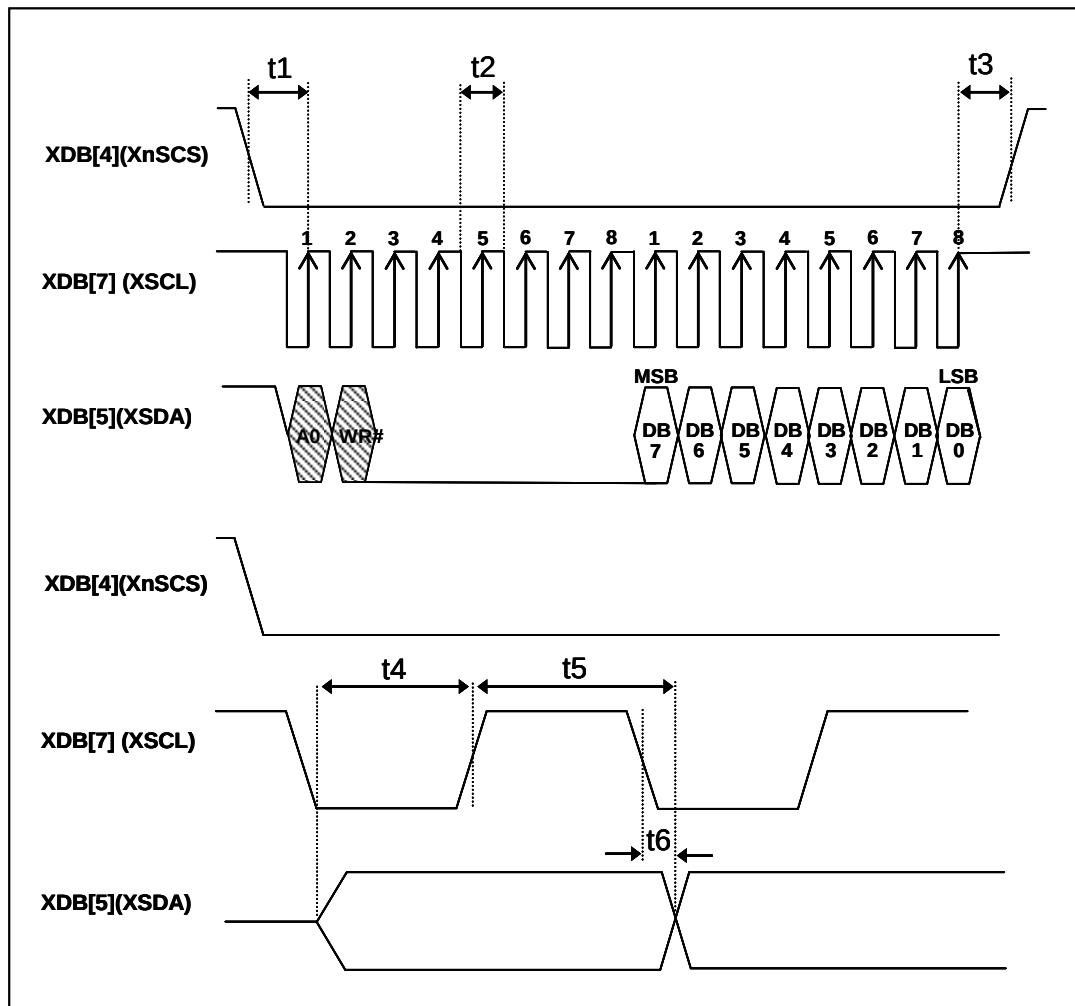


Figure 7-12 : 3-Wire SPI I/F Waveform

Table 7-4 : 3-wire SPI I/F Timing

Symbol	Parameter	Rating		Unit	Symbol
		Min.	Max.		
t ₂	Cycle time	20	10000	ns	
t ₁	CS setup time to rising edge of SCL	1/2 t ₂	--	ns	
t ₃	CS hold time from rising edge of SCL	1/2 t ₂	--	ns	
t ₄	Data setup time to rising edge of SCL	5	--	ns	
t ₅	Data hold time from rising edge of SCL	5	--	ns	
t ₆	Data output valid from falling edge of SCL	5	20	ns	

7.3.2 4-Wire SPI Interface

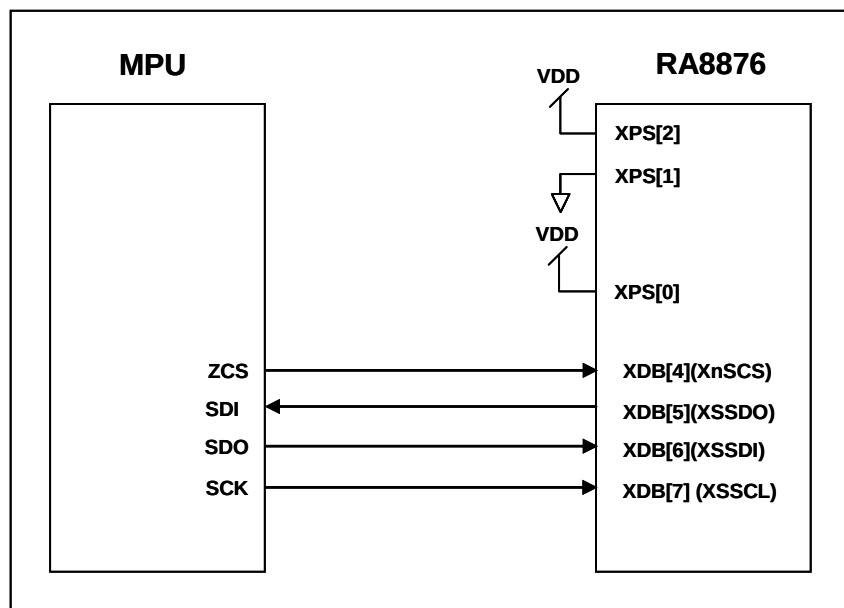


Figure 7-13 : The MPU Interface Diagram of 4-Wire SPI

The 4-wire SPI I/F is similar with 3-wire SPI I/F, the only difference is the data signal. In 3-wire SPI I/F, the bi-direction XSSD signal is used as data and can be driven by slave/master controller. In 4-wire SPI I/F, the XSSD signal function is separated into XSSDI and XSSDO signal. XSSDI is the data pin from the SPI master, XSSDO is the data output from the SPI slave. About the detail protocol, please refer to Figure 7-14~Figure 7-17.

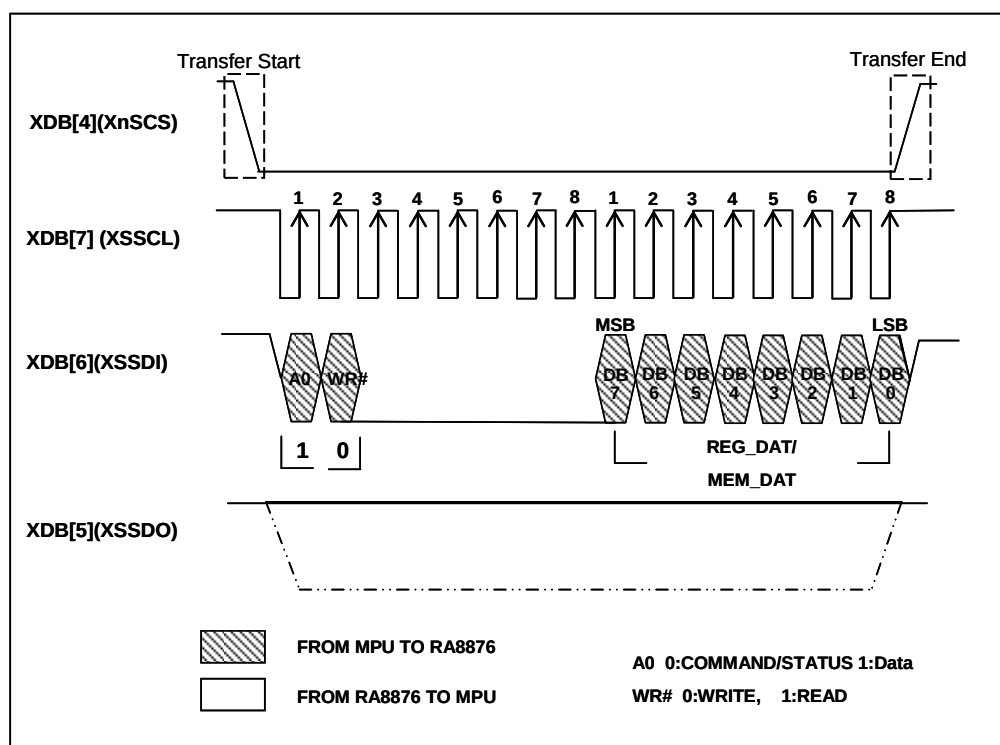


Figure 7-14 : Data Write on 4-Wire SPI-Bus

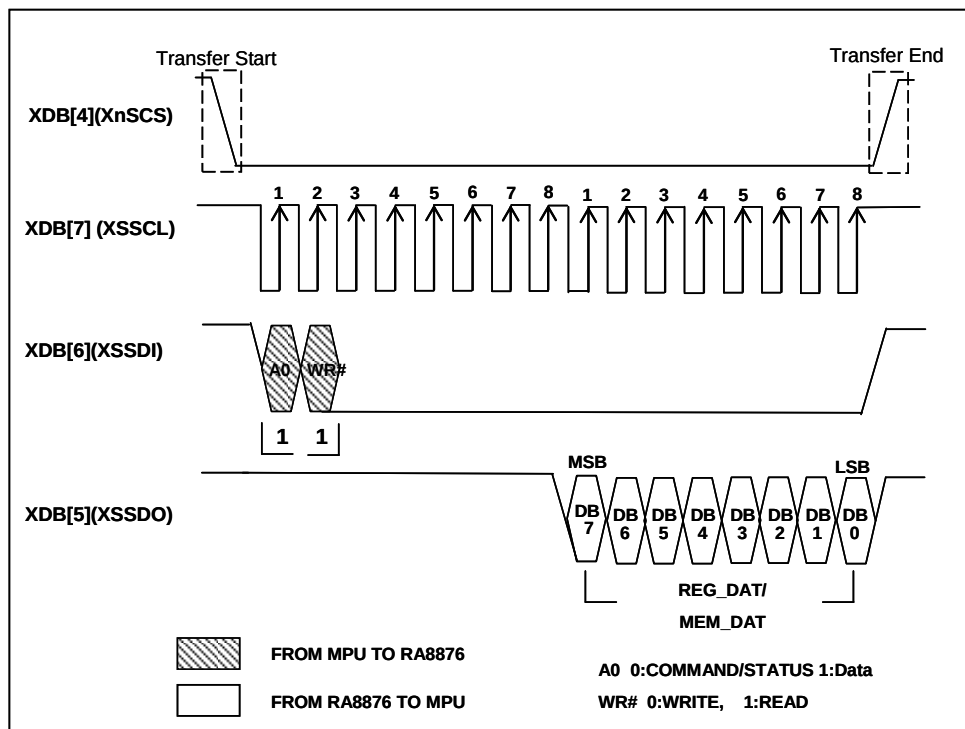


Figure 7-15 : Data Read on 4-Wire SPI-Bus

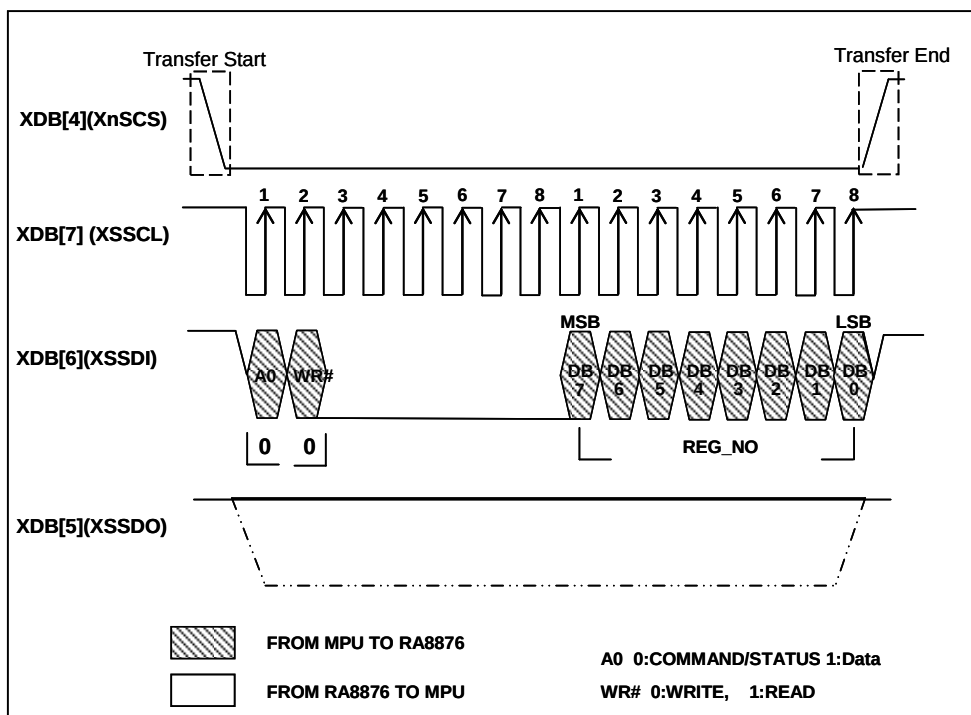


Figure 7-16 : CMD Write on 4-Wire SPI-Bus

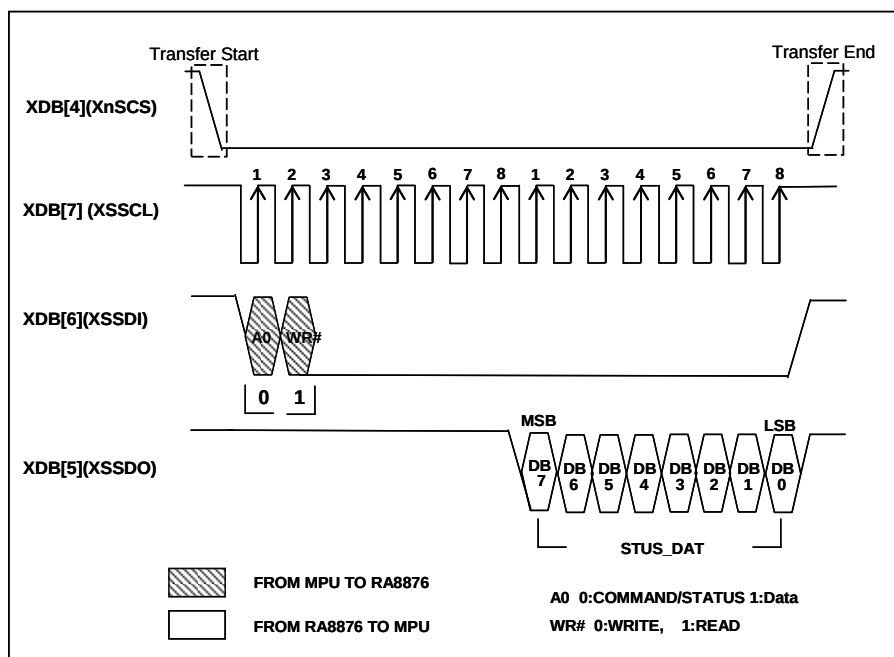


Figure 7-17 : Status Read on 4-Wire SPI-Bus

The following timing charts are used to describe the timing specification of the 4-Wire SPI Interface.

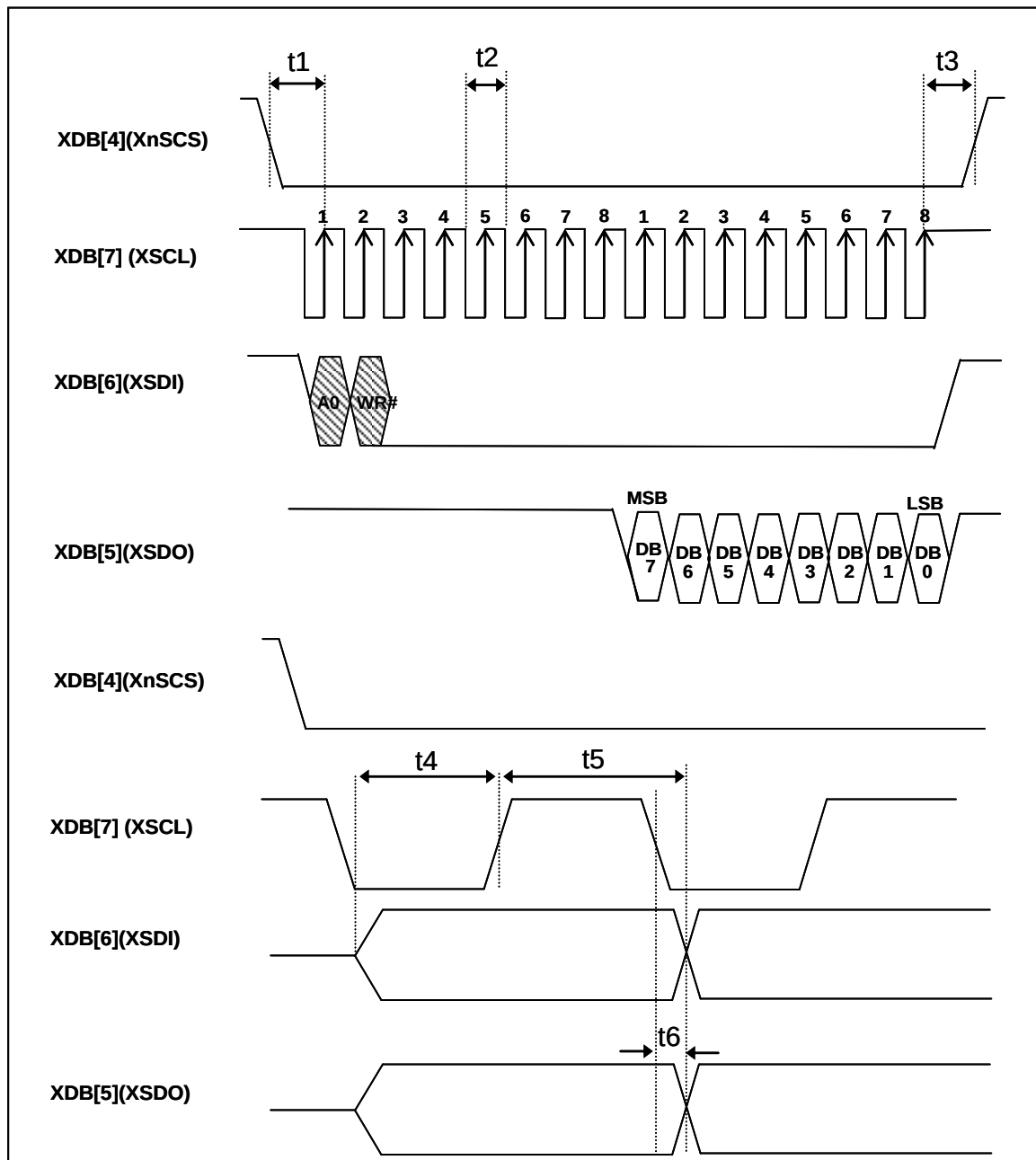
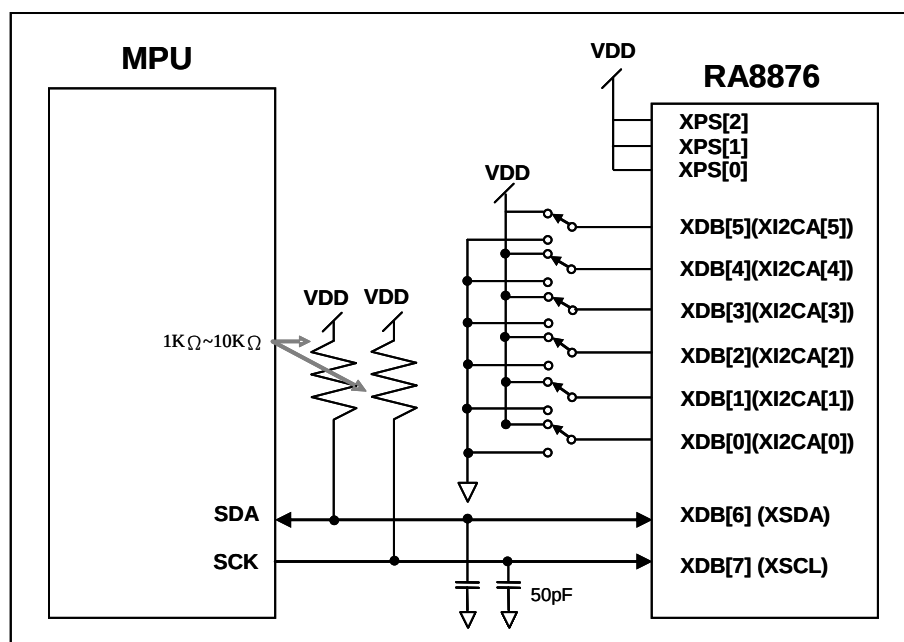


Figure 7-18 : 4-Wire SPI I/F Waveform

Table 7-5 : 4-wire SPI I/F Timing

Symbol	Parameter	Rating		Unit	Symbol
		Min.	Max.		
t ₂	Cycle time	20	10000	ns	
t ₁	CS setup time to rising edge of SCL	1/2t ₂	--	ns	
t ₃	CS hold time from rising edge of SCL	1/2t ₂	--	ns	
t ₄	Data setup time to rising edge of SCL	5	--	ns	
t ₅	Data hold time from rising edge of SCL	5	--	ns	
t ₆	Data output valid from falling edge of SCL	5	20	ns	

7.3.3 IIC I/F



IICA[5:0]					
BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
XIICA[5]	XIICA[4]	XIICA[3]	XIICA[2]	XIICA[1]	XIICA[0]

Figure 7-19 : The MPU Interface Diagram of IIC

The IIC I/F are accessed by only two bus line, XSSCL and XSSDA. It is compatible with standard IIC interface. The first 7bits of IIC protocol, indicated as the IIC slave address to program in IIC Spec. The first 6 bits indicates the IIC device ID of RA8876. The next 1bit is A0 bit which indicates the cycle type. For A0 = 1, the following cycle is a data. If A0 = 0, it's a Command/Status cycle. If the MSB 6bits of IIC address (Total 7 bits) match the RA8876 device ID, the RA8876 IIC slave is active.

The device ID of RA8876 is configurable which can be set from the XIICA[5:0]/XDB[5:0] pins directly. There are 4 types of cycles for RA8876: "Command writes cycle", "Status read cycle", "Data write cycle", and "Data read cycle". The cycle type is set by the A0 bit and WR# bit, about the detail protocol, please refer to Figure 7-20 ~ Figure 7-23.

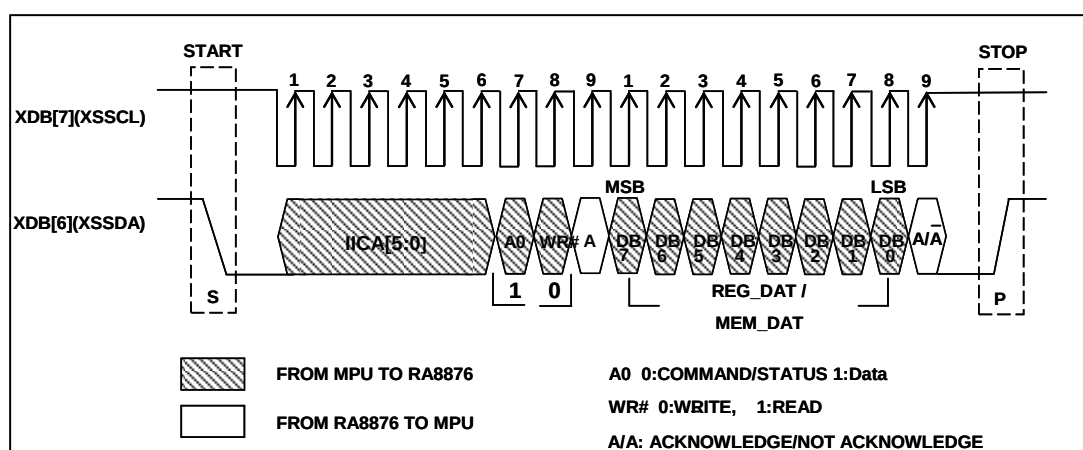


Figure 7-20 : Data Write on IIC-Bus

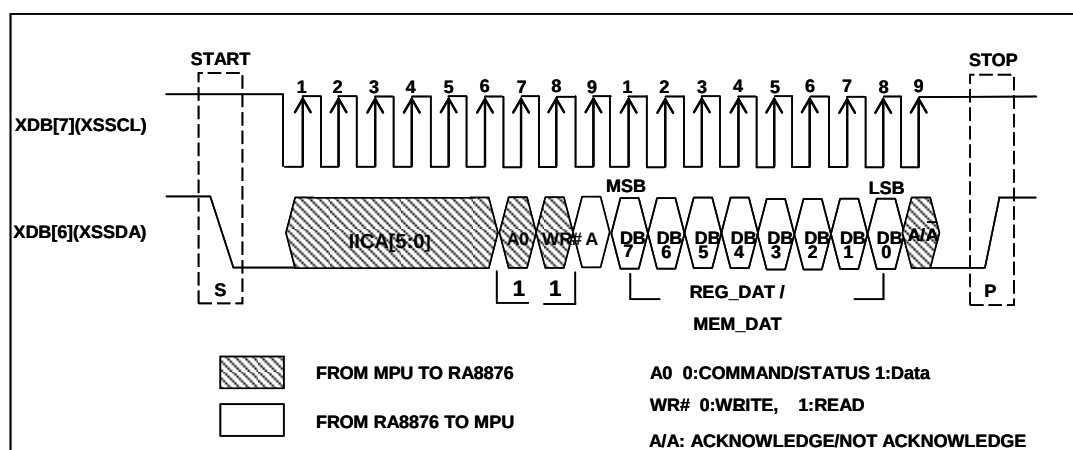


Figure 7-21 : Data Read on IIC-Bus

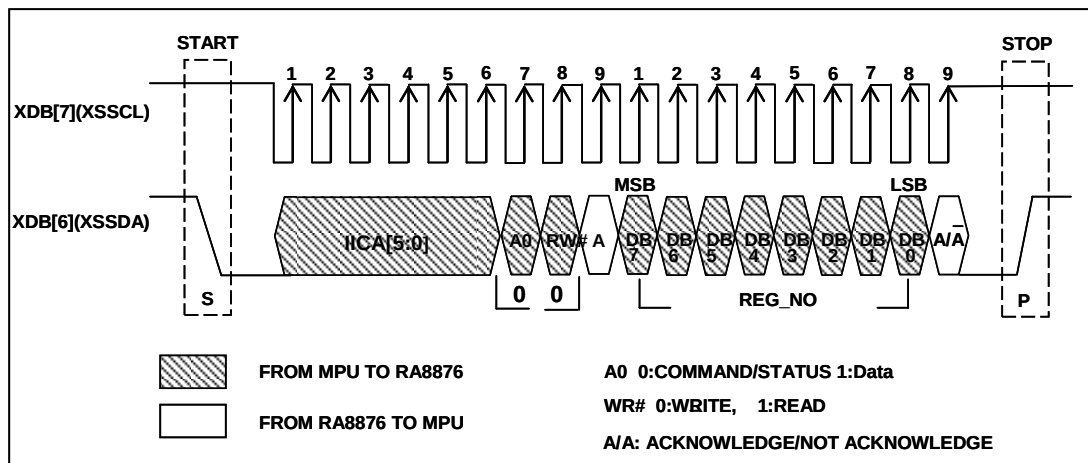


Figure 7-22 : CMD Write on IIC-Bus

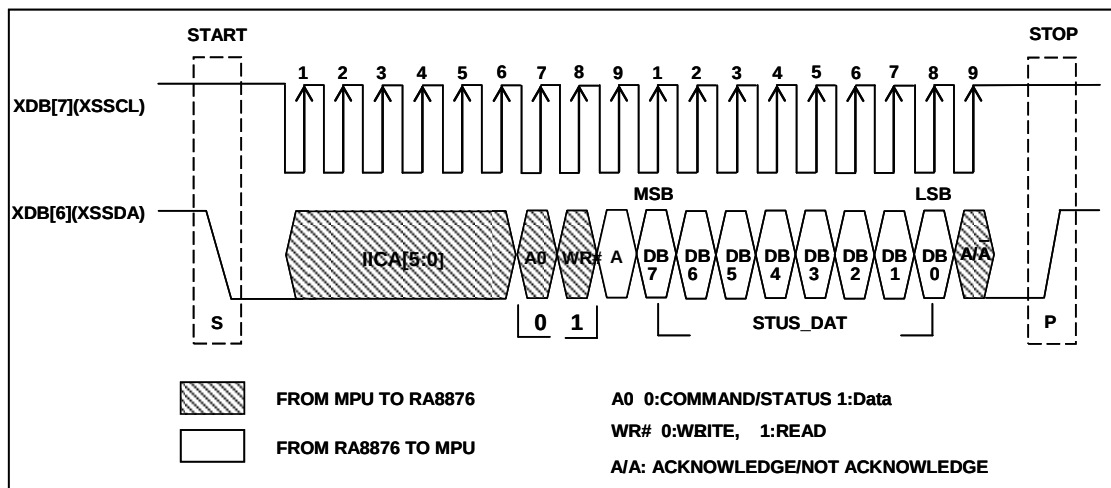


Figure 7-23 : Status Read on IIC-Bus

The following timing charts are used to describe the timing specification of the IIC Interface.

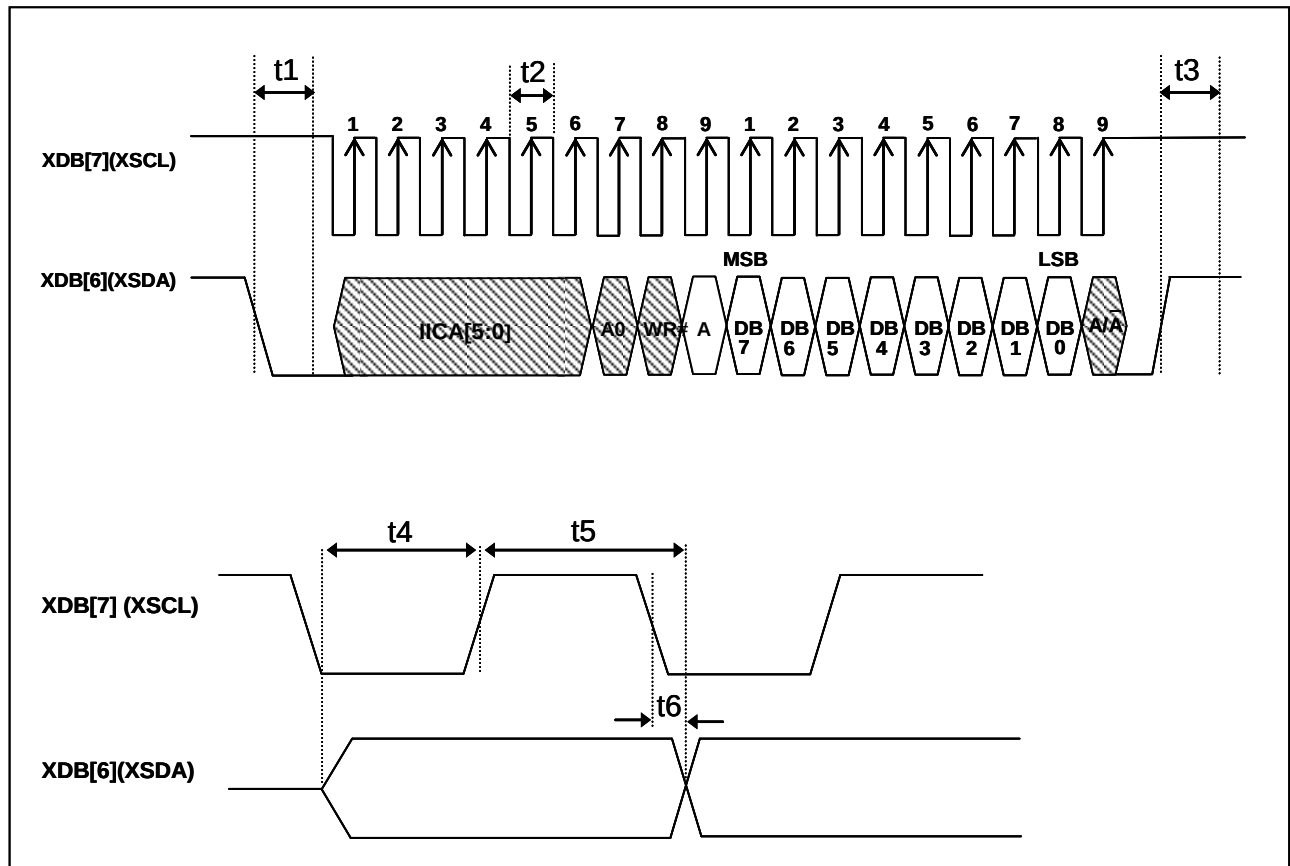


Figure 7-24 : IIC I/F Waveform

Table 7-6 : IIC I/F Timing

Symbol	Parameter	Rating		Unit	Symbol
		Min.	Max.		
t ₂	Cycle time	10000	2500	ns	
t ₁	Start Strobe Pulse width	180	--	ns	
t ₃	Stop Strobe Pulse width	180	--	ns	
t ₄	Data setup time to rising edge of SCL	5	--	ns	
t ₅	Data hold time from rising edge of SCL	5	--	ns	
t ₆	Data output valid from falling edge of SCL	5	20	ns	

7.4 Display Input Data Format

7.4.1 Input Data without Opacity (RGB)

8-bit MPU, 1bpp mode (monochrome data)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	P ₇	P ₆	P ₅	P ₄	P ₃	P ₂	P ₁	P ₀
2	P ₁₅	P ₁₄	P ₁₃	P ₁₂	P ₁₁	P ₁₀	P ₉	P ₈
3	P ₂₃	P ₂₂	P ₂₁	P ₂₀	P ₁₉	P ₁₈	P ₁₇	P ₁₆
4	P ₃₁	P ₃₀	P ₂₉	P ₂₈	P ₂₇	P ₂₆	P ₂₅	P ₂₄
5	P ₃₉	P ₃₈	P ₃₇	P ₃₆	P ₃₅	P ₃₄	P ₃₃	P ₃₂
6	P ₄₇	P ₄₆	P ₄₅	P ₄₄	P ₄₃	P ₄₂	P ₄₁	P ₄₀

*** **Note:** Only used for BTE's color expansion function. Only accept 8bits data and set canvas as 8bpp color depth then write data to memory. After write task complete then enable color expansion function and expand color depth as desired.

8-bit MPU, 8bpp mode (RGB 3:3:2)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
2	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶
3	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
4	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶
5	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
6	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶

8-bit MPU, 16bpp mode (RGB 5:6:5)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
2	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵
3	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
4	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵
5	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
6	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵

8-bit MPU, 24bpp mode (RGB 8:8:8)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
2	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰
3	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
4	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰
5	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
6	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰

16-bit MPU, 1bpp mode 1 (monochrome data)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₇	P ₆	P ₅	P ₄	P ₃	P ₂	P ₁	P ₀
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₁₅	P ₁₄	P ₁₃	P ₁₂	P ₁₁	P ₁₀	P ₉	P ₈
3	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₂₃	P ₂₂	P ₂₁	P ₂₀	P ₁₉	P ₁₈	P ₁₇	P ₁₆
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₃₁	P ₃₀	P ₂₉	P ₂₈	P ₂₇	P ₂₆	P ₂₅	P ₂₄
5	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₃₉	P ₃₈	P ₃₇	P ₃₆	P ₃₅	P ₃₄	P ₃₃	P ₃₂
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	P ₄₇	P ₄₆	P ₄₅	P ₄₄	P ₄₃	P ₄₂	P ₄₁	P ₄₀

*** **Note:** Only used for BTE's color expansion function. Only accept 8bits data and set canvas as 8bpp color depth, canvas width active window are equal to mono image width divided by 8 then write data to memory. After write task complete then enable color expansion function and expand color depth as desired.

16-bit MPU, 1bpp mode 2 (monochrome data)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	P ₁₅	P ₁₄	P ₁₃	P ₁₂	P ₁₁	P ₁₀	P ₉	P ₈	P ₇	P ₆	P ₅	P ₄	P ₃	P ₂	P ₁	P ₀
2	P ₃₁	P ₃₀	P ₂₉	P ₂₈	P ₂₇	P ₂₆	P ₂₅	P ₂₄	P ₂₃	P ₂₂	P ₂₁	P ₂₀	P ₁₉	P ₁₈	P ₁₇	P ₁₆
3	P ₄₇	P ₄₆	P ₄₅	P ₄₄	P ₄₃	P ₄₂	P ₄₁	P ₄₀	P ₃₉	P ₃₈	P ₃₇	P ₃₆	P ₃₅	P ₃₄	P ₃₃	P ₃₂
4	P ₆₃	P ₆₂	P ₆₁	P ₆₀	P ₅₉	P ₅₈	P ₅₇	P ₅₆	P ₅₅	P ₅₄	P ₅₃	P ₅₂	P ₅₁	P ₅₀	P ₄₉	P ₄₈
5	P ₇₉	P ₇₈	P ₇₇	P ₇₆	P ₇₅	P ₇₄	P ₇₃	P ₇₂	P ₇₁	P ₇₀	P ₆₉	P ₆₈	P ₆₇	P ₆₆	P ₆₅	P ₆₄
6	P ₉₅	P ₉₄	P ₉₃	P ₉₂	P ₉₁	P ₉₀	P ₈₉	P ₈₈	P ₈₇	P ₈₆	P ₈₅	P ₈₄	P ₈₃	P ₈₂	P ₈₁	P ₈₀

*** **Note:** Only used for BTE's color expansion function; user treats it like 16bpp image data, so set canvas image as 16bpp color depth and canvas width & active window are equal to mono image width divided by 16 then write data to memory. After write task complete then enable color expansion function and expand color depth as desired then set main image or PIP image with this color depth.

16-bit MPU, 8bpp mode 1 (RGB 3:3:2)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶
3	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶
5	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶

16-bit MPU, 8bpp mode 2 (RGB 3:3:2)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
2	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
3	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
4	R ₇ ⁷	R ₇ ⁶	R ₇ ⁵	G ₇ ⁷	G ₇ ⁶	G ₇ ⁵	B ₇ ⁷	B ₇ ⁶	R ₆ ⁷	R ₆ ⁶	R ₆ ⁵	G ₆ ⁷	G ₆ ⁶	G ₆ ⁵	B ₆ ⁷	B ₆ ⁶
5	R ₉ ⁷	R ₉ ⁶	R ₉ ⁵	G ₉ ⁷	G ₉ ⁶	G ₉ ⁵	B ₉ ⁷	B ₉ ⁶	R ₈ ⁷	R ₈ ⁶	R ₈ ⁵	G ₈ ⁷	G ₈ ⁶	G ₈ ⁵	B ₈ ⁷	B ₈ ⁶
6	R ₁₁ ⁷	R ₁₁ ⁶	R ₁₁ ⁵	G ₁₁ ⁷	G ₁₁ ⁶	G ₁₁ ⁵	B ₁₁ ⁷	B ₁₁ ⁶	R ₁₀ ⁷	R ₁₀ ⁶	R ₁₀ ⁵	G ₁₀ ⁷	G ₁₀ ⁶	G ₁₀ ⁵	B ₁₀ ⁷	B ₁₀ ⁶

*** **Note:** User treats it as 16bpp image data, so set canvas image as 16bpp color depth and canvas width & active window are equal to image width divided by 2, then write data to memory. Main image or PIP image's color depth also need to set as 8bpp.

16-bit MPU, 16bpp mode (RGB 5:6:5)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
2	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
3	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
4	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³
5	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	R ₄ ³	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	G ₄ ⁴	G ₄ ³	G ₄ ²	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴	B ₄ ³
6	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	R ₅ ³	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	G ₅ ⁴	G ₅ ³	G ₅ ²	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴	B ₅ ³

16-bit MPU, 24bpp mode 1 (RGB 8:8:8)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
2	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
3	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
4	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	G ₂ ¹	G ₂ ⁰	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³	B ₂ ²	B ₂ ¹	B ₂ ⁰
5	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³	B ₃ ²	B ₃ ¹	B ₃ ⁰	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	R ₂ ²	R ₂ ¹	R ₂ ⁰
6	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	R ₃ ²	R ₃ ¹	R ₃ ⁰	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	G ₃ ¹	G ₃ ⁰

16-bit MPU, 24bpp mode 2 (RGB 8:8:8)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G_0^7	G_0^6	G_0^5	G_0^4	G_0^3	G_0^2	G_0^1	G_0^0	B_0^7	B_0^6	B_0^5	B_0^4	B_0^3	B_0^2	B_0^1	B_0^0
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R_0^7	R_0^6	R_0^5	R_0^4	R_0^3	R_0^2	R_0^1	R_0^0
3	G_1^7	G_1^6	G_1^5	G_1^4	G_1^3	G_1^2	G_1^1	G_1^0	B_1^7	B_1^6	B_1^5	B_1^4	B_1^3	B_1^2	B_1^1	B_1^0
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R_1^7	R_1^6	R_1^5	R_1^4	R_1^3	R_1^2	R_1^1	R_1^0
5	G_2^7	G_2^6	G_2^5	G_2^4	G_2^3	G_2^2	G_2^1	G_2^0	B_2^7	B_2^6	B_2^5	B_2^4	B_2^3	B_2^2	B_2^1	B_2^0
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R_2^7	R_2^6	R_2^5	R_2^4	R_2^3	R_2^2	R_2^1	R_2^0

7.4.2 Input Data with opacity (α RGB)
8-bit MPU, 8bpp mode (α Index 2:6)

RA8876 provide a palette of 64 simultaneous colors from a total of 4096 different colors with opacity attribute for OSD application. User may load preferred color into embedded color palette then pick it up by index color. α value stands for opacity.

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	α_1^3	α_1^2	Index color of pixel 0					
2	α_3^3	α_3^2	Index color of pixel 1					
3	α_5^3	α_5^2	Index color of pixel 2					
4	α_7^3	α_7^2	Index color of pixel 3					
5	α_9^3	α_9^2	Index color of pixel 4					
6	α_{11}^3	α_{11}^2	Index color of pixel 5					

$\alpha_x^3 \alpha_x^2$: 0 – 100%, 1 – 20/32, 2 – 11/32, 3 – 0

8-bit MPU, 16bpp mode (α RGB 4:4:4:4)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G_0^7	G_0^6	G_0^5	G_0^4	B_0^7	B_0^6	B_0^5	B_0^4
2	α_0^3	α_0^2	α_0^1	α_0^0	R_0^7	R_0^6	R_0^5	R_0^4
3	G_1^7	G_1^6	G_1^5	G_1^4	B_1^7	B_1^6	B_1^5	B_1^4
4	α_1^3	α_1^2	α_1^1	α_1^0	R_1^7	R_1^6	R_1^5	R_1^4
5	G_2^7	G_2^6	G_2^5	G_2^4	B_2^7	B_2^6	B_2^5	B_2^4
6	α_2^3	α_2^2	α_2^1	α_2^0	R_2^7	R_2^6	R_2^5	R_2^4

$\alpha_x^3 \alpha_x^2 \alpha_x^1 \alpha_x^0$: 0 – 100%, 1 – 30/32, 2 – 28/32, 3 – 26/32, 4 – 24/32,, 12 – 8/32, 13 – 6/32, 14 – 4/32, 15 – 0.

16-bit MPU, Index mode with opacity (α Index 2:6)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_0^3	α_0^2	Index color of pixel 0					
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_1^3	α_1^2	Index color of pixel 1					
3	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_2^3	α_2^2	Index color of pixel 2					
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_3^3	α_3^2	Index color of pixel 3					
5	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_4^3	α_4^2	Index color of pixel 4					
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	α_5^3	α_5^2	Index color of pixel 5					

$\alpha_x^3 \alpha_x^2$: 0 – 0, 1 – 11/32, 2 – 20/32, 3 – 100%

16-bit MPU, 12bpp mode with opacity (α RGB 4:4:4:4)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	α_0^3	α_0^2	α_0^1	α_0^0	R_0^7	R_0^6	R_0^5	R_0^4	G_0^7	G_0^6	G_0^5	G_0^4	B_0^7	B_0^6	B_0^5	B_0^4
2	α_1^3	α_1^2	α_1^1	α_1^0	R_1^7	R_1^6	R_1^5	R_1^4	G_1^7	G_1^6	G_1^5	G_1^4	B_1^7	B_1^6	B_1^5	B_1^4
3	α_2^3	α_2^2	α_2^1	α_2^0	R_2^7	R_2^6	R_2^5	R_2^4	G_2^7	G_2^6	G_2^5	G_2^4	B_2^7	B_2^6	B_2^5	B_2^4
4	α_3^3	α_3^2	α_3^1	α_3^0	R_3^7	R_3^6	R_3^5	R_3^4	G_3^7	G_3^6	G_3^5	G_3^4	B_3^7	B_3^6	B_3^5	B_3^4
5	α_4^3	α_4^2	α_4^1	α_4^0	R_4^7	R_4^6	R_4^5	R_4^4	G_4^7	G_4^6	G_4^5	G_4^4	B_4^7	B_4^6	B_4^5	B_4^4
6	α_5^3	α_5^2	α_5^1	α_5^0	R_5^7	R_5^6	R_5^5	R_5^4	G_5^7	G_5^6	G_5^5	G_5^4	B_5^7	B_5^6	B_5^5	B_5^4

$\alpha_x^3 \alpha_x^2 \alpha_x^1 \alpha_x^0$: 0, 1 – 2/32, 2 – 4/32, 3 – 6/32, 4 – 8/32,, 12 – 24/32, 13 – 26/32, 14 – 28/32, 15 – 100%.

8. Memory

8.1 SDRAM Controller

The SDRAM controller accesses external 16/32/64/128/256/512 Mbit single data rate SDRAM effectively by using bank interleave. The initialization sequence and the auto refresh cycle are executed with hardware.

8.1.1 SDRAM Initialization

SDRAM must be initialized after resetting hardware and before any memory access. After hardware is reset, an initialization command can be executed only once. The command is ignored after initialization. The initialization sequence is as follows.

1. Set SDRAM chip's attribute; Command write 0xE0. Then according to register's definition make a data write to set bank number, row size & column size, ... etc.
2. Set SDRAM mode register parameter; Command write 0xE1 to set CAS latency.
3. Set SDRAM refresh interval to Command register 0xE2 & 0xE3. Typical SDRAM refresh interval is 15.6us.
4. Start SDRAM initial procedure. Set command register 0xE4 bit-0 as 1.
5. Check Command register 0xE4 bit-0 and wait it becomes 1 then exit initialization.

8.1.2 SDRAM Connection

SDR SDRAM Addressing

DENSITY	ADDRESSING	X16
16Mb (2 banks)	Row	A0-A10
	Column	A0-A7
32Mb (2 banks)	Row	A0-A11
	Column	A0-A7
64Mb (4 banks)	Row	A0-A11
	Column	A0-A7
128Mb (4 banks)	Row	A0-A11
	Column	A0-A8
256Mb (4 banks)	Row	A0-A12
	Column	A0-A8
512Mb (4 banks)	Row	A0-A12
	Column	A0-A9

8.2 SDRAM Data Structure

The input image is stored in memory as 1bpp, 8bpp, 16bpp or 24bpp and depends on whether have opacity bits or not.

8.2.1 8bpp Display (RGB 3:3:2 Input Data)

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
0002h	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
0004h	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
0006h	R ₇ ⁷	R ₇ ⁶	R ₇ ⁵	G ₇ ⁷	G ₇ ⁶	G ₇ ⁵	B ₇ ⁷	B ₇ ⁶	R ₆ ⁷	R ₆ ⁶	R ₆ ⁵	G ₆ ⁷	G ₆ ⁶	G ₆ ⁵	B ₆ ⁷	B ₆ ⁶
0008h	R ₉ ⁷	R ₉ ⁶	R ₉ ⁵	G ₉ ⁷	G ₉ ⁶	G ₉ ⁵	B ₉ ⁷	B ₉ ⁶	R ₈ ⁷	R ₈ ⁶	R ₈ ⁵	G ₈ ⁷	G ₈ ⁶	G ₈ ⁵	B ₈ ⁷	B ₈ ⁶
000Ah	R ₁₁ ⁷	R ₁₁ ⁶	R ₁₁ ⁵	G ₁₁ ⁷	G ₁₁ ⁶	G ₁₁ ⁵	B ₁₀ ⁷	B ₁₀ ⁶	R ₁₀ ⁷	R ₁₀ ⁶	R ₁₀ ⁵	G ₁₀ ⁷	G ₁₀ ⁶	G ₁₀ ⁵	B ₁₀ ⁷	B ₁₀ ⁶

8.2.2 16bpp Display (RGB 5:6:5 Input Data)

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
0002h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
0004h	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
0006h	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³
0008h	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	R ₄ ³	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	G ₄ ⁴	G ₄ ³	G ₄ ²	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴	B ₄ ³
000Ah	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	R ₅ ³	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	G ₅ ⁴	G ₅ ³	G ₅ ²	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴	B ₅ ³

8.2.3 24bpp Display (RGB 8:8:8 Input Data)

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
0002h	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
0004h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
0006h	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	G ₂ ¹	G ₂ ⁰	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³	B ₂ ²	B ₂ ¹	B ₂ ⁰
0008h	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³	B ₃ ²	B ₃ ¹	B ₃ ⁰	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	R ₂ ²	R ₂ ¹	R ₂ ⁰
000Ah	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	R ₃ ²	R ₃ ¹	R ₃ ⁰	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	G ₃ ¹	G ₃ ⁰

8.2.4 Index Display with opacity (αRGB 2:2:2:2)

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	α ₁ ³	α ₁ ²	Index color of pixel 1						α ₀ ³	α ₀ ²	Index color of pixel 0					
0002h	α ₃ ³	α ₃ ²	Index color of pixel 3						α ₂ ³	α ₂ ²	Index color of pixel 2					
0004h	α ₅ ³	α ₅ ²	Index color of pixel 5						α ₄ ³	α ₄ ²	Index color of pixel 4					
0006h	α ₇ ³	α ₇ ²	Index color of pixel 7						α ₆ ³	α ₆ ²	Index color of pixel 6					
0008h	α ₉ ³	α ₉ ²	Index color of pixel 9						α ₈ ³	α ₈ ²	Index color of pixel 8					
000Ah	α ₁₁ ³	α ₁₁ ²	Index color of pixel 11						α ₁₀ ³	α ₁₀ ²	Index color of pixel 10					

α_x³α_x² : 0, 1 – 11/32, 2 – 20/32, 3 – 100%

8.2.5 12bpp Display with opacity (αRGB 4:4:4:4)

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	α ₀ ³	α ₀ ²	α ₀ ¹	α ₀ ⁰	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴
0002h	α ₁ ³	α ₁ ²	α ₁ ¹	α ₁ ⁰	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴
0004h	α ₂ ³	α ₂ ²	α ₂ ¹	α ₂ ⁰	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴
0006h	α ₃ ³	α ₃ ²	α ₃ ¹	α ₃ ⁰	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴
0008h	α ₄ ³	α ₄ ²	α ₄ ¹	α ₄ ⁰	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	G ₄ ⁴	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴
000Ah	α ₅ ³	α ₅ ²	α ₅ ¹	α ₅ ⁰	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	G ₅ ⁴	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴

α_x³α_x²α_x¹α_x⁰ : 0, 1 – 2/32, 2 – 4/32, 3 – 6/32, 4 – 8/32,, 12 – 24/32, 13 – 26/32, 14 – 28/32, 15 – 100%.

8.3 Color Palette RAM

Addr	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0000h	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴
0002h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴
0004h	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴
0006h	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴
0008h	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	G ₄ ⁴	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴
000Ah	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	G ₅ ⁴	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴

*It is referenced on BTE active. And if BTE's destination image is 8bpp then Bit[1:0], Bit[4] & Bit[8] are invalid.

9. Display Data Path

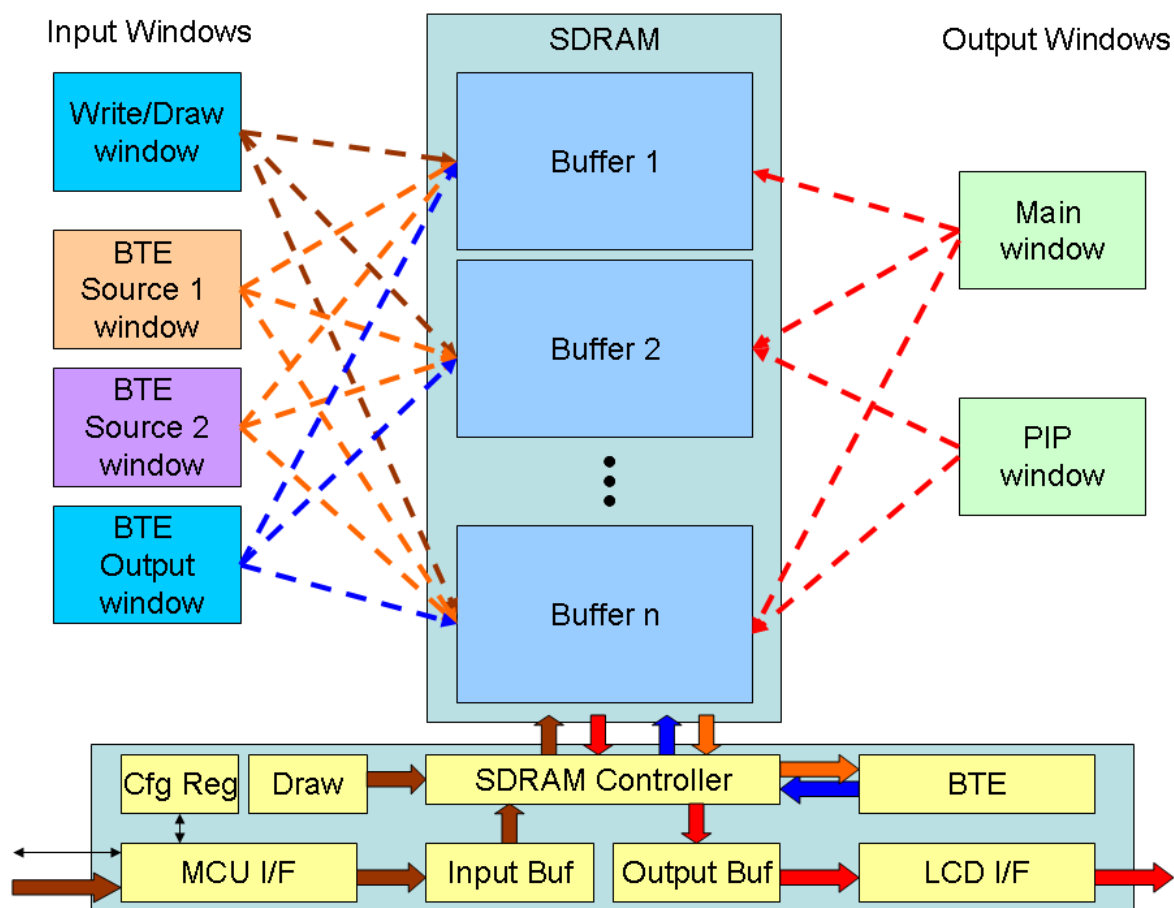


Figure 9-1 : Display Data path

10. LCD Interface

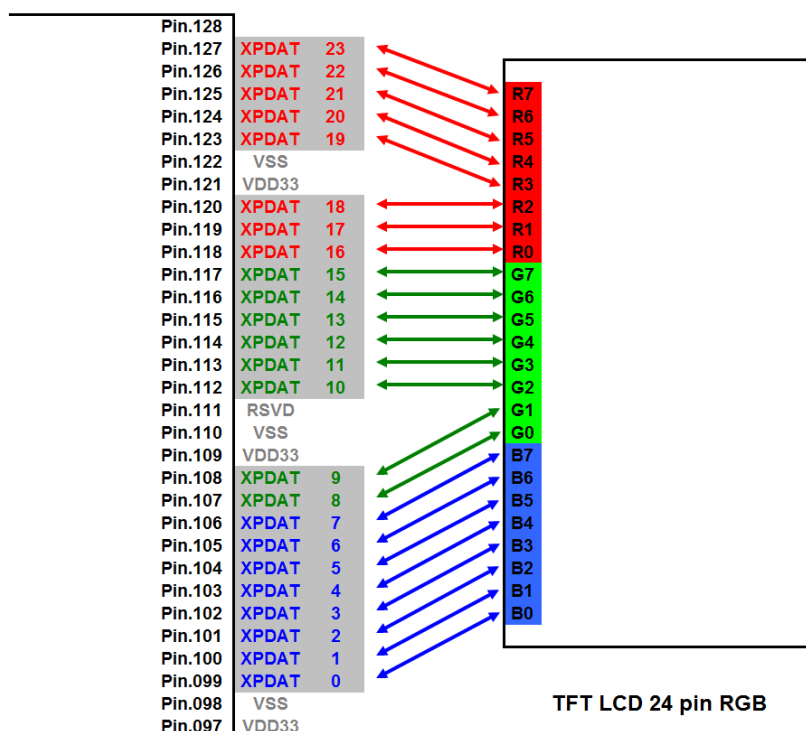
The RA8876 has the color LCD interface with the external SDRAM frame buffer. The maximum display setting supported is 2048x2048 @ 24bpp TFT. Provide 24 bits color output for 24bpp (RGB 8:8:8) color depth. RGB 3:3:2 and RGB 5:6:5 data are converted from 8/16bpp to 24bpp for output.

10.1 LCD Interface Pin Mapping

This function is controlled by REG[01h] bit 4-3.

Pin Name	TFT Panel Interface		
Color Depth	Digital Interface		
	16bpp	18bpp	24bpp
XVSYNC	XVSYNC	XVSYNC	XVSYNC
XHSYNC	XHSYNC	XHSYNC	XHSYNC
XPCLK	XPCLK	XPCLK	XPCLK
XDE	XDE	XDE	XDE
XPDAT[0]	GPIO-D0 / XKIN[1]		B0
XPDAT[1]	GPIO-D1 / XKIN[2]		B1
XPDAT[2]	GPIO-D6 / XKIN[4]	B0	B2
XPDAT[3]	B0	B1	B3
XPDAT[4]	B1	B2	B4
XPDAT[5]	B2	B3	B5
XPDAT[6]	B3	B4	B6
XPDAT[7]	B4	B5	B7
XPDAT[8]	GPIO-D2 / XKIN[3]		G0
XPDAT[9]	GPIO-D3 / XKOUT[3]		G1
XPDAT[10]	G0	G0	G2
XPDAT[11]	G1	G1	G3
XPDAT[12]	G2	G2	G4
XPDAT[13]	G3	G3	G5
XPDAT[14]	G4	G4	G6
XPDAT[15]	G5	G5	G7
XPDAT[16]	GPIO-D4 / XKOUT[1]		R0
XPDAT[17]	GPIO-D5 / XKOUT[2]		R1
XPDAT[18]	GPIO-D7 / XKOUT[4]	R0	R2
XPDAT[19]	R0	R1	R3
XPDAT[20]	R1	R2	R4
XPDAT[21]	R2	R3	R5
XPDAT[22]	R3	R4	R6
XPDAT[23]	R4	R5	R7

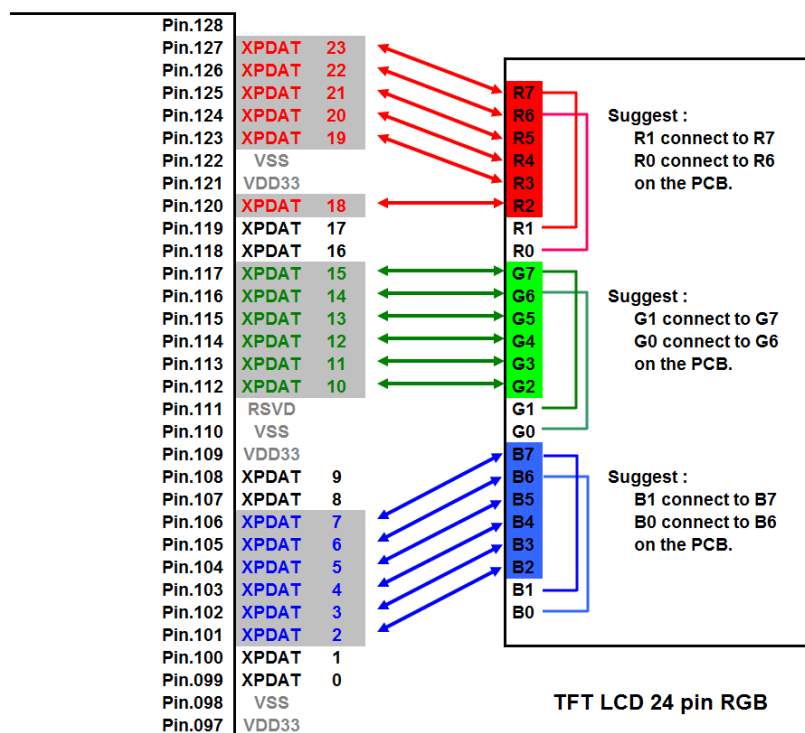
If REG[01h] bit 4-3, set 24-bits TFT output.



RA8876

Figure 10-1 : RA8876 color depth 24bit with LCD panel pin connect

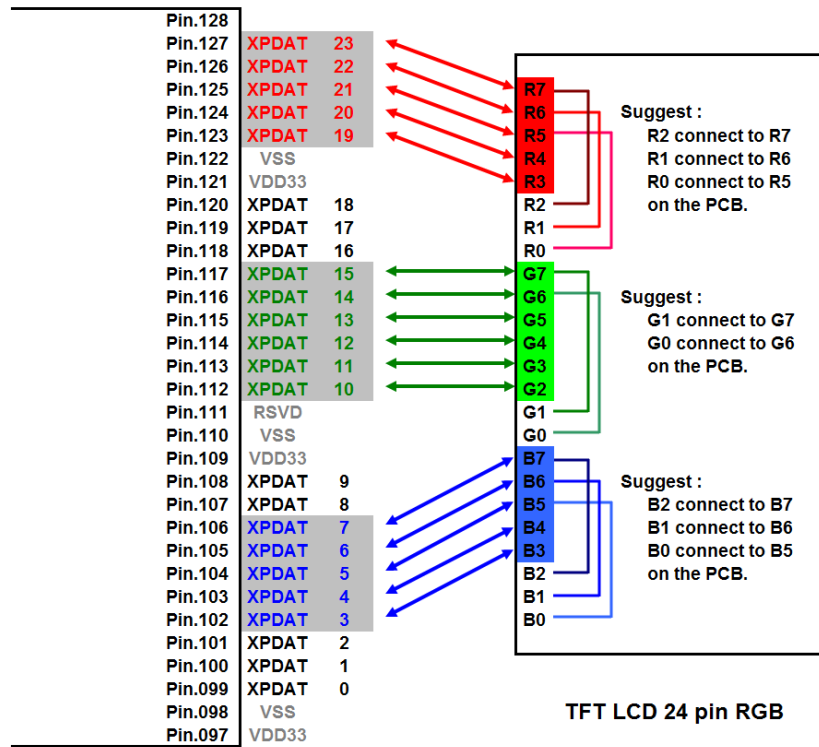
If REG[01h] bit 4-3, set 18-bits TFT output.



RA8876

Figure 10-2 : RA8876 color depth 18bit with LCD panel pin connect

If REG[01h] bit 4-3, set 16-bits TFT output.



RA8876

Figure 10-3 : RA8876 color depth 16bit with LCD panel pin connect

10.2 LCD Parallel Interface Timing

The timing parameters required to drive a flat panel display are shown below.

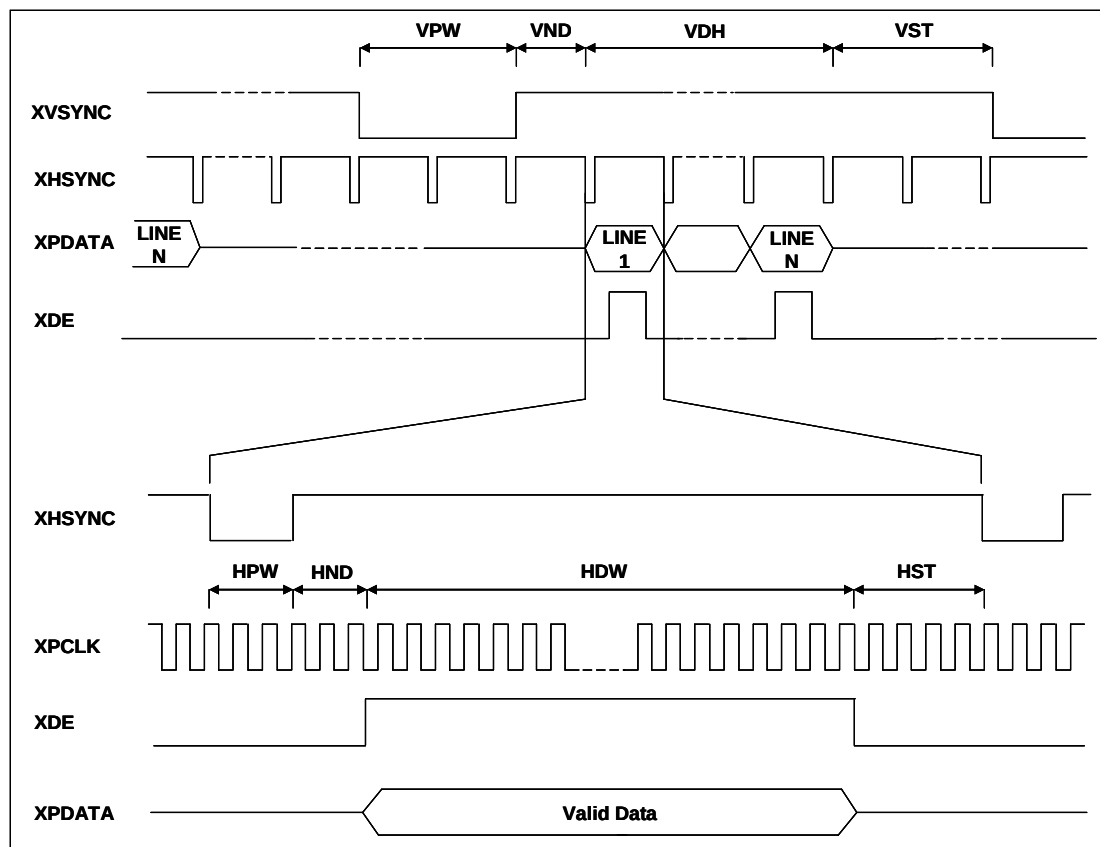


Figure 10-4 : Digital TFT Panel Timing

11. Display Function

11.1 Color Bar Display Test

The color bar display test does not use the SDRAM memory and is selected when REG[12h] bits5 = 1.

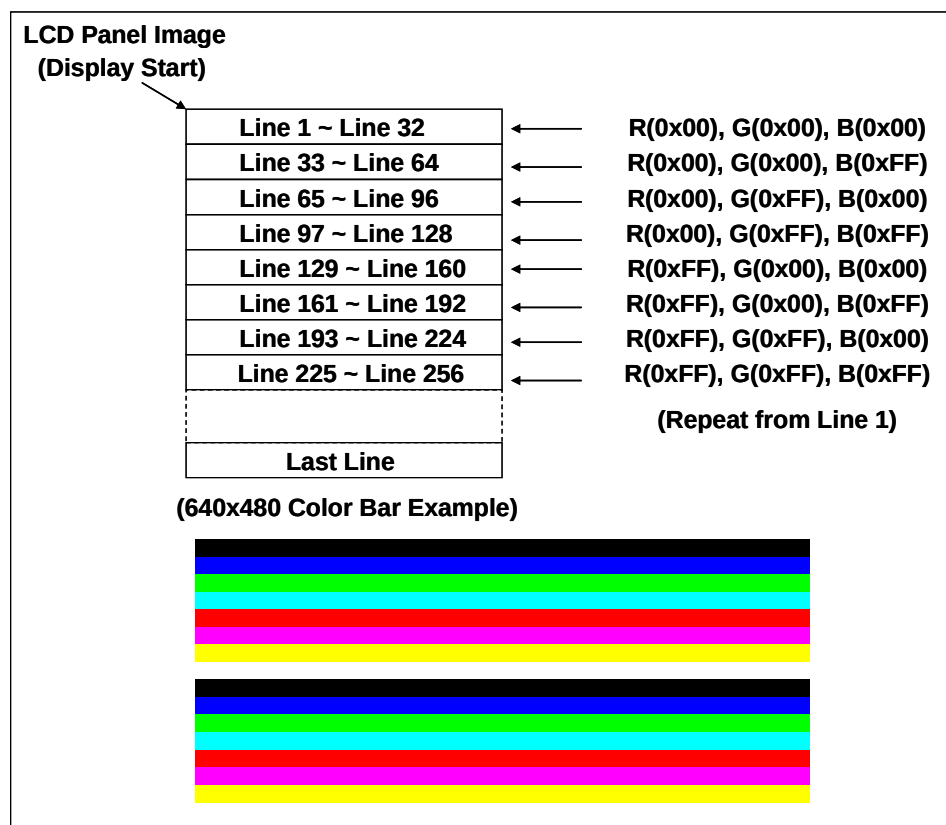


Figure 11-1: Color Bar Display Test

11.2 Main Window

Main window of the RA8876 is defined by the size of LCD panel display (refer to REG[14h] ~REG[1Fh]). The main window source can be selected from any of the available Image buffers and users just configure main window related registers (refer to REG[20h] ~REG[29h]) to display different Image buffers.

11.2.1 Configure Display Image Buffer

The SDRAM is divided into several image buffers and the maximum number of image buffers is limited by the memory size. For example : Image size is 800x600 256 color and 16 Mbits SDRAM can be divided to 4 image buffers(image width x image height x image color depth x maximum number of image buffers < SDRAM Mbytes). In order to define image size, users must configure canvas start address, canvas image width and active window range (refer to REG[50h] ~REG[5Eh]) before write data to image buffer.

11.2.2 Write Image to Display Image Buffer

Canvas is response for read/write image data to memory. Users must configure canvas start address, canvas image width (refer to REG[50h] ~REG[55h]) to decide image size and configure active window range (refer to REG[56h] ~REG[5Eh]) to write data to image buffer.

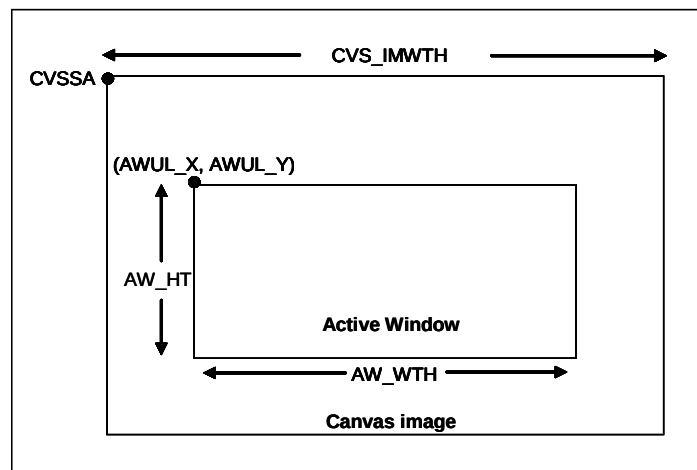


Figure 11-2

11.2.3 Display Main Window Image

Main image is in response to display image data to LCD panel. The following diagram is main window display flowchart. Users just operate step by step.

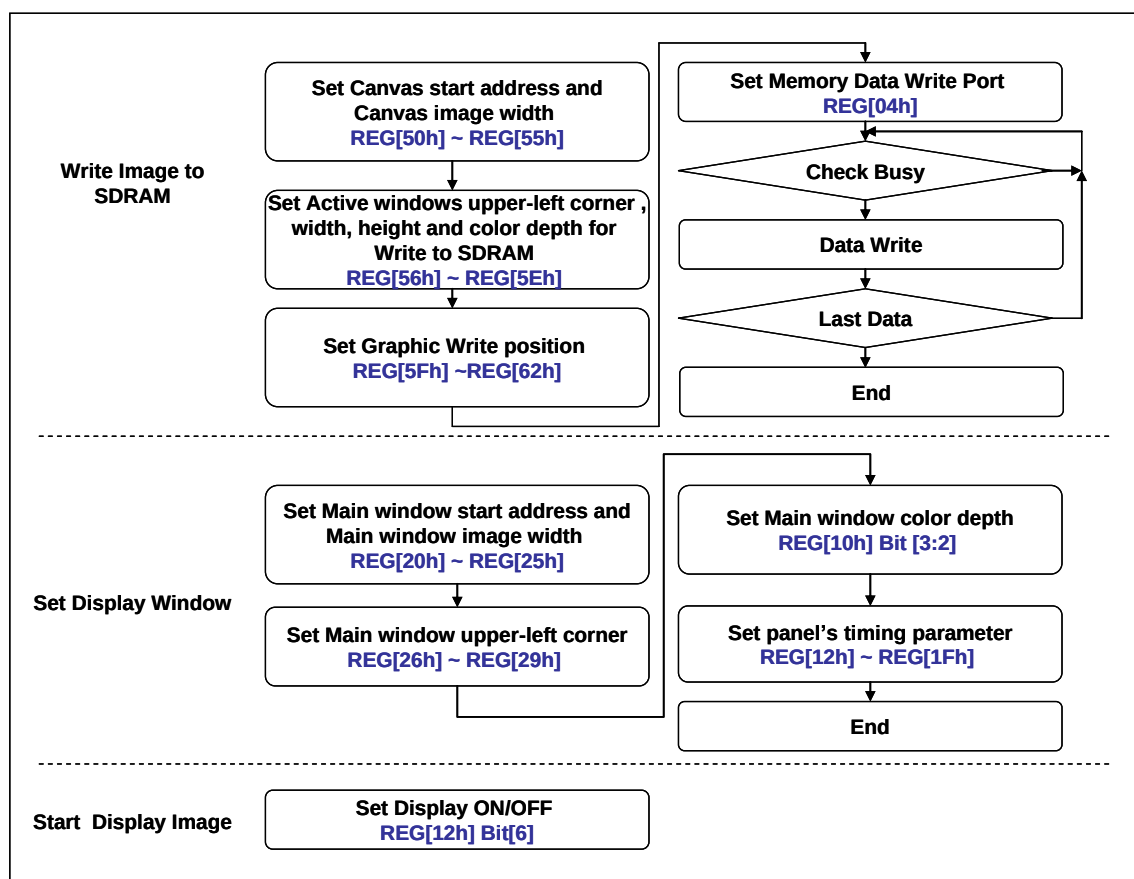


Figure 11-3

11.2.4 Switch Main Window Image

The main window source can be selected from any of the available Image buffers and users just configure main window related registers (refer to REG[20h] ~REG[29h]) to switch image buffers.

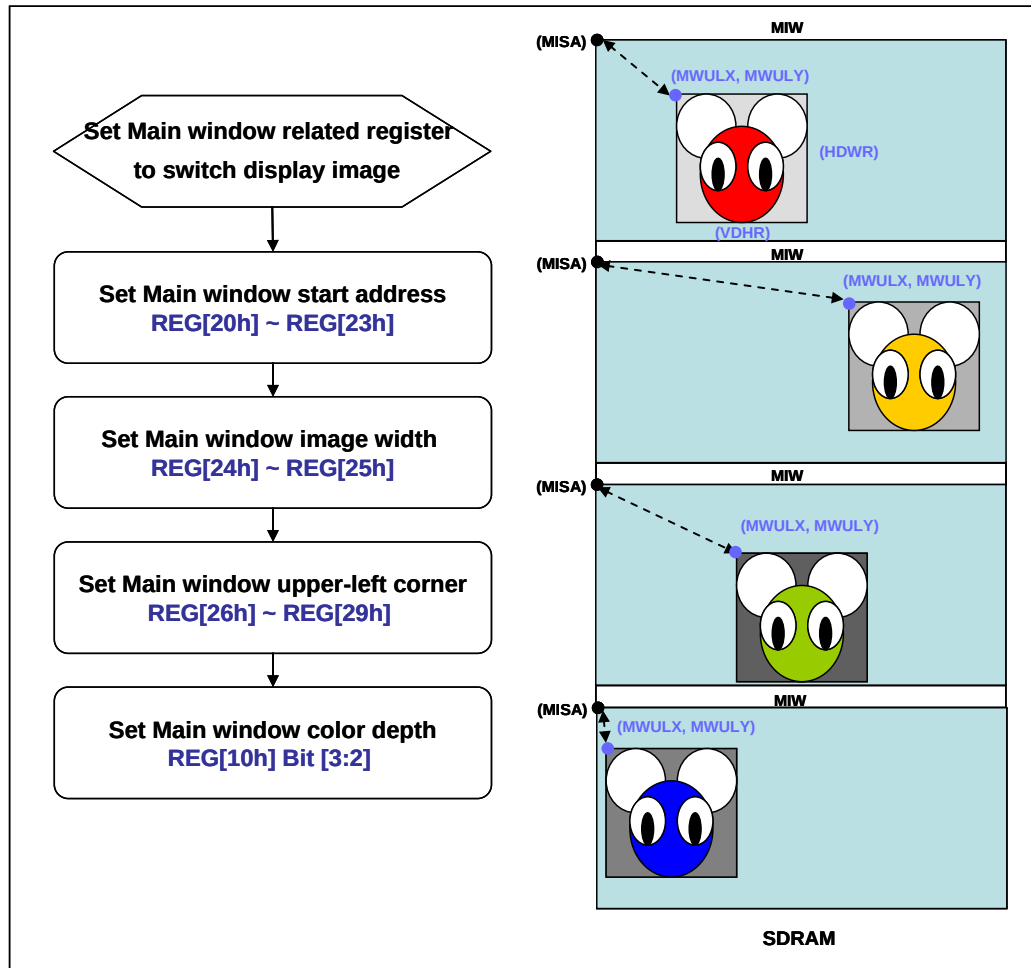


Figure 11-4

11.3 PIP Window

RA8876 supports two PIP windows that can be used with main display window. PIP windows do not support transparent overlay, it just provides users that can enable or disable without overwriting the main display window image data. If the PIP1 and PIP2 windows are overlapped, the PIP1 window is displayed over PIP2 window.

The size and position of PIP windows are specified using registers from REG[2Ah] to REG[3Bh] and REG[11h]. PIP1 and PIP2 window share same set of registers, and according REG[10h] Bit[4] to select REG [2Ah ~ 3Bh] as PIP1 or PIP2 window's parameters. Function bit will be configured for relative PIP window. PIP windows sizes and start positions are specified in 4 pixel resolution (horizontal) and 1 line resolution.

Note : When VDIR = 1 , pip window, graphic cursor and text cursor function will be disabled automatically.

11.3.1 PIP Windows Settings

A PIP window position and size is specified using PIP image start address, PIP image Width, PIP Display X/Y coordinates, PIP Image X/Y coordinates, PIP windows color depth, PIP window width and PIP window height registers. PIP1 and PIP2 window are sharing the same set of registers, and according REG[10h] Bit[4] to select REG [2Ah ~ 3Bh] as PIP1 or PIP2 window's parameters.

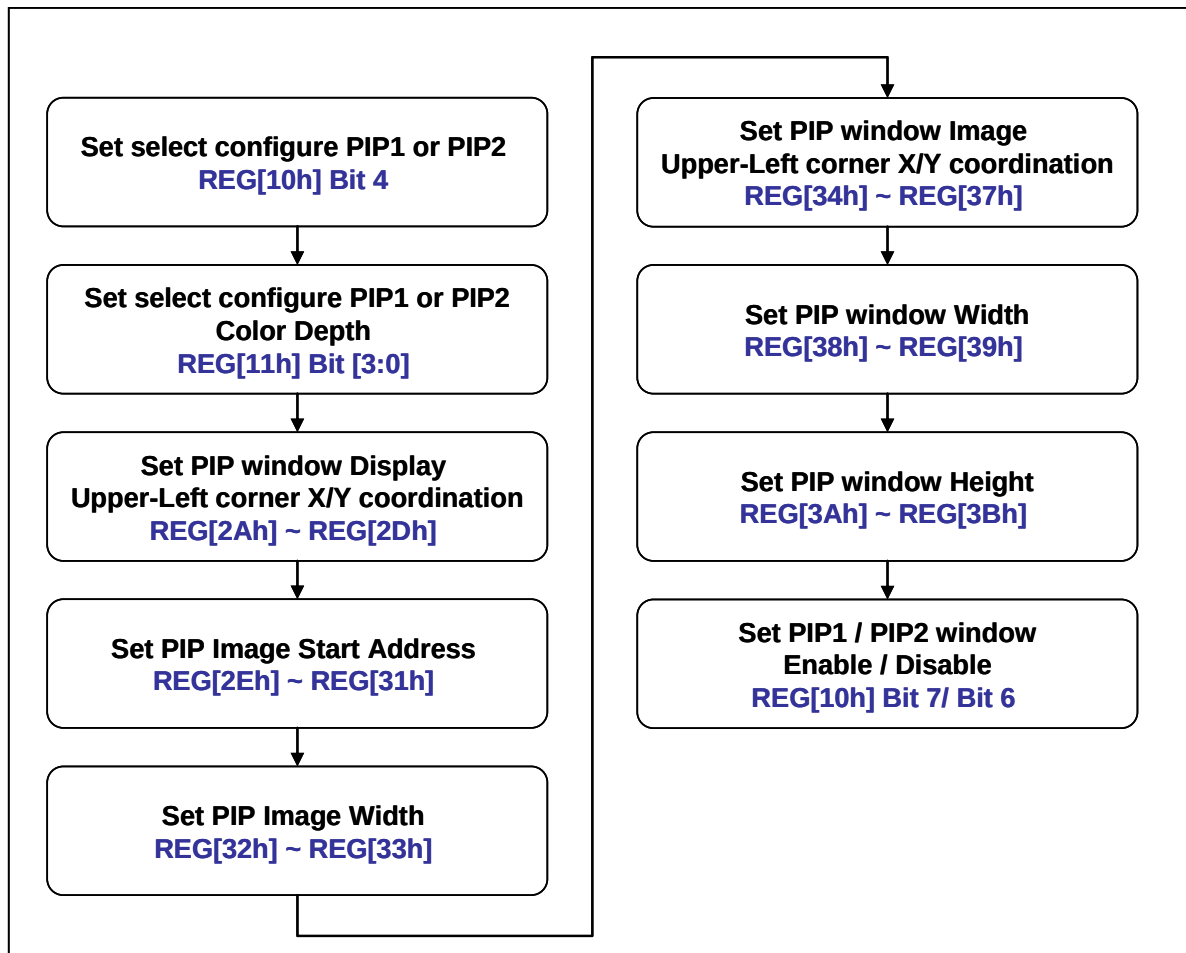


Figure 11-5

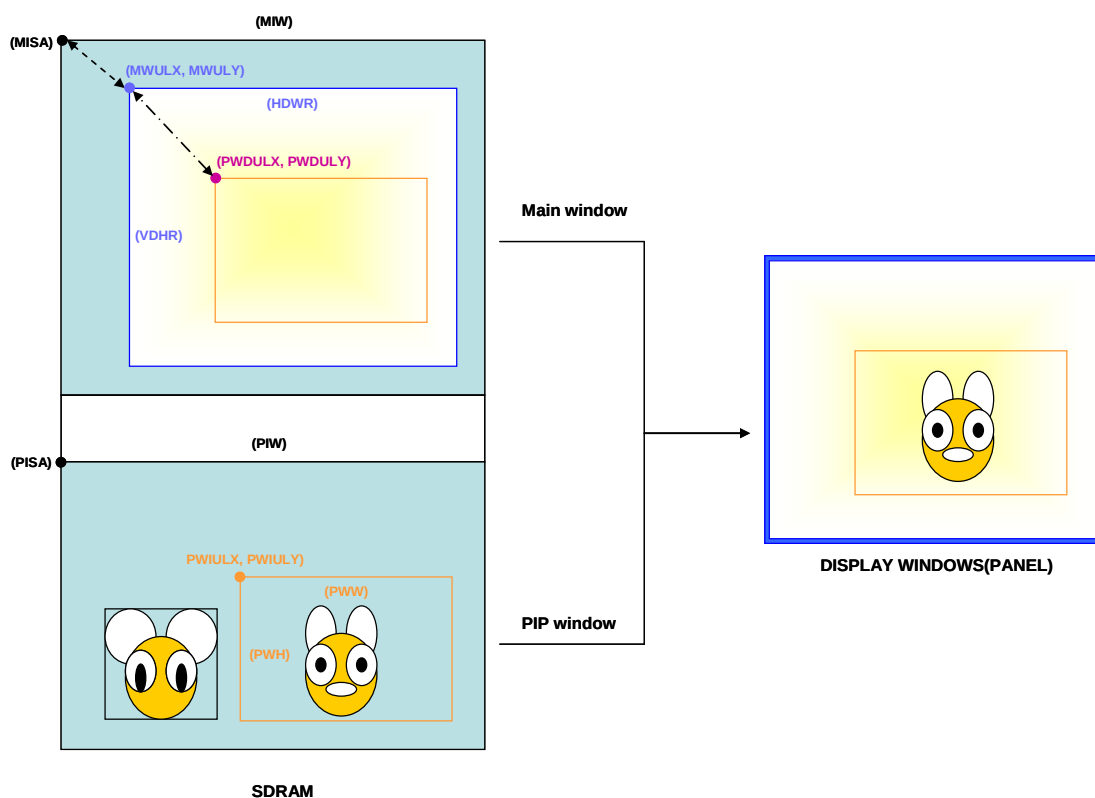


Figure 11-6

11.3.2 PIP Window Display Position and PIP Image Position

The PIP window can be changed to different display position by setting PWDULX and PWDULY. The PIP window also can be changed to different PIP image position by setting specified registers such as PISA, PIW, PWIULX and PWIULY. This method does not alter the image data in the memory, but simple changes the image data that is displayed within the PIP window.

The following example shows the main window with a single PIP window. The PIP window images are switched by changing the PIP image position.

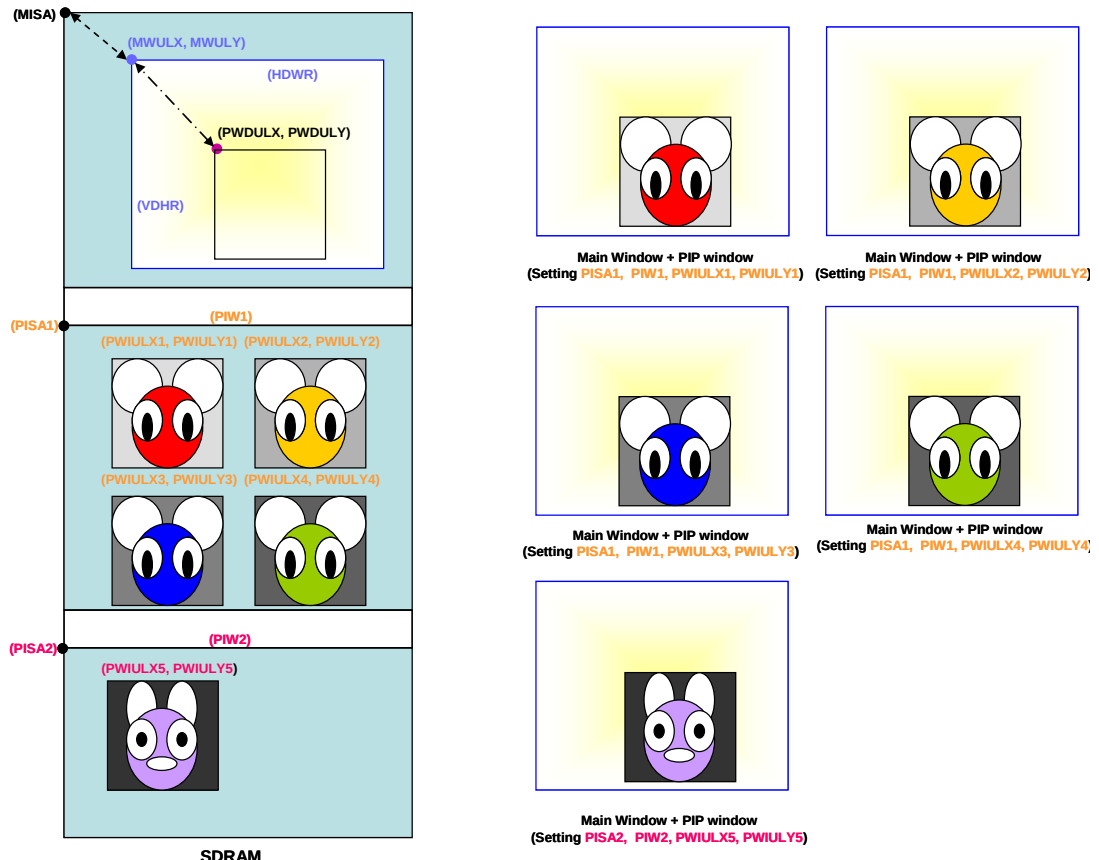


Figure 11-7

11.4 Rotate and Mirror

Most computer displays are refreshed in landscape orientation - from left to right and top to bottom. Computer images are stored in the same manner.

Rotation is designed to rotate the displayed image by 90° or 180° in a counter-clockwise direction. The rotation is done in hardware and is transparent to the user. It is accomplished by rotating the image data during display writes (see REG[02h] bit 2-1). By processing the rotation in hardware, there is a performance advantage over software rotation of the displayed image.

Mirror is designed “mirror” the displayed image from right to left. Mirror is done in the hardware and is transparent to the user. The mirror function is done during display writes (see REG[02h] bit 2-1). Mirroring the image in hardware, also offers a performance advantage over software mirroring of the same image.

Note : When VDIR = 1 , pip window, graphic cursor and text cursor function will be disabled automatically.

REG[02h] bit 2-1 provides Host Write Memory Direction control (Only for Graphic Mode)

00b: Left → Right then Top → Bottom. (Original)

01b: Right → Left then Top → Bottom. (Horizontal flip)

10b: Top → Bottom then Left → Right. (Rotate right 90° & Horizontal flip)

11b: Bottom → Top then Left → Right. (Rotate left 90°)

If accommodate with REG[12h] bit 3 (VDIR) could generate other effects.

For example, if original picture is



Figure 11-8

➔ **If VDIR (REG[12h] bit 3) == 0**

Set REG[02h]bit 2-1 as 00b, means write image data from Left to Right then Top to Bottom. It will display original picture.



Figure 11-9

Set REG[02h]bit 2-1 as 01b, means write image data from Right to Left then Top to Bottom. It will display a horizontal mirror picture.



Figure 11-10

Set REG[02h]bit 2-1 as 10b, means write image data from Top to Bottom then Left to Right. It will display a picture with rotate right 90° and horizontal mirror.



Figure 11-11

Set REG[02h]bit 2-1 as 11b, means write image data from Bottom to Top then Left to Right. It will display a picture with rotate left 90°



Figure 11-12

➔ If VDIR (REG[12h] bit 3) == 1
Set REG[02h]bit 2-1 as 00b, it will display



Figure 11-13

Set REG[02h]bit 2-1 as 01b, it will display a picture with rotate 180°



Figure 11-14

Set REG[02h]bit 2-1 as 10b, it will display a picture with rotate left 90°



Figure 11-15

Set REG[02h]bit 2-1 as 11b, it will display



Figure 11-16

12. Geometric Engine

12.1 Ellipse/Circle Input

RA8876 supports draw ellipse/circle drawing function makes user to draw ellipse/circle on the canvas and only use by few MPU cycles. By setting the center of a ellipse/circle REG[7Bh~7Eh], the major and minor radius of a ellipse REG[77h~7Ah], the color of ellipse/circle REG[D2h~D4h], the draw ellipse/circle condition REG[76h] Bit5=0 and Bit4=0, and then setting start draw REG[76h] Bit7 = 1, RA8876 will draw a corresponding ellipse/circle on the canvas. Moreover, user can fill the circle by setting REG[76h] Bit6 = 1.

Note :The center of ellipse/circle should be within active windows.
The procedure of drawing ellipse/circle just refers to the below figure:

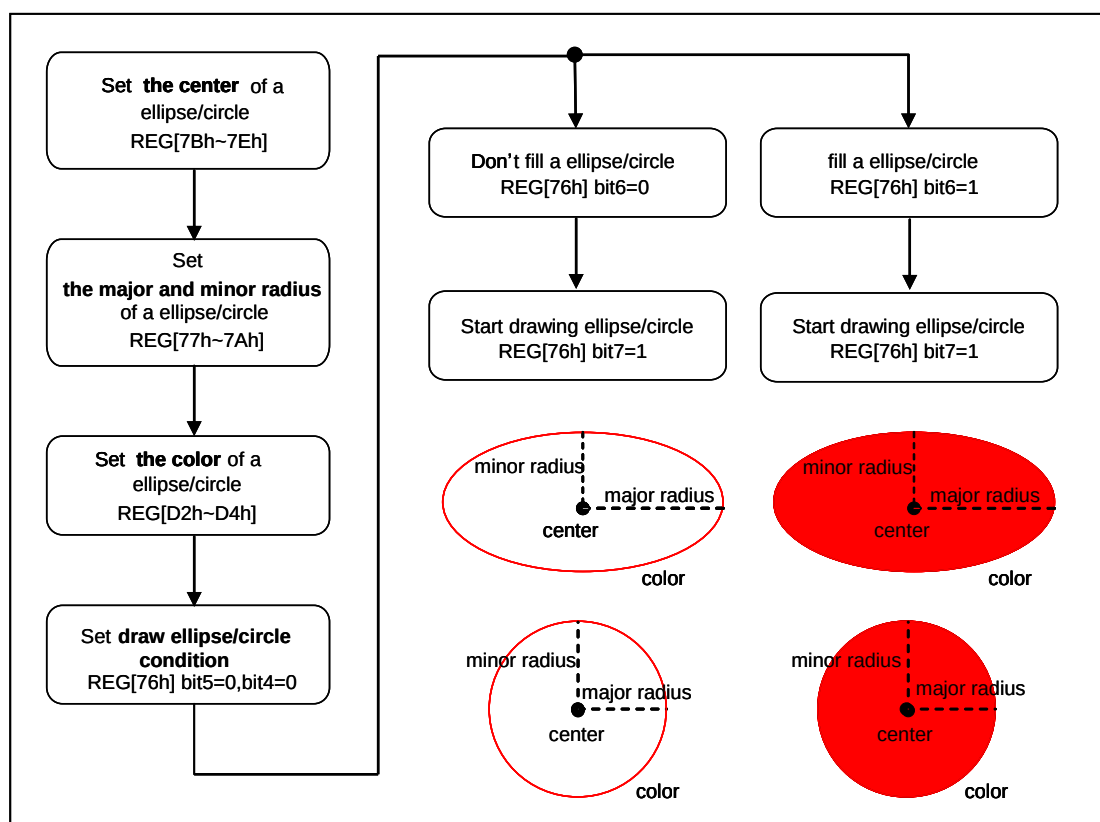


Figure 12-1

12.2 Curve Input

RA8876 supports curve drawing function for user to draw curve on the canvas and only by few MPU cycles. By setting the center of a curve REG[7Bh~7Eh], the major and minor radius of a curve REG[77h~7Ah], the color of curve REG[D2h~D4h], the draw curve condition REG[76h] Bit5=0 and Bit4=1, the curve part of the ellipse REG[76h] Bit[1:0], and then setting start draw REG[76h] Bit7 = 1, RA8876 will draw a corresponding curve on the canvas. Moreover, user can fill the curve by setting REG[76h] Bit6 = 1.

Note :the center of curve should be within active windows.
The procedure of drawing curve just refers to the below figure:

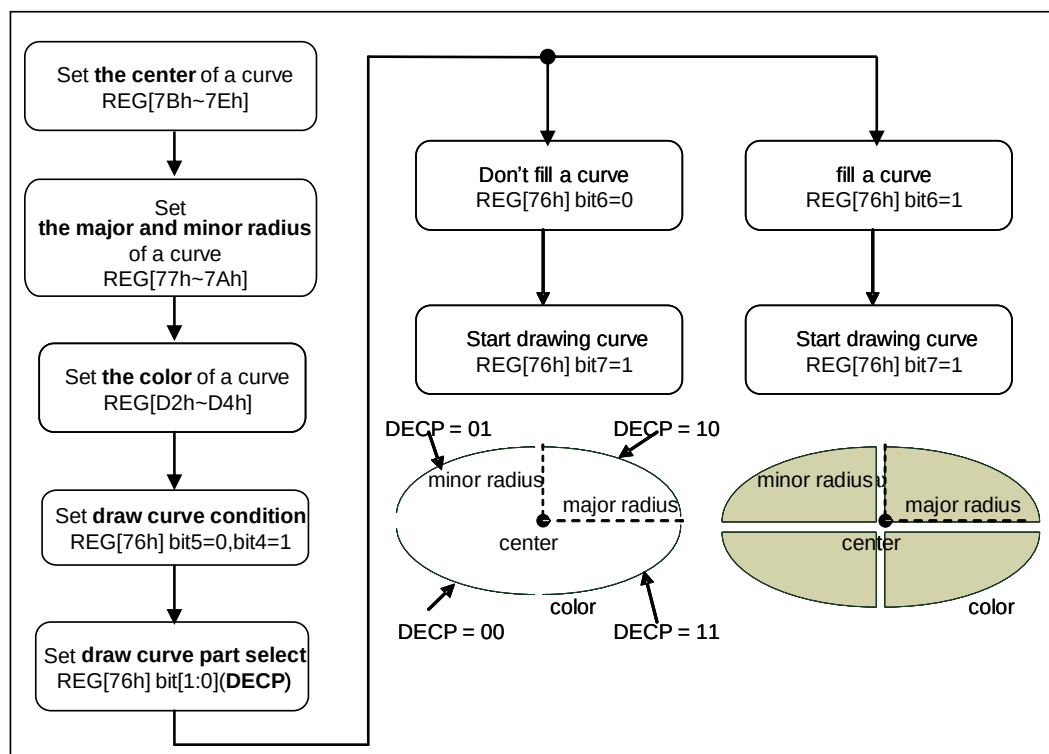


Figure 12-2

12.3 Square Input

RA8876 supports square drawing function for user to draw square on the canvas and only by few MPU cycles. By setting the start point of a square REG[68h~6Bh], the end point of a square REG[6Ch~6Fh] and the color of a square REG[D2h~D4h], then setting draw a square REG[76h] Bit4=1, Bit0=0 and start draw REG[76h] Bit7 = 1, RA8876 will draw a corresponding square on the canvas. Moreover, user can fill the square by setting REG[76h] Bit6 = 1.

Note : the start point and the end point of a square should be within active windows.
The procedure of drawing square just refers to the below figure:

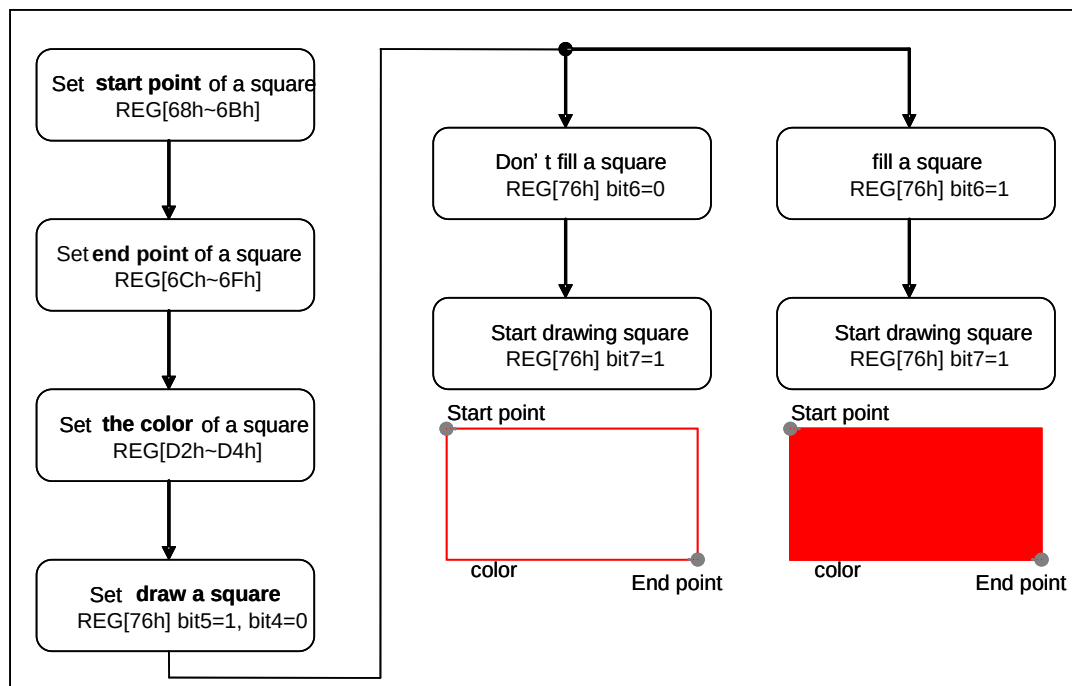


Figure 12-3 : Geometric Pattern Drawing- Draw Rectangle

12.4 Line Input

RA8876 supports line drawing function for user to draw line on the canvas and only by few MPU cycles. By setting the start point of a line REG[68h~6Bh], the end point of a line REG[6Ch~6Fh] and the color of a line REG[D2h~D4h], then setting draw a line REG[67h] Bit1 = 0 and start draw REG[67h] Bit7 = 1, RA8876 will draw a corresponding line on the canvas.

Note : the start point and the end point of line should be within active windows.
The procedure of drawing line just refers to the below figure:

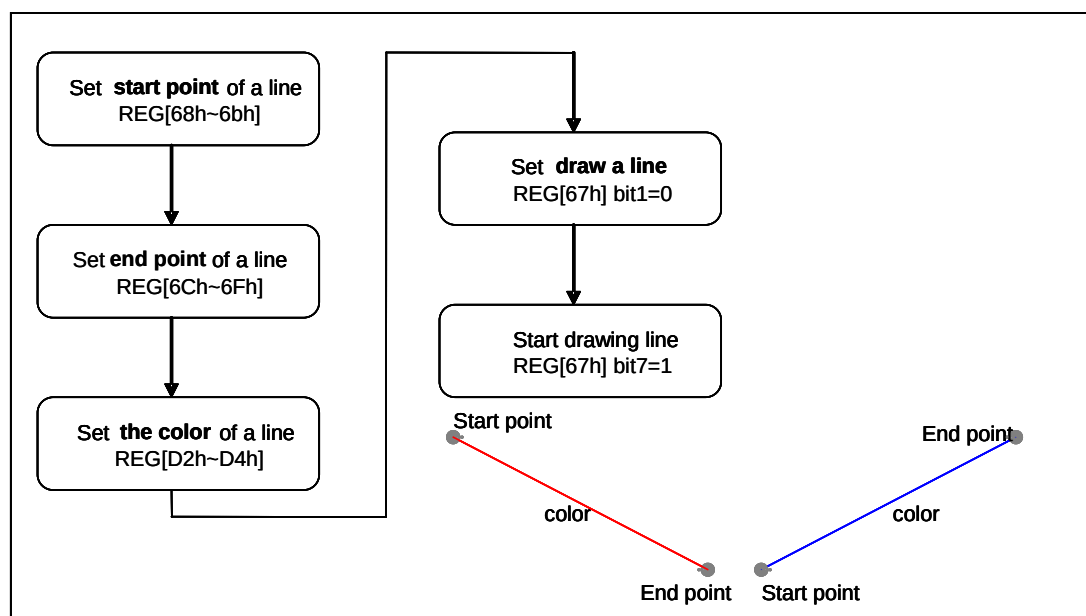


Figure 12-4 : Geometric Pattern Drawing- Draw Line

12.5 Triangle Input

RA8876 supports triangle drawing function for user to draw line on the canvas and only by few MPU cycles. By setting the point 1 of a triangle REG[68h~6Bh], the point 2 of a triangle REG[6Ch~6Fh], the point 3 of a triangle REG[70h~73h] and the color of a triangle REG[D2h~D4h], then setting draw a triangle REG[67h] Bit1 = 1 and start draw REG[67h] Bit7 = 1, RA8876 will draw a corresponding triangle on the canvas. Moreover, user can fill the triangle by setting REG[67h] Bit5 = 1.

Note : the point 1, point 2 and point 3 of triangle should be within active windows.
The procedure of drawing triangle just refers to the below figure:

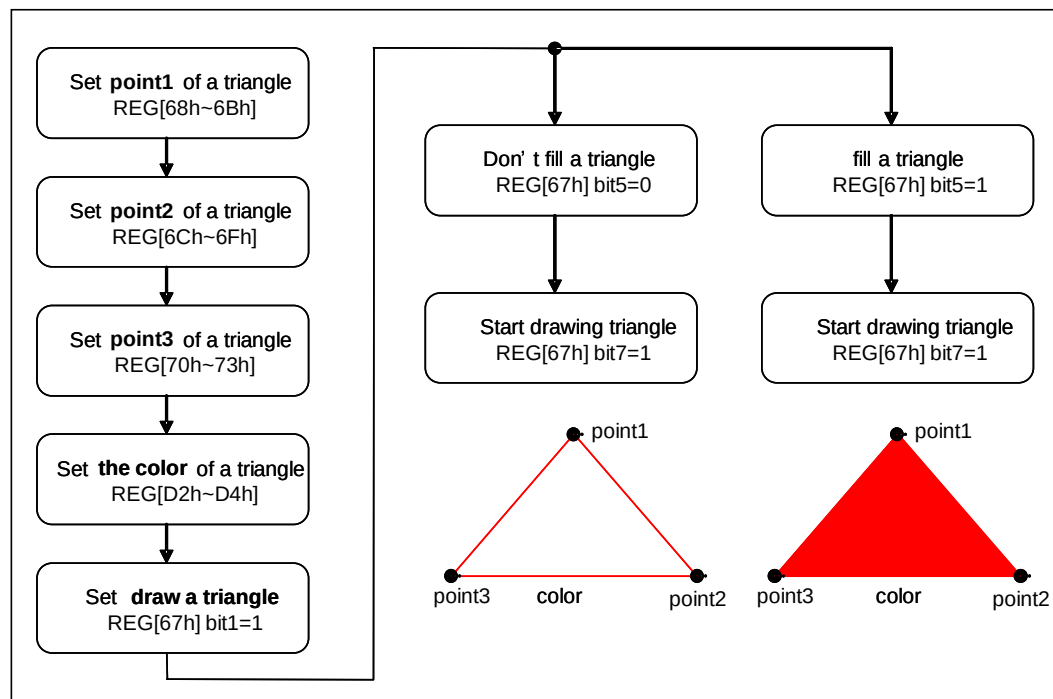


Figure 12-5 : Geometric Pattern Drawing- Draw Triangle

12.6 Square Of Circle Corner Input

RA8876 supports circle-square drawing function for user to draw circle square on the canvas and only use by few MPU cycles. By setting the start point of a square REG[68h~6Bh], the end point of a square REG[6Ch~6Fh], the major and minor radius of a ellipse/circle REG[77h~7Ah] and the color of a circle square REG[D2h~D4h], then setting draw a circle square REG[76h] Bit5=1, Bit4=1 and start draw REG[76h] Bit7 = 1, RA8876 will draw a corresponding circle square on the canvas. Moreover, user can fill the square by setting REG[76h] Bit6 = 1.

Note 1 : (End point X – Start point X) must large than (2*major radius + 1) and (End point Y – Start point Y) must large than (2*minor radius + 1)

Note 2 : the start point and the end point of a square should be within active windows.

The procedure of drawing square just refers to the below figure:

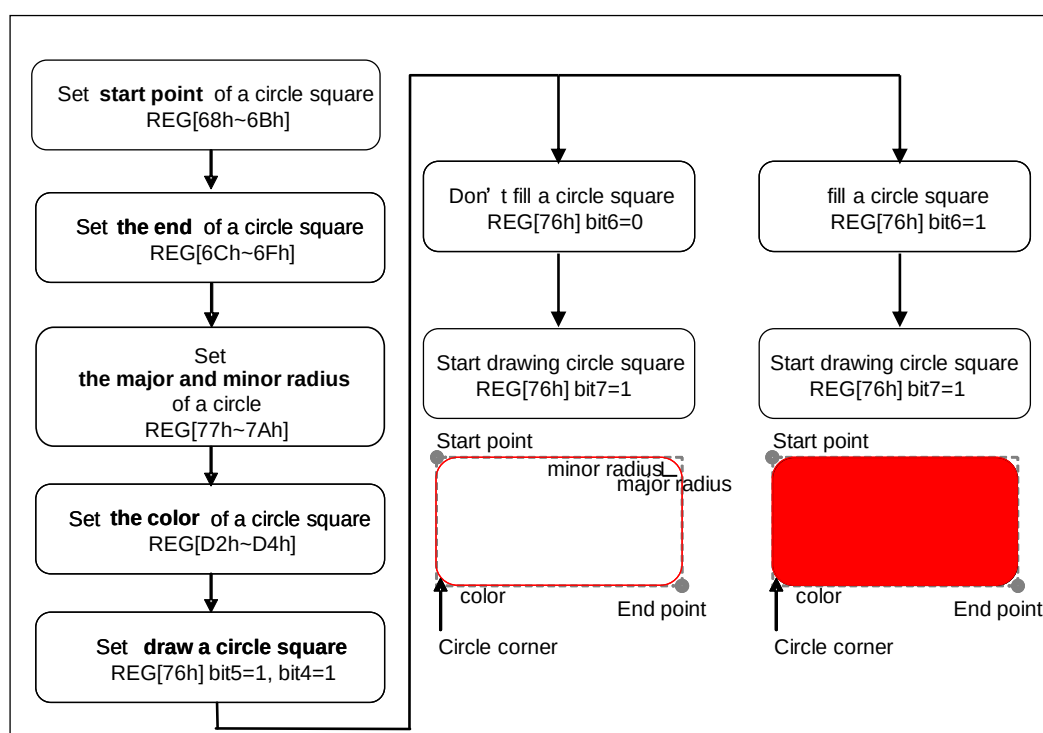


Figure 12-6 : Geometric Pattern Drawing- Draw Circle-Square

13. Block Transfer Engine (BTE)

The RA8876 embedded a built-in 2D Block Transfer Engine (BTE) which can increase the performance of block transfer operation. When a block of data needs to be moved or do some logic operation with dedicated data, RA8876 can speed up the operation by BTE hardware and also simplify the MPU program. This section will discuss the BTE engine operation and functionality.

Before using the BTE function, user must select the corresponding BTE operation. RA8876 supports 13 BTE operations. About the operation description, please refer to Table 13-1. For each BTE operation which supported ROP function, maximum 16 raster operations (ROP) are supported for different application. They could provide the different logic combinations for ROP source and ROP destination. Through the combination of the BTE operation and ROP, user can achieve many useful application operations. Please refer to the behind chapters for detail description.

User can use this function by hardware interrupt or software check busy to get BTE process status. If user checks BTE process status by software, the BTE_CTRL0(REG[90h]) Bit4 or status register (STSR) Bit3 can indicate the BTE status. By another way, user can check BTE process status by hardware interrupt, the INT# must connect to MPU and REG[0Ch] is used to check the interrupt source comes from BTE when INT# is active.

Table 13-1 : BTE Operation Function

BTE Operation REG[91h] Bits [3:0]	BTE Operation
0000b	MPU Write with ROP.
0010b	Memory Copy (move) with ROP.
0100b	MPU Write with chroma keying (w/o ROP)
0101b	Memory Copy (move) with chroma keying (w/o ROP)
0110b	Pattern Fill with ROP
0111b	Pattern Fill with chroma keying (w/o ROP)
1000b	MPU Write with Color Expansion (w/o ROP)
1001b	MPU Write with Color Expansion and chroma keying (w/o ROP)
1010b	Memory Copy with opacity (w/o ROP)
1011b	MPU Write with opacity (w/o ROP)
1100b	Solid Fill (w/o ROP)
1110b	Memory Copy with Color Expansion (w/o ROP)
1111b	Memory Copy with Color Expansion and chroma keying (w/o ROP)
Other combinations	Reserved

Table 13-2 : ROP Function

ROP Bits REG[91h] Bit[7:4]	Boolean Function Operation
0000b	0 (Blackness)
0001b	$\sim S0 \cdot \sim S1$ or $\sim (S0 + S1)$
0010b	$\sim S0 \cdot S1$
0011b	$\sim S0$
0100b	$S0 \cdot \sim S1$
0101b	$\sim S1$
0110b	$S0 \wedge S1$
0111b	$\sim S0 + \sim S1$ or $\sim (S0 \cdot S1)$
1000b	$S0 \cdot S1$
1001b	$\sim (S0 \wedge S1)$
1010b	$S1$
1011b	$\sim S0 + S1$
1100b	$S0$
1101b	$S0 + \sim S1$
1110b	$S0 + S1$
1111b	1 (Whiteness)

Note:

1. ROP Function S0: Source 0 Data, S1: Source 1 Data, D: Destination Data.
2. For pattern fill functions, the source data indicates the pattern data.

Example:

- If ROP function setting Ch, then Destination Data = Source 0 Data
 If ROP function setting Eh, then Destination Data = $S0 + S1$
 If ROP function setting 2h, then Destination Data = $\sim S0 \cdot S1$
 If ROP function setting Ah, then Destination Data = Source 1 Data

Table 13-3 : Color Expansion Function

ROP Bits REG[91h] Bit[7:4]	Start Bit Position for Color Expansion BTE operation code = 1000/1001/1110/1111	
	16-bit MPU Interface	8-bit MPU Interface
0000b	Bit0	Bit0
0001b	Bit1	Bit1
0010b	Bit2	Bit2
0011b	Bit3	Bit3
0100b	Bit4	Bit4
0101b	Bit5	Bit5
0110b	Bit6	Bit6
0111b	Bit7	Bit7
1000b	Bit8	Invalid
1001b	Bit9	Invalid
1010b	Bit10	Invalid
1011b	Bit11	Invalid
1100b	Bit12	Invalid
1101b	Bit13	Invalid
1110b	Bit14	Invalid
1111b	Bit15	Invalid

13.1 Select BTE Start Point Address and Layer

The ROP Source 0/Source 1/Destination could come from the any selected memory address. To program the ROP source or ROP destination, the start point of horizontal and vertical address is set first.

1. The source 0 address set registers are REG [93h], REG[94h], REG[95h], REG[96h], REG[97h], REG [98h], REG[99h], REG[9Ah], REG[9Bh], REG[9Ch]
2. The source 1 address set registers are REG [9Dh], REG[9Eh], REG[9Fh], REG[A0h], REG [A1h], REG[A2h], REG[A3h], REG[A4h], REG[A5h], REG[A6h]
3. The destination address set registers are REG [A7h], REG[A8h], REG[A9h], REG[AAh], REG [ABh], REG[ACH], REG[ADh], REG[A Eh], REG[AFh], REG[B0h]

13.2 Color Palette RAM

The RA8876 includes a color palette memory for 8-bit alpha blend data function. Index color is color palette RAM address to pick up real color (reference Figure 13-1). The block diagram in Figure 13-2.If user will be set this memory data. then user reference Figure 13-3.

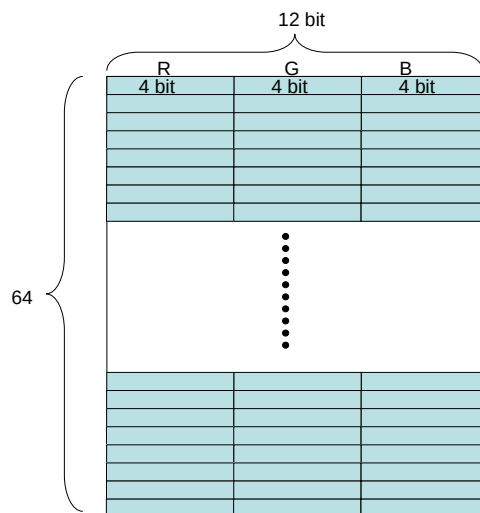


Figure 13-1 : Palette Ram Diagram

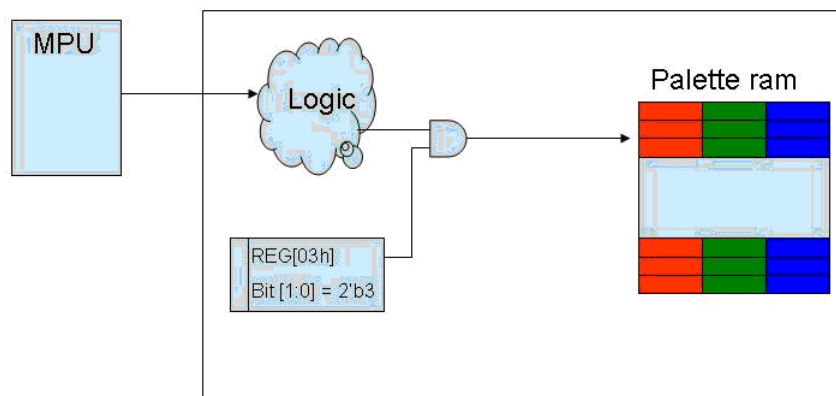


Figure 13-2 : Palette Ram Initial Data Path

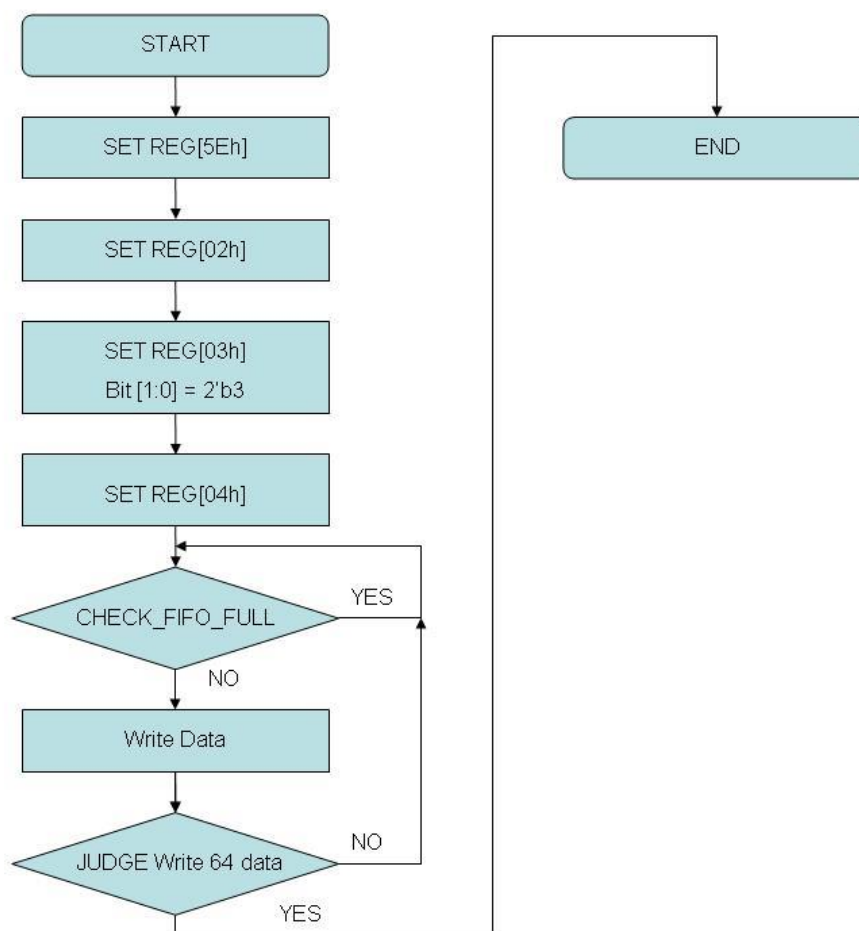


Figure 13-3 : Palette Ram Initial Flow

13.3 BTE Operations

13.3.1 MPU Write with ROP

This operation provides 16 ROP functions, where BTE engine will write the result of ROP function to the destination address.

13.3.2 Memory Copy with ROP

This operation provides 16 ROP functions, and only supports positive direction.

13.3.3 Solid Fill

This operation fills a specified BTE area (destination) with a solid color as defined in the BTE Foreground Color Register.

13.3.4 Pattern Fill

This operation fills a specified BTE area with an 8x8 / 16x16 pixel pattern.

13.3.5 Pattern Fill with Chroma Key

This operation function fills a specified BTE area with an 8x8/16x16 pixel pattern. When the pattern color is equal to the key color, which is defined in BTE Background Color Register, then the destination area is not updated. For the function no raster operation is applied.

13.3.6 MPU Write with Chomra Key

This operation supports data transfers from the host to SDRAM ram area. When the source0(S0) color is equal to the key color, which is defined in BTE Background Color Register, the destination area is not updated. The ROP (raster operation) is not supported in this function.

13.3.7 Memory Copy with Chroma Key

This operation supports data transfer in positive direction only. The source and destination data is different area of the same SDRAM. When the source0(S0) color is equal to key color, which is defined in BTE Background Color Register, the destination area is not updated. For this BTE no raster operation is applied.

13.3.8 Color Expansion

This operation expands the host's monochrome data to 8/16/24 bpp color format.

The source data "1" will expand to the color defined in the BTE Foreground Color Register.

The source data "0" will expand to the color defined in the BTE Background Color Register.

If background transparency is enabled, then the destination color will remain untouched.

Note: No matter background transparency is enabled or not, don't set the same value in foreground color register (D2h~D4h) and background color register (D5h~D7h)

13.3.9 Memory Copy with Color Expansion

This operation expands source's monochrome data to 8/16/24 bpp color format. The source data "1" will expand BTE Foreground Color to the SDRAM. The source data is "0" then expands BTE Background Color to SDRAM. If background transparency is enabled, then the destination data will remain untouched.

Note: No matter background transparency is enabled or not, don't set the same value in foreground color register (D2h~D4h) and background color register (D5h~D7h)

13.3.10 Memory Copy with Opacity

This operation deal source 0 and source 1 data to blend and write it back to destination.

The Alpha Blending has 2 mode for use -- Picture and Pixel mode.

Picture Mode is BTE area have the same alpha blending level. The level value form register.

Pixel Mode can pixel by pixel setting different alpha blending. The level form part of pixel data.

Source 0 : comes from SDRAM

Source 1 : comes from SDRAM

Destination : comes from SDRAM

13.3.11 MPU Write with Opacity

This operation deal source 0 and source 1 data to blend and write it back to destination.

The Alpha Blending has 2 mode for use -- Picture and Pixel mode.

Picture Mode is BTE area have the same opacity level. The level value form register.

Pixel Mode can pixel by pixel setting different opacity. The level form part of pixel data.

Source 0 : comes from MPU

Source 1 : comes from SDRAM

Destination : comes from SDRAM

13.4 BTE Access Memory Method

With the setting, The BTE memory source/destination data is treated as a block of display area. The below example shows source 0 / source 1 / destination address are defined as block access method:

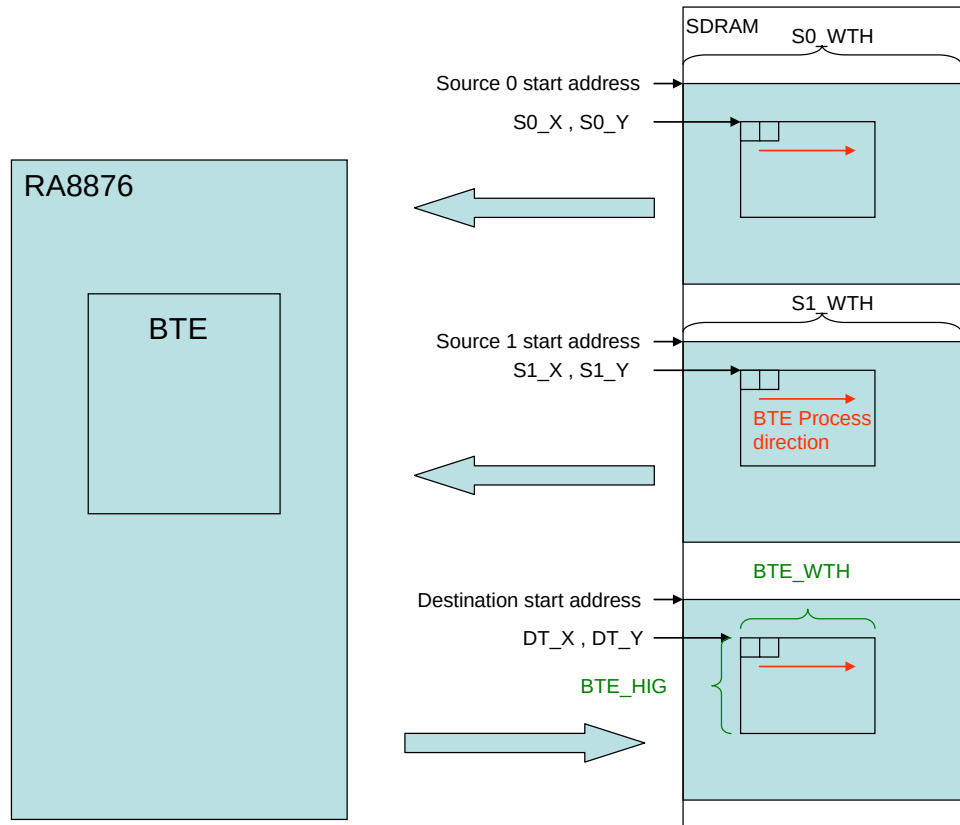


Figure 13-4 : Memory Access of BTE Function

13.5 BTE Chroma Key (Transparency Color) Compare

In BTE Chroma Key (Transparency color) function Enable, BTE process compare source 0 data and background color register data. If data equal then not change destination data otherwise write source 0 data to destination.

In source color depth = 256 color,
 Source 0 red color just only compare REG[D5h] Bit [7:5],
 Source 0 green color just only compare REG [D6h] Bit [7:5],
 Source 0 blue color just only compare REG [D7h] Bit [7:6]

In source color depth = 65k color,
 Source 0 red color just only compare REG [D5h] Bit [7:3],
 Source 0 green color just only compare REG [D6h] Bit [7:2],
 Source 0 blue color just only compare REG [D7h] Bit [7:3]

In source color depth = 16.7M color,
 Source 0 red color just only compare REG[D5h] Bit [7:0],
 Source 0 green color just only compare REG [D6h] Bit [7:0],
 Source 0 blue color just only compare REG [D7h] Bit [7:0]

13.6 BTE Function Explanation

13.6.1 Write BTE with ROP

The Write BTE increases the speed of transferring data from MPU interface to the SDRAM. The Write BTE with ROP fills a specified area of the SDRAM with data supplied by the MPU. The Write BTE supports all 16 ROPs. The Write BTE requires the MPU to provide data.

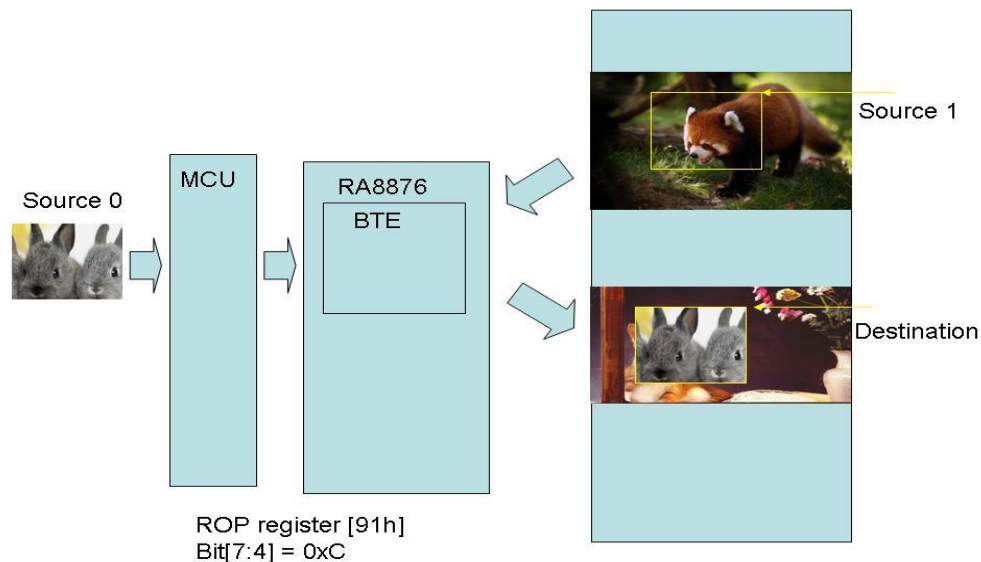


Figure 13-5 : Hardware Data Flow

The suggested programming steps and registers setting are listed below as reference.

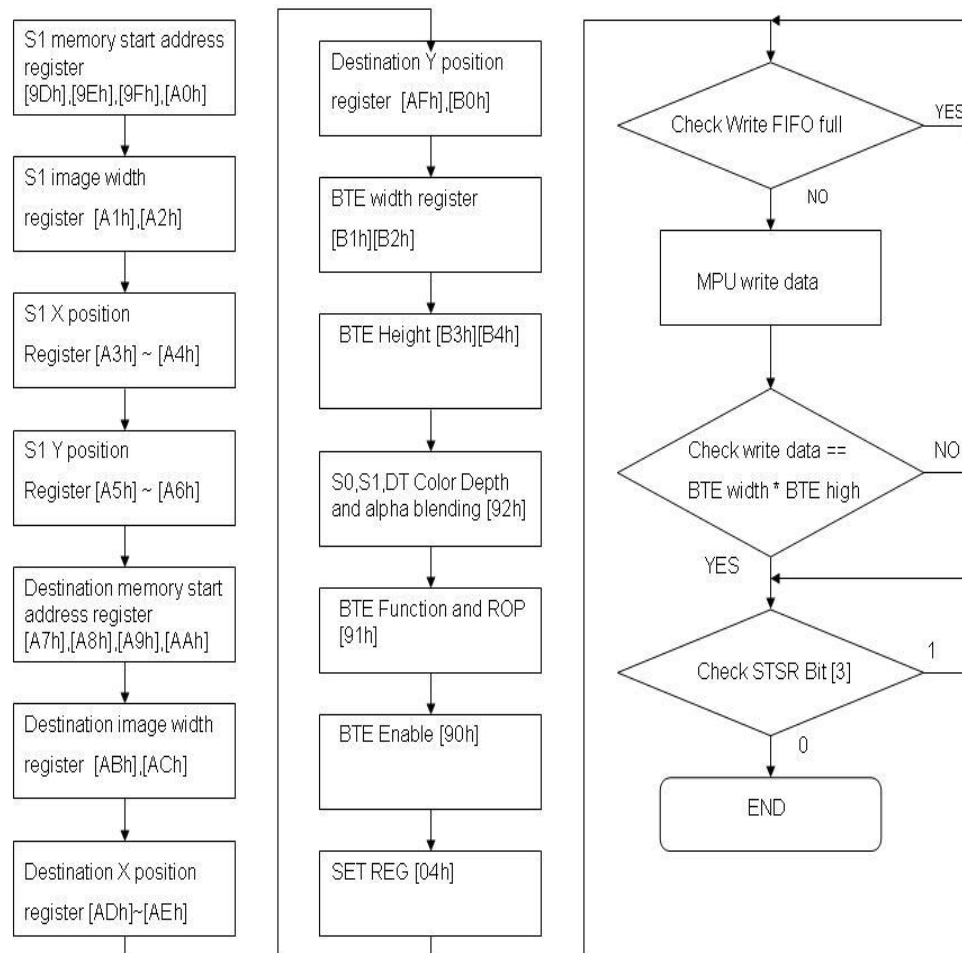


Figure 13-6 : Flow Chart

13.6.2 Memory Copy (move) BTE with ROP

This function will copy a specific area of the SDRAM to a different area of the SDRAM. This operation can speed up the data copy operation from one block to another and save a lot of MPU processing time and loading.

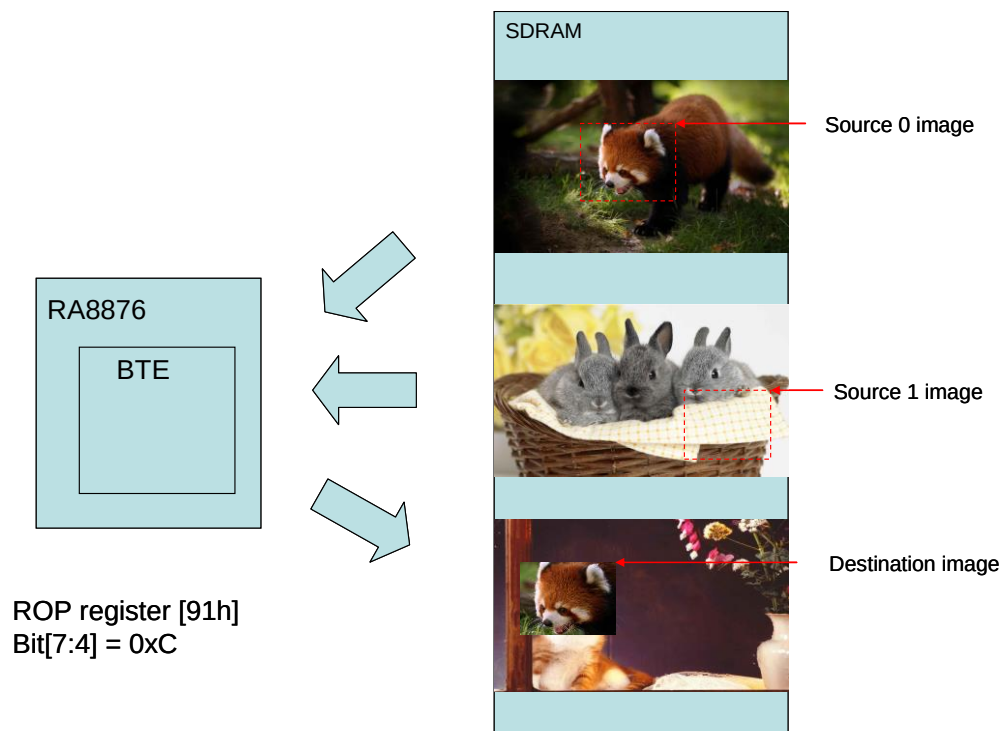


Figure 13-7 : Hardware Data Flow

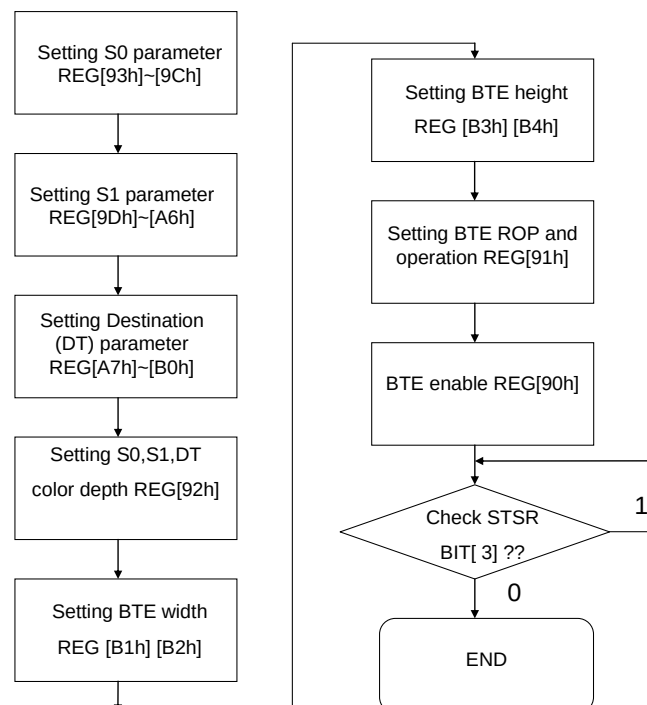


Figure 13-8 : Flow Chart

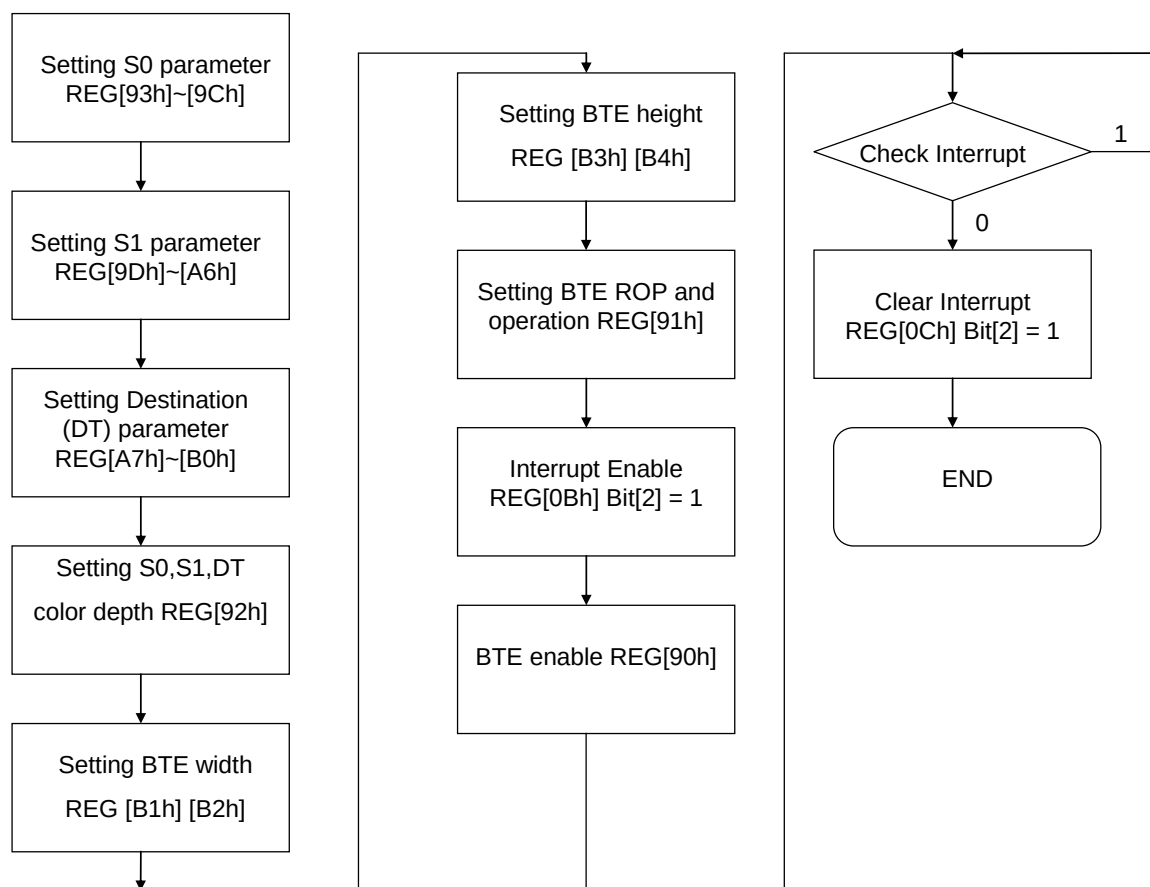


Figure 13-9 : Flow Chart – Check Int

13.6.3 MPU Write W/ Chroma Key (w/o ROP)

The MPU Write w/ chroma key function. It increases the speed of transferring data from MPU interface to the SDRAM. Once the function begins, the BTE engine remains active until all pixels have been written.

Unlike “Write BTE” operation, the “MPU Write with chroma key” will ignore the operation of a dedicated color of MPU data that is set as “Chroma key Color (Transparent Color)”. In RA8876, the “Chroma key color (Transparent Color)” is set as “BTE background Color”. For example, considering a source image has a yellow circle on a red background. By selecting the red color as the transparent color and using the MPU Write with chroma key on the whole rectangles, the effect is a BTE of the yellow circle only.

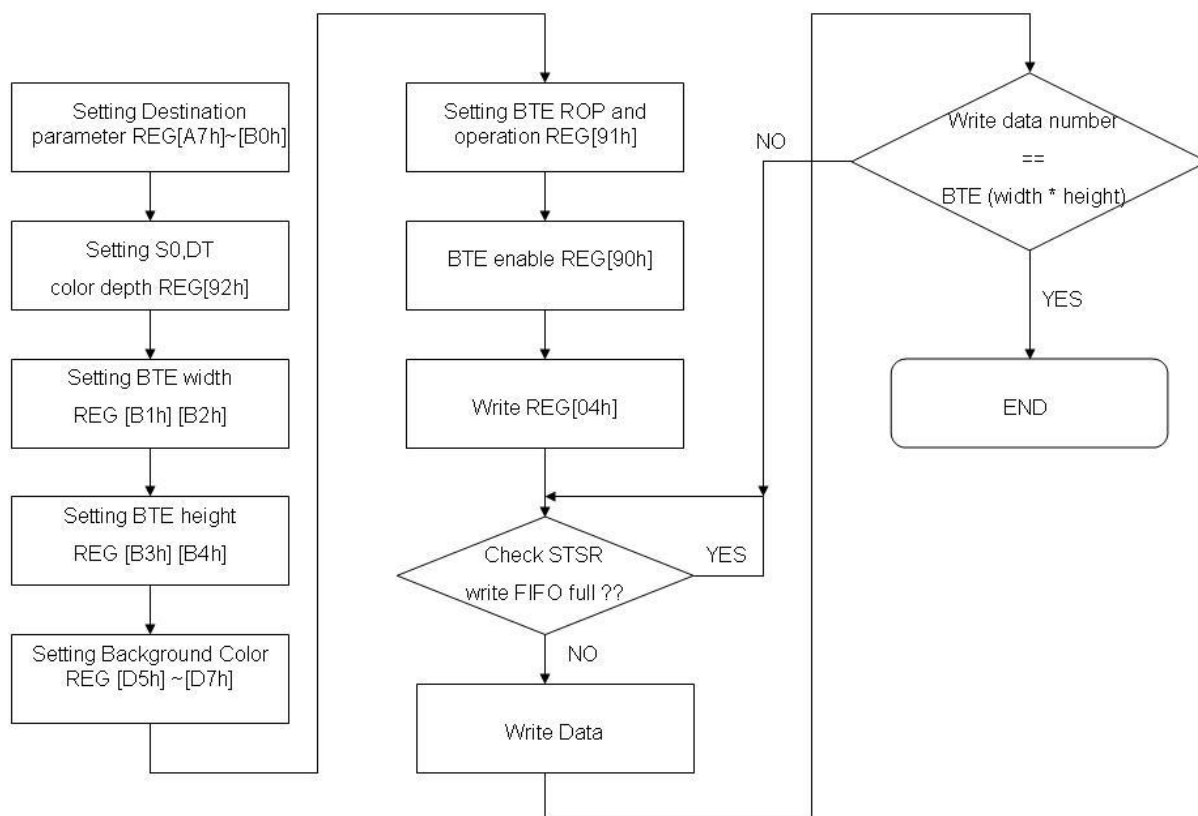


Figure 13-10 : Flow Chart

Chroma Key –
Background register
[D5h]~[D7h] = Red

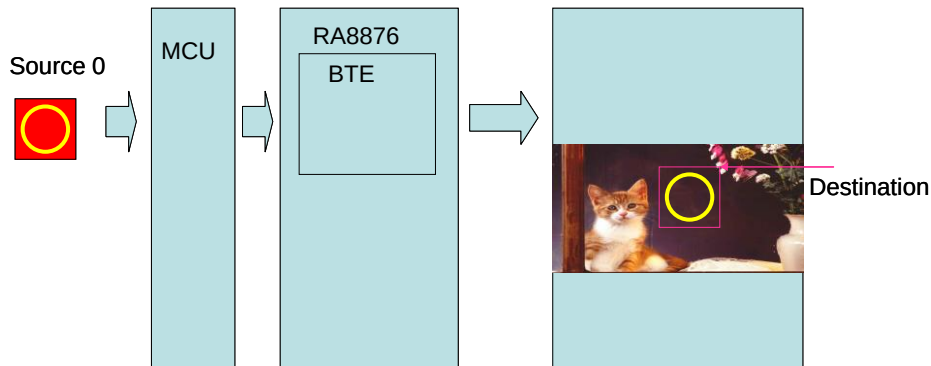


Figure 13-11 : Hardware Data Flow

13.6.4 Memory Copy W/ Chroma Key (w/o ROP)

“Memory Copy w/ chroma key” moves a specified area of the SDRAM to the different specified area of the same SDRAM with ignoring the “chroma key (Transparent Color)”. The chroma key (transparency color) setting in BTE background color register. If chroma key (transparency color) meets, then not change destination data. The source 0, destination data are in the memory.

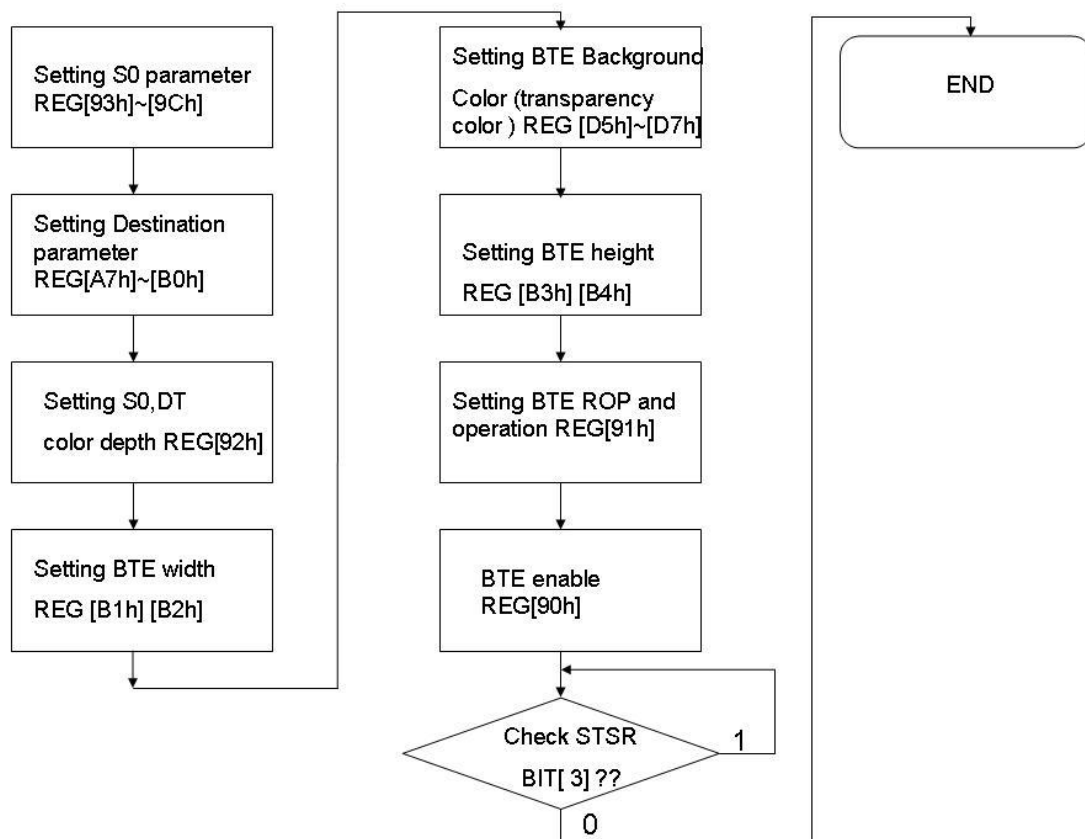


Figure 13-12 : Flow Chart

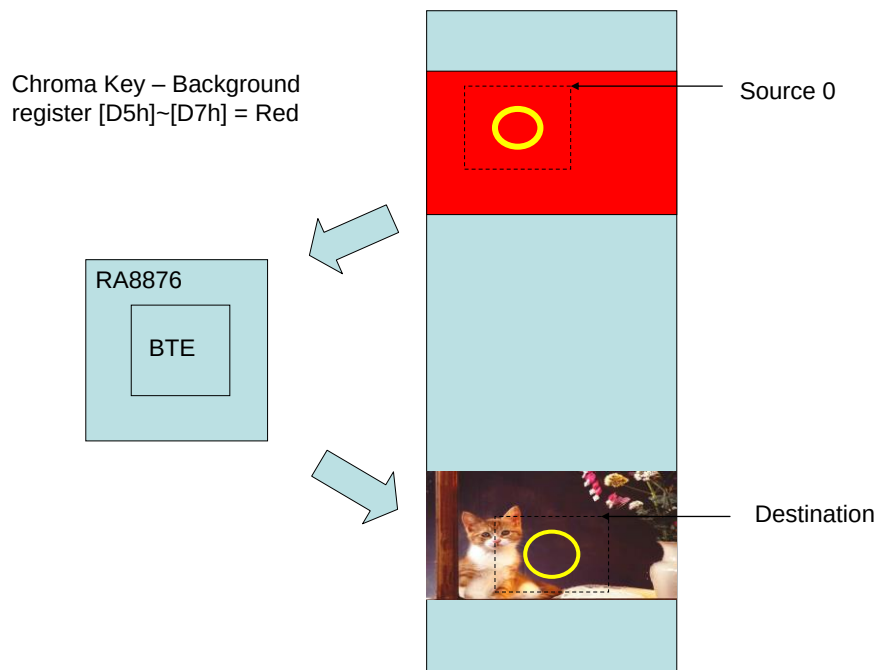


Figure 13-13 : Hardware Data Flow

13.6.5 Pattern Fill with ROP

“Pattern Fill with ROP” operation fills a specified rectangular area of the SDRAM with a dedicated pattern repeatedly. The fill pattern is an array of 8x8/16x16 pixels stored in the SDRAM. The pattern can be logically combined with the destination using one of the 16 ROP codes. The operation can be used to speed up the application with duplicate pattern write into an area, such as background paste function.

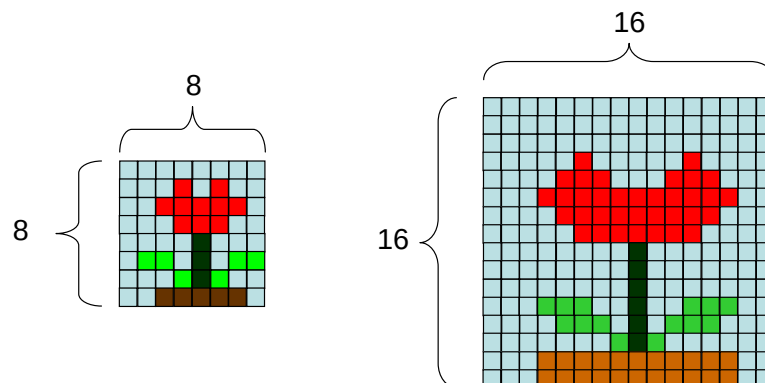


Figure 13-14 : Pattern Format

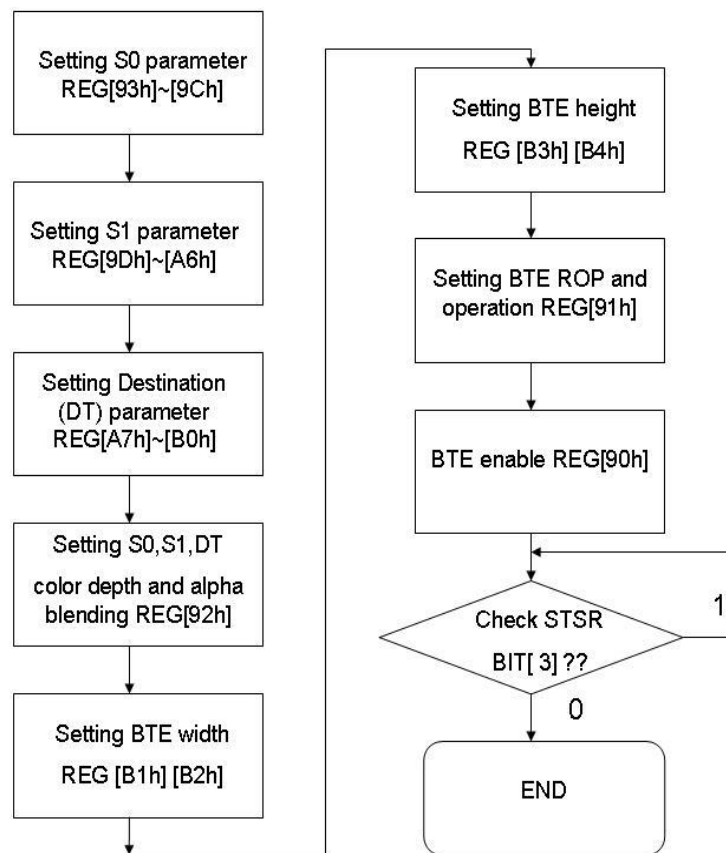


Figure 13-15 : Flow Chart

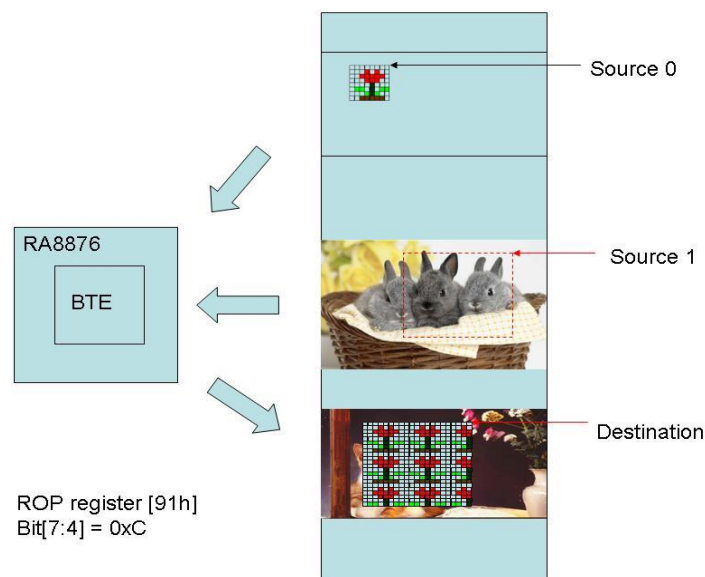


Figure 13-16 : Hardware Data Flow

13.6.6 Pattern Fill W/ Chroma Key

The Pattern Fill with chroma key fills a specified rectangular area of the SDRAM with a pattern. In the pattern fill operation, the Chroma key (transparent color) is ignored. The Chroma key setting in REG[D5h]~[D7h], When Pattern color is equal to Chroma set, then no change data in Destination.

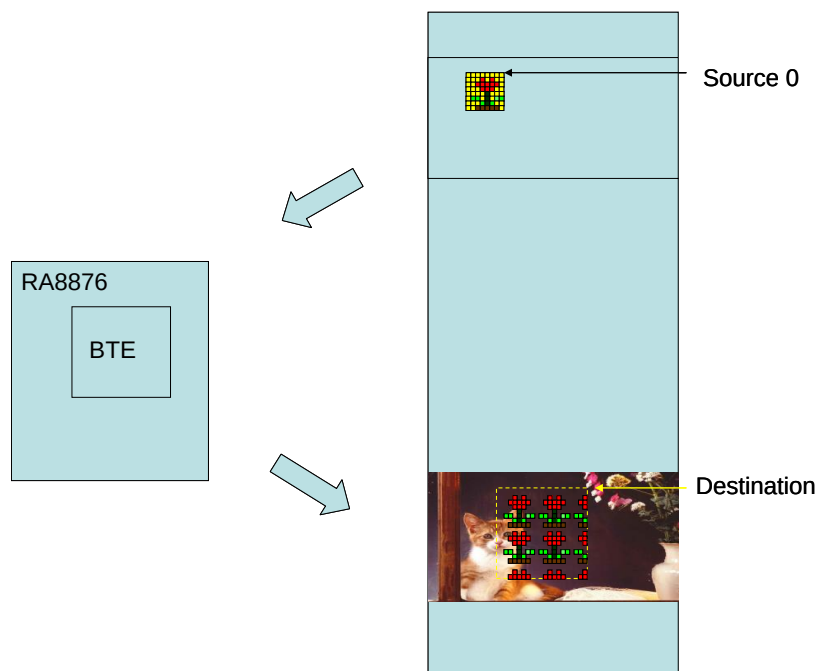


Figure 13-17 : Hardware Flow

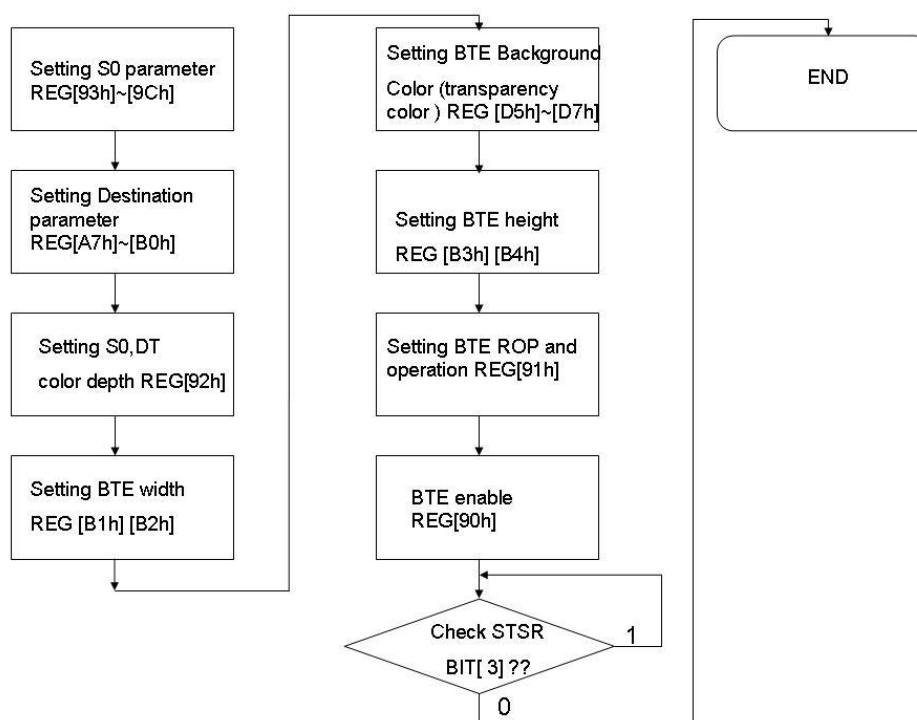


Figure 13-18 : Flow Chart

13.6.7 MPU Write w/ Color Expansion

“MPU Write w/ Color Expansion” is a useful operation to translate monochrome data of MPU interface to be colorful one. In the operation, the source data will be treated as a monochrome bit-map. The bit-wise data is translated to multi-bits per pixel color data by the setting of “BTE Foreground Color” and “BTE Background Color”. If the data on the source is equal to logical “1”, then it will be translated to “BTE Foreground Color”. If the data on the source is equal to logical “0”, then it will be translated to “BTE Background Color”. This function can largely reduce the effort of system translation from mono system to color system. When the end of the line is reached, any unused bits will be discarded. The data for the next line will be taken from the next data package. Each bit is serially expanded to the destination data starting from MSB to LSB. If MPU interface set 16bit, then ROP (start bit) - 15:0 valid., If MPU interface set 8bit, then ROP (start bit) - 7:0 valid.. Source 0 color depth REG [92h] Bit[7:6] don't care..

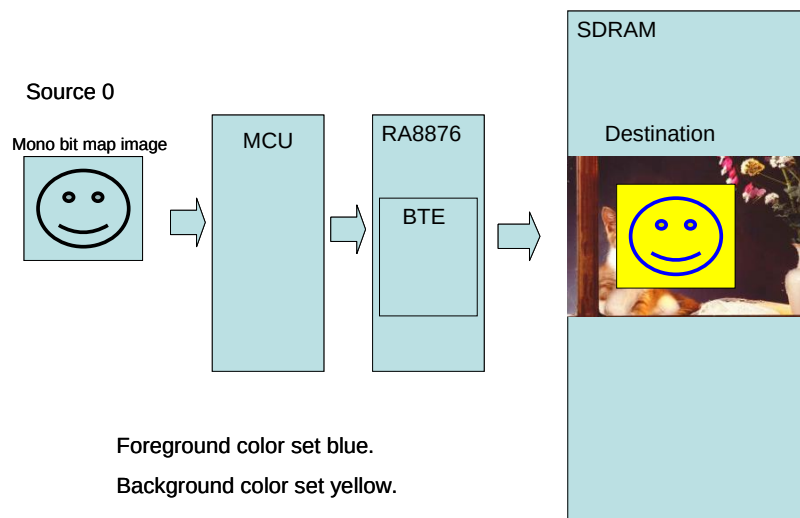


Figure 13-19 : Hardware Data Flow

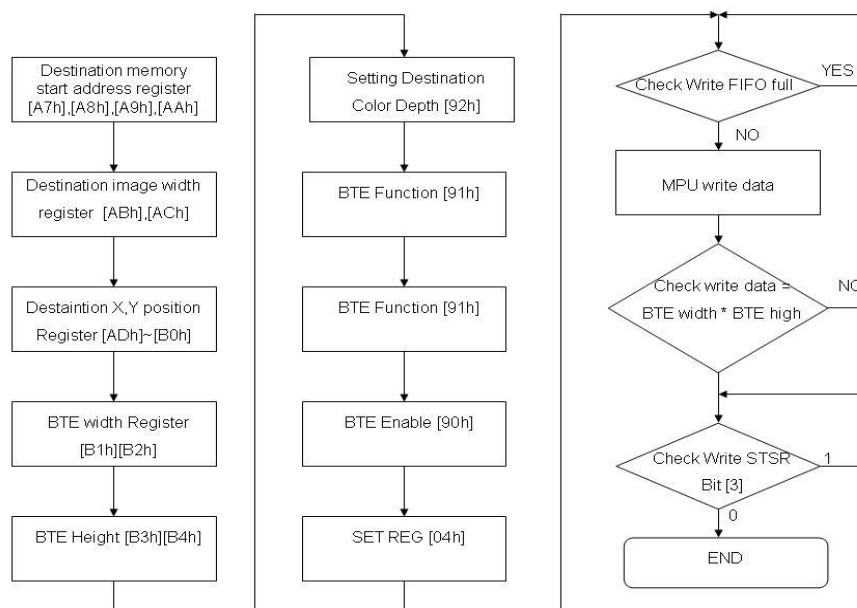


Figure 13-20 : Flow Chart

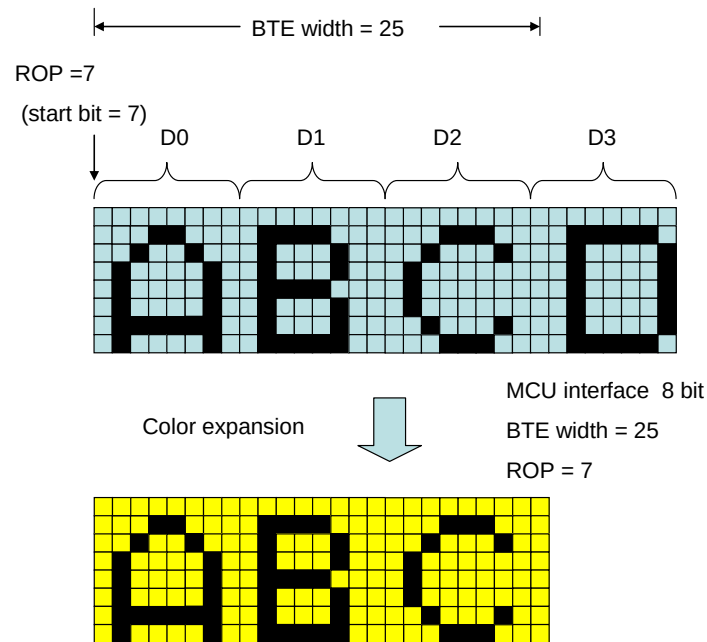


Figure 13-21 :Start Bit Example 1

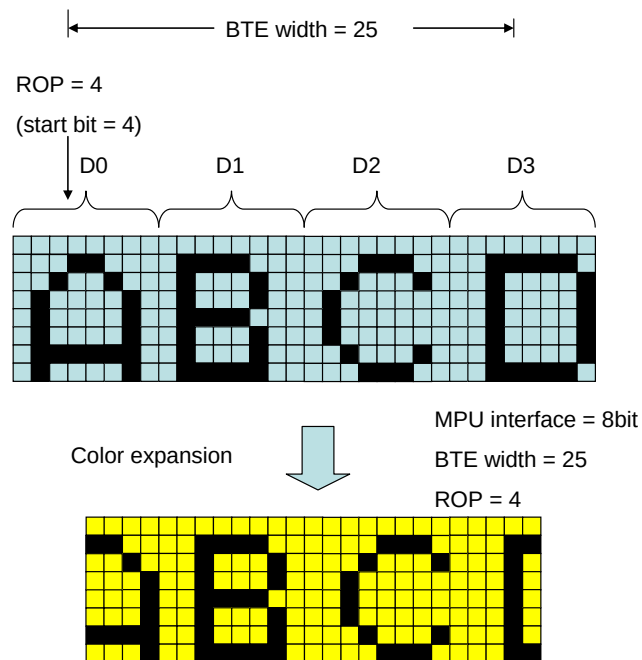


Figure 13-22 : Start bit Exapmle 2

Note:

1. Calculate sent data numbers per row = $((\text{BTE Width size REG} - (\text{MPU interface bits} - (\text{start bit} + 1))) / \text{MPU interface bits}) + ((\text{start bit} + 1) / (\text{MPU interface}))$
2. Total data number = (sent data numbers per row) x BTE Vertical REG setting

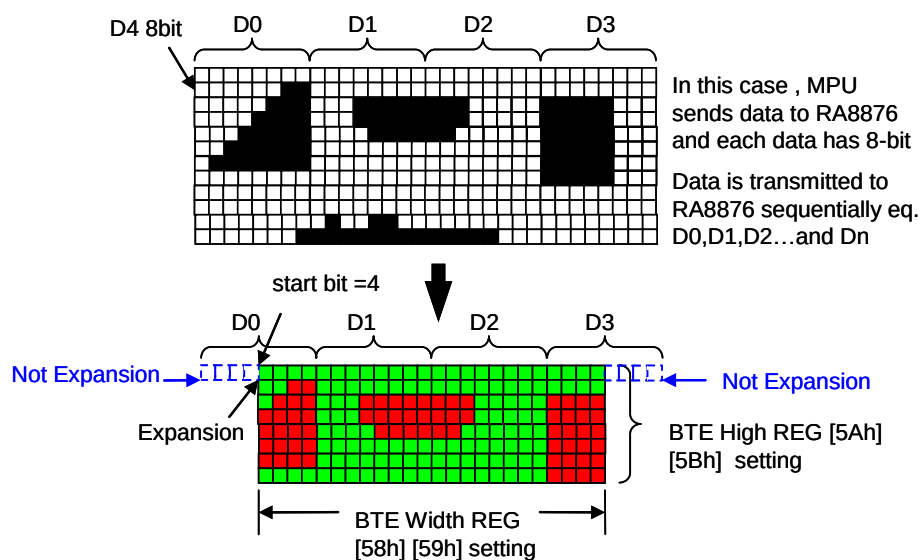


Figure 13-23 : Color Expansion Data Diagram

13.6.8 MPU Write W/ Color Expansion with Chroma key

This BTE operation is virtually identical to the Color Expand BTE, except the background color is completely ignored. All bits set to 1 in the source monochrome bitmap are color expanded to the “BTE Foreground Color”. All bits set to 0 in source monochrome bitmap that would be expanded to the “BTE Background Color” are not expanded at all.

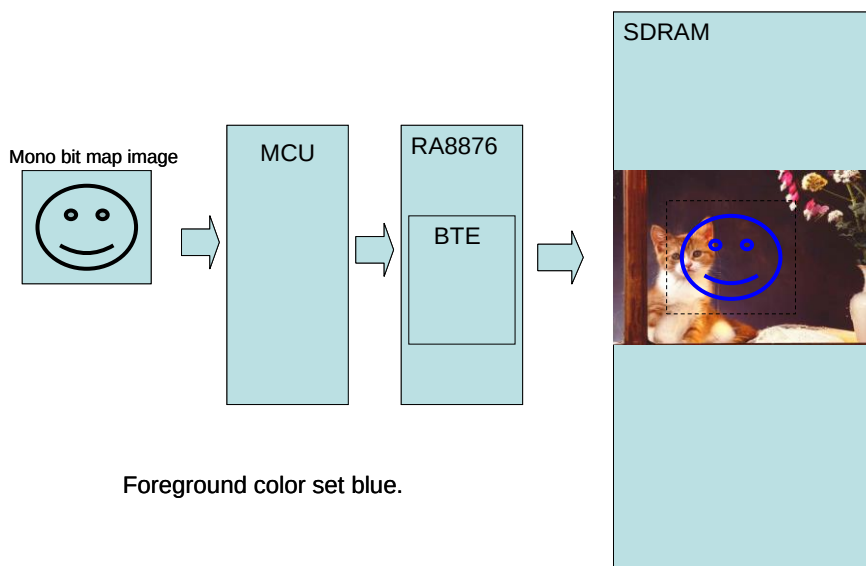
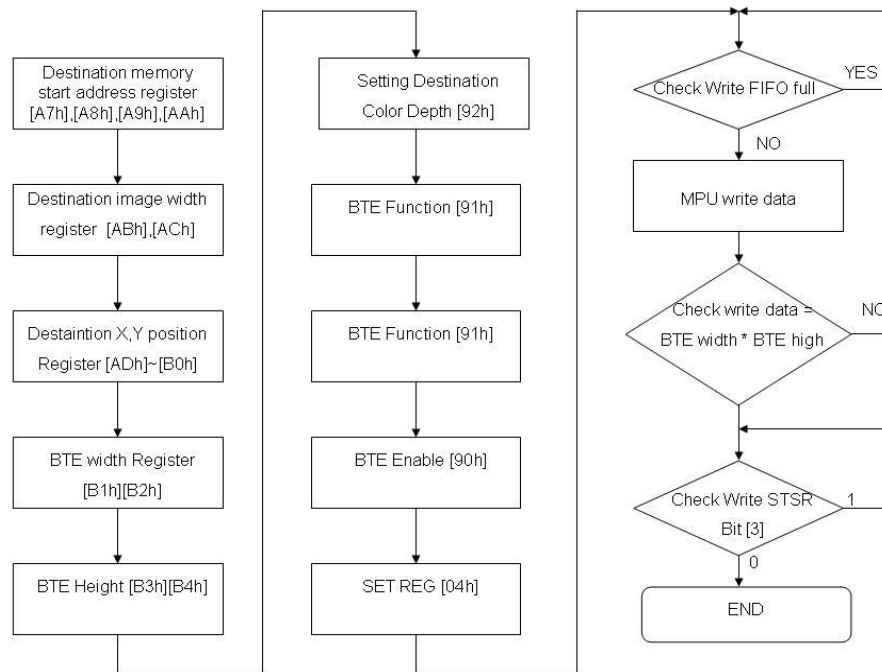


Figure 13-24 : Hardware Data Flow


Figure 13-25 : Flow Chart

13.6.9 Memory Copy with Opacity

The “Memory Copy with opacity” function can mix source 0 data and source 1 blending to Destination. That is having 2 mode - **Picture mode** and **Pixel mode**. Picture mode can be operated in 8 bpp/16bpp/24bpp mode and only has one opacity value(alpha Level) for whole picture. Opacity is specified on REG[B5h]. Pixel mode is only operated in 8bpp/16bpp mode and each pixel have its own opacity value. In the pixel mode – 16bpp the source 1 data bit [15:12] is alpha level, the other bits are color data; 8bpp the source 1 data [7:6] is alpha level. Bit [5:0] is use to destine palette color data.

Picture mode - Destination data = (Source 0 * (1 - alpha Level)) + (Source 1 * alpha Level);

Pixel mode 16bpp - Destination data = (Source 0 * (1- alpha Level)) + (Source 1 [11:0] * alpha Level)

Pixel mode 8bpp - Destination data = (Source 0 * (1- alpha Level)) + (Index palette (Source 1[5:0]) * alpha Level)

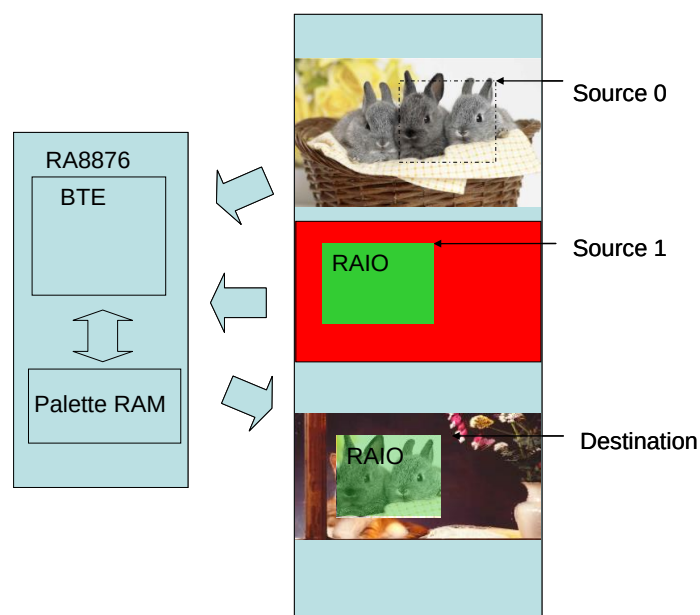


Figure 13-26 : 8bpp Pixel mode Hardware Data Flow

Table 13-4 : Alpha Blending Pixel Mode -- 8bpp

Bit [7:6]	Alpha Level
0h	0
1h	10/32
2h	21/32
3h	1

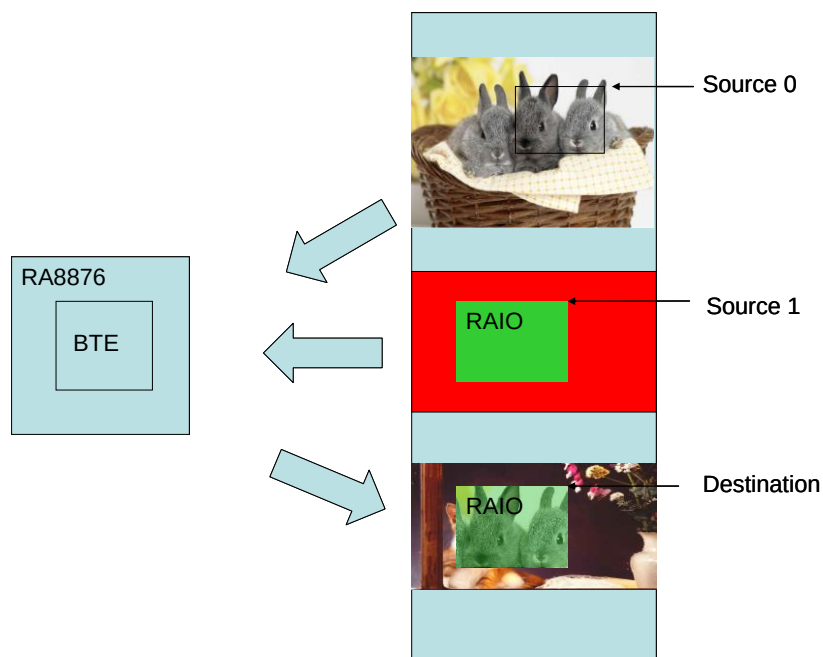
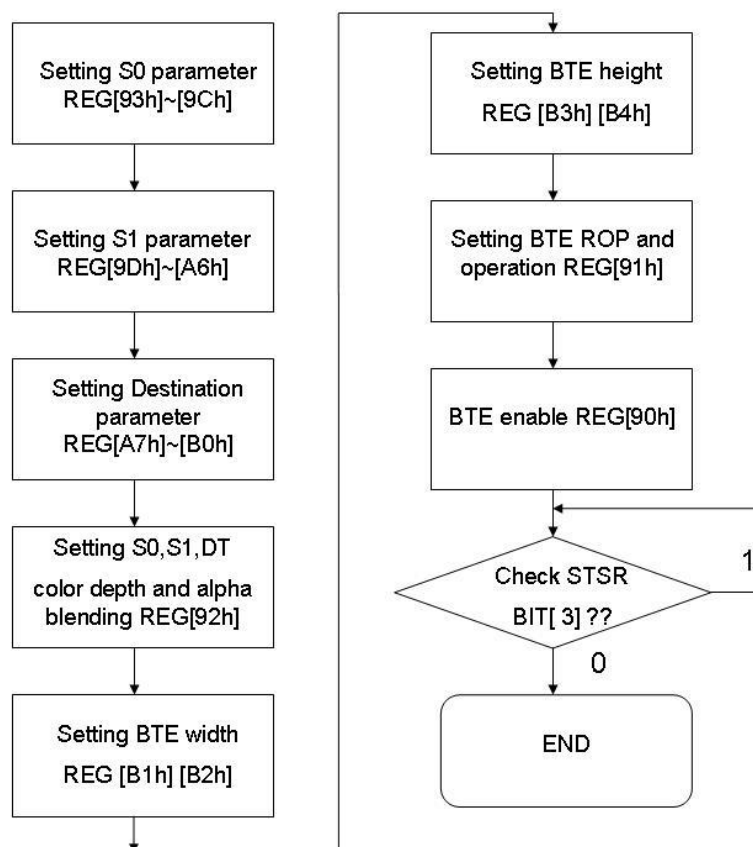


Figure 13-27 : 16bpp Pixel Mode Hardware Data Flow

Table 13-5 : Alpha Blending Pixel Mode -- 16bpp

Bit [15:12]	Alpha Level
0h	0
1h	2/32
2h	4/32
3h	6/32
4h	8/32
5h	10/32
6h	12/32
7h	14/32
8h	16/32
9h	18/32
Ah	20/32
Bh	22/32
Ch	24/32
Dh	26/32
Eh	28/32
Fh	1


Figure 13-28 : Pixel Mode Flow Chart

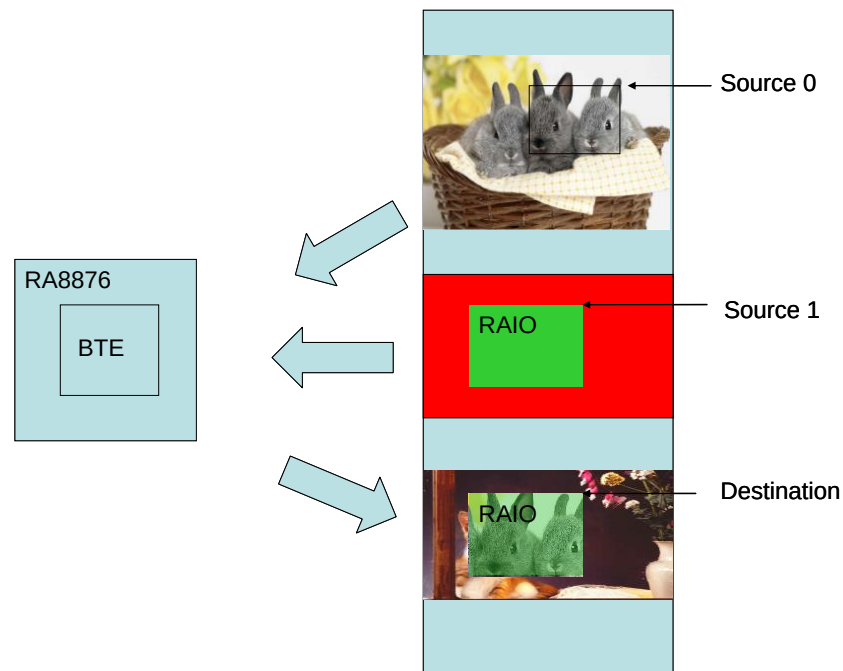


Figure 13-29 : Picture Mode Hardware Data Flow

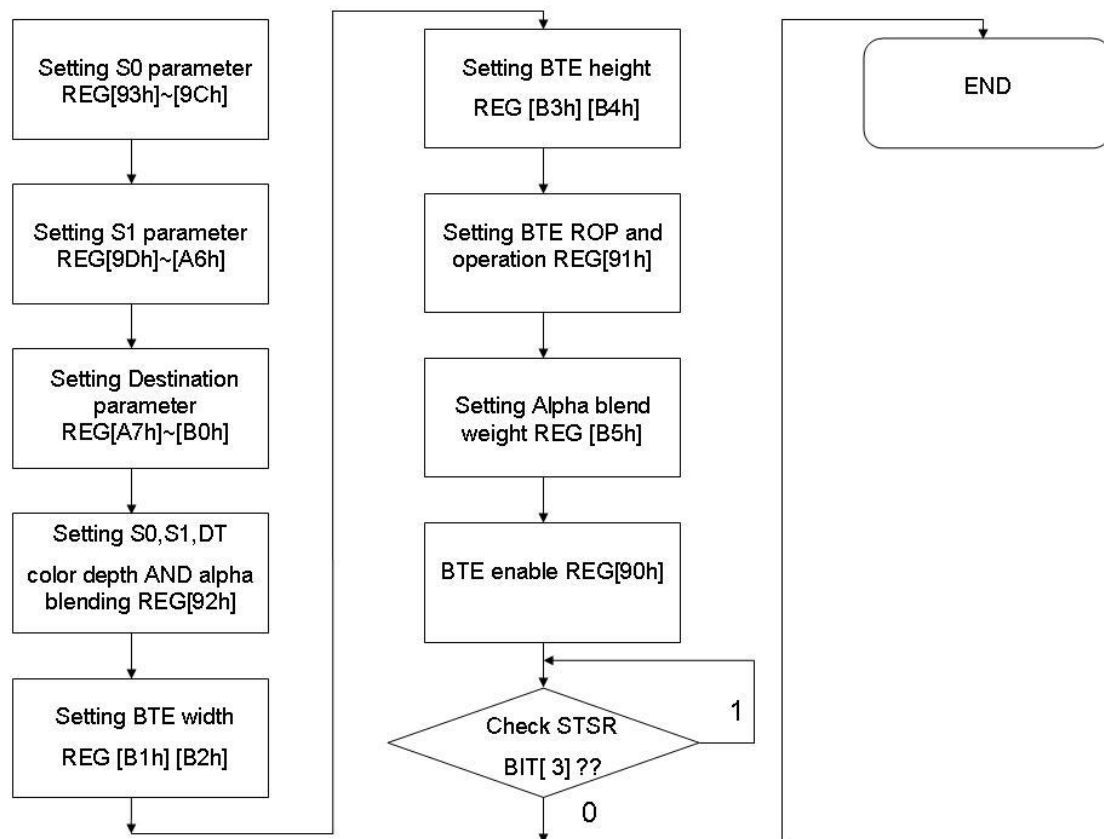


Figure 13-30 : Picture Mode Flow Chart

13.6.10 MPU Write with opacity

The “MPU Write with opacity” function mix Source 0 data and Source 1 blending to Destination. The Source 0 data form MPU. Source 1 data form SDRAM. The description of Alpha blending mode is the same with “Memory Copy with opacity”.

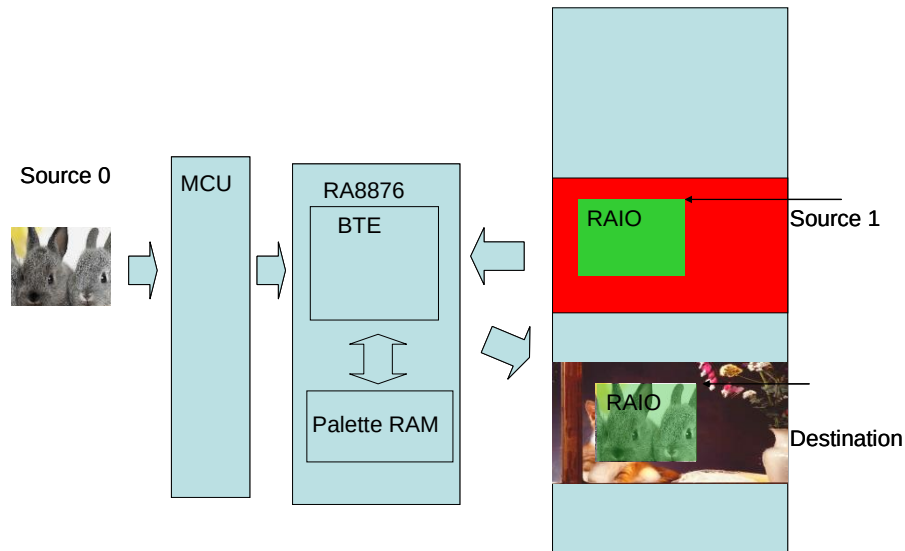


Figure 13-31 : Hardware Data Flow

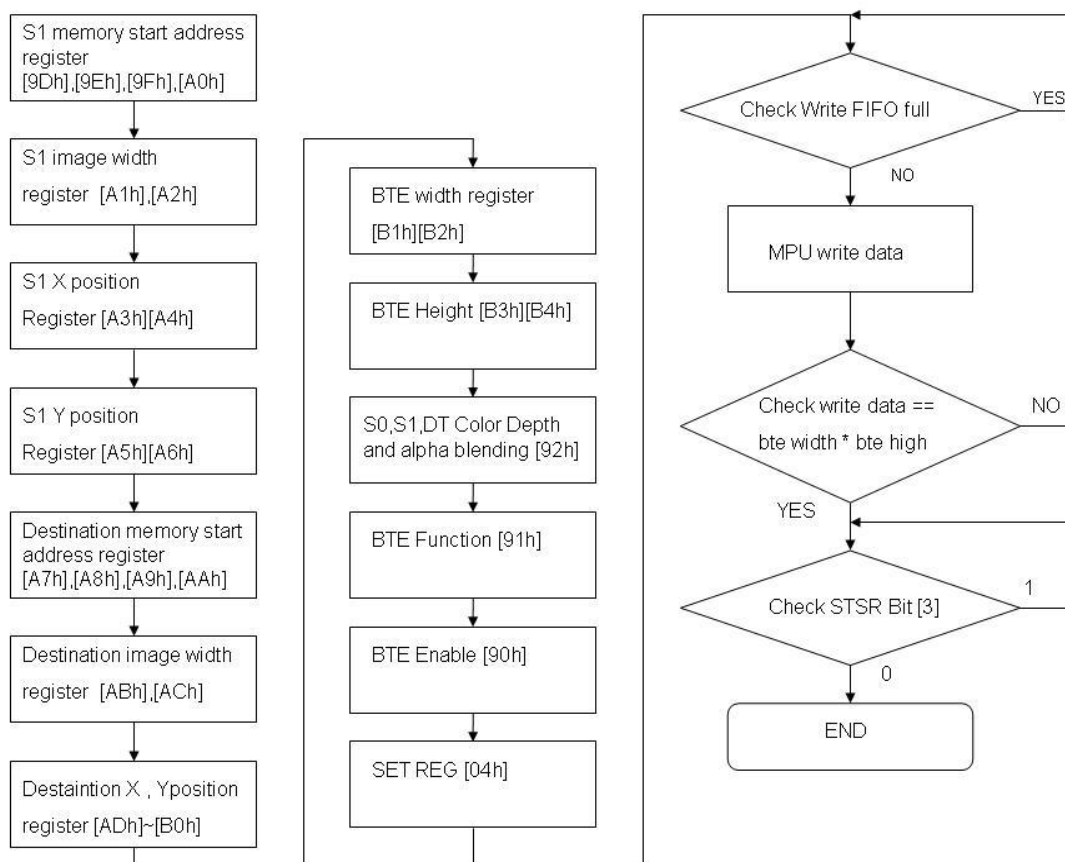


Figure 13-32 : Flow Chart

13.6.11 Memory Copy W/ Color Expansion

The function of "Memory Copy w/ Color Expansion" is to translate the mono image from the source 0 of SDRAM to be a colorful image and write into the destination of the same SDRAM. If the data on the source 0 is presented to logical "1", then it will be translated to "BTE Foreground Color". If the data on the source 0 is presented to logical "0", then it will be translated to "BTE Background Color". The data width setting of Source 0 is valid in 8bit/16bit (REG [92h]). If the data width of Source 0 = 8bit, then ROP (start bit) can be set from 7 to 0. If the data width of Source 0 = 16 bit, then ROP (start bit) can be set from 15 to 0.

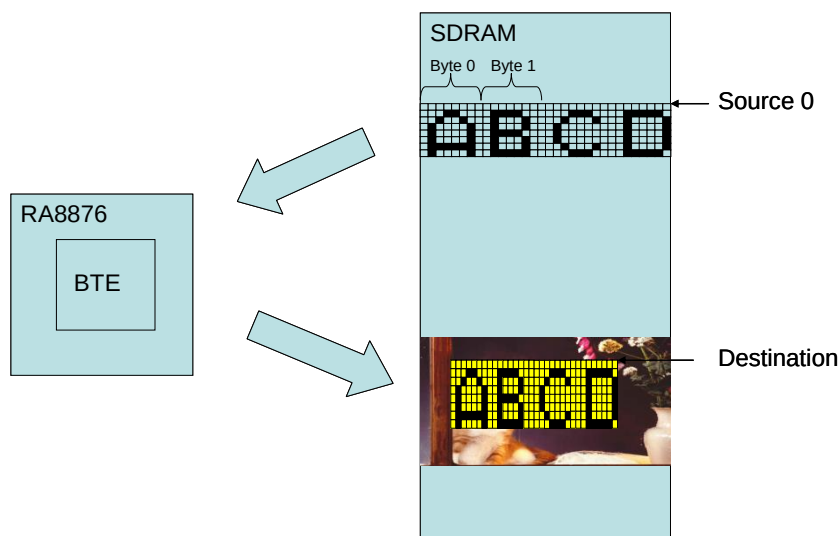


Figure 13-33 : Hardware Data Flow

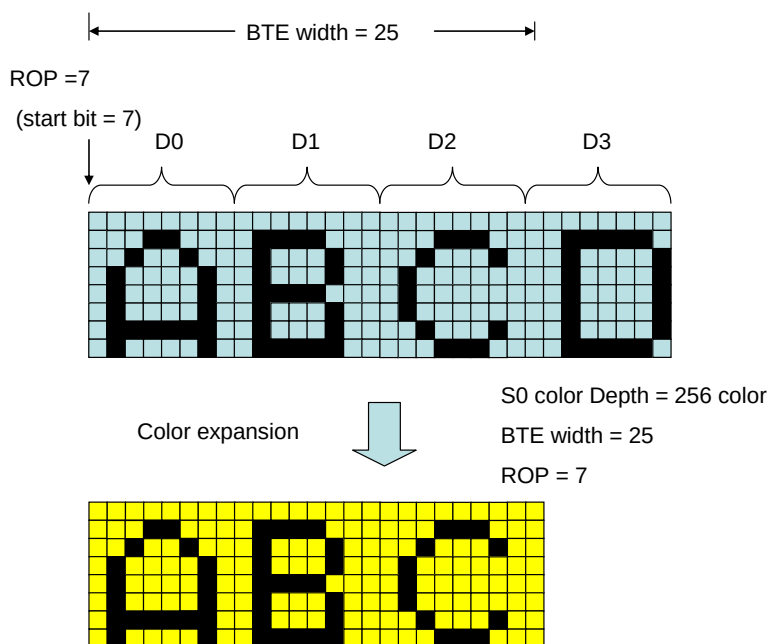


Figure 13-34 : Start Bit Example 1

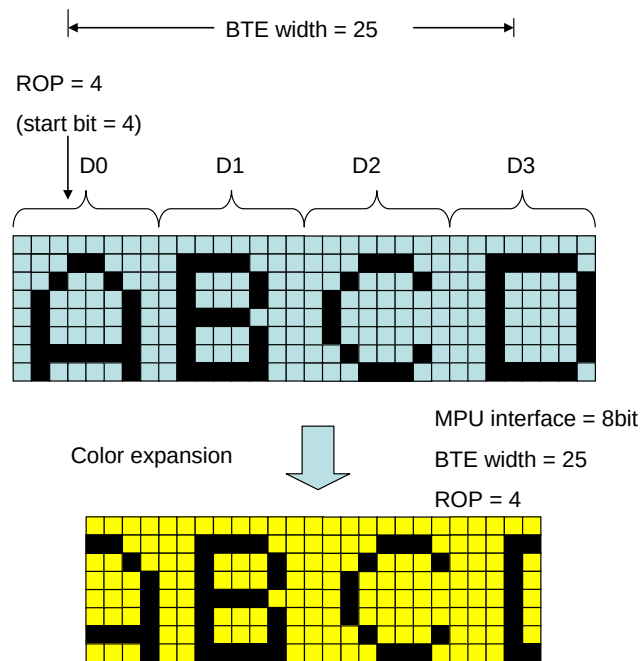


Figure 13-35 : Start Bit Example 2

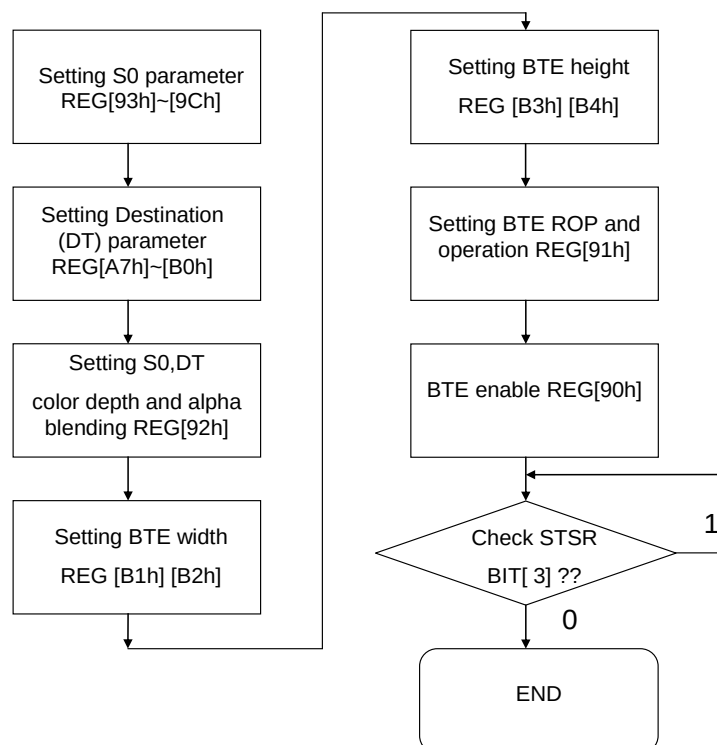


Figure 13-36 : Flow Chart

13.6.12 Memory Copy W/ Color Expansion and Chroma Keying

The function of “Memory Copy w/ Color Expansion and chroma key” is the mono image of SDRAM (Source 0) to be translated to the color image and wrote into the destination of the same SDRAM. If the data on the source 0 is presented to logical “1”, then it will be translated to “BTE Foreground Color”. If the data on the source 0 is presented to logical “0”, and then the data of destination will be kept and unchanged.

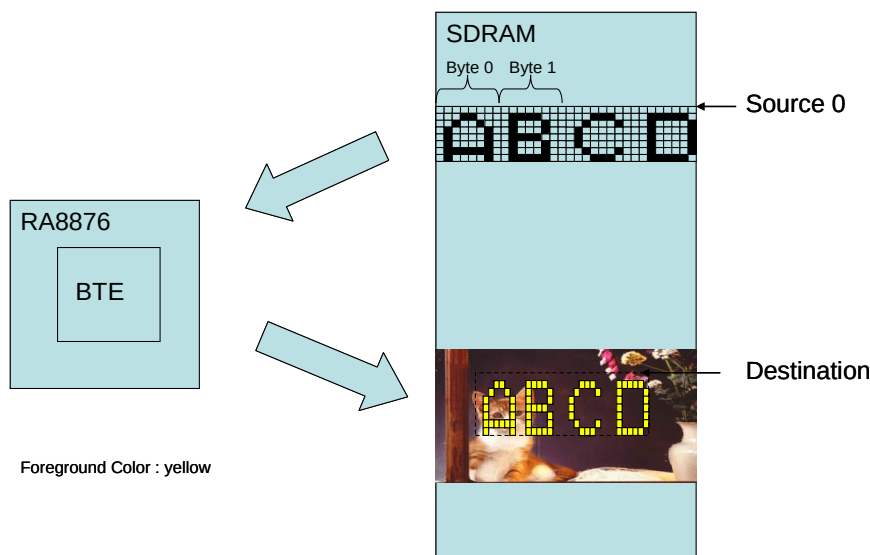


Figure 13-37 : Hardware Data Flow

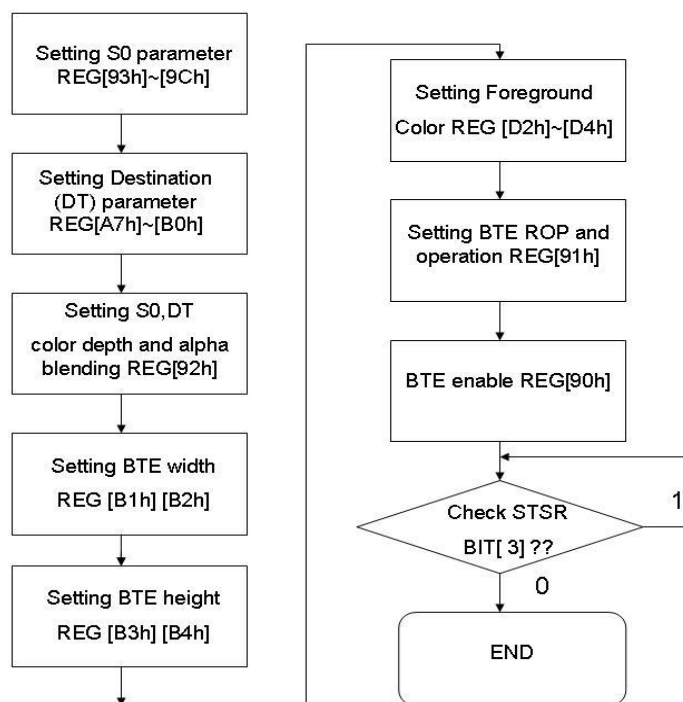


Figure 13-38 : Flow Chart

13.6.13 Solid Fill

The Solid Fill BTE fills a rectangular area of the SDRAM with a solid color. This operation is used to paint large screen areas or to set areas of the SDRAM to be a given value. The color of Solid Fill is set by "BTE Foreground Color".

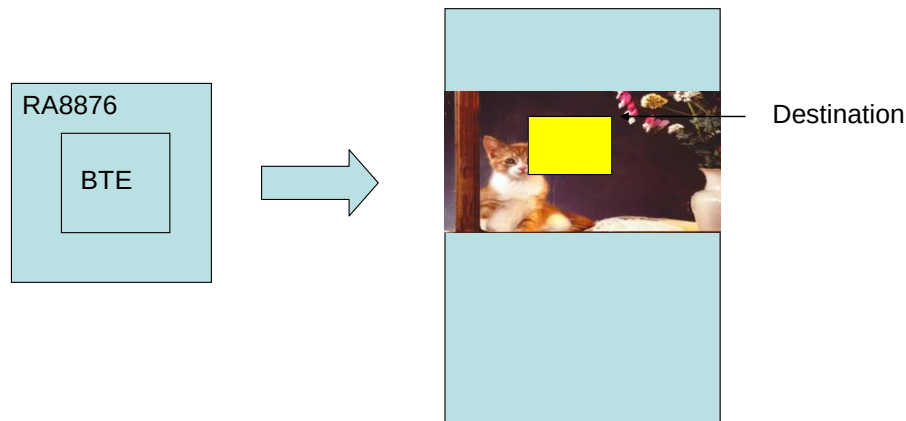


Figure 13-39 : Hardware Data Flow

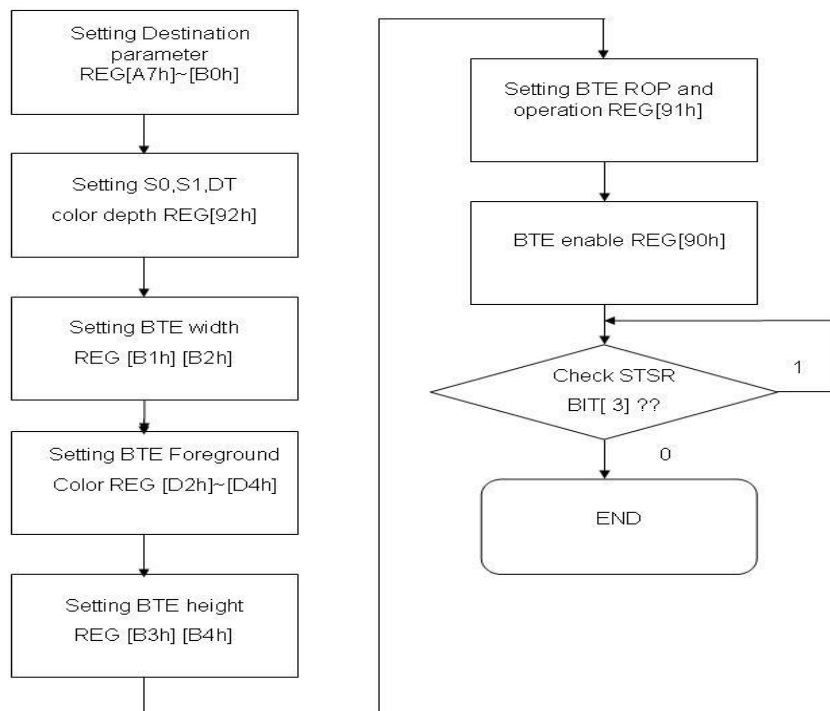


Figure 13-40 : Flow Chart

14. Text Input

RA8876 have three Text source .

2. Embedded Characters . Please reference chapter 14.1.
3. External Character ROM . Please reference chapter 14.2.
4. User Define Character (CGRAM). Please reference chapter 14.3.

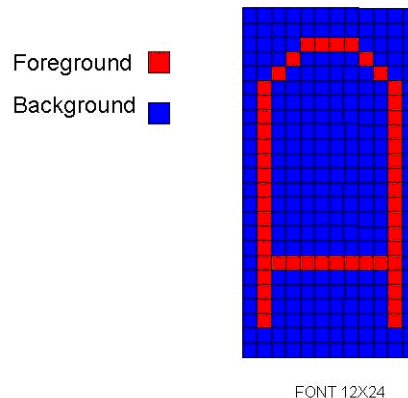


Figure 14-1 : Font Example

The Text have some parameter EX:REG[CCh]~REG[DEh],When you change any font parameter. You can reference below flow chart. The Font color set in Foreground and Background REG[D2]~[D7h].

Ex : We write 64 character as font1 and 64 characters as font2 to RA8876.

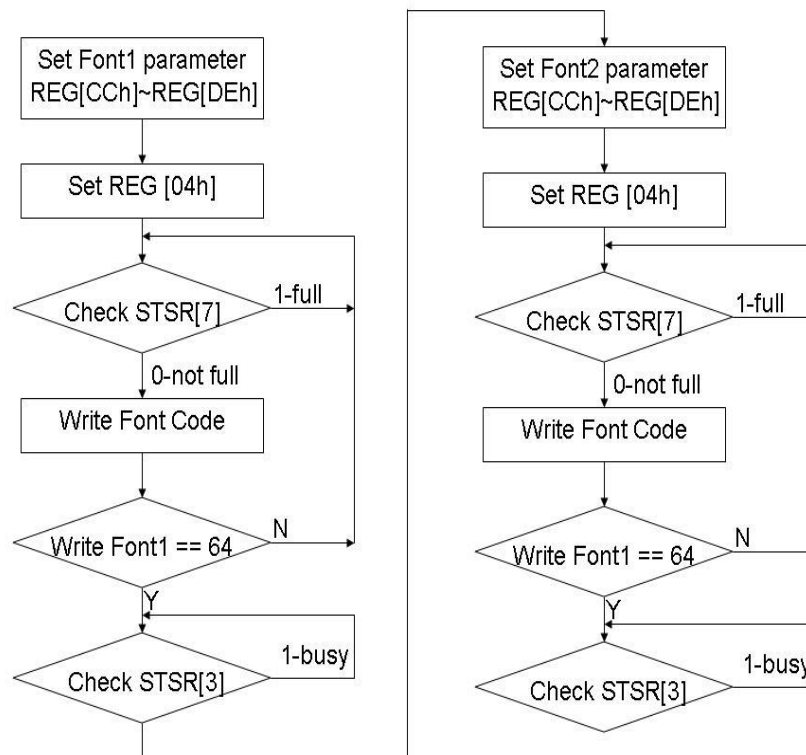


Figure 14-2

14.1 Embedded Characters

The RA8876 embedded 8x16,12x24,16x32 dots ASCII Characters ROM that provides user a convenient way to input characters by ASCII code. The embedded character set supports ISO/IEC 8859-1/2/4/5 coding standards. Besides, user can choose the character foreground color by setting the REG[D2h~D4h] and background color by setting the REG[D5h~D7h]. For the procedure of characters writing please refers to below figure:

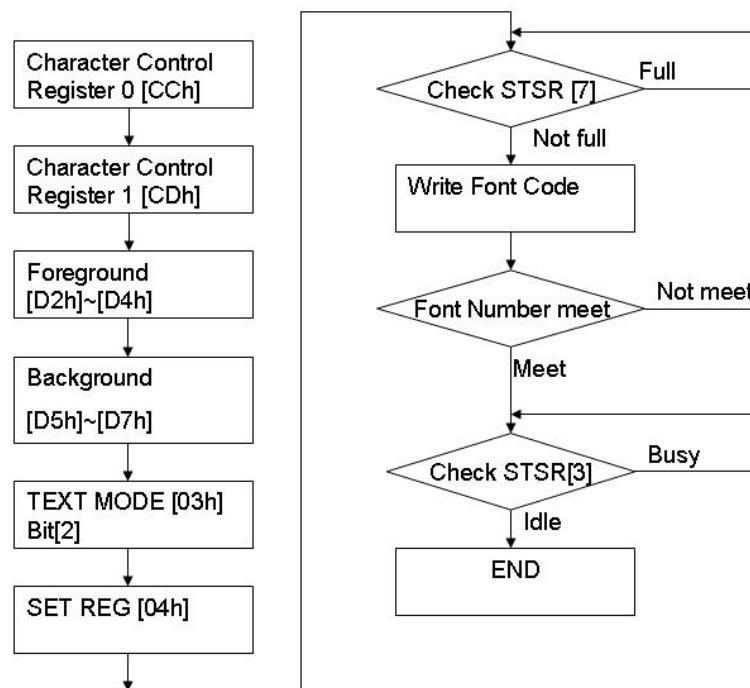


Figure 14-3 : ASCII Character ROM Programming Procedure

Table 14-1 shows the standard character encoding of ISO/IEC 8859-1. ISO means International Organization for Standardization. The ISO/IEC 8859-1, generally called "Latin-1", is the first 8-bit coded character sets that developed by the ISO. It refers to ASCII that consisting of 192 characters from the Latin script in range 0xA0-0xFF. This character encoding is used throughout Western Europe, includes Albanian, Afrikaans, Breton, Danish, Faroese, Frisian, Galician, German, Greenlandic, Icelandic, Irish, Italian, Latin, Luxembourgish, Norwegian, Portuguese, Rhaeto-Romanic, Scottish Gaelic, Spanish, Swedish. English letters with no accent marks also can use ISO/IEC 8859-1. In addition, it is also commonly used in many languages outside Europe, such as Swahili, Indonesian, Malaysian and Tagalog.

In the table, character codes 0x80-0x9F are defined by Microsoft windows, also called CP1252 (WinLatin1).

Table 14-1 : ASCII Block 1(ISO/IEC 8859-1)

L H	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0		☺	☻	♥	♦	♣	♠	●	+	○	◐	♂	♀	♪	♫	☼
1	▶	◀	↕	!!	¶	§	—	↑	↓	→	←	↲	↻	▲	▼	
2		!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	
8	€		,	f	„	…	†	‡	^	%	Š	<	Œ		Ž	
9		‘	’	“	”	•	—	~	™	š	>	œ		ž	ÿ	
A		ı	ø	£	¤	¥	¦	§	¨	©	ª	«	¬	-	®	¯
B	°	±	²	³	´	µ	¶	·	¸	¹	º	»	¼	½	¾	¿
C	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï
D	Ð	Ñ	Ò	Ó	Ô	Õ	Ö	×	Ø	Ù	Ú	Û	Ü	Ý	Þ	ß
E	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
F	ð	ñ	ò	ó	ô	õ	ö	÷	ø	ù	ú	û	ü	ý	þ	ÿ

Table 14-2 shows the standard characters of ISO/IEC 8859-2. ISO/IEC 8859-2 also cited as Latin-2 is the part 2 of the 8-bit coded character sets developed by ISO/IEC 8859. These code values can be used in almost any data interchange system to communicate in the following European languages: Croatian, Czech, Hungarian, Polish, Slovak, Slovenian, and Upper Sorbian. The Serbian, English, German, Latin can use ISO/IEC 8859-2 as well. Furthermore it is suitable to represent some western European languages like Finnish (with the exception of å used in Swedish and Finnish)

Table 14-2 : ASCII Block 2 (ISO/IEC 8859-2)

L H	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0		☺	☹	♥	♦	♣	♠	●	+	○	◊	♂	♀	♪	♫	☼
1	◀	▶	↕	!!	¶	§	■	↕	↑	↓	→	←	↔	↔	▲	▼
2		!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	
8																
9																
A	À	Á	Â	Ã	Ä	Å	Ł	Ś	Š	Ŝ	Ť	Ž	-	Ž	Ž	
B	à	á	â	ã	ä	å	ł	ś	š	ŝ	ť	ž	-	ž	ž	
C	Ř	Á	Â	Ä	Å	Ł	Ć	Č	Č	É	Ę	Ě	Ě	Í	Î	Ď
D	Đ	Ń	Ń	Ó	Ô	Ö	×	Ř	Ů	Ú	Ú	Ů	Ý	Ť	ß	
E	ř	á	â	ã	ä	å	í	ć	č	é	ę	ě	ě	í	î	ď
F	đ	ń	ń	ó	ô	ö	÷	ř	ů	ú	ú	ů	ý	ť	.	

Table 14-3 shows the standard characters of ISO/IEC 8859-4. ISO/IEC 8859-4 is known as Latin-4 or "North European" is the forth part of the ISO/IEC 8859 8-bit character encoding. It was designed originally to cover Estonian, Greenlandic, Latvian, Lithuanian, and Sami. This character set also supports Danish, English, Finnish, German, Latin, Norwegian, Slovenian, and Swedish.

Table 14-3 : ASCII Block 3 (ISO/IEC 8859-4)

L H	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0		☺	☹	♥	♦	♣	♠	●	+	○	◊	♂	♀	♪	♫	☼
1	▶	◀	↕	!!	¶	§	—	↑	↓	→	←	↵	↔	▲	▼	
2		!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	
8																
9																
A		À	Ā	Ą	Ȧ	Ȧ	İ	Ł	Š	ˆ	Š	Ē	Ġ	Ƨ	-	Ž
B	°	à	á	â	ã	ä	å	æ	ç	ē	ġ	ġ	ġ	ġ	ġ	ġ
C	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā	Ā
D	Đ	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń	Ń
E	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā	ā
F	đ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ	ņ

Table 14-4 shows the standard characters of ISO/IEC 8859-5. ISO/IEC 8859-5 is known as is the five part of the ISO/IEC 8859 8-bit character encoding. It was designed originally to cover Bulgarian , Belarusian, Russian, Serbian and Macedonian Table 14-4 : ASCII Block 4 (ISO/IEC 8859-5)

Table 14-4 : ASCII Block 4 (ISO/IEC 8859-5)

L H	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0		☺	☹	♥	♦	♣	♠	●	⊕	⊖	⊗	⊘	♂	♀	♪	♫
1	▶	◀	↕	!!	¶	§	■	↕	↑	↓	→	←	↵	↔	▲	▼
2		!	”	#	\$	%	&	'	()	*	+	,	-	.	/	
3		0	1	2	3	4	5	6	7	8	9	:	;	<	=	> ?
4		@	A	B	C	D	E	F	G	H	I	J	K	L	M	N O
5		P	Q	R	S	T	U	V	W	X	Y	Z	[\]	^	_	
6		`	a	b	c	d	e	f	g	h	i	j	k	l	m	n o
7		p	q	r	s	t	u	v	w	x	y	z	{ }	~		
8																
9																
A		Ё	Ъ	Ѓ	Є	Ѕ	І	Ї	Ј	Љ	Њ	Ћ	Ќ	-	Ў	Ц
B		А	Б	В	Г	Д	Е	Ж	З	И	Й	К	Л	М	Н	О П
C		Р	С	Т	У	Ф	Х	Ц	Ч	Ш	Щ	Ъ	Ы	Ь	Э	Ю Я
D		а	б	в	г	д	е	ж	з	и	й	к	л	м	н	о п
E		р	с	т	у	ф	х	ц	ч	ш	щ	ъ	ы	ь	э	ю я
F		Њ	ё	ђ	ѓ	є	ѕ	і	ї	ј	љ	њ	ћ	ќ	ѕ	ў ц

14.2 External Character ROM

RA8876 use External serial ROM interface to provide more characters set for different applications. It is compatible with Character ROM of Genitop Inc., which is a professional Character sets ROM vendor. The supporting product numbers are GT21L16T1W, GT30L16U2W, GT30L24T3Y, GT30L24M1Z, and GT30L32S4W, GT20L24F6Y, GT21L24S1W. According to different product, there are different character's size including 16x16, 24x24, 32x32, and variable width character size in them. Detail functionality description please refers section: "External Serial Flash/ROM Interface".

Note : The Flow chart please reference 16.3.1 .

14.2.1 GT21L16TW

- Reg[CEh][7:5]: 000b
- Character height: x16

Allowed character sets & width:

	GB12345 GB18030	BIG5	ASCII	UNI-jpn	JIS0208	Latin	Greek	Cyrillic	Arabic
Normal	V	V	V	V	V	V	V	V	
Arial			V			V	V	V	V
Roman			V						V
Bold			V						

*Arial & Roman is variable width.

14.2.2 GT30L16U2W

- Reg[CEh][7:5]: 001b
- Character height: x16

Allowed character sets & width:

	UNICODE	ASCII	Latin	Greek	Cyrillic	Arabic	GB2312 Special
Normal	V	V	V	V	V		V
Arial		V	V	V	V	V	
Roman		V				V	
Bold							

*Arial & Roman is variable width.

14.2.3 GT30L24T3Y

- Reg[CEh][7:5]: 010b
- Character height: x16

Allowed character sets & width:

	GB2312	GB12345/GB18030	BIG5	UNICODE	ASCII
Normal	V	V	V	V	V
Arial					V
Roman					
Bold					

*Arial & Roman is variable width.

- Character height: x24

Allowed character sets & width:

	GB2312	GB12345/GB18030	BIG5	UNICODE	ASCII
Normal	V	V	V	V	
Arial					V
Roman					
Bold					

*Arial & Roman is variable width.

14.2.4 GT30L24M1Z

- Reg[CEh][7:5]: 011b
- Character height: x24

Allowed character sets & width:

	GB2312 Extension	GB12345/ GB18030	ASCII
Normal	V	V	V
Arial			V
Roman			V
Bold			

*Arial & Roman is variable width.

14.2.5 GT30L32S4W

- Reg[CEh][7:5]: 100b
- Character height: x16

Allowed character sets & width:

	GB2312	GB2312 Extension	ASCII
Normal	V	V	V
Arial			V
Roman			V
Bold			

*Arial & Roman is variable width.

- Character height: x24

Allowed character sets & width:

	GB2312	GB2312 Extension	ASCII
Normal	V	V	V
Arial			V
Roman			V
Bold			

*Arial & Roman is variable width.

- Character height: x32

Allowed character sets & width:

	GB2312	GB2312 Extension	ASCII
Normal	V	V	V
Arial			V
Roman			V
Bold			

*Arial & Roman is variable width.

14.2.6 GT20L24F6Y

- Reg[CEh][7:5]: 101b
- Character height: x16

Allowed character sets & width:

	ASCII	Latin	Greek	Cyrillic	Arabic	Hebrew	Thai	ISO-8859
Normal	V	V	V	V		V	V	V
Arial	V	V	V	V	V			
Roman	V							
Bold	V							

*Arial & Roman is variable width.

- Character height: x24

Allowed character sets & width:

	ASCII	Latin	Greek	Cyrillic	Arabic
Normal		V	V	V	
Arial	V				V
Roman					
Bold					

*Arial & Roman is variable width.

14.2.7 GT21L24S1W

- Reg[CEh][7:5]: 110b
- Character height: x24

Allowed character sets & width:

	GB2312	GB2312 Extension	ASCII
Normal	V	V	V
Arial			V
Roman			
Bold			

*Arial & Roman is variable width.

14.3 User-defined Characters

User can create characters or symbols they want by using User-defined Characters. User-defined Characters support character size with half width (8x16/12x24/16x32 dots) and full width (16X16/24X24/32X32 dots). The RA8876 supports 32,768 half width User-defined Characters code or 32,768 full width User-defined Characters code. Half width character code is 0000h~7FFFh, Full width character code is 8000h~FFFFh. User just writes the character code or symbol data to the indicated space (locates in external SDRAM, also call it CGRAM space) and then writes the corresponding character code, RA8876 will write the character or symbol to the display memory. Also, user can choose the foreground color by setting the REG[D2h~D4h] and background color by setting the REG[D5h~D7h].

14.3.1 8x16 Character Format in CGRAM

CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) + ((FONT CODE) * 16)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 0001h

THEN FONT ADDR = 1010h

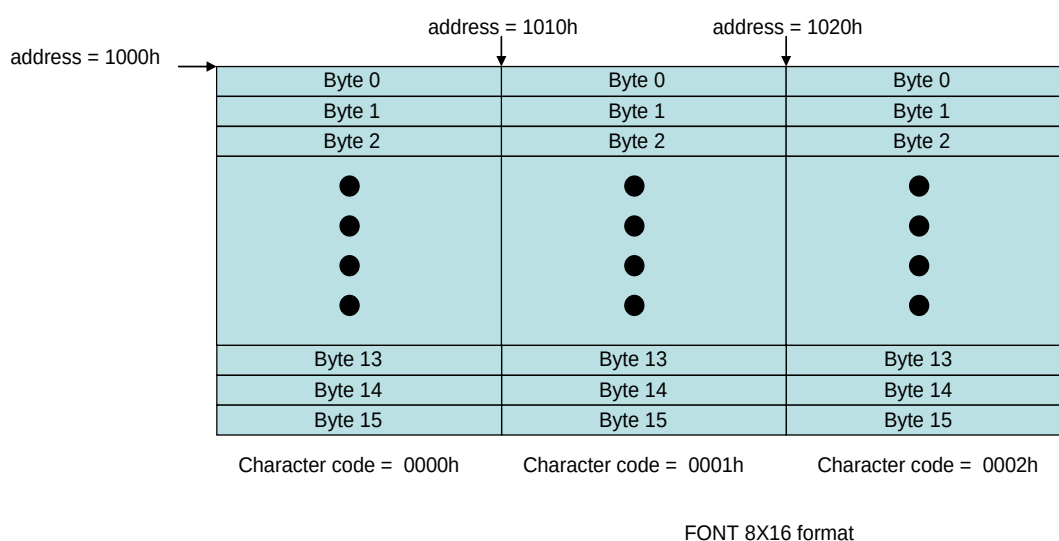


Figure 14-4 : Font 8X16 Array in SDRAM

14.3.2 16x16 Character Format in CGRAM

CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) + ((FONT CODE - 8000h) * 32)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 8001h

THEN FONT ADDR = 1020h

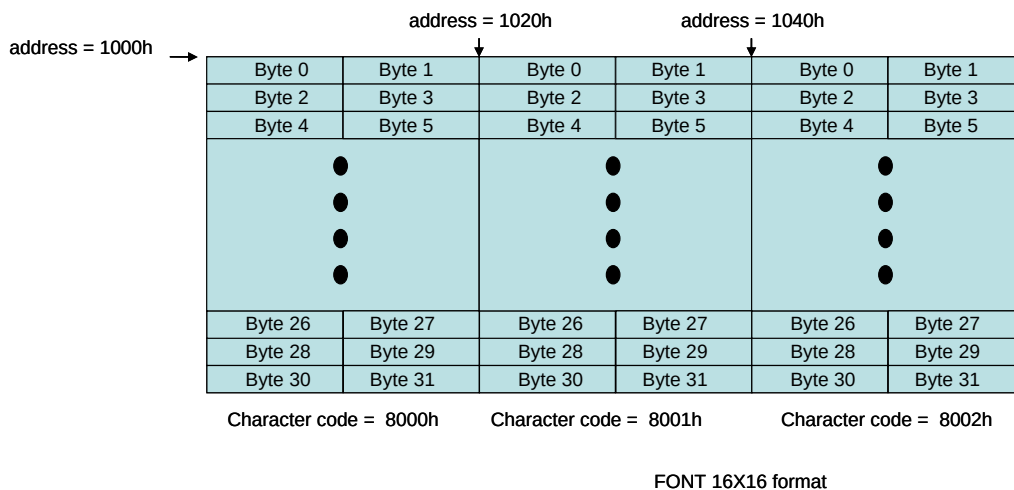


Figure 14-5 : Font Array 16x16 in SDRAM

14.3.3 12x24 Character Format in CGRAM

CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) +
((FONT CODE) * 48)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 0001h

THEN FONT ADDR = 1030h

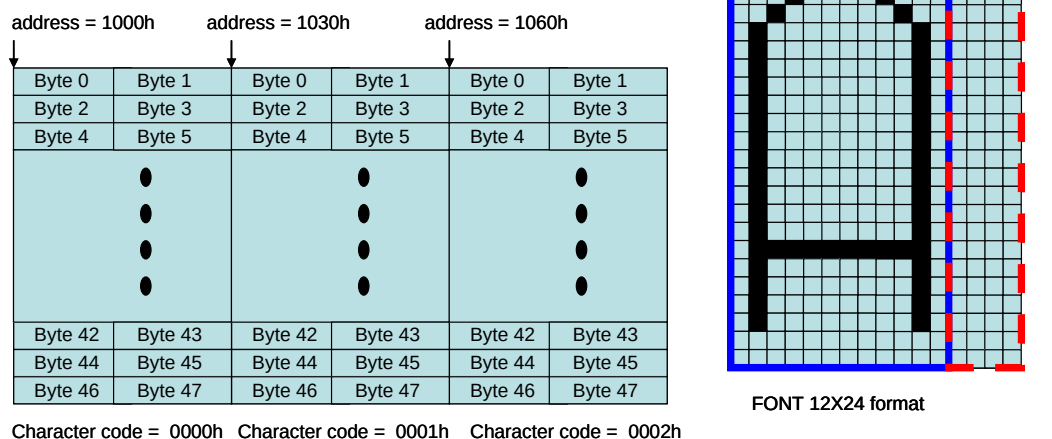


Figure 14-6 : Font Array 12x24 in SDRAM

14.3.4 24x24 Character Format in CGRAM

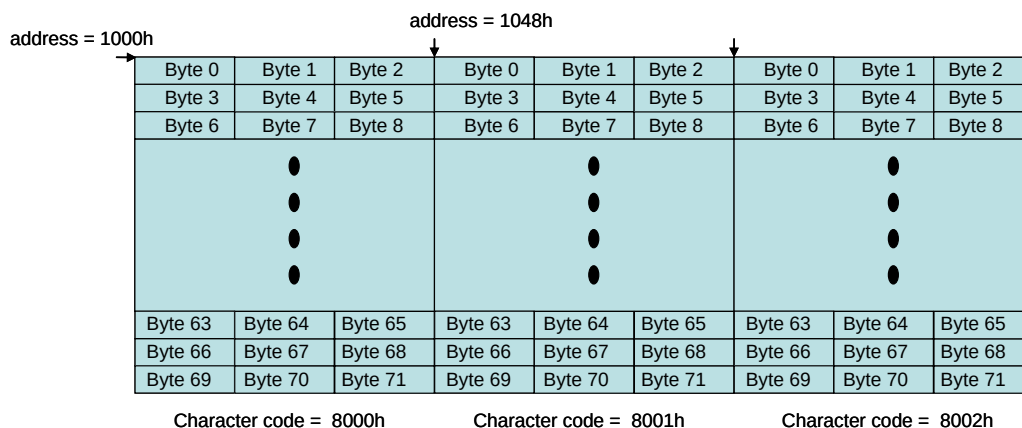
CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) + ((FONT CODE – 8000h) * 72)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 8001h

THEN FONT ADDR = 1048h



FONT 24X24 format

Figure 14-7 : Font Array 24x24 in SDRAM

14.3.5 16x32 Character Format in CGRAM

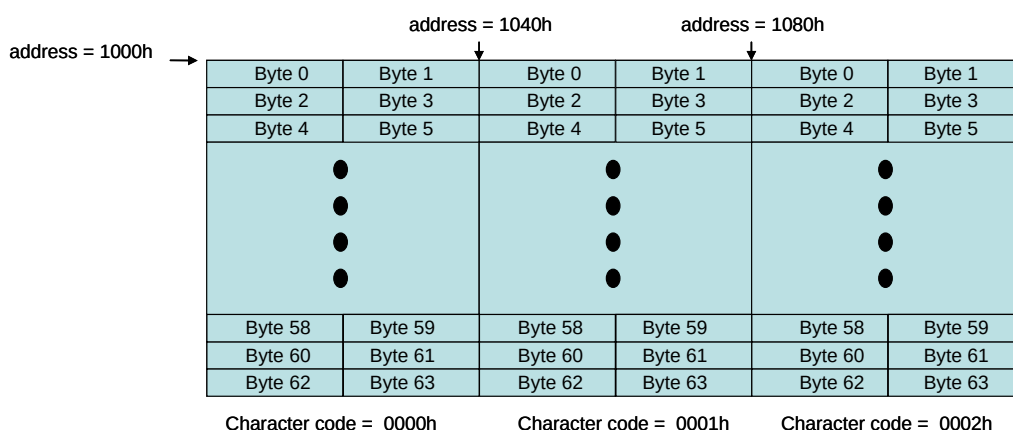
CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) + ((FONT CODE) * 64)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 0001h

THEN FONT ADDR = 1040h



FONT 16X32 format

Figure 14-8 : Font 16x32 Array in SDRAM

14.3.6 32x32 Character Format in CGRAM

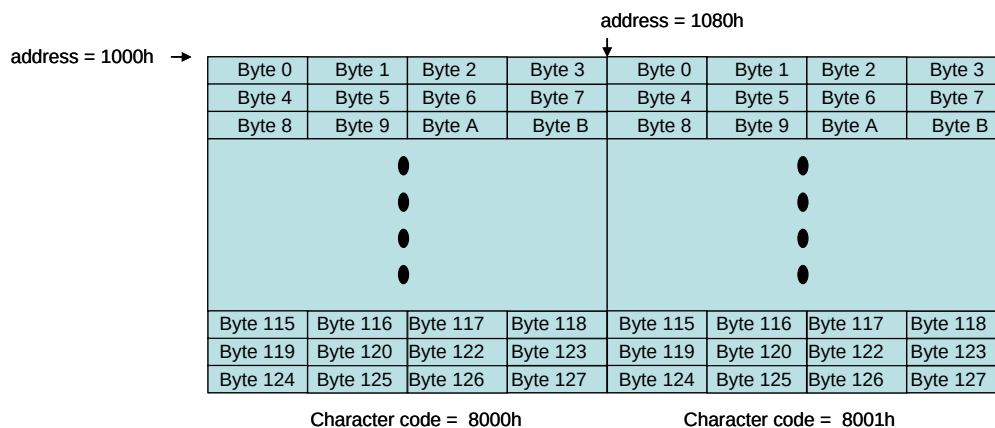
CGRAM ADDRE CALCULATE = (CGRAM_START_ADDR) + ((FONT CODE – 8000h) * 128)

EXAMPLE :

CGRAM_START_ADDR = 1000h

CHARACTER_CODE = 8001h

THEN FONT ADDR = 1080h



FONT 32X32 format

Figure 14-9 : Font 32x32 Array in SDRAM

14.3.7 Flow Chart about Initial CGRAM by MPU

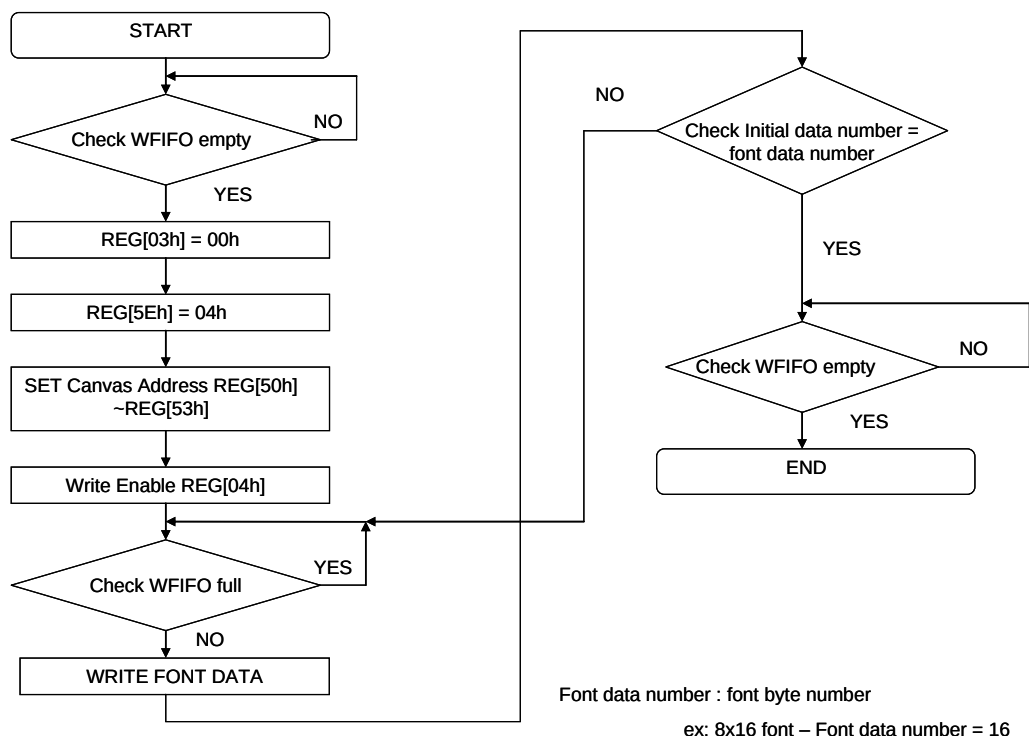


Figure 14-10 : Initial CGRAM from MPU

14.3.8 Flow Chart about Initial CGRAM by Serial Flash

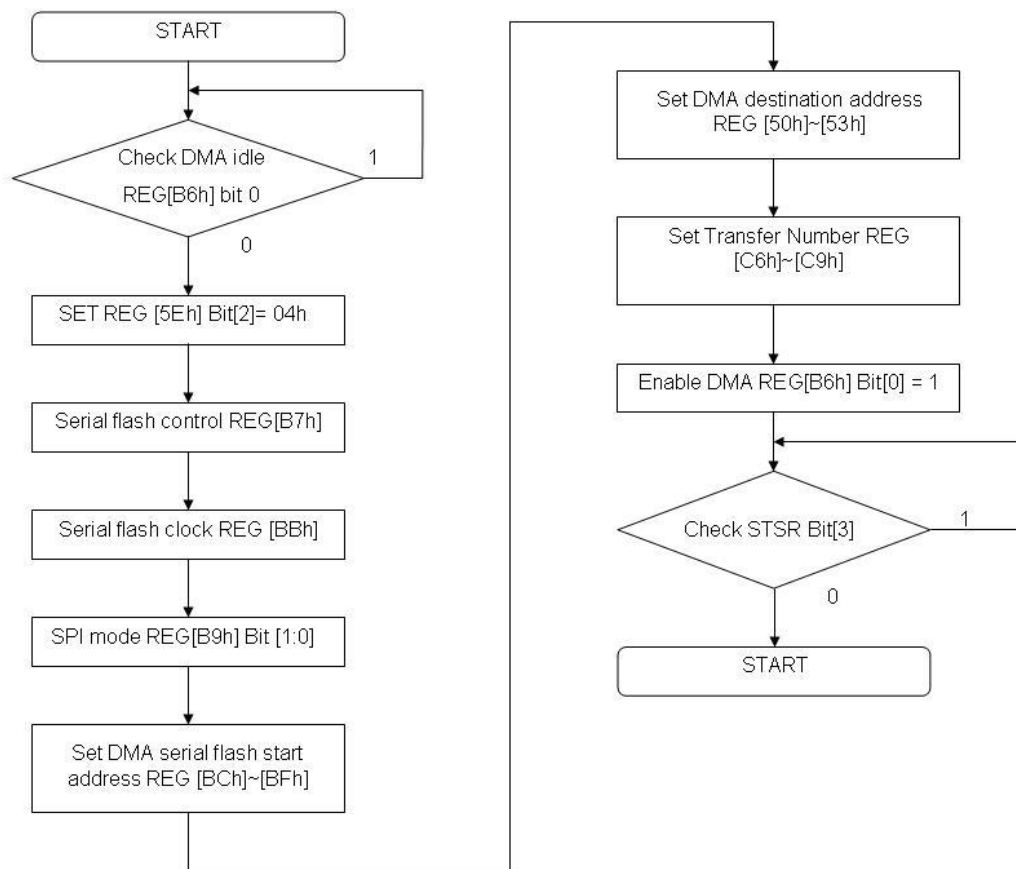


Figure 14-11 : Initial CGRAM from Serial Flash

14.4 Rotate 90 Degree's Characters

Normal text input direction is from left to right and then from top to bottom. The RA8876 supports text rotate function. Characters may rotate counterclockwise 90 degree, vertical flip by setting the REG[CDh] Bit4 = 1. And collocating the VDIR(REG[12h] Bit3), LCD module can show the 90 degree character. At text rotate mode, its text input direction is from top to bottom and then from left to right.

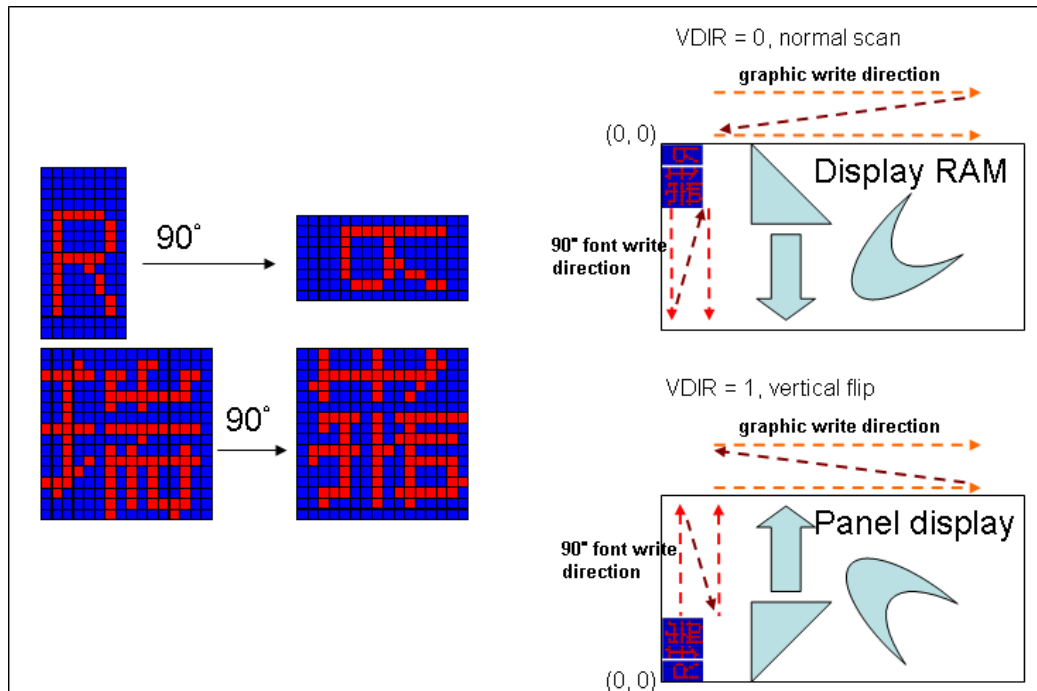


Figure 14-12: Rotation 90° Characters

When user rotate panel 90 degree in clockwise direction, user will see panel displayed like following.

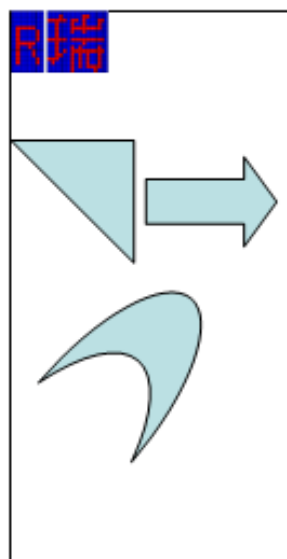


Figure 14-13

14.5 Enlargement, Transparent Characters

RA8876 also supports Character enlargement (REG[CDh] Bit[3:0]), and transparent function (REG[CDh] Bit6). Moreover, these functions can use simultaneously. The behaviors of these functions just refer to below figure:

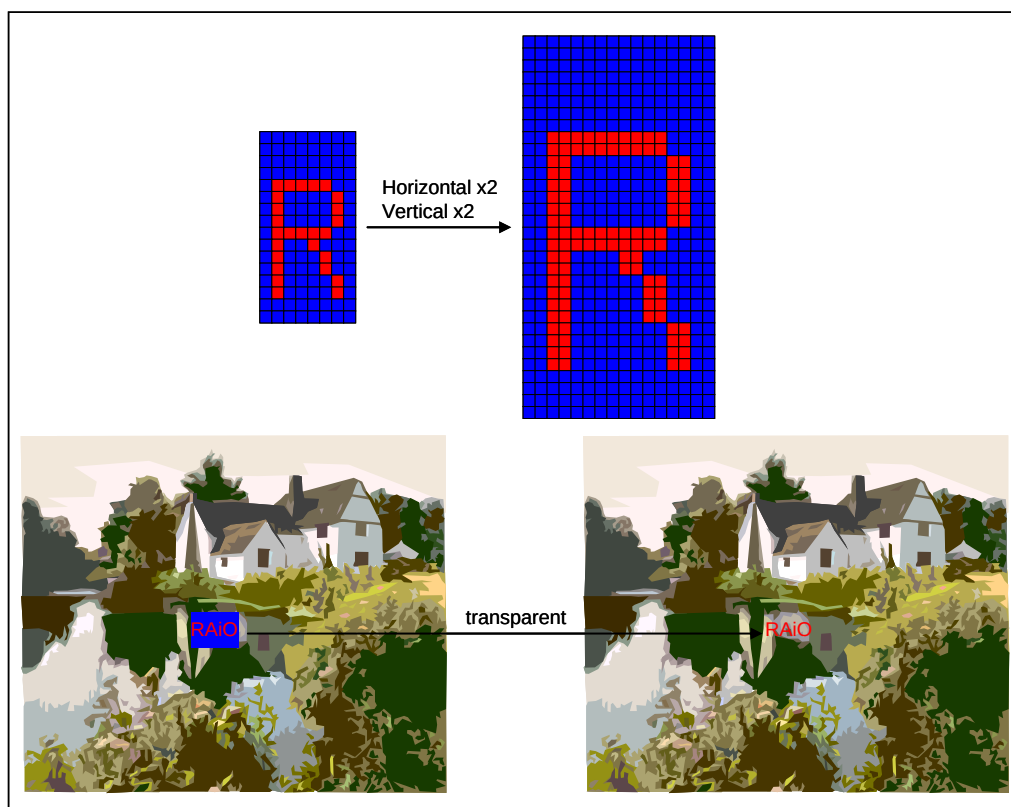


Figure 14-14 : Enlargement and Transparent Characters

14.6 Auto Line Feed When Meet Active Window Boundary

RA8876 supports the auto movement of Text write and it will auto line feed on active window boundary. In text mode, the position of text will move automatically and change line when the character over the range of horizontal or vertical active window. Refer the below figure to view the behavior of auto move.

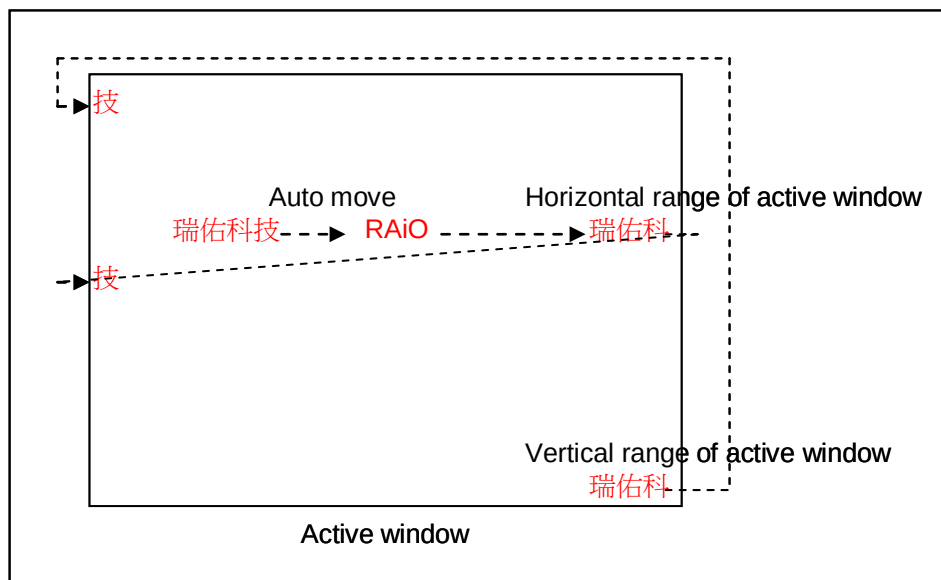


Figure 14-15 : Auto Line feed in Text Mode

14.7 Character Full-Alignment

RA8876 supports character full-alignment that makes the character to align each other when writing half or full size characters on the display memory. By setting REG[CDh] Bit7 = 1, the behavior of writing half or full size characters similar with below figure:

Note : Full-Alignment disable when GT different width font enable.

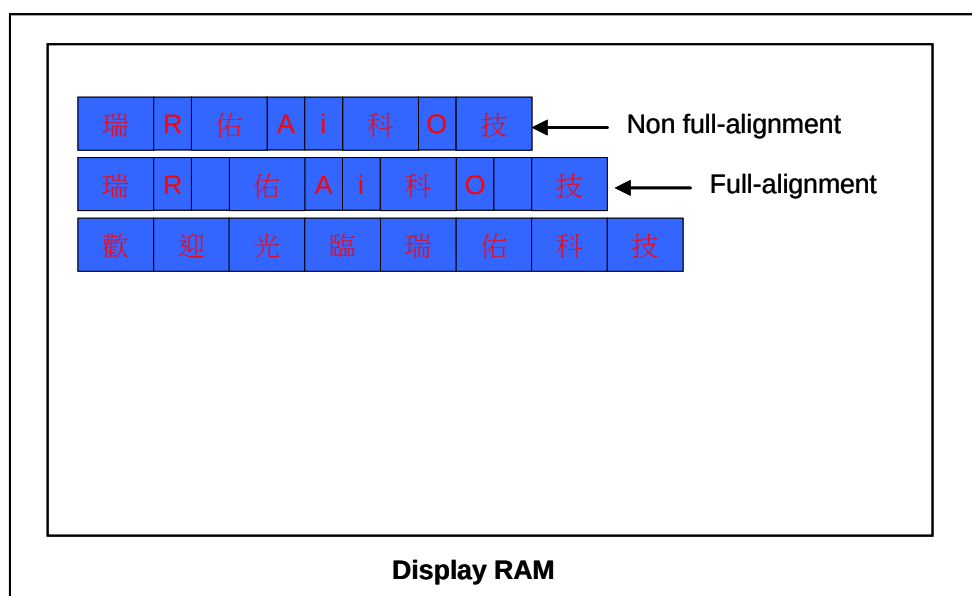


Figure 14-16: Full-Alignment Function

14.8 Cursor

There are two kinds of cursors defined in RA8876, one is graphic cursor and another is text cursor. The graphic cursor provides a 32x32 pixels graphic cursor which can be displayed at user-defined position. When the position is changed, the graphic cursor is moved. The text cursor provides a text relative cursor for text write function. The shape of it is a block and the height and width is programmable. The display location of text cursor indicates the location of text being currently written.

Note1 : When VDIR = 1 · pip window, graphic cursor and text cursor function will be disabled automatically.

Note2 : Cursors only display on main windows coordinates. PIP windows will be invisible.

14.8.1 Text Cursor

The text cursor's position can be set as blink or not. Cursor auto-move function is dominated by the active window. When a text is written, the cursor automatically moves to next position for text writing. It depends on the character size and character direction. When meeting the boundary of active window. Cursor will change to next row. The interval between the rows can also be set in pixels. Table 14-5 list the relative registers description.

Table 14-5 : Text Write Cursor Related Register Table

Register Name	Bit Num	Function Description	Address
FLDR	4-0	Character Line Gap Setting Register	D0h
F_CURX0/1	7-0 / 4-0	Text Cursor Horizontal Location	63h, 64h
F_CURY0/1	7-0 / 4-0	Text Cursor Vertical Location	65h, 66h
ICR	2	Text Mode Enable 0 : Graphic mode. 1 : Text mode.	03h
GTCCR	1	Text Cursor Enable 0 : Text cursor is not visible. 1 : Text cursor is visible.	3Ch
	0	Text Cursor Blink Enable 0 : Normal display. 1 : Blink display.	

Cursor Attribute – Cursor Blinking

The text cursor can be set as on or off or blinking with a fixed frequency. The control register is GTCCR(REG[3Ch]). The effect of blinking is repeating the cursor on(visible) and off(invisible). The blinking time of it is programmable and can be calculated as the formula below in unit of second.

$$\text{Blink Time (sec)} = \text{BTCR}[3\text{Dh}] \times (1/\text{Frame_Rate}).$$

Figure 14-17 show the example of cursor blink. The cursor position will follow the last data or character be written.

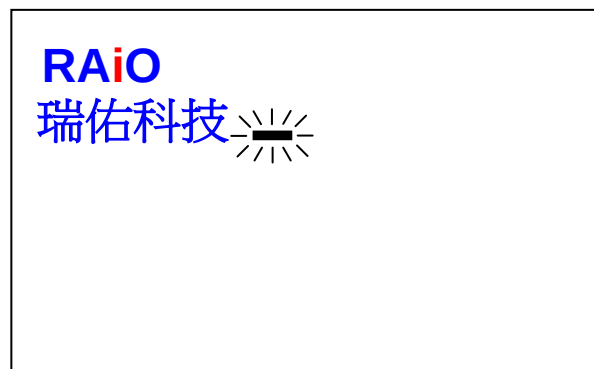


Figure 14-17: Cursor Blinking

Cursor Attribute – Cursor Height and Width

The shape of text cursor is programmable. The text cursor is a block with height and width programmable. The control register is CURHS(REG[3Eh]) and CURVS(REG[3Fh]). The text cursor in graphic mode is a line with width programmable and height fixed to 1 pixel. Please refer to Figure 14-18. The height and width of text cursor is also relative with an extra factors, the character enlargement setting(REG[CDh] Bit3~0). With the enlargement factor of 1, the width is set by CURHS/CURVS as 1~32 pixels. For enlargement factor is not 1, the real width and height of the cursor will be multiplied with the factor. Figure 14-18 is the example as character horizontal/vertical enlargement factor is 1. Note that the text cursor will not be affected by the character rotation. The shape of text cursor is still the same with the normal one. About the display please refer to Figure 14-19 below as example.

REG[3Eh] Text Cursor Horizontal Size Register (CURHR)	
Bit4-0 Text cursor horizontal size setting[4:0]	Width (Unit : Pixel)
00000 ~ 11111	1 ~ 32
REG[3Fh] Text Cursor Vertical Size Register (CURVR)	
Bit4-0 Text cursor Vertical size setting[4:0]	Height (Unit : Pixel)
00000 ~ 11111	1 ~ 32

Figure 14-18 : Text Cursor Height and Width Setting

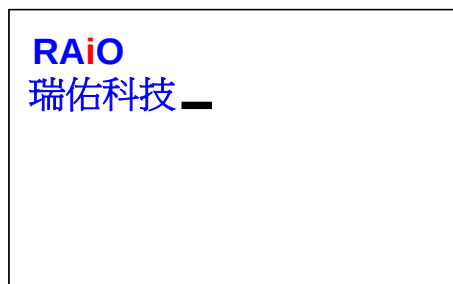


Figure 14-19 : Text Cursor Movement (without rotate)

14.8.2 Graphic Cursor

The size of graphic cursor is 32x32 pixels, each pixel is composed by 2-bit, which indicates 4 colors setting (color 0, color 1, background color, the inversion of background color). It represents that a graphic cursor takes 256 bytes(32x32x2/8). RA8876 provides four groups of graphic cursor for selection; users could use them just by setting related registers. By the way, the graphic cursor position is controlled by register GCHP0 (REG[40h]), GCHP1(REG[41h]), GCVP0(REG[42h]) and GCVP1(REG[43h]). The color of it is set by register GCC0(REG[44h])/ GCC1(REG[45h])/ Background color/ Inversion of Background color, depending on the data of it. Please refer to example for the detail explanation.

Note: The graphic cursor storage space only support 8-bit data, when initializing graphic cursor user should use graphic mode to write 256 8-bit data into graphic cursor's data space without check busy, and follow write data cycle to write data cycle interval must larger than 5 system clock if xnWait machismo hasn't be used.

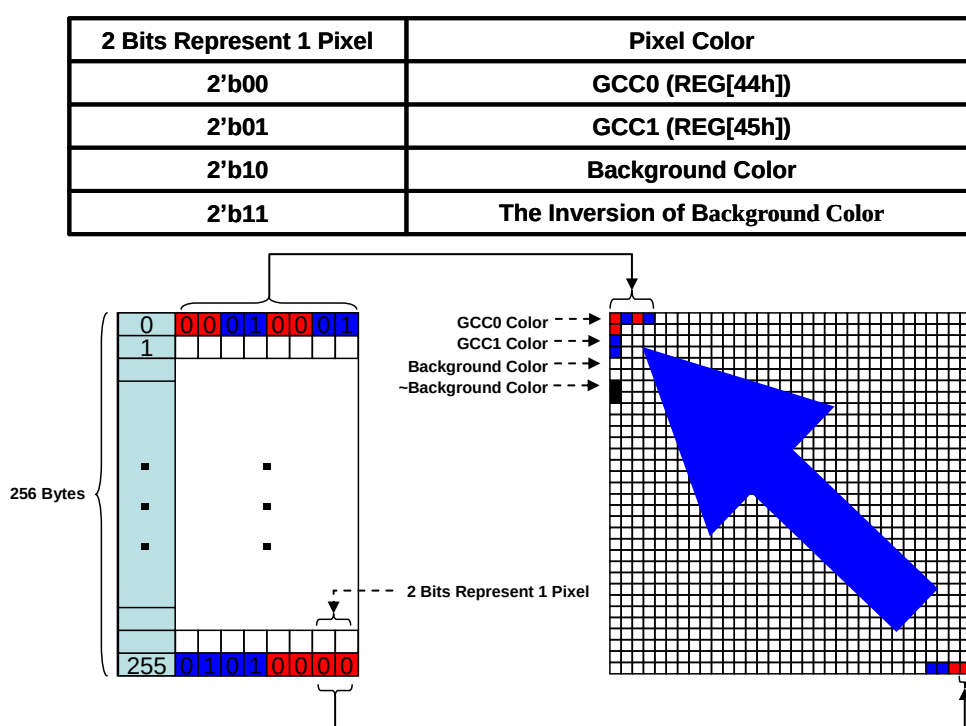


Figure 14-20 : Relation of Memory Mapping for Graphic Cursor

Procedure:

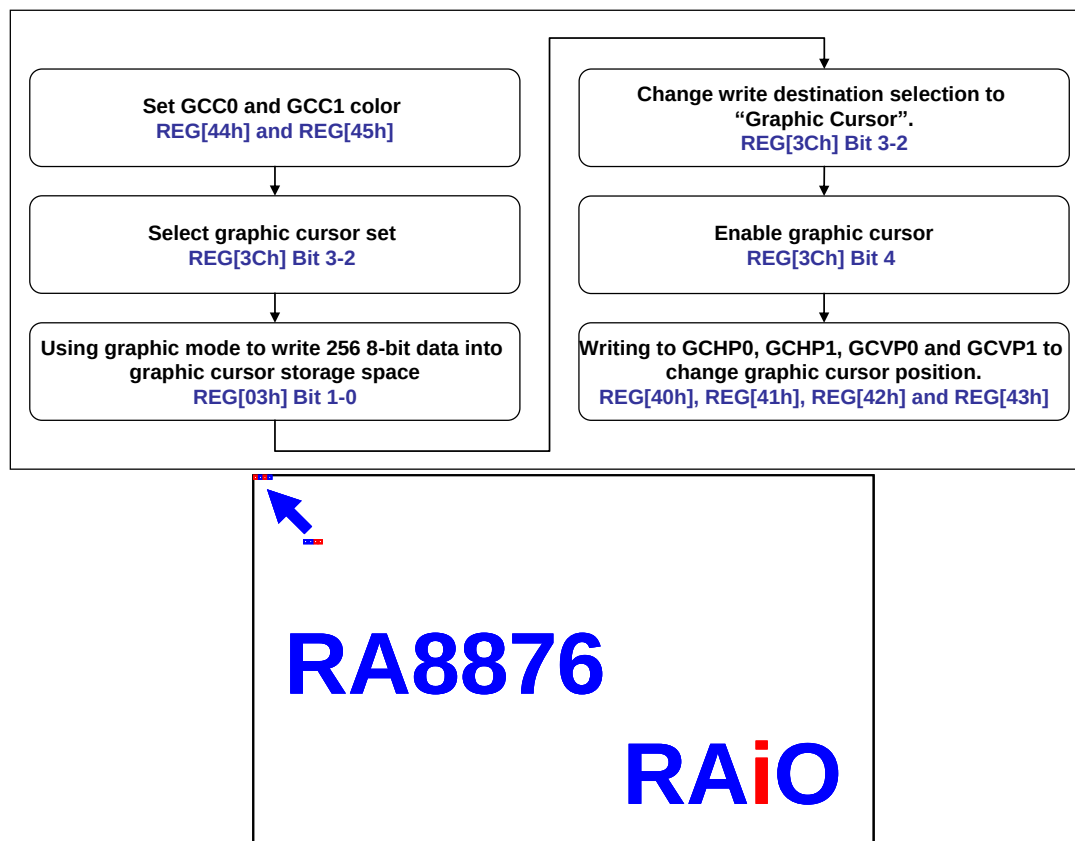


Figure 14-21 : The Display with Graphic Cursor

15. PWM Timer

The RA8876 has two 16-bit timers. Timers 0 and 1 have Pulse Width Modulation (PWM) function. The timer 0 has a dead-zone generator, which is used with a large current device.

The timer 0 and 1 share an 8-bit pre-scalar. Each timer has a clock divider, which generates 4 different divided signals (1, 1/2, 1/4 & 1/8). Each timer block receives its own clock signals from the clock divider, which receives the clock from the corresponding 8-bit pre-scalar. The 8-bit pre-scalar is programmable and divides the CCLK according to the loading value, which is stored in PSCLR and PMUXR registers.

The timer count buffer register (TCNTBn) has an initial value which is loaded into the down-counter when the timer is enabled. The timer compare buffer register (TCMPBn) has an initial value which is loaded into the compare register to be compared with the down-counter value. This double buffering feature of TCNTBn and TCMPBn makes the timer generate a stable output when the frequency and duty ratio are changed.

Each timer has its own 16-bit down counter, which is driven by the timer clock. When the down counter reaches zero, the timer interrupt request is generated to inform the CPU that the timer operation has been completed. When the timer counter reaches zero, the value of corresponding TCNTBn is automatically loaded into the down counter to continue the next operation. However, if the timer stops, for example, by clearing the timer enable bit of PCFGR during the timer running mode, the value of TCNTBn will not be reloaded into the counter.

The value of TCMPBn is used for pulse width modulation (PWM). The timer control logic changes the output level when the down-counter value matches the value of the compare register in the timer control logic. Therefore, the compare register determines the turn-on time (or turn-off time) of a PWM output.

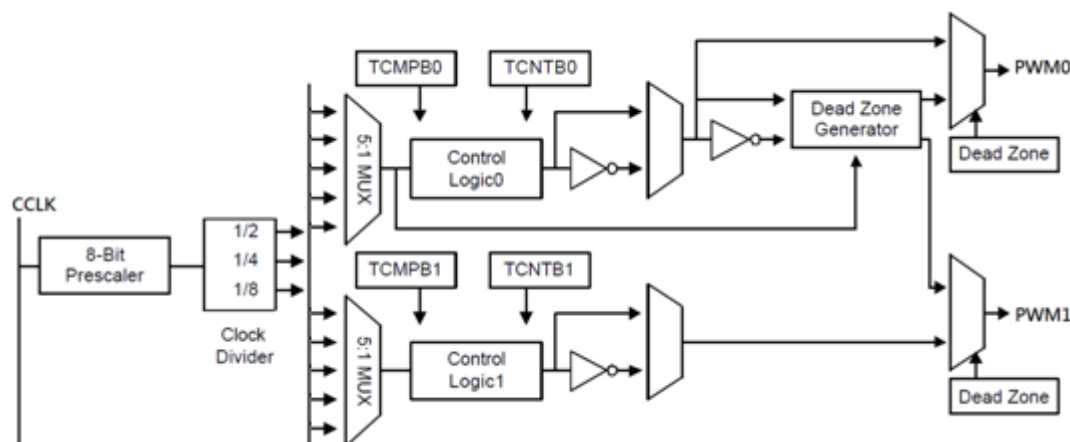


Figure 15-1 : 16-bit PWM Timer Block Diagram

15.1 Basic Timer Operation

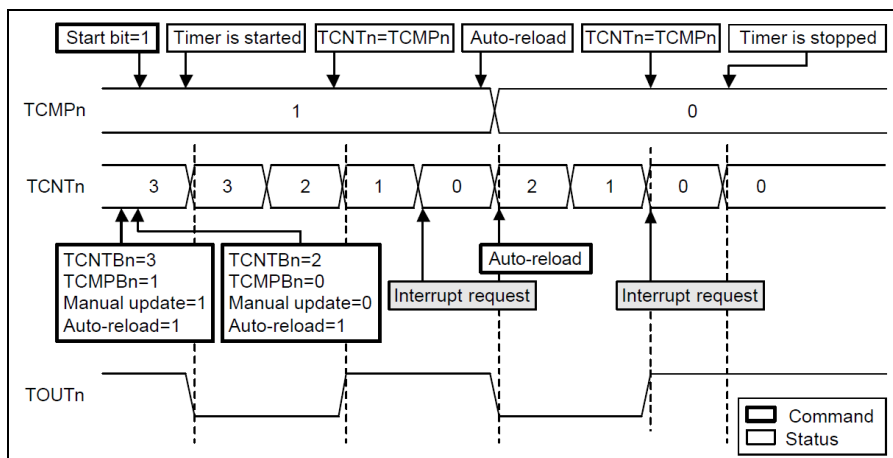


Figure 15-2 : Timer Operations

A timer has TCNTBn, TCNTn, TCMPBn and TCMPn. (TCNTn and TCMPn are the names of the internal registers. The TCNTn register can be read from the TCNTOn register) The TCNTBn and the TCMPBn are loaded into the TCNTn and the TCMPn when the timer reaches 0. When the TCNTn reaches 0, an interrupt request will occur if the interrupt is enabled.

15.2 Auto Reload & Double Buffering

PWM Timers have a double buffering function, enabling the reload value changed for the next timer operation without stopping the current timer operation. So, although the new timer value is set, a current timer operation is completed successfully.

The timer value can be written into Timer Count Buffer register (TCNTBn) and the current counter value of the timer can be read from Timer Count Observation register (TCNTOn). If the TCNTBn is read, the read value does not indicate the current state of the counter but the reload value for the next timer duration.

The auto-reload operation copies the TCNTBn into TCNTn when the TCNTn reaches 0. The value, written into the TCNTBn, is loaded to the TCNTn only when the TCNTn reaches 0 and auto reload is enabled. If the TCNTn becomes 0 and the auto reload bit is 0, the TCNTn does not operate any further.

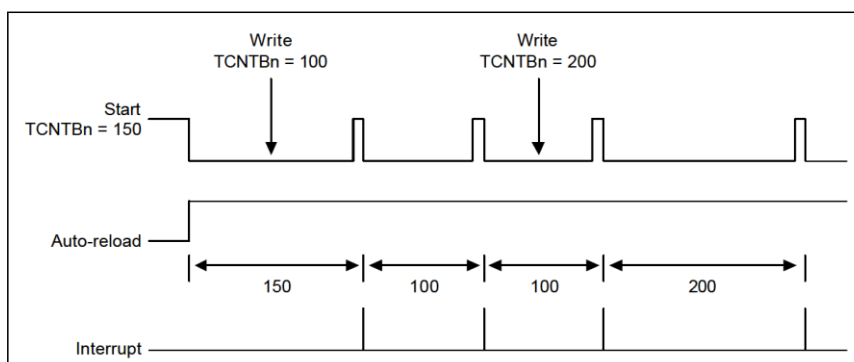


Figure 15-3 : Example of Double Buffering Function

15.3 Timer Initialization and Inverter Bit

An auto reload operation of the timer occurs when the down counter reaches 0. So, a starting value of the TCNTn has to be defined by the user in advance. In this case, the starting value has to be loaded in advance. The following steps describe how to start a timer:

- 1) Write the initial value into TCNTBn and TCMPBn.
- 2) It is recommended that you configure the inverter on/off bit. (Whether use inverter or not).
- 3) Set start bit of the corresponding timer to start the timer.

If the timer is stopped by force, the TCNTn will continue count until it reach to 0 then stop. If a new value has to be set, it will be reloaded from TCNTBn at next timer starts.

Note:

Whenever PWMn inverter on/off bit is changed, the PWMn logic value will also be changed whether the timer runs. Therefore, it is desirable that the inverter on/off bit is configured before set start bit of the corresponding timer to start the timer.

15.4 Timer Operation

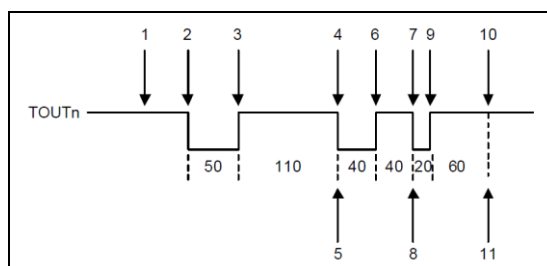


Figure 15-4 : Example of a Timer Operation

The above Figure 15-4 shows the result of the following procedure:

1. Enable the auto re-load function. Set the TCNTBn to 160 (50+110) and the TCMPBn to 110. Configure the inverter bit (on/off). The start bit sets TCNTn and TCMPn to the values of TCNTBn and TCMPBn, respectively. And then, set the TCNTBn and the TCMPBn to 80 (40+40) and 40, respectively, to determine the next reload value.
2. Set the start bit, the inverter is off and auto reload is on. The timer starts counting down after latency time within the timer resolution.
3. When the TCNTn has the same value as that of the TCMPn, the logic level of the PWMn is changed from low to high.
4. When the TCNTn reaches 0, the interrupt request is generated and TCNTBn value is loaded into a temporary register. At the next timer tick, the TCNTn is reloaded with the temporary register value (TCNTBn).
5. In Interrupt Service Routine (ISR), the TCNTBn and the TCMPBn are set to 80 (20+60) and 60, respectively, for the next duration.
6. When the TCNTn has the same value as the TCMPn, the logic level of PWMn is changed from low to high.
7. When the TCNTn reaches 0, the TCNTn is reloaded automatically with the TCNTBn, triggering an interrupt request.
8. In Interrupt Service Routine (ISR), auto reload and interrupt request are disabled to stop the timer.
9. When the value of the TCNTn is same as the TCMPn, the logic level of the PWMn is changed from low to high.
10. Even when the TCNTn reaches 0, the TCNTn is not any more reloaded and the timer is stopped because auto reload has been disabled.
11. No more interrupt requests are generated.

15.5 Pulse Width Modulation (PWM)

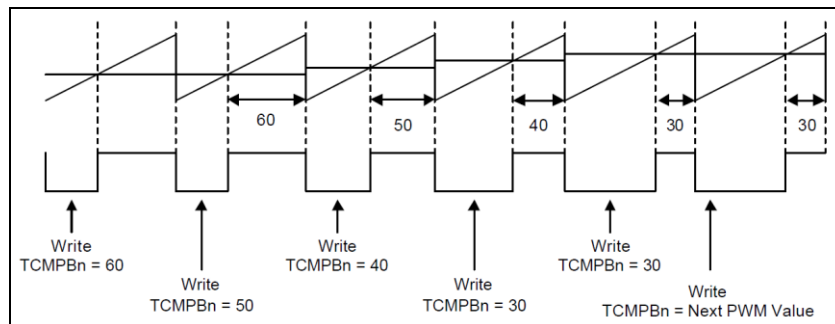


Figure 15-5 : Example of PWM

PWM function can be implemented by using the TCMPBn. PWM frequency is determined by TCNTBn. The Figure shows a PWM value determined by TCMPBn.

For a higher PWM value, decrease the TCMPBn value. For a lower PWM value, increase the TCMPBn value.

If an output inverter is enabled, the increment/decrement may be reversed. The double buffering function allows the TCMPBn, for the next PWM cycle, written at any point in the current PWM cycle by ISR or other routine.

15.6 Output Level Control

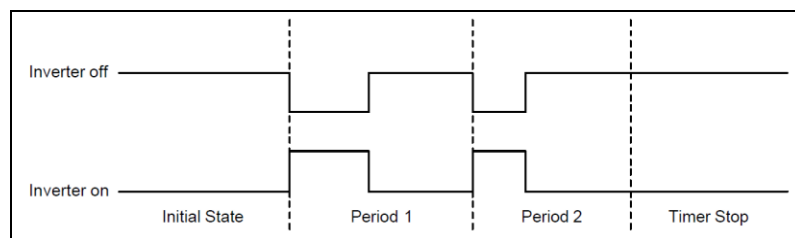


Figure 15-6 : Inverter On/Off

The following procedure describes how to maintain PWM as high or low (assume the inverter is off):

1. Turn off the auto reload bit. And then, PWMn goes to high level and the timer is stopped after the TCNTn reaches 0 (recommended).
2. Stop the timer by clearing the timer start/stop bit to 0. If TCNTn < TCMPn, the output level is high. If TCNTn > TCMPn, the output level is low.
3. The PWMn can be inverted by the inverter on/off bit in PCFGR. The inverter removes the additional circuit to adjust the output level.

15.7 Dead Zone Generator

The Dead Zone is for the PWM control in a power device. This function enables the insertion of the time gap between a turn-off of a switching device and a turn on of another switching device. This time gap prohibits the two switching devices from being turned on simultaneously, even for a very short time.

PWM0 is the PWM output. nPWM0 is the inversion of the PWM0. If the dead zone is enabled, the output wave form of PWM0 and nPWM0 will be PWM0_DZ and nPWM0_DZ, respectively. nPWM0/nPWM0_DZ is routed to the PWM1 pin.

In the dead zone interval, PWM0_DZ and nPWM0_DZ can never be turned on simultaneously.

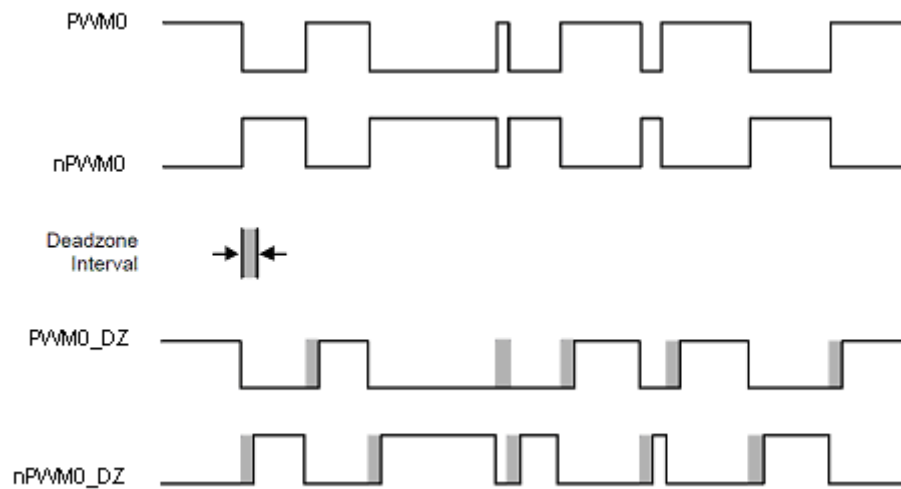


Figure 15-7 : The Wave Form When a Dead Zone Feature is Enabled

15.8 Dead Zone Application

Most of use on switching power driver is Pulse Width Modulation (PWM). Show like below:

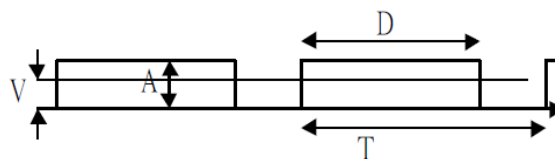


Figure 15-8

1. PWM output only have ON/OFF state. Voltage (A) is maximum voltage in ON state. Voltage in OFF state is 0.
2. If period is T, ON duration is D, then PWM has average voltage $V = (D/T) \cdot A$. i.e. We may generate any voltage in range of 0 ~ A.
3. If switching frequency too slow, it will cause motor vibration or high frequency noise from wiring. In general, reasonable switching frequency is from 4KHz to 8KHz.
4. Motor's characteristic is a low pass filter. Too high frequency cannot response it. So if switching frequency within a reasonable range, it works like a linear amplifier.

To a simple load, PWM switching circuit's block diagram like below:

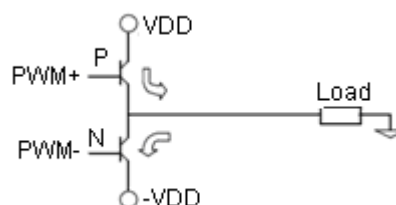


Figure 15-9

1. Use two sets of transistor to control positive power and negative power. User may choose proper transistor type according to users requirements.
2. PWM signal have positive type and negative type. PWM+ control turn on positive power or not. PWM- control turn on negative power or not.
3. Switch case have following:

P=OFF / N=OFF : separate Load & power.

P=ON / N=OFF : Load connects to positive power.

P=OFF / N=ON : Load connects to negative power.

P=ON / N=ON : short power & burn out power driver.

Basically, PWM+ & PWM- is inverted, it is impossible to ON simultaneously. But consider power MOS' transition response, response time of transistor OFF is longer than transistor ON. So it is possible to turn ON both in switching. Thus will cause two kinds of result:

- a. Long turn ON time to cause driver burn out.
- b. Short turn ON time may not burn out driver immediately but will generate heat and cause driver burn out by heat.

So PWM control must have protection.



Figure 15-10

1. Basically, PWM- is invert from PWM+.
2. In the moment of switching, it must have an OFF time on both edges. The period depends on driver's characteristics, maybe 1us ~ 4us.

16. Serial Bus Master Unit

16.1 Initial Display Unit

Initial display unit embeds a tiny processor and use to show display data and execute micro code which stored in the serial flash and need not external MPU participate. If this function enable, it will auto execute after power-on, until micro code execute complete then handover control rights to external MPU.

Initial display unit limit micro code & display data must exist in the same serial flash.

Initial display unit support 12 instructions. They are

- | | |
|--|---------------------------|
| 1. EXIT: Exit instruction (00h/FFh) | -- one byte instruction |
| 2. NOP: NOP instruction (AAh) | -- one byte instruction |
| 3. EN4B: Enter 4-Byte mode instruction (B7h) | -- one byte instruction |
| 4. EX4B: Exit 4-Byte mode instruction (E9h) | -- one byte instruction |
| 5. STSR: Status read instruction (10h) | -- two bytes instruction |
| 6. CMDW: Command write instruction (11h) | -- two bytes instruction |
| 7. DATR: Data read instruction (12h) | -- two bytes instruction |
| 8. DATW: Data write instruction (13h) | -- two bytes instruction |
| 9. REPT: Load repeat counter instruction (20h) | -- two bytes instruction |
| 10. ATTR: Fetch Attribute instruction (30h) | -- two bytes instruction |
| 11. JUMP: Jump instruction (80h) | -- five bytes instruction |
| 12. DJNZ: Decrement & Jump instruction (81h) | -- five bytes instruction |

One bytes instruction:

- **Exit instruction (EXIT) – 00h | FFh | Undefined instructions**
It without contains any extra parameters. The EXIT instruction is executed to exit the initial display function & return control right to MPU.
- **NOP instruction (NOP) – AAh**
It without contains any extra parameters. It will do nothing and then fetch next instruction.
- **Enter 4-Byte mode instruction (EN4B) – B7h**
It without contains any extra parameters. This instruction enables accessing the address length of 32-bit for the memory area of higher density (larger than 128Mb). The control unit default is in 24-bit address mode; after sending out the EN4B instruction, the address length becomes 32-bits instead of the default 24-bits. There are three methods to exit the 4-byte mode: writing exit 4-byte mode (EX4B) instruction, Hardware Reset or power-off.
- **Exit 4-Byte mode instruction (EX4B) – E9h**
It without contains any extra parameters. The EX4B instruction is executed to exit the 4-byte address mode and return to the default 3-bytes address mode. Once exiting the 4-byte address mode, the address length will return to 24-bit.

Two bytes instruction:

- **Load repeat counter instruction (REPT) – 20h + param[0]**
It contains one byte parameter. This parameter stands for repeat counter value & used by DJNZ instruction.
- **Fetch attribute instruction (ATTR) – 30h + param[0]**
It contains one byte parameter. This instruction use to configure controller how to read serial flash data when under fetch instruction period. In Param[0]:
 - bit [3:0] is clock divisor for SPI clock. Controller according to system clock set high period or low period for SPI clock. Default is 0. $F_{sck} = F_{core} / (divisor + 1) \times 2$
 - bit [4] is device mode selection for [CPOL, CPHA]. Value '0' is mode 0, value '1' is mode 3. Default is 1 for mode 3.
 - bit [5] is to define XnSFCS[1:0] deselect time or calls chip select high time (tCSH). Value '0' is 4 core clocks; value '1' is 8 clocks. Default is 8 core clocks.
 - bit [7:6] is dummy cycle number. These two bits set 4 kinds of dummy cycle. The Value 0, 1, 2 or 3 stands for 0, 8, 16 or 24 dummy cycles. Default is 0. If dummy cycle number is 0 then serial flash read command code is 03h, otherwise serial flash read command code is 0Bh.
- **Status read instruction (STSR) – 10h + param[0]**
It contains one byte parameter. This parameter stands for expected value in Status read instruction. If returned data is not match with expected value, this read instruction will repeat execution.
- **Command write instruction (CMDW) – 11h + param[0]**
It contains one byte parameter. This parameter stands for write value for Command write instruction.
- **Data read instruction (DATR) – 12h + param[0]**
It contains one byte parameter. This parameter stands for expected value in Data read instruction. If returned data is not match with expected value, this read instruction will repeat execution.
- **Data write instruction (DATW) – 13h + param[0]**
It contains one byte parameter. This parameter stands for write value for Data write instruction.

Five bytes instruction:

- **Jump instruction (JUMP) – 80h + param[3] + param[2] + param[1] + param[0]**
It contains 4-byte parameters. Parameter 3~0 are serial flash's 28-bits address information. ie. param[3] is address[27:24], param[2] is address[23:16], param[1] is address[15:8] and param[0] is address[7:0]. After execution, next instruction will fetch from this specified address.
- **Decrement & Jump while not equal to zero (DJNZ) instruction – 81h + param[3] + param[2] + param[1] + param[0]**
It contains 4-byte parameters. Parameter 3~0 are serial flash's 28-bits address information. ie. param[3] is address[27:24], param[2] is address[23:16], param[1] is address[15:8] and param[0] is address[7:0]. If repeat counter equal zero then next instruction will fetch from "current instruction address + 5", otherwise repeat counter will decrement one and jump to specified address.

After power on reset Initial display unit will search matched serial flash device over 2 SPI bus' slave select (XnSFCS[1:0]). Its first eight bytes data from address 0000h down to 0007h must as "61h, 72h, 77h, 63h, 77h, 62h, 78h, 67h". If serial flash was be recognized then continue fetch next address (0008h) otherwise return control right to MPU. Start from address 0008h the serial flash store each kind of to be executing instructions, if initial display unit fetch EXIT instruction or an undefined instruction code then return control right to MPU.

Example of initial display contents in serial flash:

It will display color bar on a 320x240 TFT panel.

```
// addr: 'h0000
61 72 77 63 77 62 78 67      // ID
AA                             // NOP
30 03                         // ATTR('h03)
AA                             // NOP
E9                             // EX4B
AA                             // NOP
20 06                         // REPT('h06)

// addr: 'h0010
10 52                         // STS_RD(52)
11 03 13 55                  // REG_WR('h03, 'h55);
11 03 12 55                  // REG_RD('h03, 'h55);
13 AA                         // DAT_WR('hAA);
12 AA                         // DAT_RD('hAA);
11 03 13 00                  // REG_WR('h03, 'h00);

// addr: 'h0022
AA                             // NOP
81 00 00 00 22              // DJNZ(32'h0000_0022);
80 00 00 00 30              // JUMP(32'h0000_0030);
78
78
78

// addr: 'h0030
// Chip configuration
11 01 13 00                  // MPU.REG_WR('h01, 'h00); // normal, Key dis, TFT-24, iicm dis, sf
dis, 8b mpu
11 13 13 03                  // MPU.REG_WR('h13, 'h03); // Panel polarity & Idle state
// Enable color bar
11 12 13 60                  // MPU.REG_WR('h12, 'h60); // sync w/ pclk rising edge, display on/off,
color bar on/off, VDIR, RGB sequence

// hdwr
11 14 13 27                  // MPU.REG_WR('h14, 'h27); // H: 320; data16 = `LCD_SEG_NO/8 - 1;
// vdhr
11 1A 13 EF                  // MPU.REG_WR('h1A, 'hEF); // V: 240; data16[7:0] =
`LCD_COM_NO - 1;
11 1B 13 00                  // MPU.REG_WR('h1B, 'h00); // V: 240; data16[15:8]=
`LCD_COM_NO - 1;

@0048
11 B9 13 23                  // MPU.REG_WR('hB9, 'h33); // select nss[1]
00                            // Exit
```

Restriction:

Initial display unit restricts micro codes & display data, fonts or required contents for micro codes must exist in the same serial flash. If user needs to switch to another serial flash then following code & display data etc. will point to that serial flash.

16.2 SPI Master Unit

During an SPI transfer, data is simultaneously transmitted and received. The serial clock line [SCK] synchronizes shifting and sampling of the information on the two serial data lines. The master places the information onto the MOSI line a half-cycle before the clock edge that the slave device uses to latch the data.

Four possible timing relationships can be chosen by using the Clock Polarity [CPOL] and Clock Phase [CPHA] bits in the Serial Peripheral Interface Control Register [SPICR]. Both master and slave devices must operate with the same timing.

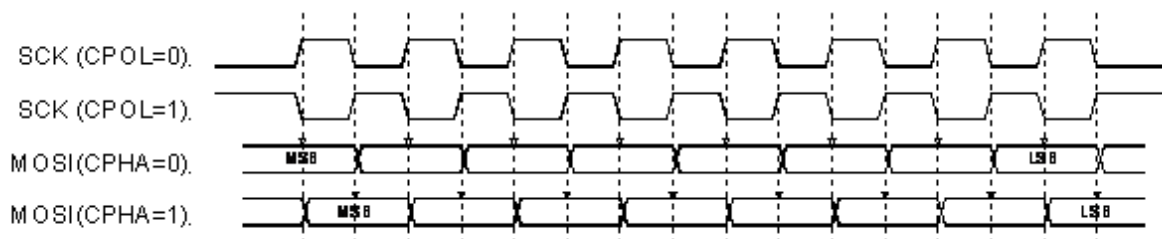


Figure 16-1

Transmitting data bytes

After programming the core's control register SPI transfers can be initiated. A transfer is initiated by writing to the Serial Peripheral Data Register [SPIDR]. Writing to the Serial Peripheral Data Register is actually writing to a 16 entries deep FIFO called the Write FIFO. Each write access adds a data byte to the Write FIFO. When the core is enabled – SS_ACTIVE is set ('1') – and the Write FIFO is not empty, the core automatically transfers the oldest data byte.

Receiving data bytes

Receiving data is done simultaneously with transmitting data; whenever a data byte is transmitted a data byte is received. For each byte that needs to be read from a device, a dummy byte needs to be written to the Write FIFO. This instructs the core to initiate an SPI transfer, simultaneously transmitting the dummy byte and receiving the desired data. Whenever a transfer is finished, the received data byte is added to the Read FIFO. The Read FIFO is the counterpart of the Write FIFO. It is an independent 16 entries deep FIFO. The FIFO contents can be read by reading from the Serial Peripheral Data Register [SPDR].

FIFO Overrun

Both the Write FIFO and the Read FIFO are FIFOs that use circular memories to simulate the infinite big memory needed for FIFOs. Because of this writing to a FIFO while it is full overwrites the oldest data byte. Writing to the Serial Peripheral Data Register [SPDR] while the Write FIFO is full sets the Overflow bit, however the damage is already done; the next byte to be transferred is not the oldest data byte, but the latest (newest).

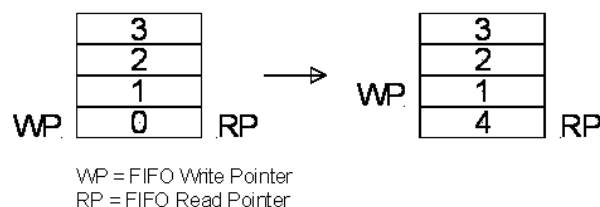


Figure 16-2

The only way to recover from this situation is to reset the Write Buffer. Both the Read FIFO and the Write FIFO are reset when the Slave Select signal active [SS_ACTIVE] bit is cleared ('0').

Read FIFO overruns might be less destructive. Especially when the SPI bus is used to transmit data only; e.g. when sending data to a DAC. The received data is simply ignored. The fact that the Read FIFO overruns is irrelevant. If the SPI bus is used to transmit and receive data, it is important to keep the Read FIFO aligned. The easiest way to do this is to perform a number of dummy reads equal to the amount of bytes transmitted modulo 16.

$$N_{dummy_reads} = N_{transmitted_bytes} \bmod 16$$

Note that a maximum sequence of 16 bytes can be stored in the Read Buffer before the oldest data byte gets overwritten. It is therefore necessary to empty (read) the Read FIFO every 16 received bytes.

Reference code for SPI master loop test (connect xmosi to xmiso)

```
REG_WR ('hBB, 8'h1f); //Divisor, configure SPI clock frequency
REG_WR ('hB9, 8'b0001_1111); // {1'b0, mask, nSS_sel, ss_active, ovfirqen, emtirqen, cpol, cpha}, nSS
low
REG_WR ('hB8, 8'h55); // TX
REG_WR ('hB8, 8'haa); // TX
REG_WR ('hB8, 8'h87); // TX
REG_WR ('hB8, 8'h78); // TX
wait (xintr);
REG_RD ('hBA, acc);
while (acc != 8'h84) begin
    $display ("wait for FIFO empty ...");
    REG_RD ('hBA, acc);
end
REG_WR ('hBA, 8'h04); // clear interrupt flag
REG_RD ('hB8, 8'h55); // RX
REG_RD ('hB8, 8'haa); // RX
REG_RD ('hB8, 8'h87); // RX
REG_RD ('hB8, 8'h78); // RX
REG_WR ('hB9, 8'b0000_1111); // {1'b0, mask, nSS_sel, ss_active, ovfirqen, emtirqen, cpol, cpha}, nSS
high.
```

16.3 Serial Flash Control Unit

RA8876 builds in a SPI master interface for Serial Flash/ROM, supporting for protocol of 4-BUS (Normal Read), 5-BUS (FAST Read), Dual mode 0, Dual mode 1 with Mode 0/Mode 3.

Serial Flash/ROM function can be used for FONT mode and DMA mode. FONT mode means that the external serial Flash/ROM is treated as a source of character bitmap. To support the most useful characters, RA8876 is compatible with the character ROM of professional font vendor—Genitop Inc. in Shanghai. DMA mode means that the external Flash/ROM is treated as the data source of DMA (Direct Memory Access). User can speed up the data transfer to display memory and need not MPU intervene by this mode.

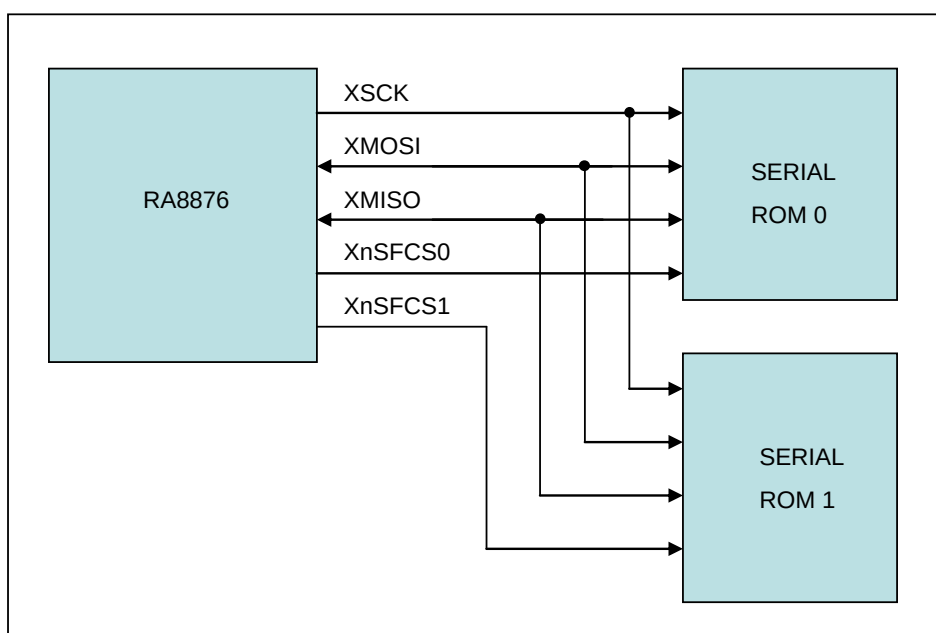


Figure 16-3 : RA8876 Serial Flash/ROM System

About Serial Flash/ROM read command protocol setting, please refer to Table 16-1 as below :

Table 16-1 : Read Command Code & Behavior Selection

REG [B7h] BIT[3:0]	Read Command code
000xb	1x read command code – 03h. Normal read speed. Single data input on xmis0. Without dummy cycle between address and data.
010xb	1x read command code – 0Bh. To some serial flash provide faster read speed. Single data input on xmis0. 8 dummy cycles inserted between address and data.
1x0xb	1x read command code – 1Bh. To some serial flash provide fastest read speed. Single data input on xmis0. 16 dummy cycles inserted between address and data.
xx10b	2x read command code – 3Bh. Interleaved data input on xmis0 & xmisi. 8 dummy cycles inserted between address and data phase. (mode 0)
xx11b	2x read command code – BBh. Address output & data input interleaved on xmis0 & xmisi. 4 dummy cycles inserted between address and data phase. (mode 1)

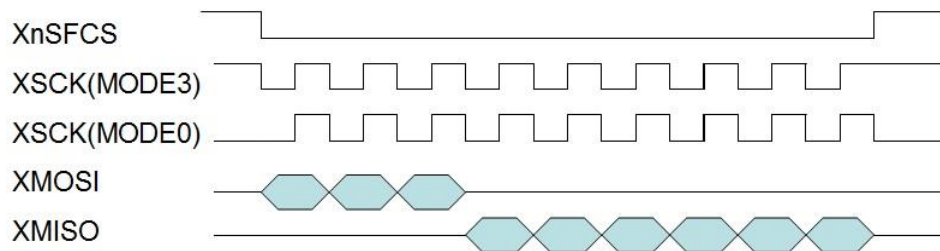
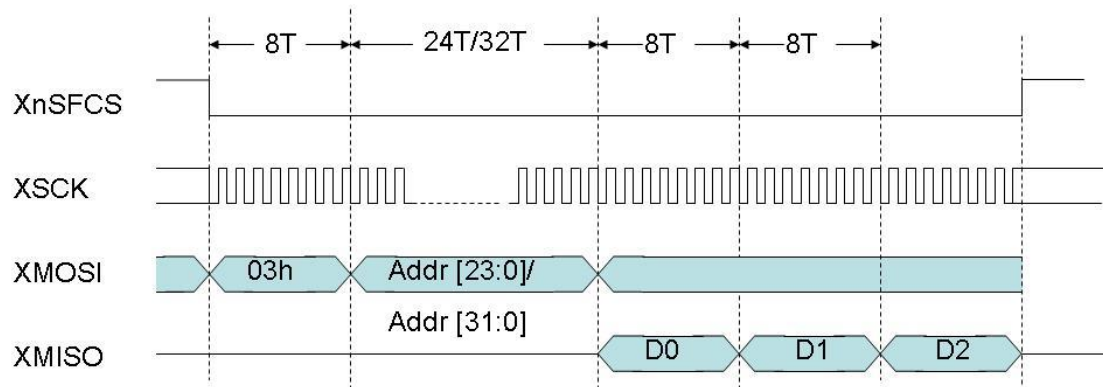


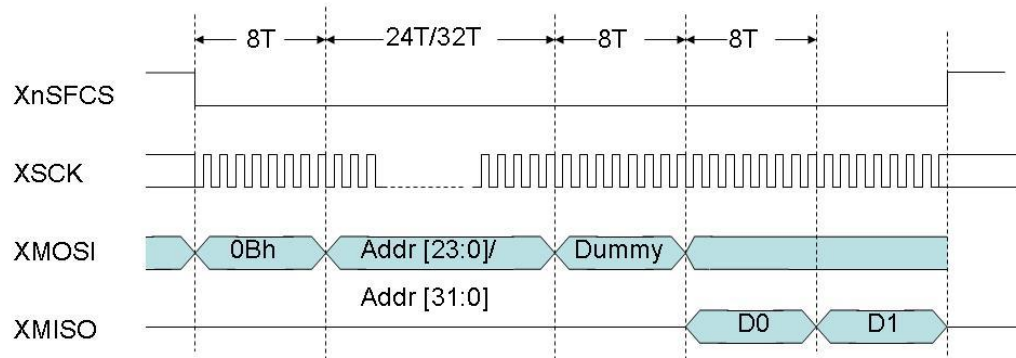
Figure 16-4 : Mode 0 and Mode 3 Protocol



If REG[B7h] Bit 5 set to 0, Then Addr state will be 24T

If REG[B7h] Bit 5 set to 1, Then Addr state will be 32T

Figure 16-5 : Normal Read Command



If REG[B7h] Bit 5 set to 0, Then Addr state will be 24T

If REG[B7h] Bit 5 set to 1, Then Addr state will be 32T

Figure 16-6 : Fast Read Command

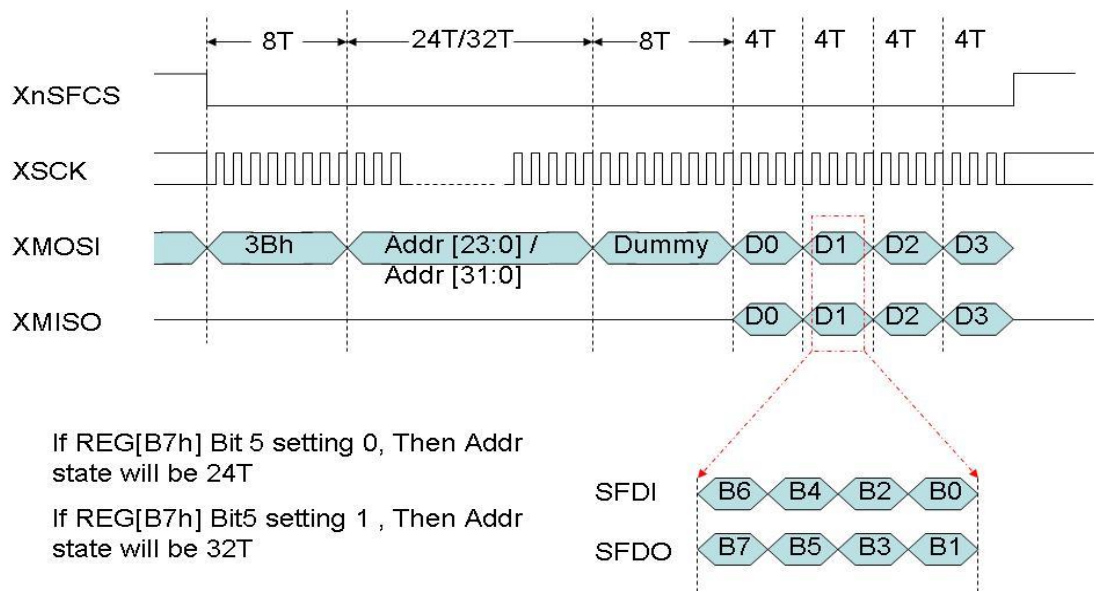
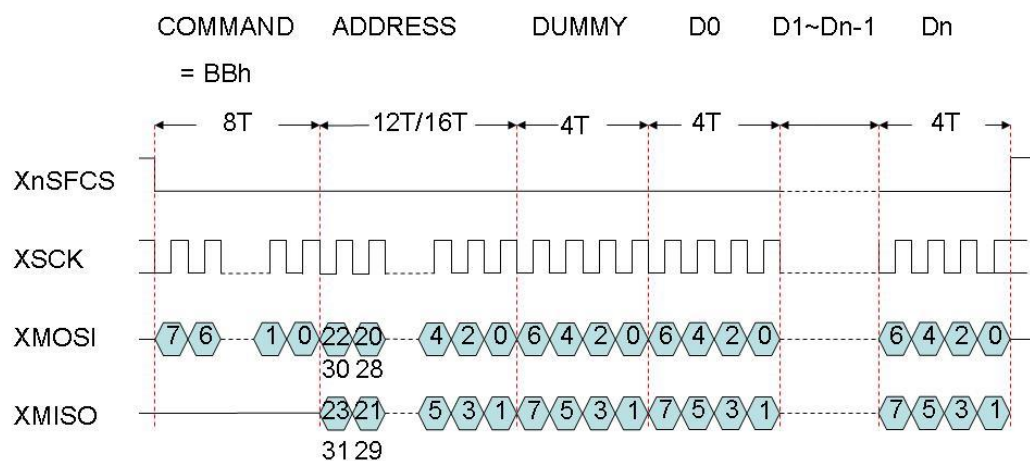


Figure 16-7 : Dual Output Read Command Mode 0



If REG[B7h] Bit 5 setting 0, Then Addr state will be 12T

If REG[B7h] Bit5 setting 1, Then Addr state will be 16T

Figure 16-8 : Dual – 1 Read (Reg need to modify)

16.3.1 External Serial Character ROM

The RA8876 supports the various fonts writing to display memory by using external Genitop Inc. serial Character ROM. RA8876 is compatible with the following products of Genitop Inc., GT21L16TW/GT21H16T1W, GT30L16U2W, GT30L24T3Y/GT30H24T3Y, GT30L24M1Z, and GT30L32S4W/GT30H32S4W. These various fonts include 16x16, 24x24, 32x32, and variable-width character size.

There are 3 types of character code format, 1 byte/2 bytes/4 bytes data, as explained below:

1. one byte character code – ASCII code for all Character ROMs
2. 2~4 bytes GB character code – The standard decoding of GB18030 in GT30L24M1Z
3. 2 bytes character code + 2 bytes Index code – Only used in Uni-code decoding of GT30L16U2W
4. Other character code length are 2 bytes only

Before adapting the specific character ROM product, it is suggested that user should know the coding rule of it first. For the detail of the mapping rule and characters set table, please contact with Genitop Inc.

To Note that in GT30L16U2W datasheet, the uni-code character code needs to refer to extra table called “ZFindex Table” to determine the actually bitmap ROM address, If user write a UNI-CODE in the range of 00A1h~33D5h or E76Ch~FFE5h, which is a special coding area, then the extra 2bytes character code (high byte first) is needed for reference of “ZFindex table”. Other UNICODE code outside the range only need 2 bytes of character code. About the detail, please also refer to the datasheet of GT30L16U2W.

EX: If user will be written UNI-CODE (00A2) with GT30L16U2W, which is located in the range of 00A1h~33D5h, then MPU must write extra 2 bytes of character code indexed from ZFindex table to RA8876.

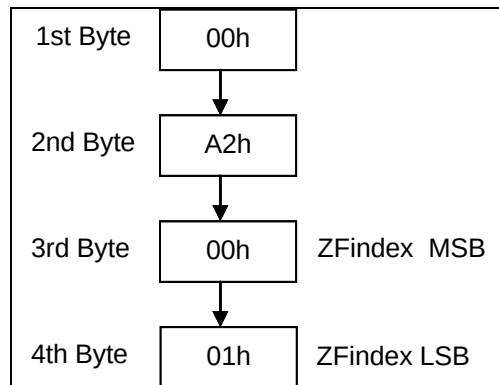


Figure 16-9 : Uni-Code Zfindex

The register of “External character ROM Cycle Speed Select” provides user modulating the speed of access external serial Flash/ROM cycle speed so that can match the ROM require access timing. The procedure of writing text just refers to below figure:

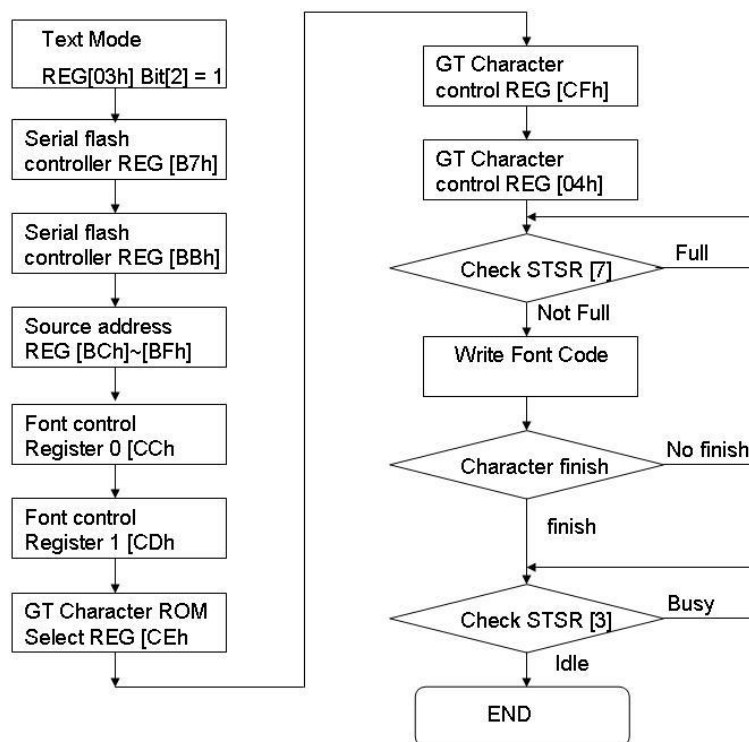


Figure 16-10 : External Character ROM Programming Procedure

16.3.2 External Serial Data ROM

External serial Flash/ROM interface can be treated as the source of data. It can be accessed by DMA (Direct Memory Access) and only operate in graphic mode.

The serial Flash/ROM interface can be used as the image source of DMA function. The Flash/ROM is treated as mass data storage. Serial flash/ROM's content formats follow structure in SDRAM memory. Image data format in serial flash as below:

8bpp data

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Addr	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0001h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶	0000h	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
0003h	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶	0002h	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
0005h	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶	0004h	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
0007h	R ₇ ⁷	R ₇ ⁶	R ₇ ⁵	G ₇ ⁷	G ₇ ⁶	G ₇ ⁵	B ₇ ⁷	B ₇ ⁶	0006h	R ₆ ⁷	R ₆ ⁶	R ₆ ⁵	G ₆ ⁷	G ₆ ⁶	G ₆ ⁵	B ₆ ⁷	B ₆ ⁶
0009h	R ₉ ⁷	R ₉ ⁶	R ₉ ⁵	G ₉ ⁷	G ₉ ⁶	G ₉ ⁵	B ₉ ⁷	B ₉ ⁶	0008h	R ₈ ⁷	R ₈ ⁶	R ₈ ⁵	G ₈ ⁷	G ₈ ⁶	G ₈ ⁵	B ₈ ⁷	B ₈ ⁶
000Bh	R ₁₁ ⁷	R ₁₁ ⁶	R ₁₁ ⁵	G ₁₁ ⁷	G ₁₁ ⁶	G ₁₁ ⁵	B ₁₀ ⁷	B ₁₀ ⁶	000Ah	R ₁₀ ⁷	R ₁₀ ⁶	R ₁₀ ⁵	G ₁₀ ⁷	G ₁₀ ⁶	G ₁₀ ⁵	B ₁₀ ⁷	B ₁₀ ⁶

16bpp data

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Addr	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0001h	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	0000h	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
0003h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	0002h	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
0005h	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	0004h	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
0007h	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	0006h	G ₃ ⁴	G ₃ ³	G ₃ ²	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³
0009h	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	R ₄ ³	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	0008h	G ₄ ⁴	G ₄ ³	G ₄ ²	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴	B ₄ ³
000Bh	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	R ₅ ³	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	000Ah	G ₅ ⁴	G ₅ ³	G ₅ ²	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴	B ₅ ³

24bpp data

Addr	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Addr	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0001h	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰	0000h	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
0003h	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰	0002h	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
0005h	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰	0004h	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
0007h	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	G ₂ ¹	G ₂ ⁰	0006h	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³	B ₂ ²	B ₂ ¹	B ₂ ⁰
0009h	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³	B ₃ ²	B ₃ ¹	B ₃ ⁰	0008h	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	R ₂ ²	R ₂ ¹	R ₂ ⁰
000Bh	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	R ₃ ²	R ₃ ¹	R ₃ ⁰	000Ah	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	G ₃ ¹	G ₃ ⁰

DMA function provides a faster method for user to update/transfer mass data to display memory. The only source of DMA function in RA8876 is external serial Flash/ROM interface. There are two kinds of data type defined for the DMA. One is linear mode and the other is block mode. It provides a flexible selection for user application. The destination of DMA function is dominated by active window in display memory. When DMA function is active, the specific data from serial Flash/ROM will be transferred one by one to display memory by RA8876 automatically. After the DMA function is completed, an interrupt will be asserted to Note host. About the detail operation, please refer to following sections.

16.3.2.1 DMA in Linear Mode for External Serial Data ROM

The DMA linear used to send CGRAM data for SDRAM. Active windows color depth must be set as 8bpp. The flow chart reference Figure 16-11.

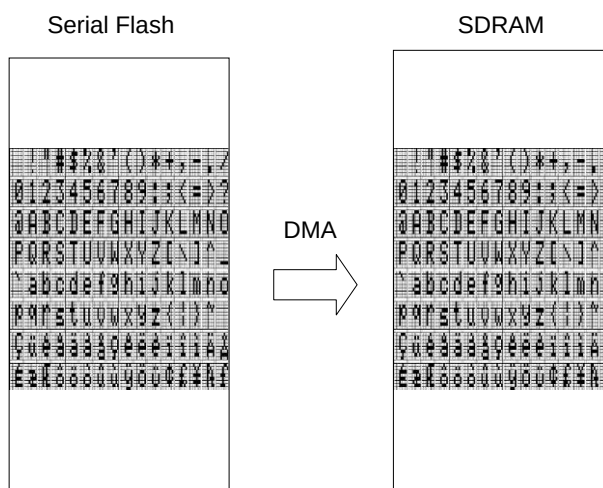


Figure 16-11

16.3.2.2 DMA in Block Mode for External Serial Data ROM

Then DMA block mode used for graphic move function. The process unit is pixel. The flow chart reference below

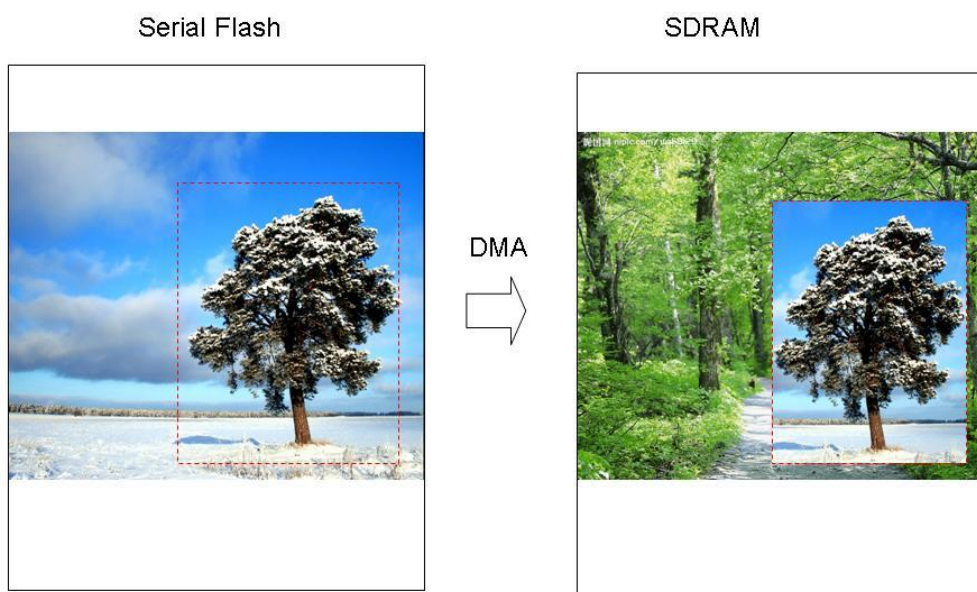


Figure 16-12 : DMA Function

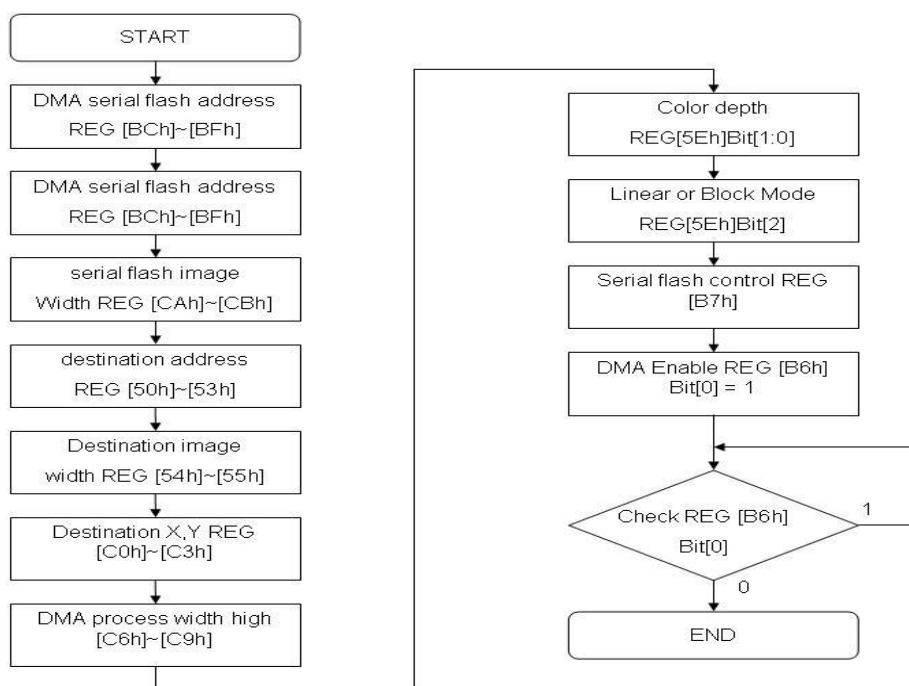


Figure 16-13 : Enable DMA Procedure – Check Flag

REG[03h] Bit[7] = 0

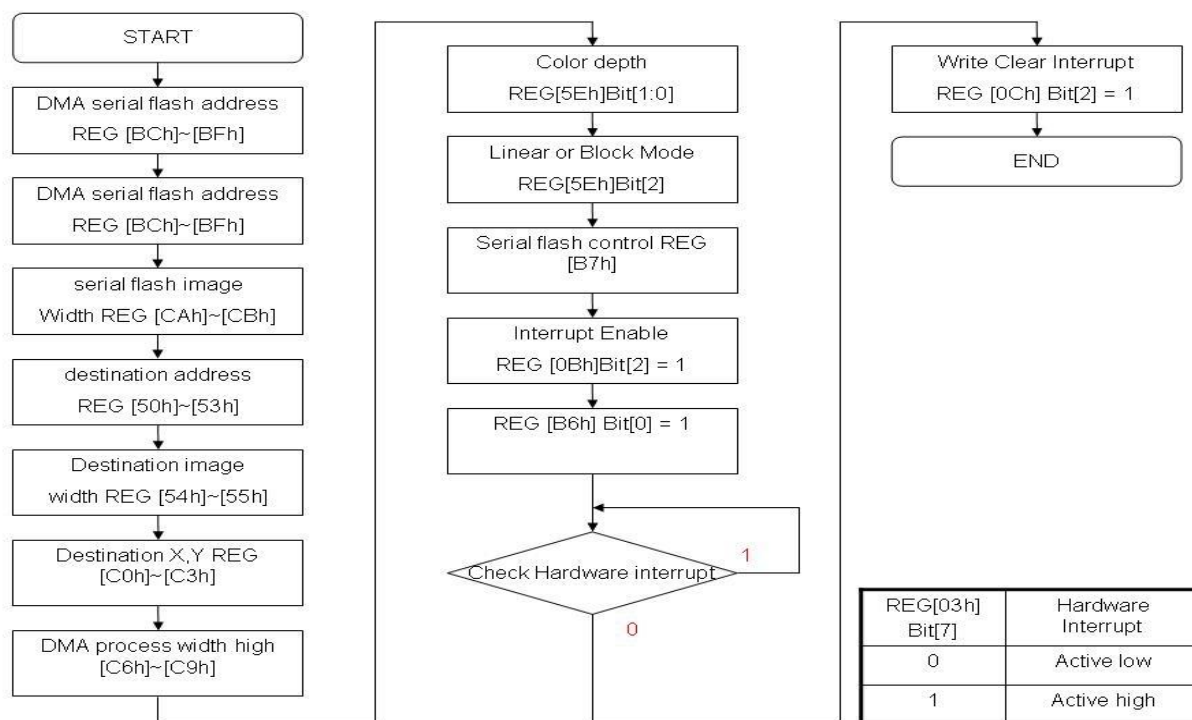


Figure 16-14 圖 : DMA Enable Procedure – Check Hardware Interrupt pin -1

REG[03h] Bit[7] = 1

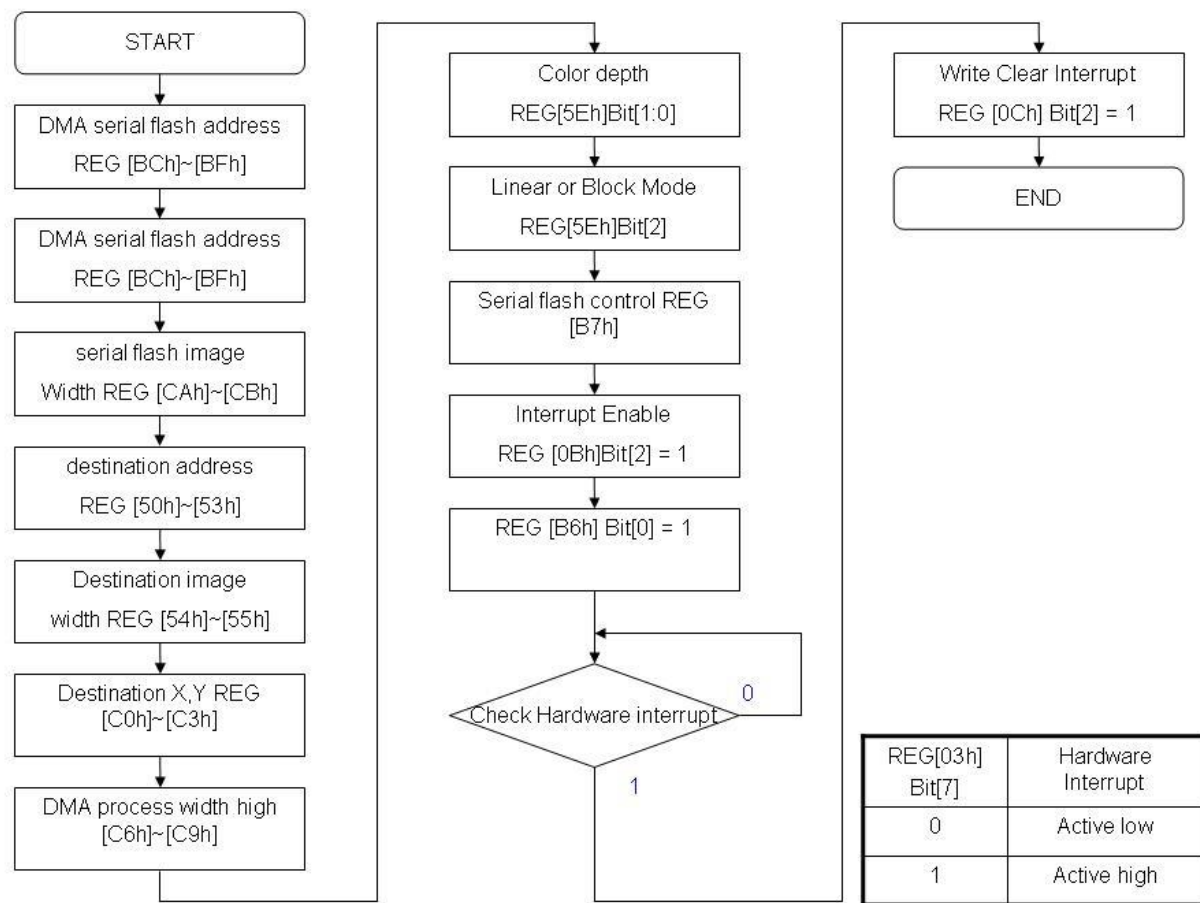


Figure 16-15 : DMA Enable Procedure –Check Hardware Interrupt pin -2

16.4 IIC Master Unit

IIC Master is a two-wire, bi-directional serial bus that provides a simple and efficient method of data exchange between devices. Only 100K bps and 400K bps modes are supported directly. The IIC Master XSCL rate formula is as the following.

$$XSCL = CCLK / (5 * (Pre-scale + 2))$$

For example : If XSCL is 100 KHz and CCLK is 100 MHz, pre-scalar (REG[E5h] & REG[E6h]) must be set to 200. Data transference between IIC Master and Slave is synchronously by XSCL on the basis of byte. Each data byte is 8-bit. There is one XSCL pulse for each XSDA bit and the MSB will be transmitted first. And then an acknowledge bit will be transmitted for each transferred byte. Each bit is processing during the high period of XSCL so that the XSDA could be change only during the low period of XSCL and must be held stable during the high period of XSCL.

Normally, a standard IIC communication protocol consists of four parts:

1. Start signal
2. Slave address transfer
3. Data transfer
4. STOP signal

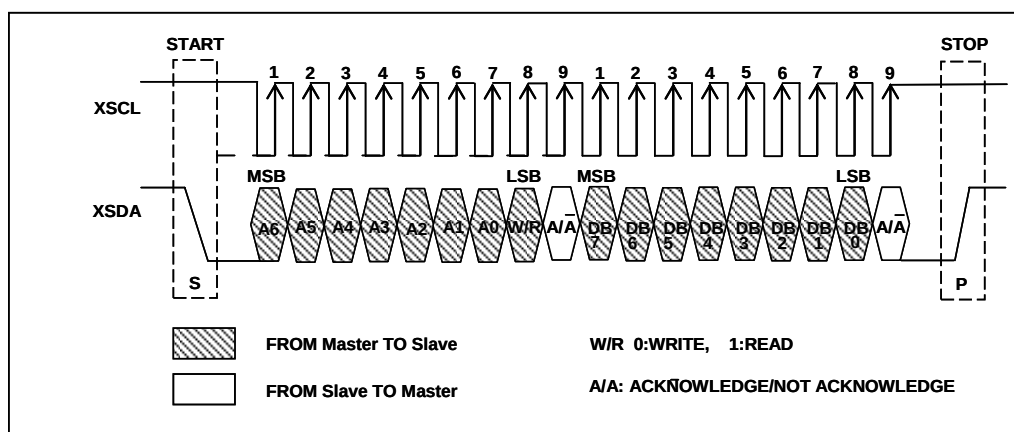


Figure 16-16

Example 1. Write 1 Byte Data to Slave

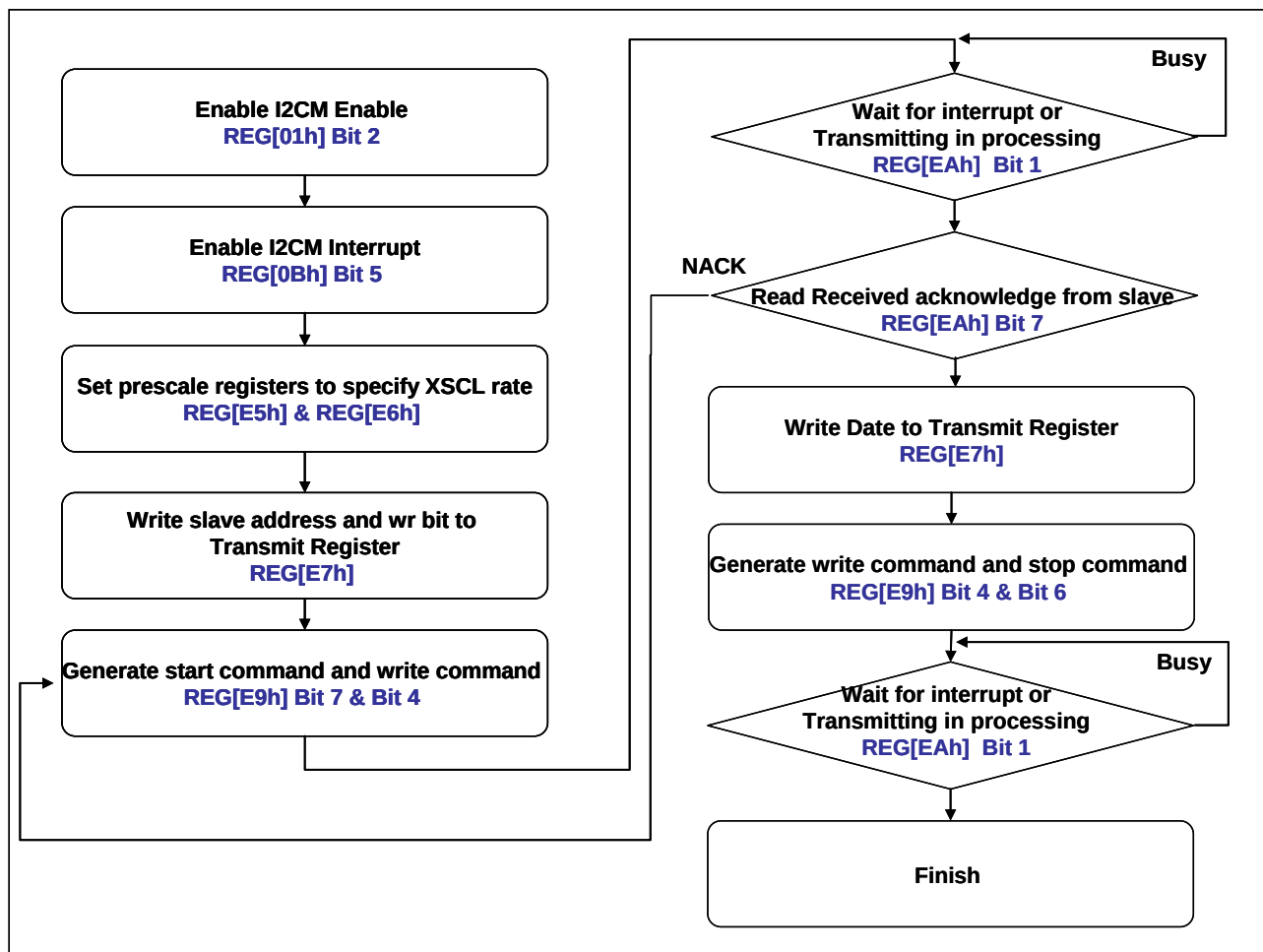


Figure 16-17 : Flow for Write 1 Byte Data to Slave

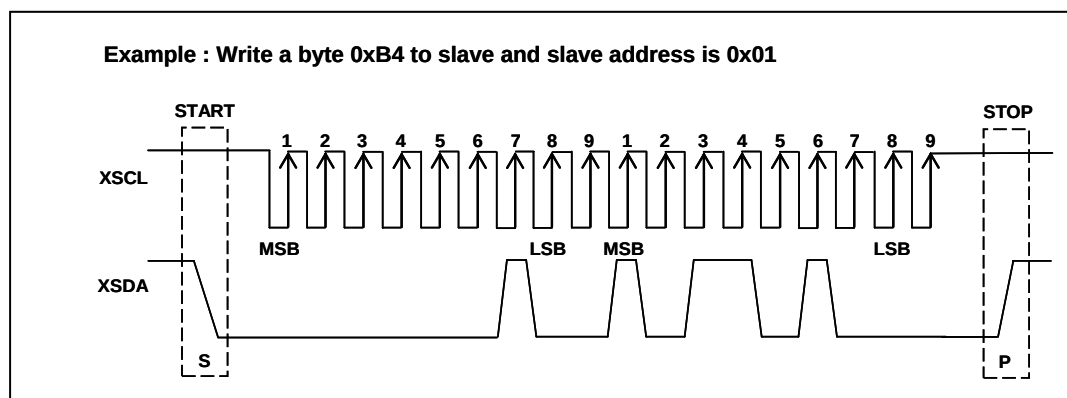


Figure 16-18 : Waveform for Write 1 Byte Data to Slave

Example 2. Read 1 Byte Data from Slave

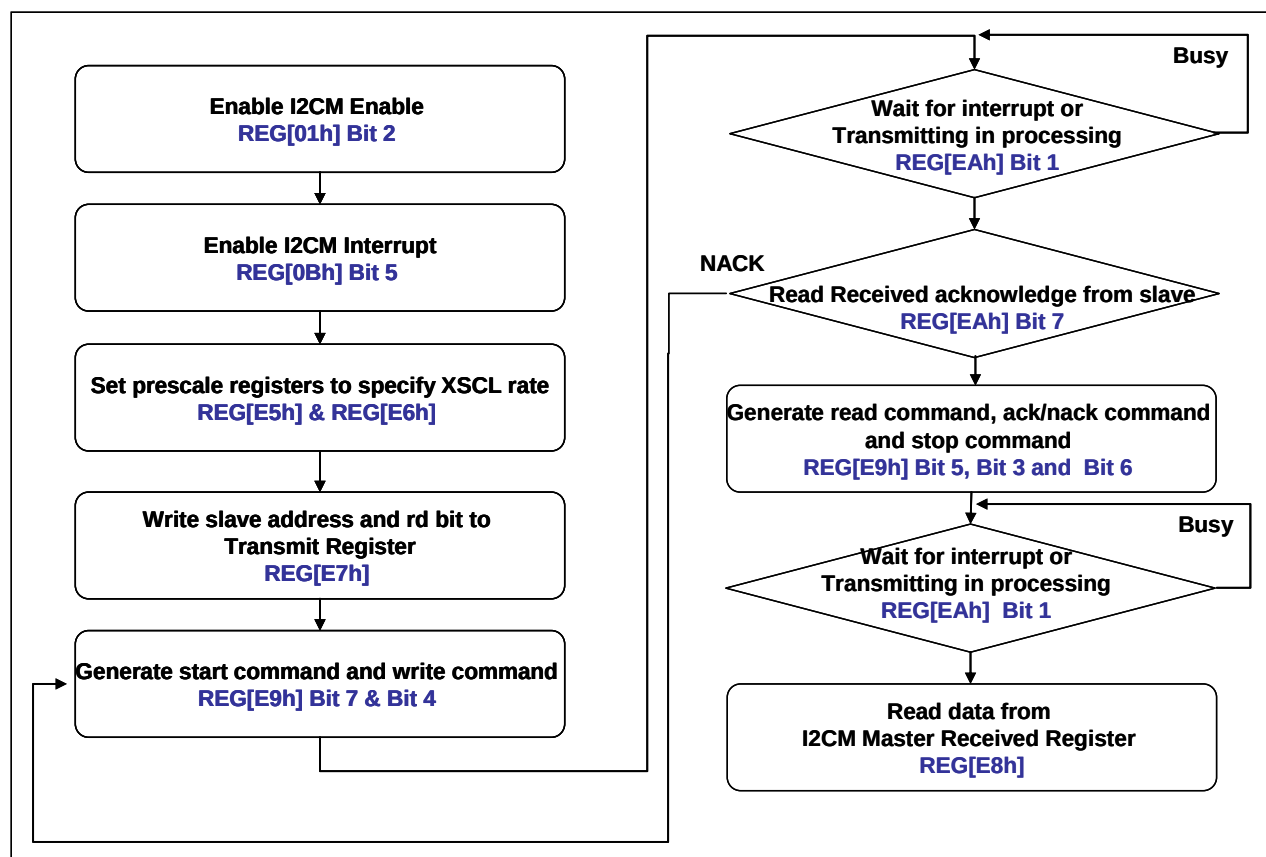


Figure 16-19 : Flow for Read 1 Byte Data from Slave

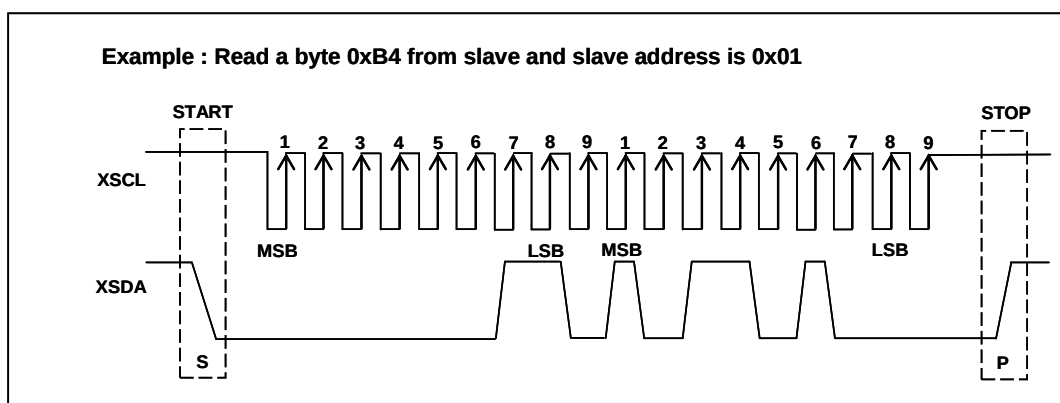


Figure 16-20 : Waveform for Read 1 Byte Data from Slave

17. Key-Scan Unit

The key-scan interface scan reads the switch data automatically by hardware scanning the key matrix switch. It will help to integrate the system circuit that includes keyboard application. Figure 17-1 shows the basic application circuit of Key-Pad on digital panel package type. RA8876 already built-in pull-up resistors in the pins “XKIN[4:0]” so no external circuit is needed.

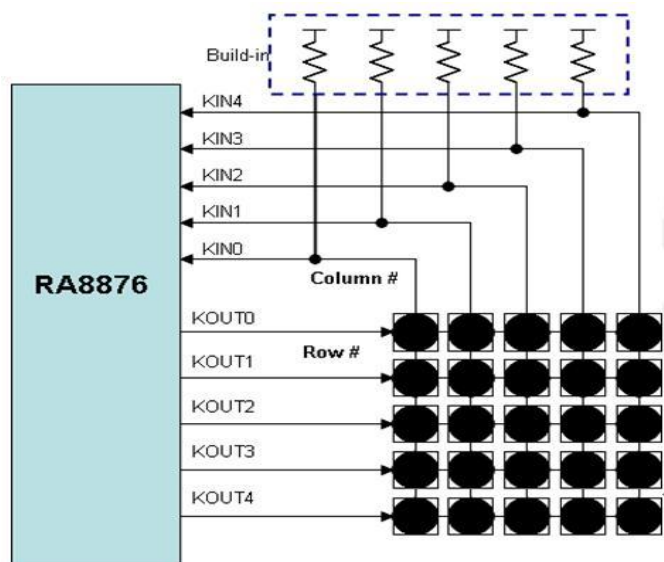


Figure 17-1 : Key-Pad Application

17.1 Operation

The RA8876 Key-Scan controller features are given as below:

1. Supporting with up-to 5x5 Key-Scan Matrix
2. Programmable setting of sampling times and scan frequency of Key-Scan
3. Adjustable long key-press timing
4. Multi-Key is available
Note: Up-to 2 keys at the same time & restricted 3 keys at the same time (3 keys cannot form 90°)
5. The function of “Key stroke to wake-up the system”

KSCR is the KEYSCAN control and status register, it is used to configure the options for KEYSCAN, such as data sample time, sample clock frequency or long key function enable etc. When key-press is active, user can sense it from the interrupt of KEYSCAN. The status bit of KSCR2 bit1~0 will update the number of current key press. Then user can get the key code directly from KSDR.

Note : “Normal key” means a key press that qualified by the sample time of RA8876. “Long Key” means a key press that keeps “pressed” for a specified long time period. That is, a “Long Key” must be a “Normal Key” first. Sometimes they need to be separated for some applications.

Table 17-1 is the key code mapping to key-pad matrix for normal press. The key code will be stored in KSDR0~2 when key was pressed. If it was a long time press, then the key code is show as Table 17-2.

Table 17-1

Note : “Normal key” means a key press that qualified by the sample time of RA8876. “Long Key” means a key press that keeps “pressed” for a specified long time period. That is, a “Long Key” must be a “Normal Key” first. Sometimes they need to be separated for some applications.

Table 17-1 is the key code mapping to key-pad matrix for normal press. The key code will be stored in KSDR0~2 when key was pressed. If it was a long time press, then the key code is show as Table 17-2.

Table 17-1: Key Code Mapping Table (Normal Key)

	Kin0	Kin1	Kin2	Kin3	Kin4
Kout0	00h	01h	02h	03h	04h
Kout1	10h	11h	12h	13h	14h
Kout2	20h	21h	22h	23h	24h
Kout3	30h	31h	32h	33h	34h
Kout4	40h	41h	42h	43h	44h

Table 17-2: Key Code Mapping Table (Long Key)

	Kin0	Kin1	Kin2	Kin3	Kin4
Kout0	80h	81h	82h	83h	84h
Kout1	90h	91h	92h	93h	94h
Kout2	A0h	A1h	A2h	A3h	A4h
Kout3	B0h	B1h	B2h	B3h	B4h
Kout4	C0h	C1h	C2h	C3h	C4h

When the multi-key function is applied, the up to 3 pressed keys data will be saved in the registers (KSDR0, KSDR1 and KSDR2). Note that the order of keys saving is determined by the position (or key code) of the keys, not the order of keys being pressed; please refer to the following example:

Press the key-code in turn of 0x34, 0x00 and 0x22, press multi-key at the same time, the key-code will be saved in KSDR0~2:

KSDR0 = 0x00

KSDR1 = 0x22

KSDR2 = 0x34

The basic features of above Key-Scan settings are introduced as follows:

Table 17-3 : Key-Scan Relative Registers

Reg.	Bit_Num	Description	Reference
KSCR1	Bit 6	Long Key Enable bit	REG[FBh]
	Bit [5:4]	Key-Scan sampling times setting	
	Bit [2:0]	Key-Scan scan frequency setting	
KSCR2	Bit [7]	Key-Scan Wakeup Function Enable Bit	REG[FCh]
	Bit [3:2]	long key timing adjustment	
	Bit [1:0]	The number of key hit	
KSDR0 KSDR1 KSDR2	Bit [7:0]	Key code for pressed key	REG[FDh ~ FFh]
CCR	Bit 5	Key-Scan enable bit	REG[01h]
INTR	Bit 4	Key-Scan interrupt enable	REG[0Bh]
INTC2	Bit 4	Key-Scan Interrupt Status bit	REG[0Ch]

Enabling the Key-Scan functions, programmer can use following methods to check keystroke.

- 1) Software check method:** to know the key be pressed from keeping check the status of Key-Scan.
- 2) Hardware check method:** to know the key be pressed from external interrupt signal.

Please be aware that when key-scan interrupt enable bit(INTEN bit[3]) set as “1” and key event of interrupt happens, the interrupt status of Key-Scan (bit[3] of INTF) is always set to “1”, no matter which method is used, programmer have to clear the status bit to 0 after reading the correct Key Code, otherwise the interrupt will be kept and no more interrupt will be generated again.

Besides, RA8876 allows the “Key-stroke wakeup function” for power saving mode. By setting the function on, any legal key-stroke event can wakeup RA8876 from sleep mode. To sense the wakeup event, RA8876 can assert hardware interrupt for MPU which can do software polling from RA8876.

The flowchart of register settings for above applications are shown as following:

1. Software Method

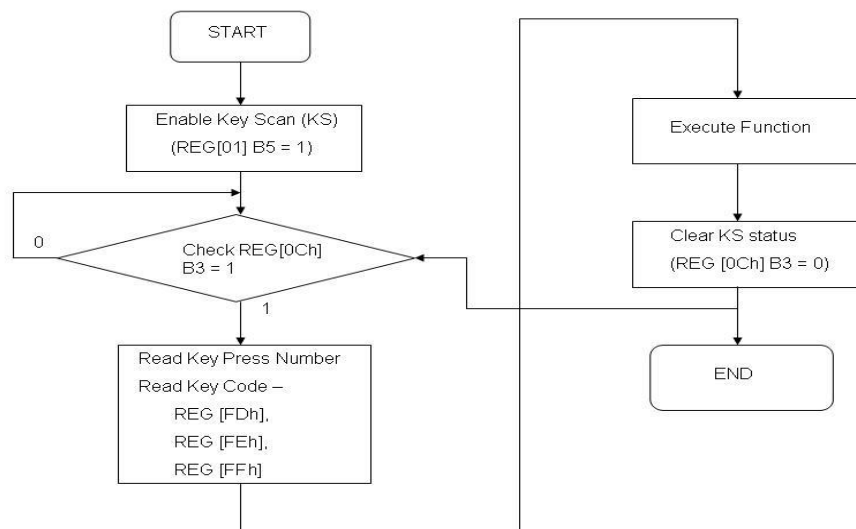


Figure 17-2 : Key-Scan Flowchart for Software Polling

2. Hardware Method

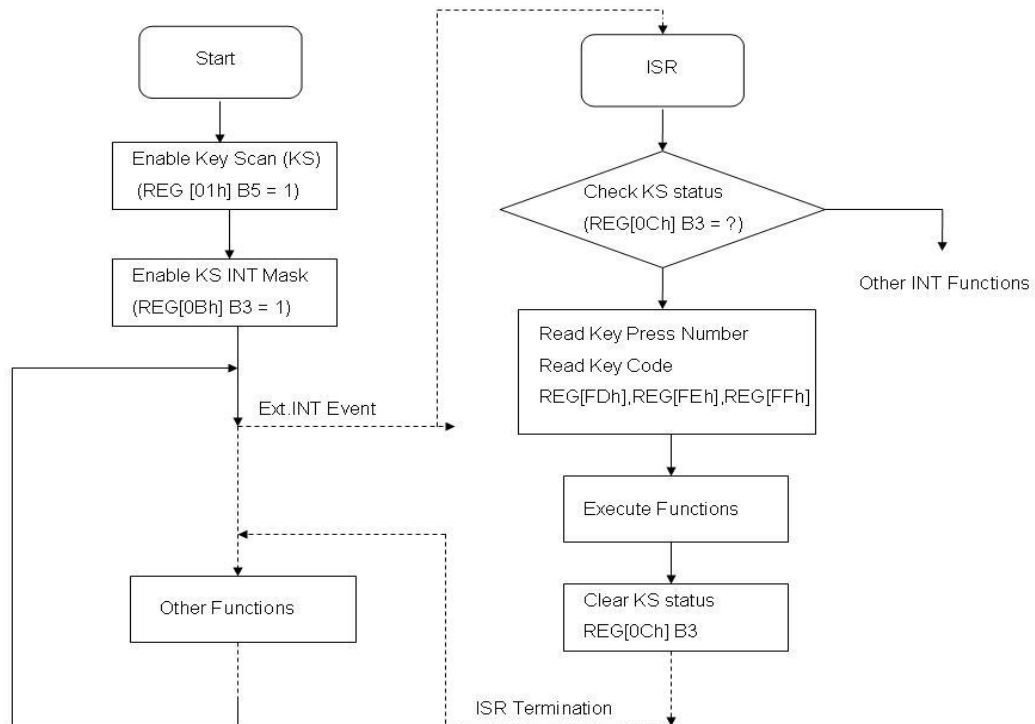


Figure 17-3 : Key-Scan for Hardware Interrupt

17.2 Restriction

		Column# (KIN#)				
		C0	C1	C2	C3	C4
Row# (KOUT#)	R0	00h	01h	02h	03h	04h
	R1	10h	11h	12h	13h	14h
	R2	20h	21h	22h	23h	24h
	R3	30h	31h	32h	33h	34h

Figure 17-4

3 key are pressed with 90°, similar above figure and circled in red or blue, it will cause wrong behavior.

18. Power Management

RA8876 have 2 operational states. One is normal state, the other is power saving state. Total have 4 power modes for these operational states. Power consumption from high to low are list as following: Normal mode, Suspend mode, Standby mode and Sleep mode. Bold characters in following procedure mean user input relative command.

Note: When RA8876 enters the power saving mode, RA8876 LCD interface will not output any signals. So before enters power saving mode, user must be display off or power off the LCD module in hardware system that is in order to avoid damage to the LCD polarization.

18.1 Normal State

18.1.1 Normal Mode

Normal Mode - User must program proper PLL parameters for each PLL (CPLL, MPLL & SPLL). User must wait PLL clock stable then start normal operation. User may check register 01h bit[7] to know PLL clock is stable or not.

18.2 Power Saving State

18.2.1 Sleep Mode

Under sleep mode, all clocks (core clock, memory clock & scan clock) will stop finally.
Enter sleep mode:

- i. **Set power saving mode as sleep mode**
- ii. **Enter power saving state (to program register DFh bit[7] as 1)**
- iii. SDRAM auto enter power down mode or self refresh mode, **depends on register E0h bit 7's setting**. (Set register E0h bit [7] as 0 will execute power down command when enter power saving state, Set register E0h bit 7 as 1 will execute self refresh command when enter power saving state.)
- iv. Internal circuit enter sleep state.
- v. Disable memory clock & scan clock
- vi. Switch core clock from CPLL clock to OSC.
- vii. If MPU I/F is parallel bus then RA8876 will stop OSC. If MPU I/F is serial bus the RA8876 will keep OSC run.
- viii. Power down all PLL (CPLL/SPLL/MPLL)
- ix. **User check status register's power saving bit and wait it becomes 1 to make sure RA8876 already enter power saving state.**

***Note:** the period between user Enter power saving state command and chip enter power saving state any wakeup event are not acceptable.

Return to normal operation mode:

- i. **Leave power saving state (to program register DFh bit[7] as 0)**
- ii. Enable OSC run if OSC is stopped in stop mode
- iii. Switch core clock back to OSC.
- iv. Resume all PLL (CPLL/SPLL/MPLL)
- v. Switch all clocks (Core clock, SDRAM clock & Scan clock) back to PLL clock.
- vi. **User checks status register's power saving bit and wait it becomes 0.**

18.2.2 Suspend Mode

Under suspend mode, core clock & scan clock will stop and memory clock will switch to OSC clock.
Enter suspend mode:

- i. **According to OSC frequency program proper SDRAM refresh rate**
- ii. **Set power saving state as suspend mode**

- iii. **Enter power saving mode (program register DFh bit[7] as 1)**
 - iv. Internal circuit enter suspend state.
 - v. Auto disable scan clock
 - vi. Auto switch core clock and memory clock from PLL clock to OSC.
 - vii. Auto disable core clock
 - viii. Keep OSC run
 - ix. Power down all PLL (CPLL/SPLL/MPLL)
 - x. **User check status register's power saving bit and wait it becomes 1 to make sure RA8876 already enter power saving state.**
- *Note:** the period between user Enter power saving state command and chip enter power saving state any wakeup event are not acceptable.

Return to normal operation mode:

- i. **Leave power saving state (program register DFh bit[7] as 0)**
- ii. Enable OSC run if OSC is stopped in suspend mode.
- iii. Switch core clock back to OSC.
- iv. Resume all PLL (CPLL/SPLL/MPLL)
- v. Switch all clocks (core clock, memory clock & scan clock) back to PLL clock.
- vi. **User checks status register's power saving bit and wait it becomes 0.**

18.2.3 Standby Mode

Under standby mode, core clock & scan clock will stop and memory clock will keep MPLL clock.
Enter standby mode:

- i. **Set power saving state as standby mode**
 - ii. **Enter power saving mode (program register DFh bit[7] as 1)**
 - iii. Internal circuit enter standby mode.
 - iv. Disable Scan clock
 - v. Switch core clock to OSC and keep memory clock on MPLL clock
 - vi. Keep OSC run
 - vii. Keep all PLL alive for rapid recovery
 - viii. **User check status register's power saving bit and wait it becomes 1 to make sure RA8876 already enter power saving state.**
- *Note:** the period between user Enter power saving state command and chip enter power saving state any wakeup event are not acceptable.

Return to normal operation mode:

- i. **Leave power saving state (program register 0 DFh bit[7] as 0)**
- ii. Switch core clock & scan clock back to PLL clock.
- iii. **User checks status register's power saving bit and wait it becomes 0.**

18.3 Power Mode Comparison Table

Item	Normal State	Power Saving State					
	Normal mode	Standby mode		Suspend mode		Sleep mode	
	PLL enable	Parallel MPU	Serial MPU	Parallel MPU	Serial MPU	Parallel MPU	Serial MPU
MCLK	MPLL clock	MPLL clock	MPLL clock	OSC	OSC	stop	stop
CCLK	CPLL clock	OSC	OSC	stop	OSC	stop	OSC
PCLK	SPLL clock	stop	stop	stop	stop	stop	stop
CPLL	On	On	On	Off	Off	Off	Off
MPLL	On	On	On	Off	Off	Off	Off
SPLL	On	On	On	Off	Off	Off	Off

19. Register

There are 4 types of cycles used in host interface of RA8876, please refer to Table 19-1 for detail. The programming or reading of the registers in RA8876 is composed by the cycles. RA8876 includes a status register and many instruction registers. The status register is read only and can be read by "Status Read" cycle. The instruction registers, that is used to program almost functions, can be programmed by "Command Write" cycle and "Data Write" cycle. The "Command Write" cycle sets the register number to program, and the "Data Write" cycle set the data of the register. When reading the specific instruction registers, host asserts a "Data read" cycle following the "Command Write cycle". The "Command Write" cycle sets the register number to program, and the "Data Read" cycle read the data of the register.

Table 19-1 : Host Cycle Type

Cycle Type	XnCS	XA0	MPU_8080		MPU_6800		Description
			XnRD_EN	XnWR_RnW	XnRD_EN	XnWR_RnW	
Command Write	0	0	1	0	1	0	Register number write cycle
Status Read	0	0	0	1	1	1	Status read cycle
Data Write	0	1	1	0	1	0	Corresponding Register data/Memory data write cycle following the Command Write cycle.
Data Read	0	1	0	1	1	1	Corresponding Register data/Memory data read cycle following the Command Write cycle.

The registers function description is listed below, for each register, a register name and register number is described upper each register function table. Each register contains up-to 8 bits data. In the register function table, detail description, default value and access attribute (RO: Read only, WO: Write only, RW: Read-able and Write-able) are described.

19.1 Status Register

Status Register (STSR)

Bit	Description	Default	Access
7	Host Memory Write FIFO full 0: Memory Write FIFO is not full. 1: Memory Write FIFO is full. Only when Memory Write FIFO is not full, MPU may write another one pixel.	0	RO
6	Host Memory Write FIFO empty 0: Memory Write FIFO is not empty. 1: Memory Write FIFO is empty. When Memory Write FIFO is empty, MPU may write 8bpp data 64 pixels, or 16bpp data 32 pixels, 24bpp data 16 pixels directly.	1	RO
5	Host Memory Read FIFO full 0: Memory Read FIFO is not full. 1: Memory Read FIFO is full. When Memory Read FIFO is full, MPU may read 8bpp data 64 pixels, or 16bpp data 32 pixels, 24bpp data 16pixels directly.	0	RO
4	Host Memory Read FIFO empty 0: Memory Read FIFO is not empty. 1: Memory Read FIFO is empty.	1	RO

Bit	Description	Default	Access
3	Core task is busy (fontwr_busy) Following task is running: BTE, Geometry engine, Serial flash DMA, Text write or Graphic write 0: task is done or idle. 1: task is busy. While User change canvas relative setting & switch text mode or graphic mode must make sure core task is done. Note: BTE, Geometry drawing & Serial flash DMA also may check each start bit. Under text mode, if user wants to change rotate attribute, character line gap, character-to-character space, foreground color, background color and Text/graphic mode setting, he must make sure core_busy(fontwr_busy) status bit is low.	0	RO
2	SDRAM ready for access 0: SDRAM is not ready for access 1: SDRAM is ready for access Before user check this bit status, user must be set "sdr_initdone" bit as 1	0	RO
1	Operation mode status 0: Normal operation state 1: Inhibit operation state Inhibit operation state means internal reset event keep running or initial display still running or chip enter power saving state. In power saving state, this bit becomes 1 until PLL clock stop. So it has a little time lag compare with power saving state bit(Reg[DFh] bit[7]).	0	RO
0	Interrupt pin state 0: without interrupt active 1: interrupt active	0	RO

Note : "RO" means read only.

19.2 Chip Configuration Registers

REG[00h] Software Reset Register (SRR)

Bit	Description	Default	Access
7-5	NA	06h	RO
4-2	NA	05h	RO
1	NA	1	RO
0	Software Reset 0: Normal operation. 1: Software Reset. Software Reset only reset internal state machine. Configuration Registers value won't be reset. So all read-only flag in the register will return to its initial value. User should have proper action to make sure flag is in desired state. Note: The bit will auto clear after reset.	0	WO
0	Warning condition flag 0: No warning operation occurred 1: Warning condition occurred. Please check REG[E4h] bit 3 for more detail.	0	RO

REG[01h] Chip Configuration Register (CCR)

Bit	Description	Default	Access
7	Reconfigure PLL frequency Write "1" to this bit will reconfigure PLL frequency. Note: a. When user change PLL relative parameters, PLL clock won't change immediately, user must set this bit as "1" again. b. User may read (check) this bit to know whether system already switch to PLL clock or not yet. Read "1" means PLL clock ready and switch successfully.	1	RW
6	Mask XnWAIT on XnCS deassert 0: No mask XnWAIT keep assert if internal state keep busy and cannot accept next R/W cycle, no matter XnCS assert/deassert. If MPU cycle cannot be extended while XnWAIT keep low, user should poll XnWAIT and wait it goes high then start next access. 1: Mask XnWAIT deassert when XnCS deassert. Use in MPU cycle can be extended by XnWAIT automatically.	1	RW
5	Key-Scan Enable/Disable 0: Disable. 1: Enable.	0	RW
4-3	For RA8876 TFT Panel I/F Output pin Setting 00b: 24-bits TFT output. 01b: 18-bits TFT output. 10b: 16-bits TFT output. 11b: w/o TFT output. Other unused TFT output pins are set as GPIO or Key function.	01b	RW
2	IIC master Interface Enable/Disable 0: Disable (GPIO function) 1: Enable (IIC master function) IIC master pins are shared with XKIN[0] & XKOUT[0]. this bit has higher priority than Key-Scan Enable bit. ie. if IIC master and Key-Scan are enable simultaneously then XKIN[0]/XKOUT[0] will become IIC function & other XKIN/XKOUT pins still keep Key-scan function.	0	RW
1	Serial Flash or SPI Interface Enable/Disable 0: Disable (GPIO function) 1: Enable (SPI master function)	0	RW
0	Host Data Bus Width Selection 0: 8-bit Host Data Bus. 1: 16-bit Host Data Bus. *** If Serial host I/F selected or in initial display operation period, chip will force this bit as 0 and only allow 8-bit access.	0	RW

REG[02h] Memory Access Control Register (MACR)

Bit	Description	Default	Access
7-6	Host Read/Write image Data Format MPU read/write data format when access memory data port. 0xb : Direct write (for all 8 bits MPU I/F, 16 bits MPU I/F with 8bpp data mode 1 & 2, 16 bits MPU I/F with 16/24-bpp data mode 1 & serial host interface) 10b : Mask high byte of each data (ex. 16 bit MPU I/F with 8-bpp data mode 1) 11b : Mask high byte of even data (ex. 16 bit MPU I/F with 24-bpp data mode 2)	0	RW
5-4	Host Read Memory Direction (Only for Graphic Mode) 00b: Left → Right then Top → Bottom. 01b: Right → Left then Top → Bottom. 10b: Top → Bottom then Left → Right. 11b: Bottom → Top then Left → Right. Ignored if canvas in linear addressing mode.	0	RW
3	NA	0	RO
2-1	Host Write Memory Direction (Only for Graphic Mode) 00b: Left → Right then Top → Bottom. (Original) 01b: Right → Left then Top → Bottom. (Horizontal flip) 10b: Top → Bottom then Left → Right. (Rotate right 90° & Horizontal flip) 11b: Bottom → Top then Left → Right. (Rotate left 90°) Ignored if canvas in linear addressing mode.	0	RW
0	NA (must keep it as 0)	0	RO

REG[03h] Input Control Register (ICR)

Bit	Description	Default	Access
7	Output to MPU Interrupt pin's active level 0 : active low. 1 : active high.	0	RW
6	External interrupt input (XPS[0] pin) de-bounce 0 : without de-bounce 1 : enable de-bounce (1024 OSC clock)	0	RW
5-4	External interrupt input (XPS[0] pin) trigger type 00 : low level trigger 01 : falling edge trigger 10 : high level trigger 11 : rising edge trigger	00b	RW
3	NA	0	RW
2	Text Mode Enable 0 : Graphic mode. 1 : Text mode. Before toggle this bit user must make sure core task busy bit in status register is done or idle. This bit always 0 (in graphic mode) if canvas' address mode is linear mode.	0	RW
1-0	Memory port Read/Write Destination Selection 00b : Image buffer (SDRAM) for image data, pattern, user-characters. Support Read-modify-Write. 01b : Gamma table for Color Red/Green/Blue. Each color's gamma table has 256 bytes. User need specify desired gamma table and continuous write 256 bytes.	0	RW

Bit	Description	Default	Access
	10b : Graphic Cursor RAM (only accept low 8-bits MPU data, similar normal register data r/w.), not support Graphic Cursor RAM read. It contains 4 graphic cursor sets. Each set has 128x16 bits. User need specify target graphic cursor set and continue write 256 bytes. 11b : Color palette RAM. It is 64x12 bits SRAM, so even address' data only low 4 bits are valid. Not support Color palette RAM read. User need continue write 128 bytes.		

REG[04h] Memory Data Read/Write Port (MRWDP)

Bit	Description	Default	Access
7-0	Write Function : Memory Write Data Data to write in memory corresponding to the setting of REG[03h][1:0]. Continuous data write cycle can be accepted in bulk data write case. Note: a. Image data in SDRAM: according MPU I/F bit width setting (8/16-bits), Host R/W image data format, canvas color depth and set canvas in block mode. b. Pattern data for BTE operation in SDRAM: according MPU I/F bit width setting (8/16-bits), Host R/W image data format, canvas color depth and set canvas in block mode. Active window's width and height should set as 8x8 or 16x16 depend on user required. c. User-characters in SDRAM: according MPU I/F bit width setting (8/16-bits), Host R/W image data format and set canvas in linear mode. d. Character code: only accept low 8-bits MPU data, similar to normal register R/W. For two bytes character code, input high byte first. To user defined Character, code < 8000h is half size, code >= 8000h is full size. e. Gamma table data: only accept low 8-bits MPU data. User must set "Select Gamma table sets([3Ch] Bit6-5)" to clear internal Gamma table's address counter then start to write data. User should program 256 bytes data to memory data port. f. Graphic Cursor RAM data: only accept low 8-bits MPU data. User must set "Select Graphic Cursor sets" bits to clear internal Graphic Cursor RAM address counter then start to write data. g. Color palette RAM data: only accept low 8-bits MPU data. User must program full Color palette RAM in a continuous 128 byte data write to memory data port and cannot change register address. Read Function : Memory Read Data Data to read from memory corresponding to the setting of REG[03h][1:0]. Continuous data read cycle can be accepted in bulk data read case. Note1: if you set this port address from different port address, must issue a dummy read, the first data read cycle is dummy read and data should be ignored. Graphic Cursor RAM & Color palette RAM data are not support data read function. Note2: read memory data is 4 bytes alignment no matter color depth setting. Note3: If user write data to SDRAM user must make sure write FIFO is empty before he change register number or core task busy status bit becomes idle.	--	RW

19.3 PLL Setting Register

REG[05h] SCLK PLL Control Register 1 (PPLLC1)

Bit	Description	Default	Access
7	RESERVED	0	RW
6	NA	0	RO
5-3	SCLK extra divider xx1b: divided by 16 000b: divided by 1. 010b: divided by 2. 100b: divided by 4. 110b: divided by 8.	0	RW
2-1	SCLK PLLDIVK[1:0] SCLK PLL Output divider 00b: divided by 1. 01b: divided by 2. 10b: divided by 4. 11b: divided by 8.	2	RW
0	SCLK PLLDIVM PCLK PLL Pre-driver parameter. 0b: divided by 1. 1b: divided by 2.	0	RW

REG[06h] SCLK PLL Control Register 2 (PPLLC2)

Bit	Description	Default	Access
7-6	NA	0	RO
5-0	SCLK PLLDIVN[5:0] SCLK PLL input parameter, the value should be 1~63. (i.e. value 0 is forbidden).	17h	RW

*PCLK is used by panel's scan clock and derived from SCLK.

REG[07h] MCLK PLL Control Register 1 (MPLLC1)

Bit	Description	Default	Access
7-3	NA	0	RO
2-1	MCLK PLLDIVK[1:0] PCLK PLL Output divider 00b: divided by 1. 01b: divided by 2. 10b: divided by 4. 11b: divided by 8.	1	RW
0	MCLK PLLDIVM MCLK PLL Pre-driver parameter. 0b: divided by 1. 1b: divided by 2.	0	RW

REG[08h] MCLK PLL Control Register 2 (MPLLC2)

Bit	Description	Default	Access
7-6	NA	0	RO
5-0	MCLK PLLDIVN[5:0] MCLK PLL input parameter, the value should be 1~63. (i.e. value 0 is forbidden).	1Dh	RW

*MCLK is used by SDRAM's clock

REG[09h] CCLK PLL Control Register 1 (SPLLC1)

Bit	Description	Default	Access
7-3	NA	0	RO
2-1	CCLK PLLDIVK[1:0] CCLK PLL Output divider 00b: divided by 1. 01b: divided by 2. 10b: divided by 4. 11b: divided by 8.	2	RW
0	CCLK PLLDIVM CCLK PLL Pre-driver parameter. 0b: divided by 1. 1b: divided by 2.	0	RW

REG[0Ah] CCLK PLL Control Register 2 (SPLLC2)

Bit	Description	Default	Access
7-6	NA	0	RO
5-0	CCLK PLLDIVN[5:0] CCLK PLL input parameter, the value should be 1~63. (i.e. value 0 is forbidden).	2Ah	RW

*CCLK is used by core's clock

The clock of RA8876 is generated by oscillator and internal xCLK PLL circuit. The following formula is used for clock calculation:

$$xCLK = \frac{\left(\frac{Fin}{2^{(xPLLDIVM)}} \right) \times (xPLLDIVN + 1)}{2^{xPLLDIVK}}$$

Note :

1. PLL's parameters can be changed only when PLL disabled.
2. After REG[05h] ~ REG[0Ah] is programmed, a lock time (< 30us) must be kept to guarantee the stability of the PLL output.
3. The input OSC frequency (F_{IN}) must greater & PLLDIVM has following restriction:

$$10MHz \leq Fin \leq 15MHz$$

&

$$10MHz \leq \frac{Fin}{2^{PLLDIVM}} \leq 40MHz$$

4. The internal multiplied clock frequency $F_{vco} = \frac{Fin}{2^{PLLDIVM}} \times (PLLDIVN + 1)$ must be equal to or greater than 250 MHz and small than 500MHz. i.e,

$$250MHz \leq F_{vco} \leq 500MHz$$

19.4 Interrupt Control Registers

The relationship about Interrupt Enable Register, Interrupt Event Flag Register & Mask Interrupt Flag Register:

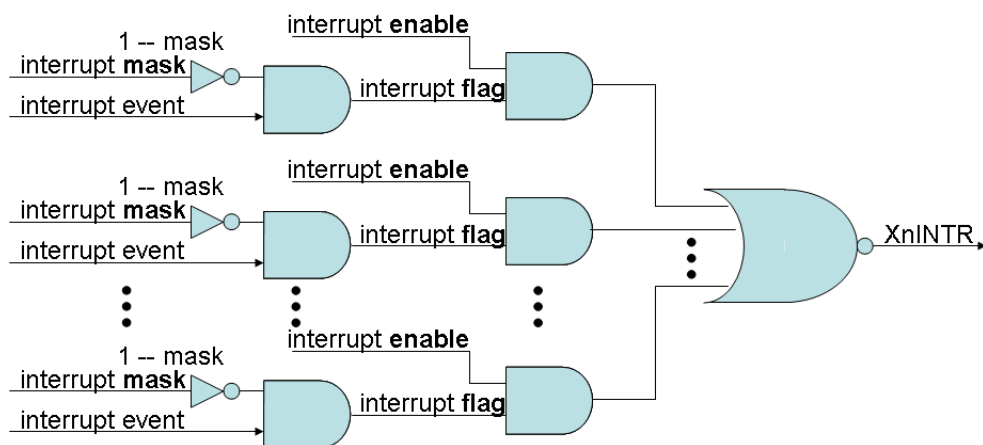


Figure 19-1

REG[0Bh] Interrupt Enable Register (INTEN)

Bit	Description	Default	Access
7	Wakeup/resume Interrupt Enable 0: Disable. 1: Enable.	0	RW
6	External Interrupt input (XPS[0] pin) Enable 0: Disable. 1: Enable	0	RW
5	IIC Master Interrupt Enable 0: Disable 1: Enable	0	RW
4	Vsync time base interrupt Enable Bit 0: Disable Interrupt 1: Enable Interrupt This interrupt event may provide the host processor with Vsync signal information for tearing effect.	0	RW
3	Key Scan Interrupt Enable Bit 0: Disable Key scan interrupt 1: Enable Key scan interrupt	0	RW
2	Serial flash DMA Complete Draw task finished BTE Process Complete etc. Interrupt Enable 0: Disable Interrupt 1: Enable Interrupt	0	RW
1	PWM timer 1 Interrupt Enable Bit 0: Disable Interrupt. 1: Enable Interrupt.	0	RW
0	PWM timer 0 Interrupt Enable Bit 0: Disable Interrupt. 1: Enable Interrupt.	0	RW

REG[0Ch] Interrupt Event Flag Register (INTF)

* If you received an interrupt but cannot identify it on Interrupt Event Flag Register, please check SPI master status register's interrupt flag bits REG[BAh].

Bit	Description	Default	Access
7	Wakeup/resume Interrupt flag Write Function → Wakeup/resume Interrupt Clear Bit 0: No operation. 1: Clear Wakeup/resume interrupt. Read Function → Wakeup/resume Interrupt Status 0: No Wakeup/resume interrupt happens. 1: Wakeup/resume interrupt happens.	0	RW
6	External Interrupt input (XPS[0] pin) flag Write Function → XPS[0] pin edge Interrupt Clear Bit 0: No operation. 1: Clear the XPS[0] pin edge interrupt. Read Function → XPS[0] pin Interrupt Status 0: No XPS[0] pin interrupt happens. 1: XPS[0] pin interrupt happens.	0	RW
5	IIC master Interrupt flag Write Function → IIC master Interrupt Clear Bit 0: No operation. 1: Clear the IIC master interrupt. Read Function → IIC master Interrupt Status 0: No IIC master interrupt happens. 1: IIC master interrupt happens.	0	RW
4	Vsync Time base interrupt flag Write Function → Vsync Interrupt Clear Bit 0: No operation. 1: Clear the Vsync interrupt. Read Function → Vsync Interrupt Status 0: No Vsync interrupt happens. 1: Vsync interrupt happens.	0	RW
3	Key Scan Interrupt flag Write Function → Key Scan Interrupt Clear Bit 0: No operation. 1: Clear the Key Scan interrupt. Read Function → Key Scan Interrupt Status 0: No Key Scan interrupt happens. 1: Key Scan interrupt happens.	0	RW
2	Serial flash DMA Complete Draw task finished BTE Process Complete etc. Interrupt flag Write Function → Interrupt Clear Bit 0: No operation. 1: Clear interrupt. Read Function → Interrupt Status 0: No interrupt happens. 1: interrupt happens.	0	RW
1	PWM 1 timer Interrupt flag Write Function → Interrupt Clear Bit 0: No operation. 1: Clear PWM1 interrupt.	0	RW

Bit	Description	Default	Access
	Read Function→Interrupt Status 0: No PWM1 interrupt happens. 1: PWM1 interrupt happens.		
0	PWM 0 timer Interrupt flag Write Function→Interrupt Clear Bit 0: No operation. 1: Clear PWM0 interrupt. Read Function→Interrupt Status 0: No PWM0 interrupt happens. 1: PWM0 interrupt happens.	0	RW

REG[0Dh] Mask Interrupt Flag Register (MINTFR)

*** If you masked certain interrupt flag, then RA8876 neither assert interrupt event to MPU nor checked it on Interrupt Flag Register. But if you unmasked certain interrupt flag and disable this interrupt then MPU won't be informed by XnINTR but you still may check it on interrupt Flag Register.

Bit	Description	Default	Access
7	Mask Wakeup/resume Interrupt Flag 0: Unmask. 1: Mask.	0	RW
6	Mask External Interrupt (XPS[0] pin) Flag 0: Unmask. 1: Mask.	0	RW
5	Mask IIC Master Interrupt Flag 0: Unmask. 1: Mask.	0	RW
4	Mask Vsync time base interrupt Flag 0: Unmask. 1: Mask.	0	RW
3	Mask Key Scan Interrupt Flag 0: Unmask. 1: Mask.	0	RW
2	Mask Serial flash DMA Complete Draw task finished BTE Process Complete etc. Interrupt Flag 0: Unmask. 1: Mask.	0	RW
1	Mask PWM timer 1 Interrupt Flag 0: Unmask. 1: Mask.	0	RW
0	Mask PWM timer 0 Interrupt Flag 0: Unmask. 1: Mask.	0	RW

REG[0Eh] Pull- high control Register (PUENR)

Bit	Description	Default	Access
7:6	NA	0	RO
5	GPIO-F[7:0] Pull-high Enable (XPDAT[23:19, 15:13]) 0: Pull-Up Disable 1: Pull-Up Enable * Only available when XPDAT configure as GPIO function	0	RW
4	GPIO-E[7:0] Pull- high Enable (XPDAT[12:10, 7:3]) 0: Pull-Up Disable 1: Pull-Up Enable * Only available when XPDAT configure as GPIO function	0	RW

Bit	Description	Default	Access
3	GPIO-D[7:0] Pull- high Enable (XPDAT[18, 2, 17, 16, 9, 8, 1,0]) 0: Pull-Up Disable 1: Pull-Up Enable * Only available when XPDAT configure as GPIO function	0	RW
2	GPIO-C[4:0] Pull- high Enable (XnSFCS1, XnSFCS0, XMISO, XMOSI , XSCK) 0: Pull-Up Disable 1: Pull-Up Enable	0	RW
1	XDB[15:8] Pull- high Enable 0: Pull-Up Disable 1: Pull-Up Enable	0	RW
0	XDB[7:0] Pull- high Enable 0: Pull-Up Disable 1: Pull-Up Enable	0	RW

REG[0Fh] PDAT for PIO/Key Function Select Register (PSFSR)

Bit	Description	Default	Access
7	XPDAT[18] – GPIO or Key function select 0: GPIO-D7 1: XKOUT[4]	0	RW
6	XPDAT[17] –GPIO or Key function select 0: GPIO-D5 1: XKOUT[2]	0	RW
5	XPDAT[16] –GPIO or Key function select 0: GPIO-D4 1: XKOUT[1]	0	RW
4	XPDAT[9] –GPIO or Key function select 0: GPIO-D3 1: XKOUT[3]	0	RW
3	XPDAT[8] –GPIO or Key function select 0: GPIO-D2 1: XKIN[3]	0	RW
2	XPDAT[2] –GPIO or Key function select 0: GPIO-D6 1: XKIN[4]	0	RW
1	XPDAT[1] –GPIO or Key function select 0: GPIO-D1 1: XKIN[2]	0	RW
0	XPDAT[0] –GPIO or Key function select 0: GPIO-D0 1: XKIN[1]	0	RW

19.5 LCD Display Control Registers

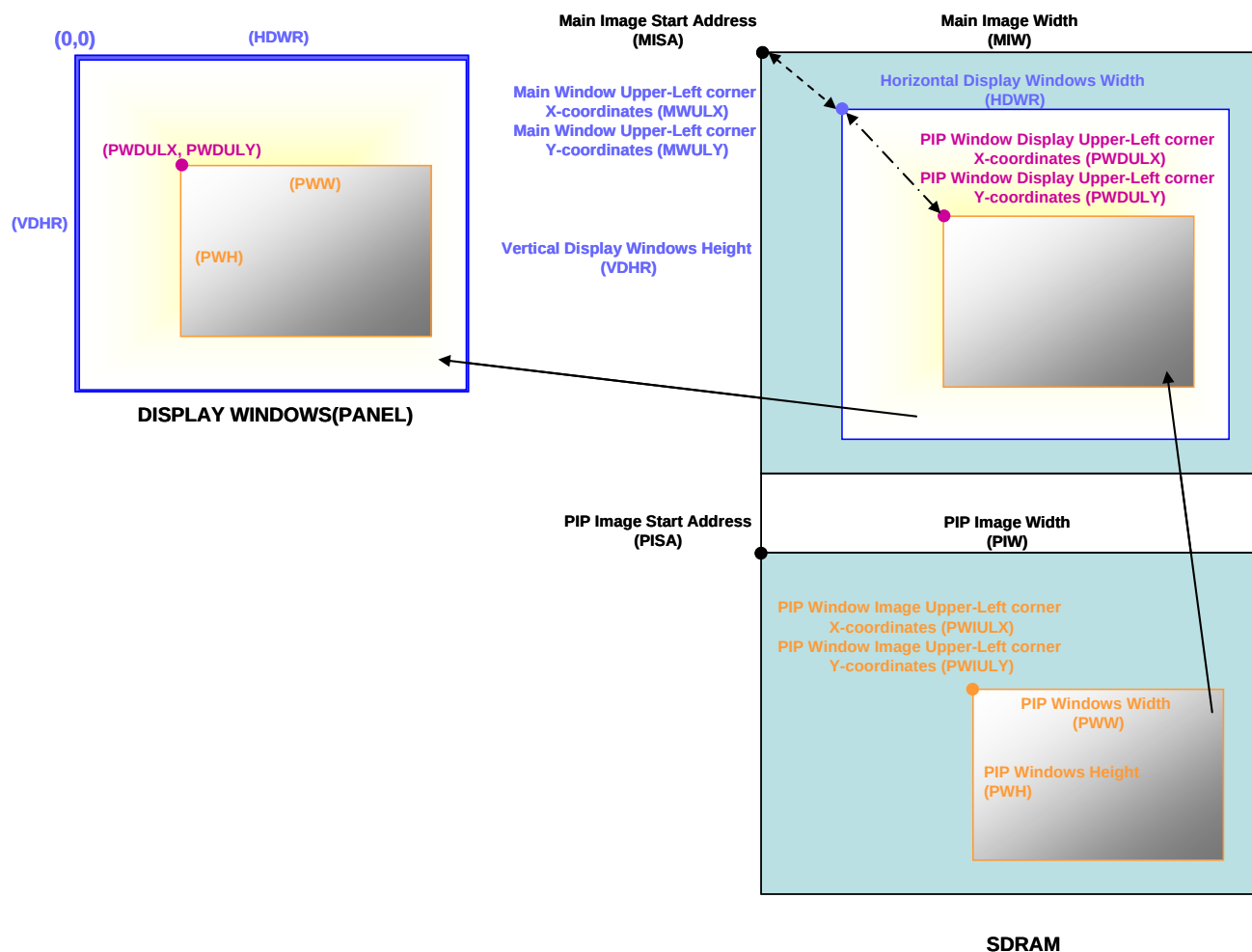


Figure 19-2 LCD Display

REG[10h] Main/PIP Window Control Register (MPWCTR)

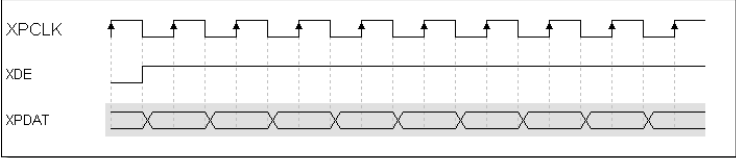
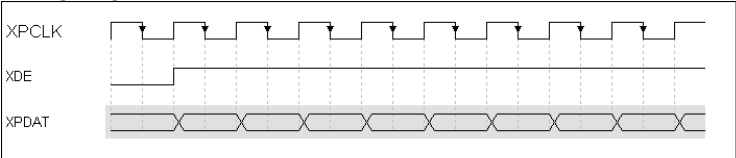
Bit	Description	Default	Access
7	PIP 1 window Enable/Disable 0 : PIP 1 window disable. 1 : PIP 1 window enable PIP 1 window always on top of PIP 2 window.	0	RW
6	PIP 2 window Enable/Disable 0 : PIP 2 window disable. 1 : PIP 2 window enable PIP 1 window always on top of PIP 2 window.	0	RW
5	NA	0	RO
4	Select Configure PIP 1 or 2 Window's parameters PIP window's parameter including Color Depth, starting address, image width, display coordinates, window coordinates, window width, window height. 0: To configure PIP 1's parameters. 1: To configure PIP 2's parameters.	0	RW

Bit	Description	Default	Access
3-2	Main Image Color Depth Setting 00b: 8-bpp generic TFT, i.e. 256 colors. 01b: 16-bpp generic TFT, i.e. 65K colors. 1xb: 24-bpp generic TFT, i.e. 1.67M colors.	1	RW
1	NA	0	RW
0	To Control panel's synchronous signals 0: Sync Mode: Enable XVSYNC, XHSYNC, XDE 1: DE Mode: Only XDE enable, XVSYNC & XHSYNC in idle state	0	RW

REG[11h] PIP Window Color Depth Setting (PIPCDEP)

Bit	Description	Default	Access
7-4	NA	0	RO
3-2	PIP 1 Window Color Depth Setting 00b: 8-bpp generic TFT, i.e. 256 colors. 01b: 16-bpp generic TFT, i.e. 65K colors. 1xb: 24-bpp generic TFT, i.e. 1.67M colors.	1	RW
1-0	PIP 2 Window Color Depth Setting 00b: 8-bpp generic TFT, i.e. 256 colors. 01b: 16-bpp generic TFT, i.e. 65K colors. 1xb: 24-bpp generic TFT, i.e. 1.67M colors.	1	RW

REG[12h] Display Configuration Register (DPCR)

Bit	Description	Default	Access
7	PCLK Inversion 0: XPDAT, XDE, XHSYNC etc. Panel fetches XPDAT at XPCLK rising edge.  1: XPDAT, XDE, XHSYNC etc. Panel fetches XPDAT at PCLK falling edge. 	0	RW
6	Display ON/OFF 0b: Display Off. 1b: Display On.	0	RW
5	Display Test Color Bar 0b: Disable. 1b: Enable.	0	RW
4	This bit must be set 0.	0	RO
3	VDIR Vertical Scan direction 0 : From Top to Bottom 1 : From bottom to Top	0	RW

Bit	Description	Default	Access
2-0	Parallel XPDAT[23:0] Output Sequence 000b : RGB 001b : RBG 010b : GRB 011b : GBR 100b : BRG 101b : BGR 110b : Gray 111b : send out idle state (all 0 or 1, black or white color).	0	RW

Note1 : When VDIR = 1 , pip window, graphic cursor and text cursor function will be disable automatically.

REG[13h] Panel scan Clock and Data Setting Register (PCSR)

Bit	Description	Default	Access
7	XHSYNC Polarity 0 : Low active. 1 : High active.	0	RW
6	XVSYNC Polarity 0 : Low active. 1 : High active.	0	RW
5	XDE Polarity 0 : High active. 1 : Low active.	0	RW
4	XDE IDLE STATE (in power saving mode or DISPLAY OFF) 0 : Pin "DE" output is low. 1 : Pin "DE" output is high.	0	RW
3	XPCLK IDLE STATE (in power saving mode or DISPLAY OFF) 0 : Pin "PCLK" output is low. 1 : Pin "PCLK" output is high.	0	RW
2	XPDAT IDLE STATE (in Vertical/horizontal non-display period or power saving mode or DISPLAY OFF) 0 : Pins "PDAT[23:0]" output is low. 1 : Pins "PDAT[23:0]" output is high.	0	RW
1	XHSYNC IDLE STATE (in power saving mode or DISPLAY OFF) 0 : Pin "HSYNC" output is low. 1 : Pin "HSYNC" output is high.	1	RW
0	XVSYNC IDLE STATE (in power saving mode or DISPLAY OFF) 0 : Pin "VSYNC" output is low. 1 : Pin "VSYNC" output is high.	1	RW

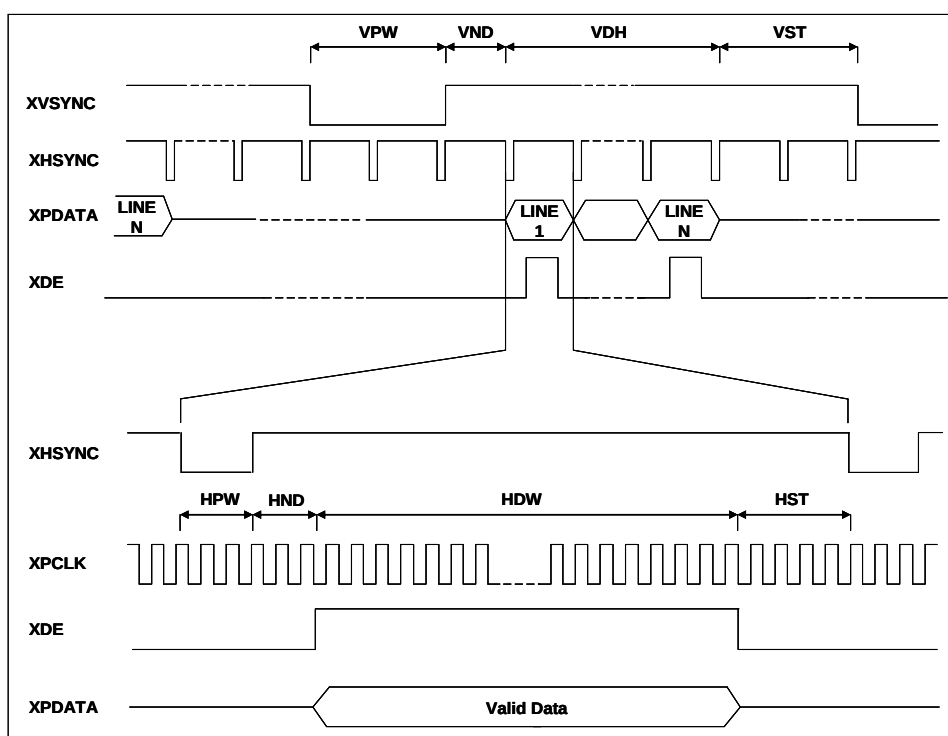


Figure 19-3 : Digital TFT Panel Timing

REG[14h] Horizontal Display Width Register (HDWR)

Bit	Description	Default	Access
7-0	Horizontal Display Width Setting Bit[7:0] The register specifies the LCD panel horizontal display width in the unit of 8 pixels resolution. Horizontal display width(pixels) = (HDWR + 1) * 8 + HDWFTR Maximum value cannot over 2048.	4Fh	RW

REG[15h] Horizontal Display Width Fine Tune Register (HDWFTR)

Bit	Description	Default	Access
7-4	NA	0	RO
3-0	Horizontal Display Width Fine Tuning (HDWFT) [3:0] The register specifies the fine tune value for horizontal display width. It is used to support panel which horizontal size is not a multiple of 8. Each level of this modulation is 1 pixel. Horizontal display width(pixels) = (HDWR + 1) * 8 + HDWFT Maximum value cannot over 2048.	0	RW

REG[16h] Horizontal Non-Display Period Register (HNDR)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Horizontal Non-Display Period(HNDR) Bit[4:0] This register specifies the horizontal non-display period. Also called back porch . Horizontal non-display period or Back porch (pixels) = $(HNDR + 1) * 8 + HNDFTR$	03h	RW

REG[17h] Horizontal Non-Display Period Fine Tune Register (HNDFTFTR)

Bit	Description	Default	Access
7-4	NA	0	RO
3-0	Horizontal Non-Display Period Fine Tuning(HNDFT) [3:0] This register specifies the fine tuning for horizontal non-display period (also called back porch); it is used to support the SYNC mode panel. Each level of this modulation is 1-pixel. Horizontal non-display period or Back porch (pixels) = $(\text{HNDR} + 1) * 8 + \text{HNDFTFTR}$	06h	RW

REG[18h] HSYNC Start Position Register (HSTR)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	HSYNC Start Position[4:0] The starting position from the end of display area to the beginning of HSYNC. Each level of this modulation is 8-pixel. Also called front porch . HSYNC Start Position or Front porch (pixels) = $(\text{HSTR} + 1) \times 8$	1Fh	RW

REG[19h] HSYNC Pulse Width Register (HPWR)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	HSYNC Pulse Width(HPW) [4:0] The period width of HSYNC. HSYNC Pulse Width(pixels) = $(\text{HPW} + 1) \times 8$	0	RW

REG[1Ah] Vertical Display Height Register 0(VDHR0)

Bit	Description	Default	Access
7-0	Vertical Display Height Bit[7:0] Vertical Display Height(Line) = $\text{VDHR} + 1$	DFh	RW

REG[1Bh] Vertical Display Height Register 1 (VDHR1)

Bit	Description	Default	Access
7-3	NA	0	RO
2-0	Vertical Display Height Bit[10:8] Vertical Display Height(Line) = $\text{VDHR} + 1$	01h	RW

REG[1Ch] Vertical Non-Display Period Register 0(VNDR0)

Bit	Description	Default	Access
7-0	Vertical Non-Display Period Bit[7:0] Vertical Non-Display Period(Line) = $(\text{VNDR} + 1)$	15h	RW

REG[1Dh] Vertical Non-Display Period Register 1(VNDR1)

Bit	Description	Default	Access
7-2	NA	0	RO
1-0	Vertical Non-Display Period Bit[9:8] Vertical Non-Display Period(Line) = $(\text{VNDR} + 1)$	0	RW

REG[1Eh] VSYNC Start Position Register (VSTR)

Bit	Description	Default	Access
7-0	VSYNC Start Position[7:0] The starting position from the end of display area to the beginning of VSYNC. VSYNC Start Position(Line) = (VSTR + 1)	0Bh	RW

REG[1Fh] VSYNC Pulse Width Register (VPWR)

Bit	Description	Default	Access
7-6	NA	0	RO
5-0	VSYNC Pulse Width[5:0] The pulse width of VSYNC in lines. VSYNC Pulse Width(Line) = (VPWR + 1)	0	RW

Note : Following multiple bytes registers from 20h to 3Bh only take effect on MSB was wrote.

Ex. To program Main Image Start Address from 20h to 23h and this address take effect on register [23h] was wrote.

REG[20h] Main Image Start Address 0 (MISA0)

Bit	Description	Default	Access
7-0	Main Image Start Address[7:0] It must be divisible by 4. Bit [1:0] tie to "0" internally.	0	RW

REG[21h] Main Image Start Address 1 (MISA1)

Bit	Description	Default	Access
7-0	Main Image Start Address[15:8]	0	RW

REG[22h] Main Image Start Address 2 (MISA2)

Bit	Description	Default	Access
7-0	Main Image Start Address [23:16]	0	RW

REG[23h] Main Image Start Address 3 (MISA3)

Bit	Description	Default	Access
7-0	Main Image Start Address [31:24]	0	RW

REG[24h] Main Image Width 0 (MIW0)

Bit	Description	Default	Access
7-0	Main Image Width [7:0] Unit: Pixel. It must be divisible by 4. MIW Bit [1:0] tie to "0" internally. The value is physical pixel number. Maximum value is 8188 pixels	0	RW

REG[25h] Main Image Width 1 (MIW1)

Bit	Description	Default	Access
7-5	NA	NA	NA
4-0	Main Image Width [12:8] Unit: Pixel. It must be divisible by 4. The value is physical pixel number. Maximum value is 8188 pixels	0	RW

REG[26h] Main Window Upper-Left corner X-coordinates 0 (MWULX0)

Bit	Description	Default	Access
7-0	Main Window Upper-Left corner X-coordinates [7:0] Reference <u>Main Image</u> coordinates. Unit: Pixel It must be divisible by 4. MWULX Bit [1:0] tie to "0" internally. X-axis coordinates plus Horizontal display width cannot large than 8188.	0	RW

REG[27h] Main Window Upper-Left corner X-coordinates 1 (MWULX1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Main Window Upper-Left corner X-coordinates [12:8] Reference <u>Main Image</u> coordinates. Unit: Pixel It must be divisible by 4. X-axis coordinates plus Horizontal display width cannot large than 8188.	0	RW

REG[28h] Main Window Upper-Left corner Y-coordinates 0 (MWULY0)

Bit	Description	Default	Access
7-0	Main Window Upper-Left corner Y-coordinates [7:0] Reference <u>Main Image</u> coordinates. Unit: Pixel Range is between 0 and 8191.	0	RW

REG[29h] Main Window Upper-Left corner Y-coordinates 1 (MWULY1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Main Window Upper-Left corner Y-coordinates [12:8] Reference <u>Main Image</u> coordinates. Unit: Pixel Range is between 0 and 8191.	0	RW

REG[2Ah] PIP 1 or 2 Window Display Upper-Left corner X-coordinates 0 (PWDULX0)

Bit	Description	Default	Access
7-0	PIP Window Display Upper-Left corner X-coordinates [7:0] Reference <u>Main Window</u> coordinates. Unit: Pixel It must be divisible by 4. PWDULX Bit [1:0] tie to "0" internally. X-axis coordinates should less than horizontal display width. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[2Bh] PIP 1 or 2 Window Display Upper-Left corner X-coordinates 1 (PWDULX1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	PIP Window Display Upper-Left corner X-coordinates [12:8] Reference <u>Main Window</u> coordinates. Unit: Pixel It must be divisible by 4. PWDULX Bit [1:0] tie to "0" internally. X-axis coordinates should less than horizontal display width. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[2Ch] PIP 1 or 2 Window Display Upper-Left corner Y-coordinates 0 (PWDULY0)

Bit	Description	Default	Access
7-0	PIP Window Display Upper-Left corner Y-coordinates [7:0] Reference <u>Main Window</u> coordinates. Unit: Pixel Y-axis coordinates should less than vertical display height. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[2Dh] PIP 1 or 2 Window Display Upper-Left corner Y-coordinates 1 (PWDULY1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	PIP Window Display Upper-Left corner Y-coordinates [12:8] Reference <u>Main Window</u> coordinates. Unit: Pixel Y-axis coordinates should less than vertical display height. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[2Eh] PIP 1 or 2 Image Start Address 0 (PISA0)

Bit	Description	Default	Access
7-0	PIP Image Start Address[7:0] According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window. It must be divisible by 4. Bit [1:0] tie to "0" internally.	0	RW

REG[2Fh] PIP 1 or 2 Image Start Address 1 (PISA1)

Bit	Description	Default	Access
7-0	PIP Image Start Address[15:8] According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[30h] PIP 1 or 2 Image Start Address 2 (PISA2)

Bit	Description	Default	Access
7-0	PIP Image Start Address [23:16] According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[31h] PIP 1 or 2 Image Start Address 3 (PISA3)

Bit	Description	Default	Access
7-0	PIP Image Start Address [31:24] According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[32h] PIP 1 or 2 Image Width 0 (PIW0)

Bit	Description	Default	Access
7-0	PIP Image Width [7:0] Unit: Pixel. It must be divisible by 4. PIW Bit [1:0] tie to "0" internally. The value is physical pixel number. This width should less than horizontal display width. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[33h] PIP 1 or 2 Image Width 1 (PIW1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	PIP Image Width [12:8] Unit: Pixel. It must be divisible by 4. PIW Bit [1:0] tie to "0" internally. The value is physical pixel number. This width should less than horizontal display width. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[34h] PIP 1 or 2 Window Image Upper-Left corner X-coordinates 0 (PWIULX0)

Bit	Description	Default	Access
7-0	PIP 1 or 2 Window Image Upper-Left corner X-coordinates [7:0] Reference <u>PIP Image</u> coordinates. Unit: Pixel It must be divisible by 4. PWIULX Bit [1:0] tie to "0" internally. X-axis coordinates plus PIP image width must less or equal to 8187. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[35h] PIP 1 or 2 Window Image Upper-Left corner X-coordinates 1 (PWIULX1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	PIP Window Image Upper-Left corner X-coordinates [12:8] Reference <u>PIP Image</u> coordinates. Unit: Pixel It must be divisible by 4. PWIULX Bit [1:0] tie to "0" internally. X-axis coordinates plus PIP image width must less or equal to 8187. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[36h] PIP 1 or 2 Window Image Upper-Left corner Y-coordinates (PWIULY0)

Bit	Description	Default	Access
7-0	PIP Windows Display Upper-Left corner Y-coordinates [7:0] Reference <u>PIP Image</u> coordinates. Unit: Pixel Y-axis coordinates plus PIP window height should less or equal to 8191. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[37h] PIP 1 or 2 Window Image Upper-Left corner Y-coordinates 1 (PWIULY1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	PIP Windows Image Upper-Left corner Y-coordinates [12:8] Reference <u>PIP Image</u> coordinates. Unit: Pixel Y-axis coordinates plus PIP window height should less or equal to 8191. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[38h] PIP 1 or 2 Window Width 0 (PWW0)

Bit	Description	Default	Access
7-0	PIP Window Width [7:0] Unit: Pixel. It must be divisible by 4. PWW Bit [1:0] tie to "0" internally. The value is physical pixel number. Maximum value is 8188 pixels. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[39h] PIP 1 or 2 Window Width 1 (PWW1)

Bit	Description	Default	Access
7-3	NA	0	RO
2-0	PIP Window Width [10:8] Unit: Pixel. It must be divisible by 4. The value is physical pixel number. Maximum value is 8188 pixels. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[3Ah] PIP 1 or 2 Window Height 0 (PWH0)

Bit	Description	Default	Access
7-0	PIP Window Height [7:0] Unit: Pixel The value is physical pixel number. Maximum value is 8191 pixels. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

REG[3Bh] PIP 1 or 2 Windows Height 1 (PWH1)

Bit	Description	Default	Access
7-3	NA	0	RO
2-0	PIP Window Height [10:8] Unit: Pixel The value is physical pixel number. Maximum value is 8191 pixels. According to bit of Select Configure PIP 1 or 2 Window's parameters. Function bit will be configured for relative PIP window.	0	RW

Note: The PIP windows sizes and start positions are specified in 8 pixel resolution (horizontal) and 1 line resolution (vertical).

Note: Above multiple bytes registers from 20h to 3Bh only take effect on MSB was wrote.

Ex. To program Main Image Start Address from 20h to 23h and this address take effect on register [23h] was wrote.

REG[3Ch] Graphic / Text Cursor Control Register (GTCCR)

Bit	Description	Default	Access
7	Gamma correction Enable 0: Disable 1: Enable Gamma correction is the last output stage.	0	RW
6-5	Gamma table select for MPU write gamma data 00b: Gamma table for Blue 01b: Gamma table for Green 10b: Gamma table for Red 11b: NA	0	RW
4	Graphic Cursor Enable 0 : Graphic Cursor disable. 1 : Graphic Cursor enable. Graphic cursor will auto disable if VDIR set as "1".	0	RW
3-2	Graphic Cursor Selection Bit Select one from four graphic cursor types. (00b to 11b) 00b : Graphic Cursor Set 1. 01b : Graphic Cursor Set 2. 10b : Graphic Cursor Set 3. 11b : Graphic Cursor Set 4.	0	RW
1	Text Cursor Enable 0 : Disable. 1 : Enable. Text cursor & Graphic cursor cannot enable simultaneously. Graphic cursor has higher priority then Text cursor if enabled simultaneously.	0	RW
0	Text Cursor Blinking Enable 0 : Disable. 1 : Enable.	0	RW

REG[3Dh] Blink Time Control Register (BTCCR)

Bit	Description	Default	Access
7-0	Text Cursor Blink Time Setting (Unit: Frame) 00h : 1 frame time. 01h : 2 frames time. 02h : 3 frames time. : FFh : 256 frames time.	0	RW

REG[3Eh] Text Cursor Horizontal Size Register (CURHS)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Text Cursor Horizontal Size Setting[4:0] Unit : Pixel Zero-based number. Value "0" means 1 pixel. Note : When character is enlarged, the cursor setting will multiply the same times as the character enlargement.	07h	RW

REG[3Fh] Text Cursor Vertical Size Register (CURVS)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Text Cursor Vertical Size Setting[4:0] Unit : Pixel Zero-based number. Value "0" means 1 pixel. Note : When character is enlarged, the cursor setting will multiply the same times as the character enlargement.	0	RW

REG[40h] Graphic Cursor Horizontal Position Register 0 (GCHP0)

Bit	Description	Default	Access
7-0	Graphic Cursor Horizontal Location[7:0] Reference <u>main Window</u> coordinates.	0	RW

REG[41h] Graphic Cursor Horizontal Position Register 1 (GCHP1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Graphic Cursor Horizontal Location[12:8] Reference <u>main Window</u> coordinates.	0	RW

REG[42h] Graphic Cursor Vertical Position Register 0 (GCVP0)

Bit	Description	Default	Access
7-0	Graphic Cursor Vertical Location[7:0] Reference <u>main Window</u> coordinates.	0	RW

REG[43h] Graphic Cursor Vertical Position Register 1 (GCVP1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Graphic Cursor Vertical Location[12:8] Reference <u>main Window</u> coordinates.	0	RW

REG[44h] Graphic Cursor Color 0 (GCC0)

Bit	Description	Default	Access
7-0	Graphic Cursor Color 0 with 256 Colors RGB Format [7:0] = RRRGGGBB.	0	RW

REG[45h] Graphic Cursor Color 1 (GCC1)

Bit	Description	Default	Access
7-0	Graphic Cursor Color 1 with 256 Colors RGB Format [7:0] = RRRGGGBB.	0	RW

REG[46h – 4Fh] – RESERVED

Bit	Description	Default	Access
7-0	NA	0	RO

19.6 Geomatric Engine Control Registers

REG[50h] Canvas Start address 0 (CVSSA0)

Bit	Description	Default	Access
7-2	Start address of Canvas [7:2] Ignored if canvas in linear addressing mode.	0	RW
1-0	Fix at 0	0	RO

REG[51h] Canvas Start address 1 (CVSSA1)

Bit	Description	Default	Access
7-0	Start address of Canvas [15:8] Ignored if canvas in linear addressing mode.	0	RW

REG[52h] Canvas Start address 2 (CVSSA2)

Bit	Description	Default	Access
7-0	Start address of Canvas [23:16] Ignored if canvas in linear addressing mode.	0	RW

REG[53h] Canvas Start address 3 (CVSSA3)

Bit	Description	Default	Access
7-0	Start address of Canvas [31:24] Ignored if canvas in linear addressing mode.	0	RW

REG[54h] Canvas image width 0 (CVS_IMWTH0)

Bit	Description	Default	Access
7-2	Canvas image width [7:2] The bits are Canvas image width. Unit: Pixel, it is 4 pixel resolutions. Width = Set Value Ignored if canvas in linear addressing mode.	0	RW
1-0	Fix at 0.	0	RO

REG[55h] Canvas image width 1 (CVS_IMWTH1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Canvas image width [12:8] The bits are Canvas image width Ignored if canvas in linear addressing mode.	0	RW

REG[56h] Active Window Upper-Left corner X-coordinates 0 (AWUL_X0)

Bit	Description	Default	Access
7-0	Active Window Upper-Left corner X-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel X-axis coordinates plus Active Window width cannot large than 8188. Ignored if canvas in linear addressing mode.	0	RW

REG[57h] Active Window Upper-Left corner X-coordinates 1 (AWUL_X1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Active Window Upper-Left corner X-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel X-axis coordinates plus Active Window width cannot large than 8188. Ignored if canvas in linear addressing mode.	0	RW

REG[58h] Active Window Upper-Left corner Y-coordinates 0 (AWUL_Y0)

Bit	Description	Default	Access
7-0	Active Window Upper-Left corner Y-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel Y-axis coordinates plus Active Window height cannot large than 8191. Ignored if canvas in linear addressing mode.	0	RW

REG[59h] Active Window Upper-Left corner Y-coordinates 1 (AWUL_Y1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Active Window Upper-Left corner Y-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel Y-axis coordinates plus Active Window height cannot large than 8191. Ignored if canvas in linear addressing mode.	0	RW

REG[5Ah] Active Window Width 0 (AW_WTH0)

Bit	Description	Default	Access
7-0	Width of Active Window [7:0] Unit: Pixel. The value is physical pixel number. Maximum value is 8188 pixels. Ignored if canvas in linear addressing mode.	0	RW

REG[5Bh] Active Window Width 1 (AW_WTH1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Width of Active Window [12:8] Unit: Pixel. The value is physical pixel number. Maximum value is 8188 pixels. Ignored if canvas in linear addressing mode.	0	RW

REG[5Ch] Active Window Height 0 (AW_HT0)

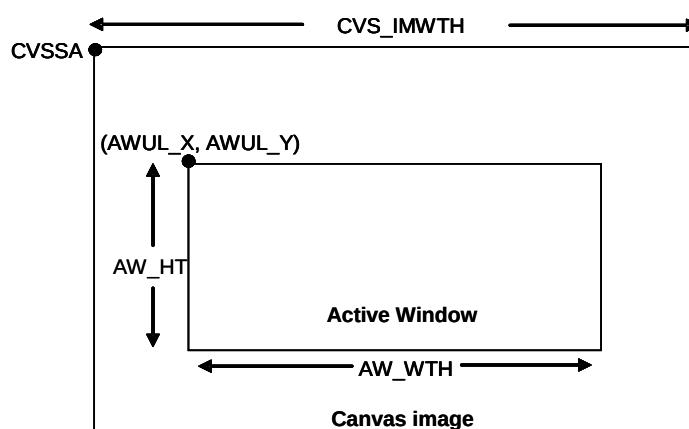
Bit	Description	Default	Access
7-0	Height of Active Window [7:0] Unit: Pixel. The value is physical pixel number. Maximum value is 8191 pixels. Ignored if canvas in linear addressing mode.	0	RW

REG[5Dh] Active Window Height 1 (AW_HT1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Height of Active Window [12:8] Unit: Pixel. The value is physical pixel number. Maximum value is 8191 pixels. Ignored if canvas in linear addressing mode.	0	RW

REG[5Eh] Color Depth of Canvas & Active Window (AW_COLOR)

Bit	Description	Default	Access
7-4	NA	0	RO
3	NA	0	RO
2	Canvas addressing mode 0: Block mode (X-Y coordinates addressing) 1: Linear mode	0	RW
1-0	Canvas image's color depth & memory R/W data width In Block Mode: 00: 8bpp 01: 16bpp 1x: 24bpp Note: monochrome data can input with any one color depth depends on proper image width. In Linear Mode: X0: 8-bits memory data read/write. X1: 16-bits memory data read/write	0	RW


Figure 19-4 : Active Window

REG[5Fh] Graphic Read/Write position Horizontal Position Register 0 (CURH0)

Bit	Description	Default	Access
7-0	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [7:0] Unit: Byte When DPRAM In Block mode: Graphic Read/Write Horizontal Position 0 [7:0] Reference Canvas image coordinates. Unit: Pixel	0	RW

Note: User should program proper active window related parameters before configure this register.

REG[60h] Graphic Read/Write position Horizontal Position Register 1 (CURH1)

Bit	Description	Default	Access
7-5	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [15:13] Unit: Byte When DPRAM In Block mode: NA Reference Canvas image coordinates. Unit: Pixel	0	RW
4-0	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [12:8] Unit: Byte When DPRAM In Block mode: Graphic Read/Write Horizontal Position 1 [12:8] Reference Canvas image coordinates. Unit: Pixel	0	RW

Note: User should program proper active window related parameters before configure this register.

REG[61h] Graphic Read/Write position Vertical Position Register 0 (CURV0)

Bit	Description	Default	Access
7-0	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [23:16] Unit: Byte When DPRAM In Block mode: Graphic Read/Write Vertical Position 0 [7:0] Reference Canvas image coordinates. Unit: Pixel	0	RW

Note: User should program proper active window related parameters before configure this register.

REG[62h] Graphic Read/Write position Vertical Position Register 1 (CURV1)

Bit	Description	Default	Access
7-5	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [31:29] Unit: Byte When DPRAM In Block mode: NA Reference Canvas image coordinates. Unit: Pixel	0	RW
4-0	Write: Set Graphic Read/Write position When DPRAM In Linear mode: Memory Read/Write address [28:24] Unit: Byte When DPRAM In Block mode: Graphic Read/Write Vertical Position 1 [12:8] Reference Canvas image coordinates. Unit: Pixel	0	RW

Note: User should program proper active window related parameters before configure this register.

REG[63h] Text Write X-coordinates Register 0 (F_CURX0)

Bit	Description	Default	Access
7-0	Write: Set Text Write position Read: Current Text Write position Text Write X-coordinates [7:0] The setting of the horizontal cursor position for text writing. Reference Canvas image coordinates. Unit: Pixel	0	RW

REG[64h] Text Write X-coordinates Register 1 (F_CURX1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Write: Set Text Write position Read: Current Text Write position Text Write X-coordinates [12:8] The setting of the horizontal cursor position for text writing. Reference Canvas image coordinates. Unit: Pixel	0	RW

REG[65h] Text Write Y-coordinates Register 0 (F_CURY0)

Bit	Description	Default	Access
7-0	Write: Set Text Write position Read: Current Text Write position Text Write Y-coordinates [7:0] The setting of the vertical cursor position for text writing. Reference Canvas image coordinates. Unit: Pixel	0	RW

REG[66h] Text Write Y-coordinates Register 1 (F_CURY1)

Bit	Description	Default	Access
7-5	NA	0	RO
4:0	Write: Set Text Write position Read: Current Text Write position Text Write Y-coordinates [12:8] The setting of the vertical cursor position for text writing. Reference Canvas image coordinates. Unit: Pixel	0	RW

REG[67h] Draw Line / Triangle Control Register 0 (DCR0)

Bit	Description	Default	Access
7	Draw Line / Triangle Start Signal Write Function 0 : Stop the drawing function. 1 : Start the drawing function. Read Function 0 : Drawing function complete. 1 : Drawing function is processing.	0	RW
6	NA	0	RO
5	Fill function for Triangle Signal 0 : Non fill. 1 : Fill.	0	RW
4-2	NA	0	RO
1	Draw Triangle or Line Select Signal 0 : Draw Line 1 : Draw Triangle	0	RW
0	NA	0	RO

REG[68h] Draw Line/Square/Triangle Point 1 X-coordinates Register0 (DLHSR0)

Bit	Description	Default	Access
7-0	Draw Line/Triangle Point 1 X-coordinates [7:0] Square diagonal Point 1 X-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[69h] Draw Line/Square/Triangle Point 1 X-coordinates Register1 (DLHSR1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Line/Triangle Point 1 X-coordinates [12:8] Square diagonal Point 1 X-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Ah] Draw Line/Square/Triangle Point 1 Y-coordinates Register0 (DLVSR0)

Bit	Description	Default	Access
7-0	Draw Line/Triangle Point 1 Y-coordinates [7:0] Square diagonal Point 1 Y-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Bh] Draw Line/Square/Triangle Point 1 Y-coordinates Register1 (DLVSR1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Line/Triangle Point 1 Y-coordinates [12:8] Square diagonal Point 1 Y-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Ch] Draw Line/Square/Triangle Point 2 X-coordinates Register0 (DLHER0)

Bit	Description	Default	Access
7-0	Draw Line/Triangle Point 2 X-coordinates [7:0] Square diagonal Point 2 X-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Dh] Draw Line/Square/Triangle Point 2 X-coordinates Register1 (DLHER1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Line/Triangle Point 2 X-coordinates [12:8] Square diagonal Point 2 X-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Eh] Draw Line/Square/Triangle Point 2 Y-coordinates Register0 (DLVER0)

Bit	Description	Default	Access
7-0	Draw Line/Triangle Point 2 Y-coordinates [7:0] Square diagonal Point 2 Y-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[6Fh] Draw Line/Square/Triangle Point 2 Y-coordinates Register1 (DLVER1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Line/Triangle Point 2 Y-coordinates [12:8] Square diagonal Point 2 Y-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel *** Note : When draw a square, start point & end point cannot locate at same point or at same X-axis or Y-axis.	0	RW

REG[70h] Draw Triangle Point 3 X-coordinates Register 0 (DTPH0)

Bit	Description	Default	Access
7-0	Draw Triangle Point 3 X-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[71h] Draw Triangle Point 3 X-coordinates Register 1 (DTPH1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Triangle Point 3 X-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[72h] Draw Triangle Point 3 Y-coordinates Register 0 (DTPV0)

Bit	Description	Default	Access
7-0	Draw Triangle Point 3 Y-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[73h] Draw Triangle Point 3 Y-coordinates Register 1 (DTPV1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Triangle Point 3 Y-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

*** **Note**: for triangle's three endpoint setting:

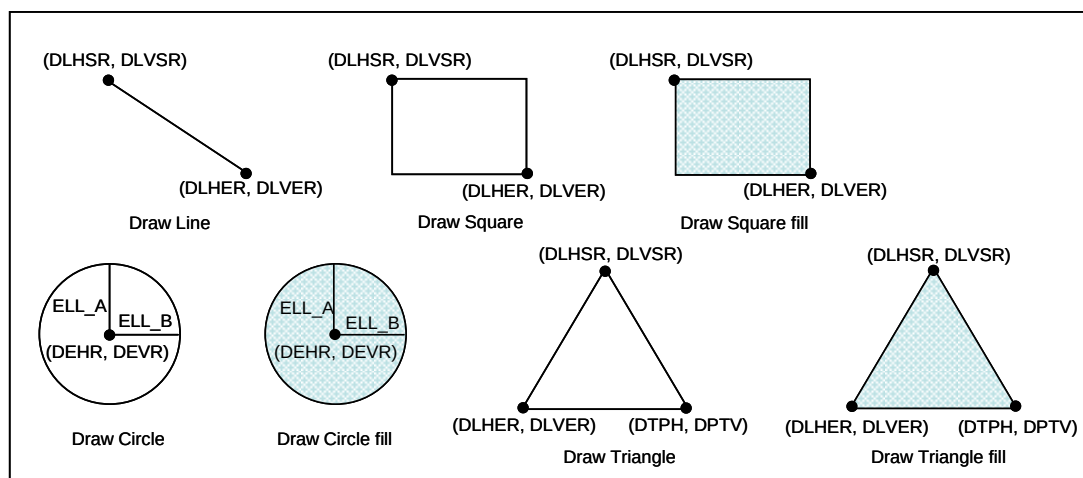
1. Any two endpoints overlap will draw a line.
2. Three endpoints overlap will draw a dot.

REG[74h – 75h] RESERVED

Bit	Description	Default	Access
7-0	NA	0	RO

REG[76h] Draw Circle/Ellipse/Ellipse Curve/Circle Square Control Register 1 (DCR1)

Bit	Description	Default	Access
7	Draw Circle / Ellipse / Square / Circle Square Start Signal Write Function 0 : Stop the drawing function. 1 : Start the drawing function. Read Function 0 : Drawing function complete. 1 : Drawing function is processing.	0	RW
6	Fill the Circle / Ellipse / Square / Circle Square Signal 0 : Non fill. 1 : fill.	0	RW
5-4	Draw Circle / Ellipse / Square / Ellipse Curve / Circle Square Select 00 : Draw Circle / Ellipse 01 : Draw Circle / Ellipse Curve 10 : Draw Square. 11 : Draw Circle Square.	0	RW
3-2	NA	0	RO
1-0	Draw Circle / Ellipse Curve Part Select(DECP) 00: bottom-left Ellipse Curve 01: upper-left Ellipse Curve 10: upper-right Ellipse Curve 11: bottom-right Ellipse Curve	0	RW


Figure 19-5 : Drawing Function Parameter

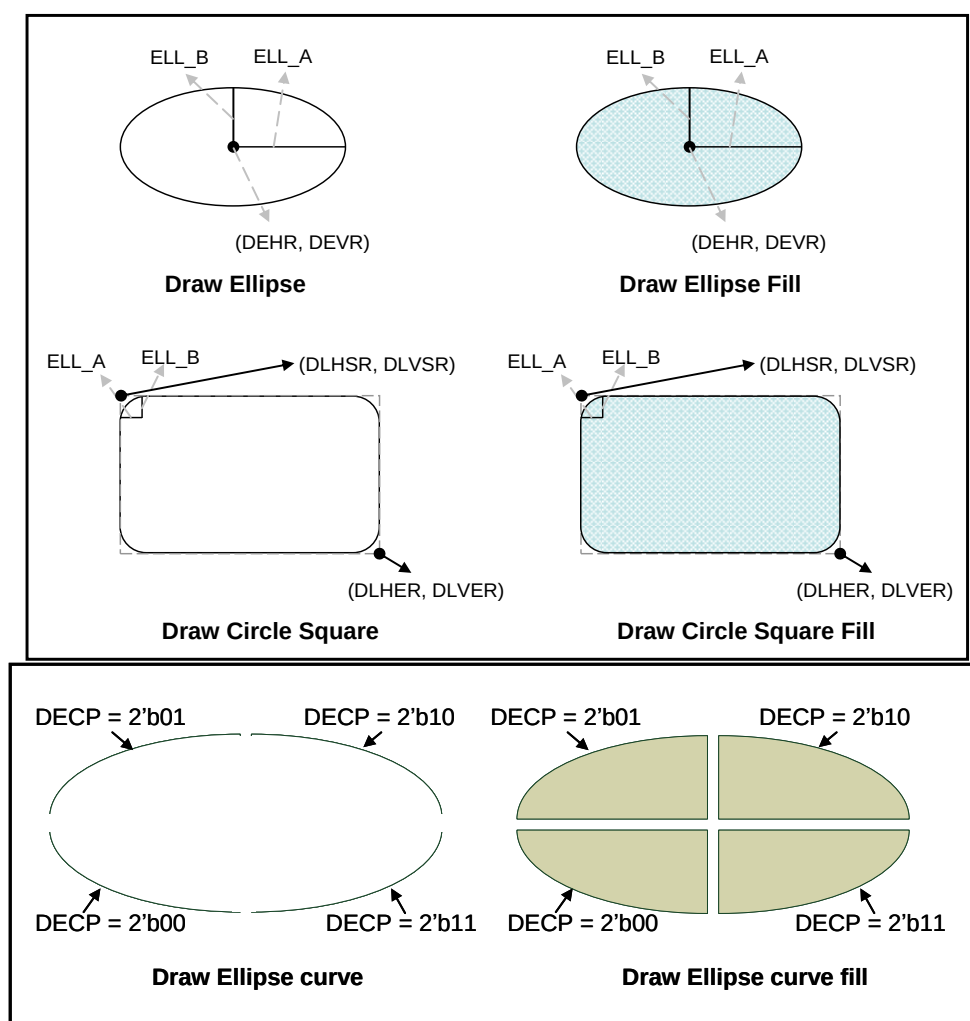


Figure 19-6 : The Drawing Function

REG[77h] Draw Circle/Ellipse/Circle Square Major radius Setting Register (ELL_A0)

Bit	Description	Default	Access
7-0	Draw Circle/Ellipse/Circle Square Major radius [7:0] Unit: Pixel Draw circle need to set major axis equal to minor radius.	0	RW

REG[78h] Draw Circle/Ellipse/Circle Square Major radius Setting Register (ELL_A1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Circle/Ellipse/Circle Square Major radius [12:8] Unit: Pixel Draw circle need to set major axis equal to minor radius.	0	RW

REG[79h] Draw Circle/Ellipse/Circle Square Minor radius Setting Register (ELL_B0)

Bit	Description	Default	Access
7-0	Draw Circle/Ellipse/Circle Square Minor radius [7:0] Unit: Pixel Draw circle need to set major axis equal to radius axis.	0	RW

REG[7Ah] Draw Circle/Ellipse/Circle Square Minor radius Setting Register (ELL_B1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Circle/Ellipse/Circle Square Minor radius [12:8] Unit: Pixel Draw circle need to set major axis equal to minor radius.	0	RW

REG[7Bh] Draw Circle/Ellipse/Circle Square Center X-coordinates Register0 (DEHR0)

Bit	Description	Default	Access
7-0	Draw Circle/Ellipse/Circle Square Center X-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[7Ch] Draw Circle/Ellipse/Circle Square Center X-coordinates Register1 (DEHR1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Circle/Ellipse/Circle Square Center X-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[7Dh] Draw Circle/Ellipse/Circle Square Center Y-coordinates Register0 (DEVRO)

Bit	Description	Default	Access
7-0	Draw Circle/Ellipse/Circle Square Center Y-coordinates [7:0] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[7Eh] Draw Circle/Ellipse/Circle Square Center Y-coordinates Register1 (DEVR1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Draw Circle/Ellipse/Circle Square Center Y-coordinates [12:8] Reference <u>Canvas image</u> coordinates. Unit: Pixel	0	RW

REG[7Fh] – RESERVED

Bit	Description	Default	Access
7-0	NA	0	RO

REG[D2h] Foreground Color Register - Red (FGCR)

Bit	Description	Default	Access
7-0	Foreground Color – Red; for draw, text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

REG[D3h] Foreground Color Register - Green (FGCG)

Bit	Description	Default	Access
7-0	Foreground Color - Green; for draw, text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:2]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

REG[D4h] Foreground Color Register - Blue (FGCB)

Bit	Description	Default	Access
7-0	Foreground Color - Blue; for draw, text or color expansion 256 colors, the register only uses Bit[7:6]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

19.7 PWM Timer Control Registers

REG[84h] PWM Prescaler Register (PSCLR)

Bit	Description	Default	Access
7-0	PWM Prescaler Register These 8 bits determine prescaler value for Timer 0 and 1. Time base is "Core_Freq / (Prescaler + 1)"	0	RW

REG[85h] PWM clock Mux Register (PMUXR)

Bit	Description	Default	Access
7-6	Select 2nd clock divider's MUX input for PWM Timer 1 00 = 1 01 = 1/2 10 = 1/4 11 = 1/8	0	RW
5-4	Select 2nd clock divider's MUX input for PWM Timer 0 00 = 1 01 = 1/2 10 = 1/4 11 = 1/8	0	RW
3-2	XPWM[1] pin function control 0X: XPWM[1] output system error flag (Scan FIFO pop error or Memory access out of range) 10: XPWM[1] output PWM timer 1 event or invert of PWM timer 0 11: XPWM[1] output oscillator clock If XTEST[0] set high, then XPWM[1] will become panel scan clock input.	0	RW
1-0	XPWM[0] pin function control 0X: XPWM[0] becomes GPIO-C[7] 10: XPWM[0] output PWM timer 0 11: XPWM[0] output core clock	0	RW

REG[86h] PWM Configuration Register (PCFGR)

Bit	Description	Default	Access
7	NA	0	RO
6	PWM Timer 1 output inverter on/off Determine the output inverter on/off for Timer 1. 0 = Inverter off 1 = Inverter on for PWM1	0	RW
5	PWM Timer 1 auto reload on/off Determine auto reload on/off for Timer 1. 0 = One-shot 1 = Interval mode(auto reload)	1	RW
4	PWM Timer 1 start/stop Determine start/stop for Timer 1. 0 = Stop 1 = Start -- In Interval mode (auto reload), user need program it as 0 to stop PWM timer. -- In One-shot, this bit will auto clear. User may read this bit to know current PWMx is running or stopped.	0	RW
3	PWM Timer 0 Dead zone enable Determine the dead zone operation. 0 = Disable 1 = Enable	0	RW

Bit	Description	Default	Access
2	PWM Timer 0 output inverter on/off Determine the output inverter on/off for Timer 0. 0 = Inverter off 1 = Inverter on for PWM0	0	RW
1	PWM Timer 0 auto reload on/off Determine auto reload on/off for Timer 0. 0 = One-shot 1 = Interval mode(auto reload)	1	RW
0	PWM Timer 0 start/stop Determine start/stop for Timer 0. 0 = Stop 1 = Start -- In Interval mode (auto reload), user need program it as 0 to stop PWM timer. -- In One-shot, this bit will auto clear. User may read this bit to know current PWMx is running or stopped.	0	RW

REG[87h] Timer 0 Dead zone length register [DZ_LENGTH]

Bit	Description	Default	Access
7-0	Timer 0 Dead zone length register These 8 bits determine the dead zone length. The 1 unit time of the dead zone length is equal to that of timer 0.	0	RW

REG[88h] Timer 0 compare buffer register [TCMPB0L]

Bit	Description	Default	Access
7-0	Timer 0 compare buffer register --- Low Byte Compare buffer register total has 16 bits. When timer counter equal or less than compare buffer register will cause PWM 0 out high level if PWM Timer 0 output inverter on/off bit is off.	0	RW

REG[89h] Timer 0 compare buffer register [TCMPB0H]

Bit	Description	Default	Access
7-0	Timer 0 compare buffer register --- High Byte Compare buffer register total has 16 bits. When timer counter equal or less than compare buffer register will cause PWM 0 out high level if PWM Timer 0 output inverter on/off bit is off.	0	RW

REG[8Ah] Timer 0 count buffer register [TCNTB0L]

Bit	Description	Default	Access
7-0	Timer 0 count buffer register --- Low Byte Count buffer register total has 16 bits. When timer counter equal to 0 will cause PWM timer reload Count buffer register if reload_en bit set as enable. It may read back timer counter's real time value (TCNT0) when PWM timer start.	0	RW

REG[8Bh] Timer 0 count buffer register [TCNTB0H]

Bit	Description	Default	Access
7-0	Timer 0 count buffer register --- High Byte Count buffer register total has 16 bits. When timer counter equal to 0 will cause PWM timer reload Count buffer register if reload_en bit set as enable. It may read back timer counter's real time value (TCNT0) when PWM timer start.	0	RW

REG[8Ch] Timer 1 compare buffer register [TCMPB1L]

Bit	Description	Default	Access
7-0	Timer 1 compare buffer register --- Low Byte Compare buffer register total has 16 bits. When timer counter equal or less than compare buffer register will cause PWM out high level if inv_on bit is off.	0	RW

REG[8Dh] Timer 1 compare buffer register [TCMPB1H]

Bit	Description	Default	Access
7-0	Timer 1 compare buffer register --- High Byte Compare buffer register total has 16 bits. When timer counter equal or less than compare buffer register will cause PWM out high level if inv_on bit is off.	0	RW

REG[8Eh] Timer 1 count buffer register [TCNTB1L]

Bit	Description	Default	Access
7-0	Timer 1 count buffer register --- Low Byte Count buffer register total has 16 bits. When timer counter equal to 0 will cause PWM timer reload Count buffer register if reload_en bit set as enable. It may read back timer counter's real time value (TCNT1) when PWM timer start.	0	RW

REG[8Fh] Timer 1 count buffer register [TCNTB1H]

Bit	Description	Default	Access
7-0	Timer 1 count buffer register --- High Byte Count buffer register total has 16 bits. When timer counter equal to 0 will cause PWM timer reload Count buffer register if reload_en bit set as enable. It may read back timer counter's real time value (TCNT1) when PWM timer start.	0	RW

19.8 Block Transfer Engine (BTE) Control Registers

REG[90h] BTE Function Control Register 0 (BTE_CTRL0)

Bit	Description	Default	Access
7-5	NA	0	RO
4	BTE Function Enable / Status Write 0 : No action. 1 : BTE function enable. Read 0 : BTE function is idle. 1 : BTE function is busy. *** When BTE function enable, Normal MPU R/W memory through canvas[active window] doesn't allow.	0	RW
3-1	NA	0	RO
0	PATTERN Format 0: 8X8 1: 16X16	0	RW

REG[91h] BTE Function Control Register1 (BTE_CTRL1)

Bit	Description	Default	Access																																		
7-4	BTE ROP Code Bit[3:0] or Color expansion starting bit a. ROP is the acronym for Raster Operation. Some of BTE operation code has to collocate with ROP for the detailed function. (Please refer to the Section 2.7)	0	RW																																		
	<table><tr><th>Code</th><th>Description</th></tr><tr><td>0000b</td><td>0 (Blackness)</td></tr><tr><td>0001b</td><td>$\sim S0 \cdot \sim S1$ or $\sim (S0+S1)$</td></tr><tr><td>0010b</td><td>$\sim S0 \cdot S1$</td></tr><tr><td>0011b</td><td>$\sim S0$</td></tr><tr><td>0100b</td><td>$S0 \cdot \sim S1$</td></tr><tr><td>0101b</td><td>$\sim S1$</td></tr><tr><td>0110b</td><td>$S0 \wedge S1$</td></tr><tr><td>0111b</td><td>$\sim S0 + \sim S1$ or $\sim (S0 \cdot S1)$</td></tr><tr><td>1000b</td><td>$S0 \cdot S1$</td></tr><tr><td>1001b</td><td>$\sim (S0 \wedge S1)$</td></tr><tr><td>1010b</td><td>$S1$</td></tr><tr><td>1011b</td><td>$\sim S0 + S1$</td></tr><tr><td>1100b</td><td>$S0$</td></tr><tr><td>1101b</td><td>$S0 + \sim S1$</td></tr><tr><td>1110b</td><td>$S0 + S1$</td></tr><tr><td>1111b</td><td>1 (Whiteness)</td></tr></table>			Code	Description	0000b	0 (Blackness)	0001b	$\sim S0 \cdot \sim S1$ or $\sim (S0+S1)$	0010b	$\sim S0 \cdot S1$	0011b	$\sim S0$	0100b	$S0 \cdot \sim S1$	0101b	$\sim S1$	0110b	$S0 \wedge S1$	0111b	$\sim S0 + \sim S1$ or $\sim (S0 \cdot S1)$	1000b	$S0 \cdot S1$	1001b	$\sim (S0 \wedge S1)$	1010b	$S1$	1011b	$\sim S0 + S1$	1100b	$S0$	1101b	$S0 + \sim S1$	1110b	$S0 + S1$	1111b	1 (Whiteness)
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b. If BTE operation code function are color expansion with or without chroma key (08h / 09h / Eh / Fh), then these bits stand for starting bit on BTE window left boundary. MSB stands for left most pixel. For 8-bits MPU, value should within 0 to 7. For 16-bits MPU, value should within 0 to 15.																																					
3-0	BTE Operation Code Bit[3:0] RA8876 includes a 2D BTE Engine, it can execute 13 BTE functions. Some of BTE Operation Code has to accommodate	0	RW																																		

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REG[92h] Source 0/1 & Destination Color Depth (BTE_COLR)

Bit	Description	Default	Access
7	N/A	0	RO
6-5	S0 Color Depth 00: 256 Color (8bpp) 01: 64k Color (16bpp) 1x: 16M Color (24bpp)	0	RW
4-2	S1 Color Depth 000: 256 Color (8bpp) 001: 64k Color (16bpp) 010: 16M Color (24bpp) 011: Constant color (S1 memory start address' setting definition change as S1 constant color definition) 100: 8 bit pixel alpha blending 101: 16 bit pixel alpha blending	0	RW
1-0	Destination Color Depth 00: 256 Color (8bpp) 01: 64k Color (16bpp) 1x: 16M Color (24bpp)	0	RW

REG[93h] Source 0 memory start address 0 (S0_STR0)

Bit	Description	Default	Access
7-2	Source 0 memory start address [7:2]	0	RW
1-0	Fix at 0	0	RO

REG[94h] Source 0 memory start address 1 (S0_STR1)

Bit	Description	Default	Access
7-0	Source 0 memory start address [15:8]	0	RW

REG[95h] Source 0 memory start address 2 (S0_STR2)

Bit	Description	Default	Access
7-0	Source 0 memory start address [23:16]	0	RW

REG[96h] Source 0 memory start address 3 (S0_STR3)

Bit	Description	Default	Access
7-0	Source 0 memory start address [31:24]	0	RW

REG[97h] Source 0 image width 0 (S0_WTH0)

Bit	Description	Default	Access
7-2	Source 0 image width [7:2] Unit: Pixel. It must be divisible by 4. S0_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW
1-0	Fix at 0.	0	RO

REG[98h] Source 0 image width 1 (S0_WTH1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Source 0 image width [12:8] Unit: Pixel. It must be divisible by 4. S0_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW

REG[99h] Source 0 Window Upper-Left corner X-coordinates 0 (S0_X0)

Bit	Description	Default	Access
7-0	Source 0 Window Upper-Left corner X-coordinates [7:0] The bits are Source 0 Window Upper-Left corner X-coordinates	0	RW

REG[9Ah] Source 0 Window Upper-Left corner X-coordinates 1 (S0_X1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Source 0 Window Upper-Left corner X-coordinates [12:8] The bits are Source 0 Window Upper-Left corner X-coordinates	0	RW

REG[9Bh] Source 0 Window Upper-Left corner Y-coordinates 0 (S0_Y0)

Bit	Description	Default	Access
7-0	Source 0 Window Upper-Left corner Y-coordinates [7:0] The bits are Source 0 Window Upper-Left corner Y-coordinates	0	RW

REG[9Ch] Source 0 Window Upper-Left corner Y-coordinates 1 (S0_Y1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Source 0 Window Upper-Left corner Y-coordinates [12:8] The bits are Source 0 Window Upper-Left corner Y-coordinates	0	RW

**REG[9Dh] Source 1 memory start address 0 (S1_STR0) /
S1 constant color – Red element (S1_Red)**

Bit	Description	Default	Access
7-0	Source 1 memory start address [7:2] If source 1 set as Constant color then S1 memory start address' setting definition change as S1 constant color definition. This is red element. When it is source 1 memory start address, bit [1:0] should set as 0.	0	RW

**REG[9Eh] Source 1 memory start address 1 (S1_STR1) /
S1 constant color – Green element (S1_GREEN)**

Bit	Description	Default	Access
7-0	Source 1 memory start address [15:8] If source 1 set as Constant color then S1 memory start address' setting definition change as S1 constant color definition. This is green element.	0	RW

**REG[9Fh] Source 1 memory start address 2 (S1_STR2) /
S1 constant color – Blue element (S1_BLUE)**

Bit	Description	Default	Access
7-0	Source 1 memory start address [23:16] If source 1 set as Constant color then S1 memory start address' setting definition change as S1 constant color definition. This is blue element.	0	RW

REG[A0h] Source 1 memory start address 3 (S1_STR3)

Bit	Description	Default	Access
7-0	Source 1 memory start address [31:24] If source 1 set as Constant color then S1 memory start address' setting definition change as S1 constant color definition. This is not used for color definition.	0	RW

REG[A1h] Source 1 image width 0 (S1_WTH0)

Bit	Description	Default	Access
7-2	Source 1 image width [7:2] Unit: Pixel. It must be divisible by 4. S1_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW
1-0	Fix at 0.	0	RO

REG[A2h] Source 1 image width 1 (S1_WTH1)

Bit	Description	Default	Access
7-5	N/A	0	RO
4-0	Source 1 image width [12:8] Unit: Pixel. It must be divisible by 4. S1_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW

REG[A3h] Source 1 Window Upper-Left corner X-coordinates 0 (S1_X0)

Bit	Description	Default	Access
7-0	Source 1 Window Upper-Left corner X-coordinates [7:0] The bits are Source 1 Window Upper-Left corner X-coordinates	0	RW

REG[A4h] Source 1 Window Upper-Left corner X-coordinates 1 (S1_X1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Source 1 Window Upper-Left corner X-coordinates [12:8] The bits are Source 1 Window Upper-Left corner X-coordinates	0	RW

REG[A5h] Source 1 Window Upper-Left corner Y-coordinates 0 (S1_Y0)

Bit	Description	Default	Access
7-0	Source 1 Window Upper-Left corner Y-coordinates [7:0] The bits are Source 1 Window Upper-Left corner Y-coordinates	0	RW

REG[A6h] Source 1 Window Upper-Left corner Y-coordinates 1 (S1_Y1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Source 1 Window Upper-Left corner Y-coordinates [12:8] The bits are Source 1 Window Upper-Left corner Y-coordinates	0	RW

REG[A7h] Destination memory start address 0 (DT_STR0)

Bit	Description	Default	Access
7-2	Destination memory start address [7:2]	0	RW
1-0	Fix at 0	0	RO

REG[A8h] Destination memory start address 1 (DT_STR1)

Bit	Description	Default	Access
7-0	Destination memory start address [15:8]	0	RW

REG[A9h] Destination memory start address 2 (DT_STR2)

Bit	Description	Default	Access
7-0	Destination memory start address [23:16]	0	RW

REG[AAh] Destination memory start address 3 (DT_STR3)

Bit	Description	Default	Access
7-0	Destination memory start address [31:24]	0	RW

Note: Destination memory start address cannot set within from source 0|1 start address to source 0|1's (image width) * (image height) * (color depth[1|2|3])

REG[ABh] Destination image width 0 (DT_WTH0)

Bit	Description	Default	Access
7-2	Destination image width [7:2] Unit: Pixel. It must be divisible by 4. DT_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW
1-0	Fix at 0	0	RO

REG[ACh] Destination image width 1 (DT_WTH1)

Bit	Description	Default	Access
7-5	N/A	0	RO
4-0	Destination image width [12:8] Unit: Pixel. It must be divisible by 4. DT_WTH Bit [1:0] tie to "0" internally. The value is physical pixel number.	0	RW

REG[ADh] Destination Window Upper-Left corner X-coordinates 0 (DT_X0)

Bit	Description	Default	Access
7-0	Destination Window Upper-Left corner X-coordinates [7:0]	0	RW

REG[AEh] Destination Window Upper-Left corner X-coordinates 1 (DT_X1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Destination Window Upper-Left corner X-coordinates [12:8]	0	RW

REG[AFh] Destination Window Upper-Left corner Y-coordinates 0 (DT_Y0)

Bit	Description	Default	Access
7-0	Destination Window Upper-Left corner Y-coordinates [7:0]	0	RW

REG[B0h] Destination Window Upper-Left corner Y-coordinates 1 (DT_Y1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Destination Window Upper-Left corner Y-coordinates [12:8]	0	RW

REG[B1h] BTE Window Width 0 (BTE_WTH0)

Bit	Description	Default	Access
7-0	BTE Window Width Setting[7:0] Unit: Pixel. The value is physical pixel number.	0	RW

REG[B2h] BTE Window Width 1 (BTE_WTH1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	BTE Window Width Setting [12:8] Unit: Pixel. The value is physical pixel number.	0	RW

REG[B3h] BTE Window Height 0 (BTE_HIG0)

Bit	Description	Default	Access
7-0	BTE Window Height Setting[7:0] Unit: Pixel. The value is physical pixel number.	0	RW

REG[B4h] BTE Window Height 1 (BTE_HIG1)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	BTE Window Height Setting [12:8] Unit: Pixel. The value is physical pixel number.	0	RW

REG[B5h] Alpha Blending (APB_CTRL)

Bit	Description	Default	Access
7-4	N/A	0	RO
5-0	Window Alpha Blending effect for S0 & S1 The value of alpha in the color code ranges from 1.0 down to 0.0, where 1.0 represents a fully opaque color, and 0.0 represents a fully transparent color. 00h: 0 01h: 1/32 02h: 2/32 : 1Eh: 30/32 1Fh: 31/32 2Xh: 1 Output Effect = (S0 image x (1 - alpha setting value)) + (S1 image x alpha setting value)	0	RW

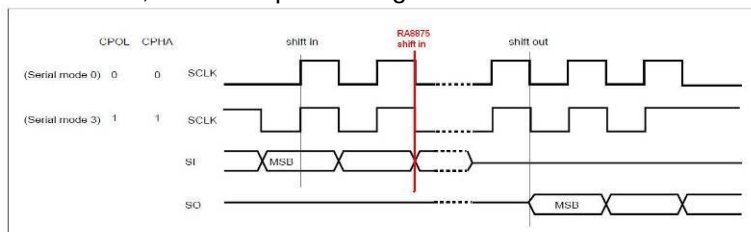
19.9 Serial Flash & SPI Master Control Registers

REG[B6h] Serial flash DMA Controller REG (DMA_CTRL)

Bit	Description	Default	Access
7-1	NA	0	RO
0	Write Function: DMA Start Bit Set to 1 by MPU and reset to 0 automatically It cannot start with fontwr_busy as 1. And if DMA enabled also cannot set as text mode & send character code. Read Function: DMA Busy Check Bit 0: Idle 1: Busy *** about serial flash's DMA transfer, its destination starting address, destination image width, color depth & address mode in SDRAM are followed by Canvas' setting and only operate in graphic mode.	0	RW

REG[B7h] Serial Flash/ROM Controller Register (SFL_CTRL)

Bit	Description	Default	Access
7	Serial Flash/ROM I/F # Select 0: Serial Flash/ROM 0 I/F is selected. 1: Serial Flash/ROM 1 I/F is selected.	0	RW
6	Serial Flash /ROM Access Mode 0: Font mode – for external CGROM 1: DMA mode – for CGRAM , pattern , boot start image or OSD	0	RW
5	Serial Flash/ROM Address Mode 0: 24 bits address mode 1: 32 bits address mode If user wants to use 32 bits address mode, user must manual send EX4B command (B7h) to serial flash then set high this bit. User also may check this bit to know whether serial flash had enter 32 bits address mode by initial display function.	0	RW
4	RA8875 compatible mode 0: standard SPI mode 0 or mode 3 timing 1: Follow RA8875 mode 0 & mode 3 timing In the RA8875 compatible mode, Data are read on the clock's falling edge (high->low transition) and data are changed on a falling edge (high->low transition). For Mode 0, SPI clock park on low when idle. For Mode 3, SPI clock park on high when idle.	0	RW
3-0	Read Command code & behavior selection 000xb: 1x read command code – 03h. Normal read speed. Single data input on xmis0. Without dummy cycle between address and data. 010xb: 1x read command code – 0Bh. To some serial flash provide faster read speed. Single data input on xmis0. 8 dummy	0	R/W



Bit	Description	Default	Access
	cycles inserted between address and data. 1x0xb: 1x read command code – 1Bh. To some serial flash provide fastest read speed. Single data input on xmis0. 16 dummy cycles inserted between address and data. xx10b: 2x read command code – 3Bh. Interleaved data input on xmis0 & xmosi. 8 dummy cycles inserted between address and data phase. (dual mode 0, reference Figure 16-7) xx11b: 2x read command code – BBh. Address output & data input interleaved on xmis0 & xmosi. 4 dummy cycles inserted between address and data phase. (dual mode 1, reference Figure 16-8) Note: Not serial flash support above read command, please according to serial flash's datasheet to select proper read command.		

REG[B8h] SPI master Tx /Rx FIFO Data Register (SPIDR)

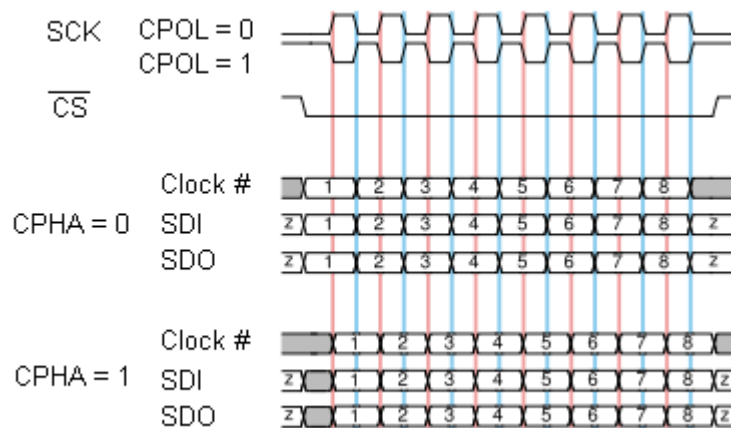
Bit	Description	Default	Access
7-0	SPI master Tx /Rx FIFO Data Register After programming the core's control register SPI transfers can be initiated. A transfer is initiated by writing to the Serial Peripheral Data Register [SPIDR]. Writing to the Serial Peripheral Data Register is actually writing to a 16 entries deep FIFO called the Write FIFO. Each write access adds a data byte to the Write FIFO. When the core is enabled – SS_ACTIVE is set ('1') – and the Write FIFO is not full, the core automatically transfers the oldest data byte. Receiving data is done simultaneously with transmitting data; whenever a data byte is transmitted a data byte is received. For each byte that needs to be read from a device, a dummy byte needs to be written to the Write FIFO. This instructs the core to initiate an SPI transfer, simultaneously transmitting the dummy byte and receiving the desired data. Whenever a transfer is finished, the received data byte is added to the Read FIFO. The Read FIFO is the counterpart of the Write FIFO. It is an independent 16 entries deep FIFO. The FIFO contents can be read by reading from the Serial Peripheral Data Register [SPIDR]	NA	RW

REG[B9h] SPI master Control Register (SPIMCR2)

Bit	Description	Default	Access
7	NA	0	RO
6	SPI Master Interrupt enable 0: Disable interrupt. 1: Enable interrupt. *** If you disable SPIM interrupt flag, then RA8876 won't assert interrupt event to inform MPU but you still may check interrupt event on SPIMSR.	0	RW
5	Control Slave Select drive on which xnsfcs 0: nSS drive on xnsfcs[0] 1: nSS drive on xnsfcs[1]	0	RW
4	Slave Select signal active [SS_ACTIVE] 0: inactive (nSS port will goes high) 1: active (nSS port will goes low) While Slave Select signal in inactive, FIFO will clear and engine will stay in Idle state. Note: Suggest don't change CPOL/CPHA and active Slave	0	RW

Bit	Description	Default	Access															
	Select signal simultaneously.																	
3	Mask interrupt for FIFO overflow error [OVFIRQMSK] 0: unmask 1: mask	1	RW															
2	Mask interrupt for while Tx FIFO empty & SPI engine/FSM idle [EMTIRQMSK] 0: unmask 1: mask	1	RW															
1:0	SPI operation mode Only support mode 0 & mode 3, when enable serial flash's DMA or access Getop's character serial ROM device. <table><tr><td>mode</td><td>CPOL: Clock Polarity bit</td><td>CPHA: Clock Phase bit</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>2</td><td>1</td><td>0</td></tr><tr><td>3</td><td>1</td><td>1</td></tr></table>	mode	CPOL: Clock Polarity bit	CPHA: Clock Phase bit	0	0	0	1	0	1	2	1	0	3	1	1	0	RW
mode	CPOL: Clock Polarity bit	CPHA: Clock Phase bit																
0	0	0																
1	0	1																
2	1	0																
3	1	1																

- At CPOL=0 the base value of the clock is zero
 - For CPHA=0, data are read on the clock's rising edge (low->high transition) and data are changed on a falling edge (high->low clock transition).
 - For CPHA=1, data are read on the clock's falling edge and data are changed on a rising edge.
- At CPOL=1 the base value of the clock is one (inversion of CPOL=0)
 - For CPHA=0, data are read on clock's falling edge and data are changed on a rising edge.
 - For CPHA=1, data are read on clock's rising edge and data are changed on a falling edge.


Figure 19-7
Table 19-2 : SPI MODES

SPI MODE	CPOL	CPHA
0	0	0
1	0	1
2	1	0
3	1	1

REG[BAh] SPI master Status Register (SPIMSR)

Bit	Description	Default	Access
7	Tx FIFO empty flag 0: not empty 1: empty	1	RO
6	Tx FIFO full flag 0: not full 1: full	0	RO
5	Rx FIFO empty flag 0: not empty 1: empty	1	RO
4	Rx FIFO full flag 0: not full 1: full	0	RO
3	1: Overflow interrupt flag Write 1 will clear this flag	0	RW
2	1: Tx FIFO empty & SPI engine/FSM idle interrupt flag Write 1 will clear this flag	0	RW
1-0	NA	0	RO

REG[BBh] SPI Clock period (SPI_DIVSOR)

Bit	Description	Default	Access
7-0	SPI Clock period According to system clock to set low & high period for SPI clock. $F_{sck} = F_{core} / (divisor + 1) \times 2$	3	RW

REG[BCh] Serial flash DMA Source Starting Address 0 (DMA_SSTR0)

Bit	Description	Default	Access
7-0	Serial flash DMA Source START ADDRESS [7:0] This bits index serial flash address [7:0] Direct point to exact source image's transfer starting position.	0	RW

REG[BDh] Serial flash DMA Source Starting Address 1 (DMA_SSTR1)

Bit	Description	Default	Access
7-0	Serial flash DMA Source START ADDRESS [15:8] This bits index serial flash address [15:8] Direct point to exact source image's transfer starting position.	0	RW

REG[BEh] Serial flash DMA Source Starting Address 2 (DMA_SSTR2)

Bit	Description	Default	Access
7-0	Serial flash DMA Source START ADDRESS [23:16] This bits index serial flash address [23:16] Direct point to exact source image's transfer starting position.	0	RW

REG[BFh] Serial flash DMA Source Starting Address 3 (DMA_SSTR3)

Bit	Description	Default	Access
7-0	Serial flash DMA Source START ADDRESS [31:24] This bits index serial flash address [31:24] Direct point to exact source image's transfer starting position.	0	RW

REG[C0h] DMA Destination Window Upper-Left corner X-coordinates 0 (DMA_DX0)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) This register defines DMA Destination Window Upper-Left corner X-coordinates [7:0] on Canvas area. When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) This register defines Destination address [7:2] in SDRAM.	0	RW

REG[C1h] DMA Destination Window Upper-Left corner X-coordinates 1 (DMA_DX1)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) This register defines DMA Destination Window Upper-Left corner X-coordinates [12:8] on Canvas area. When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) This register defines Destination address [15:8] in SDRAM.	0	RW

REG[C2h] DMA Destination Window Upper-Left corner Y-coordinates 0 (DMA_DY0)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) This register defines DMA Destination Window Upper-Left corner Y-coordinates [7:0] on Canvas area. When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) This register defines Destination address [23:16] in SDRAM.	0	RW

REG[C3h] DMA Destination Window Upper-Left corner Y-coordinates 1 (DMA_DY1)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) This register defines DMA Destination Window Upper-Left corner Y-coordinates [12:8] on Canvas area. When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) This register defines Destination address [31:24] in SDRAM.	0	RW

REG[C4h] – REG[C5h] : RESERVED

Bit	Description	Default	Access
7-0	NA	0	RO

REG[C6h] DMA Block Width 0 (DMAW_WTH0)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) DMA Block Width [7:0] When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) DMA Transfer Number [7:0]	0	RW

REG[C7h] DMA Block Width 1 (DMAW_WTH1)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) DMA Block Width [15:8] When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) DMA Transfer Number [15:8]	0	RW

REG[C8h] DMA Block Height 0 (DMAW_HIGH0)

Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) DMA Block Height [7:0] When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) DMA Transfer Number [23:16]	0	RW

REG[C9h] DMA Block Height 1 (DMAW_HIGH1)

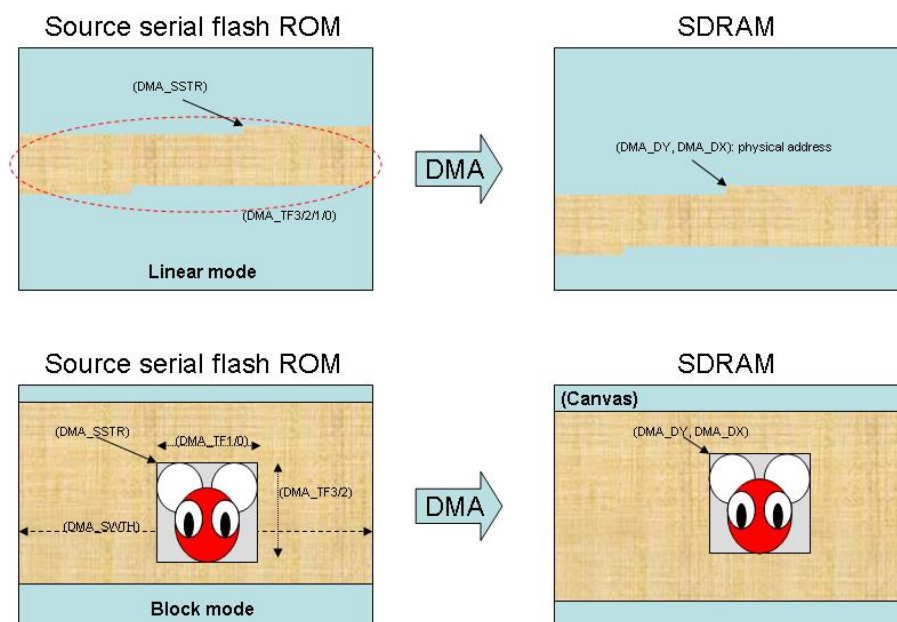
Bit	Description	Default	Access
7-0	When REG 5Eh (AW_COLOR) bit 2 = 0 (Block Mode) DMA Block Height [15:8] When REG 5Eh (AW_COLOR) bit 2 = 1 (Linear Mode) DMA Transfer Number [31:24]	0	RW

REG[CAh] DMA Source Picture Width 0(DMA_SWTH0)

Bit	Description	Default	Access
7-0	DMA Source Picture Width [7:0] Unit: pixel	0	RW

REG[CBh] DMA Source Picture Width 0(DMA_SWTH1)

Bit	Description	Default	Access
4-0	DMA Source Picture Width [12:8]	0	RW


Figure 19-8 : DMA Linear and Block Mode

19.10 Text Engine

REG[CCh] Character Control Register 0 (CCR0)

Bit	Description	Default	Access
7:6	Character source selection 00: Select internal CGROM Character. 01: Select external CGROM Character. (Genitop serial flash) 10: Select user-defined Character. 11: NA	0	RW
5-4	Character Height Setting for external CGROM & user-defined Character 00b : 16; ex. 8x16 / 16x16 / variable character width x 16 01b : 24; ex. 12x24 / 24x24 / variable character width x 24 10b : 32; ex. 16x32 / 32x32 / variable character width x 32 Note: 1. User-defined character width is decided by character code; width for code < 8000h is 8/12/16. width for code >=8000h is 16/24/32. 2. Genitop serial flash's character width is decided by chosen character sets and need program into GT Font ROM register (CEh, CFh). 3. Internal CGROM supports size 8x16 / 12x24 / 16x32.	0	RW
3-2	NA	0	RO
1-0	Character Selection for internal CGROM When FNCR0 B7 = 0 and B6 = 0, Internal CGROM supports character sets with the standard coding of ISO/IEC 8859-1,2,4,5, which supports English and most of European country languages 00b : ISO/IEC 8859-1. 01b : ISO/IEC 8859-2. 10b : ISO/IEC 8859-4. 11b : ISO/IEC 8859-5.	0	RW

REG[CDh] Character Control Register 1 (CCR1)

Bit	Description	Default	Access
7	Full Alignment Selection Bit 0 : Full alignment disable. 1 : Full alignment enable. When Full alignment enable, displayed character width is equal to (Character Height)/2 if character width equal or small than (Character Height)/2, otherwise displayed font width is equal to Character Height.	0	RW
6	Chroma keying enable on Text input 0 : Character's background displayed with specified color. 1 : Character's background displayed with original canvas' background.	0	RW
5	NA	0	RO
4	Character Rotation 0 : Normal Text direction from left to right then from top to bottom 1 : Counterclockwise 90 degree & vertical flip Text direction from top to bottom then from left to right (it should accommodate with set VDIR as 1) This attribute can be changed only when previous Text write finished (core_busy = 0)	0	RW

Bit	Description	Default	Access
3-2	Character width enlargement factor 00b : X1. 01b : X2. 10b : X3. 11b : X4.	0	RW
1-0	Character height enlargement factor 00b : X1. 01b : X2. 10b : X3. 11b : X4.	0	RW

REG[CEh] GT Character ROM Select (GTFNT_SEL)

Bit	Description	Default	Access
7-5	GT Serial Character ROM Select 000b: GT21L16T1W 001b: GT30L16U2W 010b: GT30L24T3Y 011b: GT30L24M1Z 100b: GT30L32S4W 101b: GT20L24F6Y 110b: GT21L24S1W	0	RW
4-0	N/A	0	RO

REG[CFh] GT Character ROM Control register (GTFNT_CR)

Bit	Description	Default	Access
7-3	Character sets For specific GT serial Character ROM, the coding method must be set for decoding. a. Single byte character code for following character sets: 00100b: ASCII only (00h-1Fh, 80-FFh will send "blank space") 10001b: ISO-8859-1 + ASCII code 10010b: ISO-8859-2 + ASCII code 10011b: ISO-8859-3 + ASCII code 10100b: ISO-8859-4 + ASCII code 10101b: ISO-8859-5 + ASCII code 10110b: ISO-8859-7 + ASCII code 10111b: ISO-8859-8 + ASCII code 11000b: ISO-8859-9 + ASCII code 11001b: ISO-8859-10 + ASCII code 11010b: ISO-8859-11 + ASCII code 11011b: ISO-8859-13 + ASCII code 11100b: ISO-8859-14 + ASCII code 11101b: ISO-8859-15 + ASCII code 11110b: ISO-8859-16 + ASCII code b. Two byte character code for following character sets: 00000b: GB2312 00001b: GB12345/GB18030 00010b: BIG5 00011b: UNICODE 00101b: UNI-Japanese 00110b: JIS0208 00111b: Latin / Greek / Cyrillic / Arabic / Thai / Hebrew Note: While character sets are not 00011b, 00101b, 00110b, 00111b (UNICODE, UNI-Japanese, JIS0208, Latin / Greek / Cyrillic / Arabic / Thai / Hebrew then if 1 st character code under 80h will treat as ASCII code.	0	RW

Bit	Description	Default	Access
2	N/A	0	RO
1-0	GT Character width setting 00b: for fix width's font sets. Its width is half of character height. Ex. ISO-8859, GB2312, GB12345/GB18030, BIG5, UNI-Japanese, JIS0208, Thai. Others: variable width for following character sets: ASCII, Latin, Greek, Cyrillic & Arabic.	0	RW

Relationship of Character sets & GT Character width as following:

Char. set Width	ASCII Code/ ISO-8859-x (00100b /1xxxxb)	Latin / Greek / Cyrillic (00111b)	Arabic (00111b)	Others
00b	Fixed width	Fixed width	NA	Fixed width (auto set by chip)
01b	variable-width for Arial	variable-width	variable-width for Presentation Forms-A	NA
10b	variable-width for Roman	NA	variable-width for Presentation Forms-B	NA
11b	Bold	NA	NA	NA

REG[D0h] Character Line gap Setting Register (FLDR)

Bit	Description	Default	Access
7-5	NA	0	RO
4-0	Character Line gap Setting Setting the character line gap when meet active window boundary. (Unit: pixel) Color of gap will fill-in background color. *** It won't be enlarged by character enlargement function.	0	RW

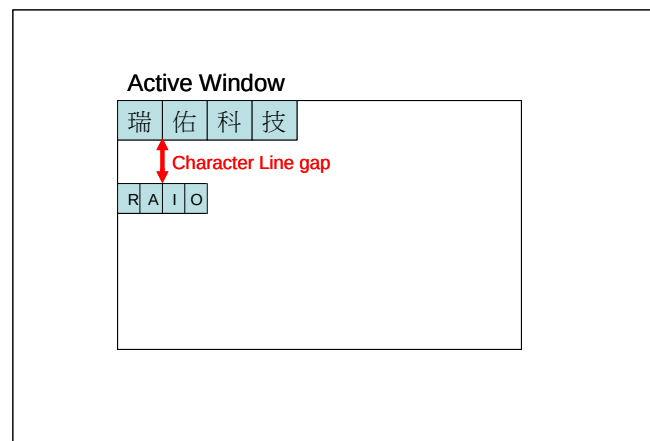
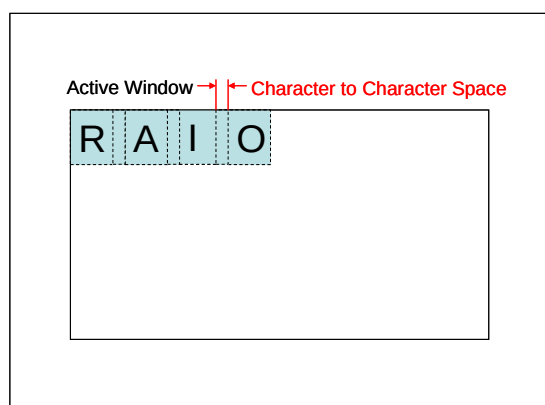


Figure 19-9 : Character Line Gap

REG[D1h] Character to Character Space Setting Register (F2FSSR)

Bit	Description	Default	Access
7-6	NA	0	RW
5-0	Character to Character Space Setting 00h : 0 pixel 01h : 1 pixel 02h : 2 pixels : 3Fh : 63 pixels Color of space will fill-in background color. *** It won't be enlarged by character enlargement function.	0	RW


Figure 19-10 : Character to Character Space
REG[D2h] Foreground Color Register - Red (FGCR)

Bit	Description	Default	Access
7-0	Foreground Color - Red; for draw, text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

REG[D3h] Foreground Color Register - Green (FGCG)

Bit	Description	Default	Access
7-0	Foreground Color - Green; for draw, text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:2]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

REG[D4h] Foreground Color Register - Blue (FGCB)

Bit	Description	Default	Access
7-0	Foreground Color - Blue; for draw, text or color expansion 256 colors, the register only uses Bit[7:6]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0].	FFh	RW

REG[D5h] Background Color Register - Red (BGCR)

Bit	Description	Default	Access
7-0	Background Color - Red; for Text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0]. *** Note: No matter background transparency is enabled or not, don't set same value with Foreground Color otherwise image or text will become a square with Foreground Color even BTE function.	00h	RW

REG[D6h] Background Color Register - Green (BGCG)

Bit	Description	Default	Access
7-0	Background Color - Green; for Text or color expansion 256 colors, the register only uses Bit[7:5]. 65K colors, the register uses Bit[7:2]. 16.7M colors, the register uses Bit[7:0]. *** Note: No matter background transparency is enabled or not, don't set same value with Foreground Color otherwise image or text will become a square with Foreground Color even BTE function.	00h	RW

REG[D7h] Background Color Register - Blue (BGCB)

Bit	Description	Default	Access
7-0	Background Color - Blue; for Text or color expansion 256 colors, the register only uses Bit[7:6]. 65K colors, the register uses Bit[7:3]. 16.7M colors, the register uses Bit[7:0]. *** Note: No matter background transparency is enabled or not, don't set same value with Foreground Color otherwise image or text will become a square with Foreground Color even BTE function.	00h	RW

REG[D8h] – REG[DAh] : RESERVED

Bit	Description	Default	Access
7-0	NA	0	RO

REG[DBh] CGRAM Start Address 0 (CGRAM_STR0)

Bit	Description	Default	Access
7-0	CGRAM START ADDRESS [7:0] User-defined Characters space User must use canvas image setting to organize CGRAM data and set CGRAM address to tell engine where to fetch CGRAM data.	0	RW

REG[DCh] CGRAM Start Address 1 (CGRAM_STR1)

Bit	Description	Default	Access
7-0	CGRAM START ADDRESS [15:8] User-defined Characters space User must use canvas image setting to organize CGRAM data and set CGRAM address to tell engine where to fetch CGRAM data.	0	RW

REG[DDh] CGRAM Start Address 2 (CGRAM_STR2)

Bit	Description	Default	Access
7-0	CGRAM START ADDRESS [23:16] User-defined Characters space User must use canvas image setting to organize CGRAM data and set CGRAM address to tell engine where to fetch CGRAM data.	0	RW

REG[DEh] CGRAM Start Address 3 (CGRAM_STR3)

Bit	Description	Default	Access
7-0	CGRAM START ADDRESS [31:24] User-defined Characters space User must use canvas image setting to organize CGRAM data and set CGRAM address to tell engine where to fetch CGRAM data.	0	RW

*** **Note:** If user wants to change rotate attribute, character line gap, character-to-character space, foreground color, background color and Text/graphic mode setting, he must make sure core_busy(fontwr_busy) status bit is low.

19.11 Power Management Control Register

REG[DFh] : Power Management register (PMU)

Bit	Description	Default	Access
7	Enter Power saving state 0: Normal state or wakeup from power saving state 1: Enter power saving state. Note: There are 3 ways to wakeup from power saving state: External interrupt event, Key Scan wakeup, Software wakeup. Write this bit to 0 will cause Software wakeup. It will be cleared until chip resume. MPU must wait until system quit from power saving state than allow to write other registers. User may check this bit or check status bit [1] (power saving status bit) to know whether chip back to normal operation.	0	RW
6-2	NA	0	RO
1-0	Power saving Mode definition 00: NA 01: Standby Mode CCLK & PCLK will stop, MCLK keep MPLL clock 10: Suspend Mode CCLK & PCLK will stop, MCLK switch to OSC clock 11: Sleep Mode All clock & PLL will stop	3	RW

19.12 SDRAM Control Register

REG[E0h] SDRAM attribute register (SDRAR)

Bit	Description	Default	Access
7	SDRAM Power Saving type 0: Execute power down command to enter power saving mode 1: Execute self refresh command to enter power saving mode	0	RW
6	SDRAM memory type (sdr_type) 0b: SDR SDRAM 1b: mobile SDR SDRAM	0	RW
5	SDRAM Bank number (sdr_bank) 0b: 2 banks (column addressing size only support 256 words) 1b: 4 banks	1	RW
4-3	SDRAM Row addressing (sdr_row) 00b: 2K (A0-A10) 01b: 4K (A0-A11) 1Xb: 8K (A0-A12)	1	RW
2-0	SDRAM Column addressing (sdr_col) 000b: 256 (A0-A7) 001b: 512 (A0-A8) 010b: 1024 (A0-A9) 011b: 2048 (A0-A9, A11) 1XXb: 4096 (A0-A9, A11-A12)	0	RW

Reference setting:

16Mb, 2MB, 1Mx16: 0x00; bank no: 2, row size: 2048, col size: 256
 32Mb, 4MB, 2Mx16: 0x08; bank no: 2, row size: 4096, col size: 256
 64Mb, 8MB, 4Mx16: 0x28; bank no: 4, row size: 4096, col size: 256
 128Mb, 16MB, 8Mx16: 0x29; bank no: 4, row size: 4096, col size: 512
 256Mb, 32MB, 16Mx16: 0x31; bank no: 4, row size: 8192, col size: 512
 512Mb, 64MB, 32Mx16: 0x32; bank no: 4, row size: 8192, col size: 1024

REG[E1h] SDRAM mode register & extended mode register (SDRMD)

Bit	Description	Default	Access
7-5	Partial-Array Self Refresh (sdr_pasr) *Only for mobile SDR SDRAM 000b: Full array 001b: Half array (1/2) 010b: Quarter array (1/4) 011b: Reserved 100b: Reserved 101b: One-eighth array (1/8) 110b: One-sixteenth array (1/16) 111b: Reserved	0	RW
4-3	To select the driver strength of the DQ outputs (sdr_drv) *Only for mobile SDR SDRAM 00b: Full-strength driver 01b: Half-strength driver 10b: Quarter-strength driver 11b: One eighth-strength driver	0	RW
2-0	SDRAM CAS latency (sdr-caslat) 010b: 2 SDRAM clock 011b: 3 SDRAM clock Other: reserved	03h	RW

***NOTE:** This register was locked after sdr_initdone bit was set as 1.

REG[E2h] SDRAM auto refresh interval (SDR_REF_ITVL0)

Bit	Description	Default	Access
7-0	Refresh interval (Low byte) SDRAM auto refresh time interval. Counted by SDRAM clock. *** If Refresh interval register set as 0000h then SDRAM Auto Refresh is disabled. Refresh time interval should according to SDRAM's Refresh period specification (tREF) & its row size. Ex. If SDRAM clock is 100MHz, SDRAM's Refresh period tREF is 64ms and row size is 8192, then refresh interval should small than : $64e-3 / 8192 * 100e6 \approx 781 = 30Dh$	00h	RW

REG[E3h] SDRAM auto refresh interval (SDR_REF_ITVL1)

Bit	Description	Default	Access
7-0	Refresh interval (High byte) SDRAM auto refresh time interval. Counted by SDRAM clock. *** If Refresh interval register set as 0000h then SDRAM Auto Refresh is disabled.	00h	RW

REG[E4h] SDRAM Control register (SDRCR)

Bit	Description	Default	Access
7-6	Length to break a burst transfer 00: 256 01: 128 10: 64 11: 32	0	RW
5	This bit must set to 0	0	RW
4	XMCKE pin state Current XMCKE pin state. 0: SDR memory clock disable 1: SDR memory clock enable	1	RO
3	Report warning condition 0: Disable or Clear warning flag 1: Enable warning flag Warning condition are memory read cycle close to SDRAM maximum address boundary (may over maximum address minus 512 bytes) or out of range or SDRAM bandwidth insufficient to fulfill panel's frame rate, then this warning event will be latched, user could check this bit to do some judgments. That warning flag could be cleared by set this bit as 0.	0	RW
2	SDRAM timing parameter register enable (SDR_PARAMEN) 0: Disable SDRAM timing parameter registers 1: Enable SDRAM timing parameter registers	0	RW
1	SDRAM enter power saving mode (sdr_psaving) 0 to 1 transition will enter power saving mode 1 to 0 transition will exit power saving mode	0	RW
0	Start SDRAM initialization procedure (sdr_initdone) 0 to 1 transition will execute SDRAM initialization procedure. Read value '1' means SDRAM is initialized and ready for access. Once it was written as 1, it cannot be rewrite as 0. 1 to 0 transition without have any operation.	0	RW

*** Following SDRAM timing register only valid when SDR_PARAMEN (REG[E4], b2) set.
REG[E0h] SDRAM timing parameter 1

Bit	Description	Default	Access
7	NA	0	RO
6	NA	0	RW
5	NA	0	RW
4	NA	0	RW
3-0	tMRD : Load Mode Register command to Active or Refresh command 00h – 0Fh: 1 ~ 16 SDRAM clock	2	RW

REG[E1h] SDRAM timing parameter 2

Bit	Description	Default	Access
7-4	tRFC : Auto refresh period 00h – 0Fh: 1 ~ 16 SDRAM clock	8	RW
3-0	tXSR : Exit SELF REFRESH-to-ACTIVE command 00h – 0Fh: 1 ~ 16 SDRAM clock	7	RW

REG[E2h] SDRAM timing parameter 3

Bit	Description	Default	Access
7-4	tRP : Time of PRECHARGE command period (15/20ns) 00h – 0Fh: 1 ~ 16 SDRAM clock	2	RW
3-0	tWR : Time of WRITE recovery time 00h – 0Fh: 1 ~ 16 SDRAM clock	0	RW

REG[E3h] SDRAM timing parameter 4

Bit	Description	Default	Access
7-4	tRCD : Time of ACTIVE-to-READ or WRITE delay 00h – 0Fh: 1 ~ 16 SDRAM clock	2	RW
3-0	tRAS : Time of ACTIVE-to-PRECHARGE 00h – 0Fh: 1 ~ 16 SDRAM clock	6	RW

19.13 IIC Master Registers

REG[E5h] IIC Master Clock Pre-scale Register 0 (IICMCPR0)

Bit	Description	Default	Access
7-0	IIC Master Clock Pre-scale [7:0] $XSCL = CCLK / (5 * (Pre-scale + 2))$	0	RW

REG[E6h] IIC Master Clock Pre-scale Register 1 (IICMCPR1)

Bit	Description	Default	Access
7-0	IIC Master Clock Pre-scale [15:8] $XSCL = CCLK / (5 * (Pre-scale + 2))$	0	RW

REG[E7h] IIC Master Transmit Register (IICMTXR)

Bit	Description	Default	Access
7-0	IIC Master Transmit [7:0]	0	RW

REG[E8h] IIC Master Receiver Register (IICMRXR)

Bit	Description	Default	Access
7-0	IIC Master Receiver [7:0]	0	RW

REG[E9h] IIC Master Command Register (IICMCMDR)

Bit	Description	Default	Access
7	START Generate (repeated) start condition and be cleared by hardware automatically Note : This bit is always read as 0.	0	RW
6	STOP Generate stop condition and be cleared by hardware automatically Note : This bit is always read as 0.	0	RW
5	READ(READ and WRITE can't be used simultaneously) Read from slave and be cleared by hardware automatically Note : This bit is always read as 0.	0	RW
4	WRITE(READ and WRITE can't be used simultaneously) Write to slave and be cleared by hardware automatically Note : This bit is always read as 0.	0	RW
3	ACKNOWLEDGE When as a IIC master receiver 0 : Sent ACK. 1 : Sent NACK. Note : This bit is always read as 0.	0	RW
2-1	NA	0	RO
0	Noise Filter 0 : Disable. 1 : Enable	0	RW

REG[EAh] IIC Master Status Register (IICMSTUR)

Bit	Description	Default	Access
7	Received acknowledge from slave 0 : Acknowledge received. 1 : No Acknowledge received.	0	RO
6	IIC Bus is Busy 0 : Idle. '0' after STOP signal detected 1 : Busy. '1' after START signal detected	0	RO
5-2	NA	0	RO
1	Transfer in progress 0 : when transfer complete 1 : when transferring data	0	RO
0	Arbitration lost This bit is set 1 when the core lost arbitration. Arbitration is lost when: a STOP signal is detected, but non requested The master drives SDA high, but SDA is low.	0	RO

19.14 GPI & GPO Register

REG[F0h] GPIO-A direction (GPIOAD)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port A GPIO-A_dir[7:0] : General Purpose I/O direction control. 0: Output 1: Input	FFh	RW

REG[F1h] GPIO-A (GPIOA)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port A Only available in parallel 8-bits MPU I/F & serial MPU I/F For Write, Port A's General Purpose Output GPO-A[7:0] : Port A's General Purpose Output, share with DB[15:8] For Read, Port A's General Purpose Input GPI-A[7:0] : Port A's General Purpose Input, share with DB[15:8]	NA	RW

REG[F2h] GPIO-B (GPIOB)

Bit	Description	Default	Access
7-5	NA	NA	NA
4	For Write. Port B's General Purpose Output The output data pin share with KOUT[0] For Read, Port B's General Purpose Input The input data pin share with KIN[0]	NA	RW
3-0	For Read, Port B's General Purpose Input This bit not writable. Only valid on serial host interface. {XA0, XnWR, XnRD, XnCS}	NA	R

REG[F3h] GPIO-C direction (GPIOCD)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port C GPIO-C_dir[7:0] : General Purpose I/O direction control. 0: Output 1: Input	FFh	RW

REG[F4h] GPIO-C (GPIOC)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port C GPIO-C[7] & GPIO_C[4:0] : General Purpose Input / Output share with {XPWM0, XnSFCS1, XnSFCS0, XMISO, XMOSI, XSCK} GPIO function valid only when relative function disabled. (ex. PWM, SPI master disabled). *** GPIO_C[6:5] are not available.	NA	RW

REG[F5h] GPIO-D direction (GPIODD)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port D GPIO-D_dir[7:0] : General Purpose I/O direction control. 0: Output 1: Input	FFh	RW

REG[F6h] GPIO-D (GPIOD)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port D Only available on digital display package type GPIO-D[7:0] : General Purpose Input/Output share with PDAT[18, 2, 17, 16, 9, 8, 1, 0]	NA	RW

REG[F7h] GPIO-E direction (GPIOED)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port E GPIO-E_dir[7:0] : General Purpose I/O direction control. 0: Output 1: Input	FFh	RW

REG[F8h] GPIO-E (GPIOE)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port E Only available on digital display package type GPIO-E[7:0] : General Purpose Input/Output. share with XPDAT[12, 11, 10, 7, 6, 5, 4, 3]	NA	RW

REG[F9h] GPIO-F direction (GPIOFD)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port F GPIO-F_dir[7:0] : General Purpose I/O direction control. 0: Output 1: Input	FFh	RW

REG[FAh] GPIO-F (GPIOF)

Bit	Description	Default	Access
7-0	General Purpose I/O, Port F Only available on digital display package type GPIO-F[7:0] : General Purpose Input/Output. share with XPDAT[23, 22, 21, 20, 19, 15, 14, 13]	NA	RW

19.15 Key-Scan Control Registers

REG[FBh] Key-Scan Control Register 1 (KSCR1)

Bit	Description	Default	Access
7	Reserved Must set as 0.	0	0
6	Long Key Enable Bit 1 : Enable. Long key period is set by KSCR2 bit4-2. 0 : Disable.	0	RW
5-4	Short Key de-bounce Times De-bounce times of keypad scan frequency. 00b : 4 01b : 8 10b : 16 11b : 32	0	RW
3	Repeatable Key enable 0: Disable Repeatable Key 1: Enable Repeatable Key ie, if key is always pressed, then controller will repeat issue key interrupt in every short key de-bounce time (long key disable) or long key recognition time (long key enable) after user clear interrupt flag.	0	RW
2-0	Row Scan Time Period of Key scan controller to scan one row. $T_{KEYCLK} = \frac{1}{F_{SYSCLK}} \times 2048$ 000: ROW_SCAN_Time = T_{KEYCLK} 001: ROW_SCAN_Time = $T_{KEYCLK} \times 2$ 010: ROW_SCAN_Time = $T_{KEYCLK} \times 4$ 011: ROW_SCAN_Time = $T_{KEYCLK} \times 8$ 100: ROW_SCAN_Time = $T_{KEYCLK} \times 16$ 101: ROW_SCAN_Time = $T_{KEYCLK} \times 32$ 110: ROW_SCAN_Time = $T_{KEYCLK} \times 64$ 111: ROW_SCAN_Time = $T_{KEYCLK} \times 128$ This key pad controller supports 5x5 keys. Total Key pad scan time = Row Scan Time * 5	0	RW

REG[FCh] Key-Scan Controller Register 2 (KSCR2)

Bit	Description	Default	Access
7	Key-Scan Wakeup Function Enable Bit 0: Key-Scan Wakeup function is disabled. 1: Key-Scan Wakeup function is enabled.	0	R/W
6	Key released interrupt enable 0: Without interrupt event when all key released 1: Generate an interrupt when all key released	0	RW
5	NA	0	RO
4-2	Long Key Recognition Factor It determines long key recognition time since short key was recognized. Value from 0 to 7.	0	RW

Bit	Description	Default	Access
	<i>LongKey RecognitionTime</i> $= RowScanTime \times 5 \times (LongKey RecognitionFactor + 1) \times 1024$		
1-0	Numbers of Key Hit. 0 : No key is pressed 1 : One key is pressed, REG[FDh] for the key code. 2 : Two keys are pressed, REG[FEh] for the 2 nd key code. 3 : Three keys are pressed, REG[FFh] for the 3 rd key code. It will auto return to 0 if w/o any keys are pressed for a debounce time.	0	RO

REG[FDh] Key-Scan Data Register (KSDR0)

Bit	Description	Default	Access
7-0	Key Strobe Data0 The corresponding key code 0 that is pressed. It will auto return to FFh if w/o any keys are pressed for a debounce time.	TBD	RO

REG[FEh] Key-Scan Data Register (KSDR1)

Bit	Description	Default	Access
7-0	Key Strobe Data1 The corresponding key code 1 that is pressed. It will auto return to FFh if w/o any keys are pressed for a debounce time.	TBD	RO

REG[FFh] Key-Scan Data Register (KSDR2)

Bit	Description	Default	Access
7-0	Key Strobe Data2 The corresponding key code 2 that is pressed. It will auto return to FFh if w/o any keys are pressed for a debounce time.	TBD	RO

Table 19-3 : Key Code Mapping Table (Normal Key)

	Kin0	Kin1	Kin2	Kin3	Kin4
Kout0	00h	01h	02h	03h	04h
Kout1	10h	11h	12h	13h	14h
Kout2	20h	21h	22h	23h	24h
Kout3	30h	31h	32h	33h	34h
Kout4	40h	41h	42h	43h	44h

Table 19-4 : Key Code Mapping Table (Long Key)

	Kin0	Kin1	Kin2	Kin3	Kin4
Kout0	80h	81h	82h	83h	84h
Kout1	90h	91h	92h	93h	94h
Kout2	A0h	A1h	A2h	A3h	A4h
Kout3	B0h	B1h	B2h	B3h	B4h
Kout4	C0h	C1h	C2h	C3h	C4h

20. Summary for GENITOP's Character Supported by RA8876

Table A- 1

● : Supported, — : Not supported

GT21L16T1W supports font	RA8876 Supported Status	Remarks
15X16 dots GB12345 font	●	
15X16 dots BIG5 basic font	●	
15X16 dots JIS0208 basic font	●	The RA8876 can not support the particular fonts which are illustrated in the Table A-2, caused by the designing bug from GENITOP, but this problem could be solved through the software modification when needed.
15X16 dots Unicode font (Japanese)	●	
5X7 dots ASCII font	—	
7X8 dots ASCII font	—	
6X12 dots ASCII font	—	
8X16 dots ASCII font	●	
8X16 dots bold ASCII font	●	
12 dots ASCII font (Arial)	—	
16 dots ASCII font (Arial)	●	
8X16 dots Latin font	●	
8X16 dots Greek font	●	
8X16 dots Cyril font	●	
12 dots Unicode font (Latin)	—	
12 dots Unicode font (Greek)	—	
12 dots Unicode font (Cyril)	—	
16 dots Unicode font (Latin)	●	
16 dots Unicode font (Greek)	●	
16 dots Unicode font (Cyril)	●	
12 dots Arabia font	—	
12 dots Arabia extendable font	—	
16 dots Arabia font	●	
16 dots Arabia extendable font	●	

Table A- 2 : Character code for JIS0208 (RA8876 can not support)

	≤	≥	♂	▽	▼	o	き	ぎ	翹
0135	0169	0170	0173	0206	0207	0379	0413	0414	3344
墮	陳	梯	屆	汎	篋	墨	冀	寫	幕
3436	3636	3680	3847	4038	4247	4347	4935	4948	4949
剗	𠂔	哈	營	塙	幫	憇	揶	斛	哲
4974	5036	5093	5159	5229	5483	5660	5756	5847	5881
桿	淦	箏	紉	繃	閭	霖	騙	熙	
5969	6232	6823	6913	6962	7967	8035	8157	8406	° 8503
	≤	≥	♂	¥					
8565	8569	8570	8573	8579					

Table A- 3

GT30L24M1Z supports font	RA8876 Supported Status	Remarks
24X24 dots GB18030 basic font	•	
12X24 dots GB2312 extension font	•	
12X24 dots ASCII font	•	
24 dots ASCII font (Arial)	•	
24 dots ASCII font (Times New Roman)	•	

Table A- 4

GT30L32S4W supports font	RA8876 Supported Status	Remarks
11X12 dots GB2312 basic font	—	
15X16 dots GB2312 basic font	•	
24X24 dots GB2312 basic font	•	
32X32 dots GB2312 basic font	•	
6X12 dots GB2312 extension font	—	
8X16 dots GB2312 extension font	•	
8X16 dots GB2312 special font	•	
12X24 dots GB2312 extension font	•	
16X32 dots GB2312 extension font	•	
5X7 dots ASCII font	—	
7X8 dots ASCII font	—	
6X12 dots ASCII font	—	
8X16 dots ASCII font	•	
12X24 dots ASCII font	•	
16X32 dots ASCII font	•	
12 dots ASCII font (Arial)	—	
12 dots ASCII font (Times New Roman)	—	

GT30L32S4W supports font	RA8876 Supported Status	Remarks
16 dots ASCII font (Arial)	•	
16 dots ASCII font (Times New Roman)	•	
24 dots ASCII font (Arial)	•	
24 dots ASCII font (Times New Roman)	•	
32 dots ASCII font (Arial)	•	
32 dots ASCII font (Times New Roman)	•	

Table A- 5

GT30L16U2W supports font	RA8876 Supported Status	Remarks
11X12 dots Unicode font	—	
15X16 dots Unicode font	•	
8X16 dots Special font	•	
5X7 dots ASCII font	—	
7X8 dots ASCII font	—	
6X12 dots ASCII font	—	
8X16 dots ASCII font	•	
12 dots ASCII font (Arial)	—	
12 dots ASCII font (Times New Roman)	—	
16 dots ASCII font (Arial)	•	
16 dots ASCII font (Times New Roman)	•	
8X16 dots Latin font	•	
8X16 dots Greek font	•	
8X16 dots Cyril font	•	
12 dots Latin font (Arial)	—	
12 dots Greek font (Arial)	—	
12 dots Cyril font (Arial)	—	
12 dots Arabia font (Arial)	—	
12 dots Arabia extendable font (Arial)	—	
16 dots Latin font (Arial)	•	
16 dots Greek font (Arial)	•	
16 dots Cyril font (Arial)	•	
16 dots Arabia font (Arial)	•	
16 dots Arabia extendable font (Arial)	•	

Table A- 6

GT30L24T3Y supports font	RA8876 Supported Status	Remarks
11X12 dots GB2312 basic font	—	
15X16 dots GB2312 basic font	•	
24X24 dots GB2312 basic font	•	
11X12 dots GB12345 basic font	—	
15X16 dots GB12345 basic font	•	
24X24 dots GB12345 basic font	•	
11X12 dots BIG5 basic font	—	
15X16 dots BIG5 basic font	•	
24X24 dots BIG5 basic font	•	
11X12 dots Unicode font	—	
15X16 dots Unicode font	•	
24X24 dots Unicode font	•	
5X7 dots ASCII font	—	
7X8 dots ASCII font	—	
6X12 dots ASCII font	—	
8X16 dots ASCII font	•	
12 dots ASCII font (Arial)	—	
16 dots ASCII font (Arial)	•	
24 dots ASCII font (Arial)	•	

Table A- 7

GT20L24F6Y supports font	RA8876 Supported Status	Remarks
5X7 dots ASCII font	—	
7X8 dots ASCII font	—	
6X12 dots ASCII font	—	
8X16 dots ASCII font	•	
8X16 dots bold ASCII font	•	
12 dots ASCII font (Arial)	—	
12 dots ASCII font (Times New Roman)	—	
16 dots ASCII font (Arial)	•	
16 dots ASCII font (Times New Roman)	•	
24 dots ASCII font (Arial)	•	
8X16 dots Latin font	•	

GT20L24F6Y supports font	RA8876 Supported Status	Remarks
8X16 dots Greek font	•	
8X16 dots Cyril font	•	
8X16 dots Hebrew font	•	
8X16 dots Thai font	•	
12X24 dots Latin font	•	
12X24 dots Greek font	•	
12X24 dots Cyril font	•	
16 dots Arabia font (Arial)	•	
16 dots Latin font (Arial)	•	
16 dots Greek font (Arial)	•	
16 dots Cyril font (Arial)	•	
12 dots Latin font (Arial)	—	
12 dots Greek font (Arial)	—	
12 dots Cyril font (Arial)	—	
24 dots Arabia font (Arial)	•	
8x16 ISO8859-1	•	
8x16 ISO8859-2	•	
8x16 ISO8859-3	•	
8x16 ISO8859-4	•	
8x16 ISO8859-5	•	
8x16 ISO8859-7	•	
8x16 ISO8859-8	•	
8x16 ISO8859-9	•	
8x16 ISO8859-10	•	
8x16 ISO8859-11	•	
8x16 ISO8859-13	•	
8x16 ISO8859-14	•	
8x16 ISO8859-15	•	
8x16 ISO8859-16	•	
5x7 ISO8859-1	—	
5x7 ISO8859-2	—	
5x7 ISO8859-3	—	
5x7 ISO8859-4	—	
5x7 ISO8859-5	—	
5x7 ISO8859-7	—	

GT20L24F6Y supports font	RA8876 Supported Status	Remarks
5x7 ISO8859-8	—	
5x7 ISO8859-9	—	
5x7 ISO8859-10	—	
5x7 ISO8859-11	—	
5x7 ISO8859-13	—	
5x7 ISO8859-14	—	
5x7 ISO8859-15	—	
5x7 ISO8859-16	—	
5x10 LCM Area 0	—	
5x10 LCM Area 1	—	
5x10 LCM Area 2	—	
5x10 LCM Area 3	—	
5x10 LCM Area 8	—	
5x10 LCM Area 11	—	
5x10 LCM Area 12	—	
5x10 LCM Area 13	—	

Table A- 8

GT21L24S1W supports font	RA8876 Supported Status	Remarks
24X24 dots GB2312 basic font	•	
12X24 dots GB2312 extension font	•	
12X24 dots ASCII font	•	
24 dots ASCII font (Arial)	•	

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